Gauge-Covariant Variational Free Energy
Supplement G

The Multivariate-Gaussian Realization

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Chapter G0

Introduction

This supplement asks what the single-point theory of the main manuscript becomes when every belief and model law is multivariate Gaussian. It cites the main manuscript and adopts its standing conventions (Section 1.4), and it states no result that the main manuscript uses. The belief fiber has dimension K and the model fiber dimension d_m, which are not assumed equal. The represented frame groups are $\Gamma_b = \rho_b(G)$ and $\Gamma_m = \rho_m(G)$, with the specialization $\Gamma_b = \mathrm{GL}^+(K)$ named wherever it is used (Hypothesis G2.1); both channels of an agent use one common frame, the gauge choice of the main manuscript. The belief-sector precision studied throughout is that of $p(k \mid m, o, X)$, the fast belief at frozen model latents (Section G2.1); its membership in the interaction family requires $A_i + O_i \succeq 0$ at every agent, so the balance condition on the prior alone is sufficient but not necessary (Proposition G3.10).

The Gaussian supplies explicit determinant gaps, Fisher charts, matrix-weighted interaction operators, aggregation formulas, Kron counterexamples, and generalized spectra. The Laplacian-plus-self-term interaction family is adopted as Hypothesis G3.13; it does not follow from a general linear-Gaussian directed model, since an orientation-reversing gain, and for odd K a v-structure, leave it in every chart reached by orientation-preserving frames (Proposition G3.9). Within the family, hard aggregation is closed (Proposition G7.3) and is the fine law conditioned on zero within-cluster disagreement (Proposition G7.4), although closure selects neither the family nor a nondegenerate partition. Unrestricted matrix-weighted Kron reduction leaves the family (Proposition G7.7); the fixed congruence-diagonal cone is closed (Theorem G7.10), the maximal pair-tested families at a fixed hidden anchor are the full anchor-congruence-diagonal cones (Theorem G7.12), and on a fixed flat graph with independent positive-definite weights and $K \geq 2$, closure for every weight assignment holds exactly when every hidden component has at most two retained terminals (Theorem G7.15).

Coarse families in this supplement admit newly generated positive-semidefinite self terms. The strict family in which cutting a pair creates no new anchor is a named subfamily, Definition 11.23 of the main manuscript, and the completed-square criterion that separates the two is Lemma 11.24.

Operator coarse-graining is narrower than law-level coarse-graining. Aggregation is a positive semigroup on the interaction parameters (Proposition G9.2), so a rescaling must be declared before any flow is typed, and on the interior of the cone the rescaled step is a map of normalized laws (Proposition G9.3). Bi-additive parameters are closed under additive pushforward (Section G9.2), and under a measurability hypothesis positive additive rules are forced to be linear (Proposition G9.5). Fixed rays, their attraction, and independence of the blocking scheme need a hierarchical identification

and a limiting protocol; they belong to Supplement B, and this supplement states no fixed point of a scale map.

Chapter G1 develops exponential families, their operator layer, and the information projection. Chapter G2 realizes the directed law in information form and proves the Schur-complement lemma on which the later chapters rest. Chapter G3 characterizes when the belief precision lies in the interaction family, and Chapter G4 computes the costs of recognition restrictions. Chapter G5 and Chapter G6 treat Fisher geometry, invariants, and the Hermite spectrum of score tangents. Chapter G7 and Chapter G8 treat aggregation, Kron closure, nonflat links, relaxation, and quotient volume; Chapter G9 treats the operator arrow; and Chapter G10 treats reciprocal pairs, anchored stars, and loop holonomy. Chapter G11 realizes the latent Lie-algebra links of Chapter 9 in the Gaussian setting: the law-level divergence in generator coordinates, the delta-rule current of a comparison record and the attainment of its infimum over the exponential chart, expected loop defects under Gaussian recognition of the generators, and the action that undirected Gaussian matter induces on the generators, with its hopping expansion over weighted Wilson loops, the joint normalization of unanchored matter, and the linearized link operator on a ring. Appendix GA holds the decision metatheory for semialgebraic families and the treatment of singular Gaussian laws, and Appendix GB holds the notation, open problems, and numerical index of this supplement.

This supplement asserts nothing for a non-Gaussian belief fiber beyond the exponential-family statements of Chapter G1, and it states no fixed point, attraction, or asymptotic scaling result for a scale map. [NOT-CLAIMED]

Chapter G1

Exponential Families and the Operator Layer

The belief and model law fibers of the main manuscript are collections of normalized laws (Chapter 2, (2.3)), acted on by pushforward ((2.5)); coordinates, densities, and convex parameter sets are further choices. This chapter asks what a finite-dimensional exponential family adds to a law fiber and what happens when an algebraic map of its natural parameters leaves the natural domain. The answer is that a natural-domain boundary point carries no law (Proposition G1.7) and that a fixed point of a map on an operator cone need not be a law (Proposition G1.8), so that pinning, quotienting, and retaining a mass coordinate are separate constructions (Section G1.3); inside the domain, relative entropy is a Bregman divergence and affine mean constraints have a unique information projection with a Pythagorean identity (Proposition G1.16). The Gaussian realization of Chapter G2 is the case used throughout this supplement, the Fisher charts of Chapter G5 are its dual coordinates, and the scale-free boundary operator of Chapter G9 is an instance of Proposition G1.8.

This chapter derives no coarse-graining statement, and no Fisher or evidence reading of one, from exponential-family structure alone: an operator arrow needs a preserved parameter set, scale comparisons, and a declared rescaling, while a law-level arrow needs normalized laws, compatible reference measures, and finite normalizers at every scale. The boundary results below do not exclude a quotient, a pin, a new reference measure, or a boundary stratum; each is a different object that must be constructed. [NOT-CLAIMED] Two questions remain open: the scaling of a retained mass coordinate (Open Problem G1.14) and the commutation of the information projection with gauge actions and coarse parameter maps (Open Problem G1.18).

G1.1 Regular finite-dimensional exponential families

Definition G1.1 (Canonical family). Let ν be a nonzero sigma-finite measure on a standard Borel space $(\mathsf{Z}, \mathscr{Z})$, and let $\mathsf{T} : \mathsf{Z} \to \mathsf{V}$ be a measurable statistic into a finite-dimensional inner-product space V. The log partition and the natural domain are

$$\mathsf{A}(\lambda) = \log \int_{\mathsf{Z}} e^{\langle \lambda, \mathsf{T}(z) \rangle} \nu(dz), \qquad \mathsf{N} = \{\lambda \in \mathsf{V} : \mathsf{A}(\lambda) < \infty\}, \tag{G1.1}$$

and for $\lambda \in \mathsf{N}$ the family member is

$$q_\lambda(z) = \exp\{\langle \lambda, \mathsf{T}(z)\rangle - \mathsf{A}(\lambda)\}, \qquad Q_\lambda(dz) = q_\lambda(z)\nu(dz). \tag{G1.2}$$

The vector λ is the natural parameter. No law is attached to a parameter outside N.

Hypothesis G1.2 (Regularity and minimality). Where a result names this hypothesis, the family is regular, so that N is open, and minimal, so that no nonzero $a \in \mathsf{V}$ makes $\langle a, \mathsf{T}\rangle$ constant ν-almost everywhere. Regularity excludes natural-domain boundary points from the parameter set; minimality excludes affine redundancy of the statistic.

Proposition G1.3 (Log partition, cumulants, and Fisher information). Assume $\mathsf{N} \neq \varnothing$. (i) The extended-real function A is convex and lower semicontinuous on V, and N is convex. (ii) On $\operatorname{int}\mathsf{N}$ the function A is infinitely differentiable, its derivatives are computed by differentiating under the integral sign, and T has moments of every order, with

$$\nabla\mathsf{A}(\lambda) = \mathbb{E}_\lambda[\mathsf{T}], \qquad \nabla^2\mathsf{A}(\lambda) = \operatorname{Cov}_\lambda(\mathsf{T}). \tag{G1.3}$$

Since the score of the family is $\mathsf{T} - \nabla\mathsf{A}(\lambda)$, the Hessian $\nabla^2\mathsf{A}(\lambda)$ is the Fisher information in the natural chart. (iii) If the family is minimal, then A is strictly convex on N, $\nabla^2\mathsf{A}(\lambda)$ is positive definite at every interior parameter, and $\lambda \mapsto Q_\lambda$ is injective on N [1, Theorems 7.1, 7.2 and 8.1 and Section 8.1]. [STANDARD]

Proof. The cited results concern a sigma-finite measure on a Euclidean space; they apply to the image measure $\mathsf{T}_{\#}\nu$ on V, because A is the logarithm of its Laplace transform and, for any $\lambda_0 \in \mathsf{N}$, it has the positive density $e^{-\langle\lambda_0,t\rangle}$ with respect to the finite measure $e^{\langle\lambda_0,t\rangle}\mathsf{T}_{\#}\nu(dt)$. Minimality is the statement that $\mathsf{T}_{\#}\nu$ is not concentrated on an affine hyperplane, and positive definiteness in (iii) then follows from $a^\top\nabla^2\mathsf{A}(\lambda)a = \operatorname{Var}_\lambda\langle a, \mathsf{T}\rangle$. The domination behind (ii) is the elementary bound

$$|x|^r \le \mathrm{c}_{r,\varepsilon}\left(e^{\varepsilon x} + e^{-\varepsilon x}\right), \qquad r \in \{0,1,2\},\ x \in \mathbb{R}, \tag{G1.4}$$

applied to each coordinate of T at the vertices of a cube of side 4ε inside N around an interior parameter. □

Proposition G1.4 (Finite-domain interface to evidence-bound coordinates). The map $(\lambda, z) \mapsto q_\lambda(z)$ is jointly measurable on $\mathsf{N} \times \mathsf{Z}$, and every law, density ratio, Fisher tensor, or evidence-bound coordinate built from the canonical representation is defined only for parameters in N. If two canonical representations share one sigma-finite base measure ν, their density ratio is defined at every pair of parameters taken each from its own natural domain. Consequently an algebraic map that leaves N ceases to define a probability law or an evidence-bound coordinate, even when it remains in an operator cone. [ESTABLISHED]

Proof. Lower semicontinuity of A (Proposition G1.3(i)) makes it Borel measurable, and measurability of T makes $(\lambda, z) \mapsto \langle\lambda, \mathsf{T}(z)\rangle$ jointly measurable; exponentiation proves the first claim. The

normalizer is finite exactly on N by (G1.1). With a common base measure, both densities in a ratio are strictly positive ν-densities of the form (G1.2), so the ratio is defined at every pair of parameters in the respective domains; representations with different statistics have domains in different parameter spaces, so no intersection of domains is formed. The last statement is a domain check against Definition G1.5 and the extended evidence bound of Definition 5.6, which needs a normalized posterior slice of finite positive mass. □

An evidence-bound update additionally needs a positive finite evidence slice at every compared parameter, and its classical split into two expected log densities needs clauses (H3) and (H4) of Hypothesis 5.3.

G1.2 Operator and probability layers

Definition G1.5 (Partial realization). A parametrized model consists of a parameter set P, a law domain $\mathsf{P}_{\mathrm{law}} \subseteq \mathsf{P}$, and a map

$$\mathfrak{r} : \mathsf{P}_{\mathrm{law}} \longrightarrow \mathcal{B}(\mathsf{Z}), \qquad \mathsf{p} \longmapsto Q_{\mathsf{p}}, \tag{G1.5}$$

into a collection $\mathcal{B}(\mathsf{Z})$ of normalized laws on Z. A parameter outside $\mathsf{P}_{\mathrm{law}}$ may still denote an algebraic operator, but it denotes no law through $\mathfrak{r}$.

Definition G1.6 (Operator layer). Let $\mathcal{W} \subseteq \mathsf{V}$ be a closed convex cone preserved by the algebraic maps under study. Its probability layer is $\mathcal{W} \cap \mathsf{N}$. The cone is chosen for algebraic closure, not for normalization, and a general law fiber carries no such cone.

Proposition G1.7 (A natural-domain boundary carries no law). Assume the family is regular and let $\lambda^\star \in \partial\mathsf{N}$. Then

$$\int e^{\langle \lambda^\star, \mathsf{T}(z)\rangle} \nu(dz) = +\infty. \tag{G1.6}$$

For $\lambda_0 \in \mathsf{N}$, $w = \lambda^\star - \lambda_0$, and $\lambda_t = (1-t)\lambda_0 + t\lambda^\star$ with $t \uparrow 1$,

$$\mathsf{A}(\lambda_t) \to +\infty, \qquad \langle w, \nabla\mathsf{A}(\lambda_t)\rangle \to +\infty, \qquad \int_0^1 \mathrm{Var}_{Q_{\lambda_t}}(\langle w, \mathsf{T}\rangle)dt = +\infty. \tag{G1.7}$$

Hence the representation supplies at $\lambda^\star$ no finite log partition, mean parameter, Fisher metric, relative entropy from an endpoint law, or evidence bound. [ESTABLISHED]

Proof. Regularity puts $\lambda^\star$ outside the open set N, which is (G1.6). Convexity of N (Proposition G1.3(i)) puts every λ_t, $t < 1$, in N, and lower semicontinuity forces the first limit. For $g(t) = \mathsf{A}(\lambda_t)$, convexity makes g' nondecreasing, and if g' were bounded above then g would remain bounded, a contradiction. The identities $g' = \langle w, \nabla\mathsf{A}(\lambda_t)\rangle$ and $g'' = \mathrm{Var}_{Q_{\lambda_t}}(\langle w, \mathsf{T}\rangle)$ of (G1.3), and integration of g'', prove the remaining limits. The last sentence restates that each listed object is built from $\mathsf{A}(\lambda^\star)$ or from $Q_{\lambda^\star}$, neither of which exists. □

Proposition G1.8 (An operator fixed point need not be a law). There is a regular canonical family with an operator cone $\mathcal{W}$ and a continuous map $\psi : \mathcal{W} \to \mathcal{W}$ whose iterates from a point of $\mathcal{W} \cap \mathsf{N}$ all lie in $\mathcal{W} \cap \mathsf{N}$, while its unique fixed point lies on $\partial\mathsf{N}$. One such family is $\mathsf{Z} = \mathbb{R}$, $\mathsf{T}(z) = -z^2/2$, Lebesgue ν, for which

$$\mathsf{A}(\lambda) = -\tfrac{1}{2}\log\lambda + \tfrac{1}{2}\log 2\pi, \qquad \mathsf{N} = (0,\infty), \qquad \mathcal{W} = [0,\infty), \tag{G1.8}$$

with $\psi(\lambda) = \lambda/2$. [ESTABLISHED]

Proof. Starting at one gives $\lambda_\ell = 2^{-\ell} > 0$, so every iterate is a law, while $\lambda_\ell \to 0$, the unique fixed point, and $\mathsf{A}(\lambda_\ell) \to +\infty$. The constant integrand at $\lambda = 0$ has infinite Lebesgue integral. □

G1.3 Repairs of the boundary

Definition G1.9 (Pinning). Choose $\lambda_{\mathrm{m}} \in \mathsf{N}$ and replace $\mathcal{W}$ by $\lambda_{\mathrm{m}} + \mathcal{W}$; the pin does not flow.

Proposition G1.10 (Pinning under a recession condition). Write $\operatorname{rec}\mathsf{N} = \{\delta \in \mathsf{V} : \mathsf{N} + \delta \subseteq \mathsf{N}\}$ for the recession cone of the convex, possibly nonclosed set N [26, Theorem 8.1]. If $\lambda_{\mathrm{m}} \in \operatorname{int}\mathsf{N}$ and $\mathcal{W} \subseteq \operatorname{rec}\mathsf{N}$, then $\lambda_{\mathrm{m}} + \mathcal{W} \subseteq \operatorname{int}\mathsf{N}$. [ESTABLISHED]

Proof. Let $\delta \in \mathcal{W}$ and let $\mathcal{O} \subseteq \mathsf{N}$ be an open ball around λ_{m}. Since $\delta \in \operatorname{rec}\mathsf{N}$, the translated ball $\mathcal{O} + \delta$ lies in N, so its center $\lambda_{\mathrm{m}} + \delta$ is interior. □

A pin changes the family and adds a parameter that a map on the cone does not move; removing it afterwards is a separate limit.

Definition G1.11 (Quotienting). Suppose $\mathsf{Z} = \mathbb{R}^n$, ν is Lebesgue measure, and the energy is invariant along a nonzero linear subspace $\mathcal{V}$. The quotient takes the sample space $\mathcal{V}^\perp$, its Lebesgue measure, and the restricted statistic.

Proposition G1.12 (Quotient normalization is not full-space normalization). Writing $z = v + w$ with $v \in \mathcal{V}$ and $w \in \mathcal{V}^\perp$,

$$\int_{\mathbb{R}^n} e^{\langle\lambda,\mathsf{T}(z)\rangle}\,dz = \operatorname{Leb}_{\mathcal{V}}(\mathcal{V}) \int_{\mathcal{V}^\perp} e^{\langle\lambda,\mathsf{T}(w)\rangle}\,dw = +\infty. \tag{G1.9}$$

The quotient law is normalized exactly when the second factor is finite. In particular, a Moore–Penrose inverse and a pseudodeterminant inserted into the full-space density formula compute the finite quotient expression, whose dimension and reference measure differ, and leave the full-space integral infinite. [ESTABLISHED]

Proof. The orthogonal change of variables has unit Jacobian and the integrand does not depend on v, so Tonelli's theorem gives (G1.9), whose first factor is infinite. A pseudoinverse and a pseudodeterminant evaluate the Gaussian integral on $\mathcal{V}^\perp$ only, which is the second factor. □

Definition G1.13 (Retaining a mass sector). Enlarge the state by a coercive parameter that remains in the natural domain under the scale map. This is neither pinning nor quotienting: the mass is a running coordinate, and its scaling law is part of the declared scale map.

Open Problem G1.14 (Scaling of a retained mass sector). Pinning keeps an orbit interior under the recession condition of Proposition G1.10, and quotienting normalizes exactly when the quotient integral of (G1.9) is finite. For a retained mass coordinate (Definition G1.13) nothing here fixes its law under a scale map, and so nothing decides whether its iterates stay in N or approach ∂N. The question is settled for a given scale map by an explicit induced law on the mass coordinate together with a proof that its orbit stays in N, or reaches the boundary, under stated hypotheses on the map.

G1.4 Relative entropy and affine information projection

For a regular minimal family write $\boldsymbol{\mu}_\lambda = \nabla \mathsf{A}(\lambda)$ for the mean parameter and A^* for the convex conjugate of A. For a convex function ϕ differentiable at y, let $D_\phi(x,y) = \phi(x) - \phi(y) - \langle \nabla\phi(y), x - y\rangle$ be its Bregman divergence [8].

Proposition G1.15 (Bregman form of relative entropy). Under Hypothesis G1.2, for $\lambda, \lambda' \in \mathsf{N}$,

$$D_{\mathrm{KL}}(Q_{\lambda'} \| Q_\lambda) = \mathsf{A}(\lambda) - \mathsf{A}(\lambda') - \langle \nabla\mathsf{A}(\lambda'), \lambda - \lambda'\rangle = D_{\mathsf{A}}(\lambda, \lambda'), \tag{G1.10}$$

and

$$D_{\mathrm{KL}}(Q_{\lambda'} \| Q_\lambda) = D_{\mathsf{A}^*}(\boldsymbol{\mu}_{\lambda'}, \boldsymbol{\mu}_\lambda). \tag{G1.11}$$

This is the exact divergence of the family; it is not in general a finite-displacement Fisher quadratic. [ESTABLISHED]

Proof. The log density ratio is $\log(q_{\lambda'}/q_\lambda) = \langle \lambda' - \lambda, \mathsf{T}\rangle - \mathsf{A}(\lambda') + \mathsf{A}(\lambda)$, and its expectation under $Q_{\lambda'}$, with $\mathbb{E}_{\lambda'}\mathsf{T} = \nabla\mathsf{A}(\lambda')$ from (G1.3), is (G1.10); this is Brown [9, Proposition 6.3]. The function A is closed by Proposition G1.3(i), has the open domain N, is differentiable there, and is strictly convex by Proposition G1.3(iii). The subgradients of A^* at $\boldsymbol{\mu}_\lambda$ are therefore exactly the solutions of $\nabla\mathsf{A}(\lambda'') = \boldsymbol{\mu}_\lambda$, namely λ alone, so A^* is differentiable there with $\nabla\mathsf{A}^*(\boldsymbol{\mu}_\lambda) = \lambda$ and $\mathsf{A}^*(\boldsymbol{\mu}_\lambda) = \langle \lambda, \boldsymbol{\mu}_\lambda\rangle - \mathsf{A}(\lambda)$ [26, Theorems 23.5 and 25.1]. Substituting these identities at λ and λ' into the definition of D_{A^*} gives $D_{\mathsf{A}}(\lambda, \lambda')$, which is (G1.11). □

Proposition G1.16 (Affine mean constraints attain a unique projection). Assume the extended A is Legendre, as in a regular minimal canonical family, and let $\mathcal{M} = \operatorname{int}\operatorname{dom}\mathsf{A}^*$. If $\mathscr{A}$ is a closed affine subspace with $\mathscr{A} \cap \mathcal{M} \neq \varnothing$, then for every $\boldsymbol{\mu}_\lambda \in \mathcal{M}$ there is a unique $\widehat{\boldsymbol{\mu}} \in \mathscr{A} \cap \mathcal{M}$ minimizing

$$\boldsymbol{\mu} \longmapsto D_{\mathsf{A}^*}(\boldsymbol{\mu}, \boldsymbol{\mu}_\lambda). \tag{G1.12}$$

Its natural parameter is $\widehat{\lambda} = \nabla \mathsf{A}^*(\widehat{\boldsymbol{\mu}})$, and every natural parameter λ' with $\boldsymbol{\mu}_{\lambda'} \in \mathscr{A} \cap \mathcal{M}$ satisfies the Pythagorean identity

$$D_{\mathrm{KL}}(Q_{\lambda'} \| Q_\lambda) = D_{\mathrm{KL}}(Q_{\lambda'} \| Q_{\widehat{\lambda}}) + D_{\mathrm{KL}}(Q_{\widehat{\lambda}} \| Q_\lambda), \tag{G1.13}$$

so $Q_{\widehat{\lambda}}$ is the unique minimizer of $D_{\mathrm{KL}}(\cdot \| Q_\lambda)$ over the members of the family whose mean parameters lie in $\mathscr{A}$. [ESTABLISHED]

Proof. The objective in (G1.12) is the Bregman divergence of the Legendre function A^* from the interior point $\boldsymbol{\mu}_\lambda$. For such a function and a closed convex feasible set meeting the interior of its domain, the minimizer exists, is unique, and lies in that interior [4, Theorem 3.12(iii)]. Write $\mathscr{A} = \{\boldsymbol{\mu} : \mathsf{D}\boldsymbol{\mu} = \mathsf{b}\}$. The objective is differentiable at the interior minimizer with gradient $\nabla \mathsf{A}^*(\widehat{\boldsymbol{\mu}}) - \lambda$, which must annihilate the direction space $\ker \mathsf{D}$; conversely, by convexity, a feasible interior point at which the gradient annihilates $\ker \mathsf{D}$ is a minimizer. Since $(\ker \mathsf{D})^\perp = \operatorname{im} \mathsf{D}^\top$, the solution is characterized by

$$\mathsf{D}\widehat{\boldsymbol{\mu}} = \mathsf{b}, \qquad \nabla \mathsf{A}^*(\widehat{\boldsymbol{\mu}}) - \lambda + \mathsf{D}^\top \omega = 0 \tag{G1.14}$$

for some multiplier ω, which proves the natural-parameter formula; Bauschke [3] gives the dual characterization of Bregman projections onto affine subspaces. For the Pythagorean identity, write $\boldsymbol{\mu}' = \boldsymbol{\mu}_{\lambda'}$ and expand the three divergences by (G1.10); the values of A cancel and

$$D_{\mathrm{KL}}(Q_{\lambda'} \| Q_\lambda) - D_{\mathrm{KL}}(Q_{\lambda'} \| Q_{\widehat{\lambda}}) - D_{\mathrm{KL}}(Q_{\widehat{\lambda}} \| Q_\lambda) = \langle \widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}', \lambda - \widehat{\lambda} \rangle.$$

By (G1.14), $\lambda - \widehat{\lambda} = \mathsf{D}^\top \omega$, while feasibility gives $\mathsf{D}(\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}') = 0$, so the pairing vanishes. The middle divergence is nonnegative and vanishes only when $Q_{\lambda'} = Q_{\widehat{\lambda}}$, which gives the minimizing property, and injectivity of $\lambda \mapsto Q_\lambda$ (Proposition G1.3(iii)) gives uniqueness. For a general closed convex feasible set only the inequality form of (G1.13) holds [4, Proposition 3.16]. □

Remark G1.17 (Feasibility, closedness, and convexity are needed). Take the Bernoulli family with mean domain $(0, 1)$ and target mean $\frac{1}{2}$. The feasible set $\{0\}$ misses the mean domain, so no projection exists; on the open feasible interval $(0, \frac{1}{4})$ the divergence is strictly decreasing and its infimum, attained only at the excluded endpoint $\frac{1}{4}$, is finite and not attained; and the closed nonconvex set $\{\frac{1}{4}, \frac{3}{4}\}$ has two minimizers, because $D_{\mathrm{KL}}(\mathrm{Bern}(\mu) \| \mathrm{Bern}(\frac{1}{2}))$ is symmetric under $\mu \mapsto 1 - \mu$. [ESTABLISHED]

Proof. The divergence $\mu \log(2\mu) + (1 - \mu) \log(2(1 - \mu))$ has derivative $\log(\mu/(1 - \mu)) < 0$ for $\mu < \frac{1}{2}$, which gives the interval claim, and the symmetry is visible in the formula. □

Open Problem G1.18 (Commutation of the information projection). Proposition G1.16 proves existence, uniqueness, and the Pythagorean identity of the projection onto an affine mean constraint. It is not proved whether the projection is equivariant under a gauge action on the natural parameters, or whether it commutes with a coarse parameter map. The question is settled by an intertwining theorem that states hypotheses on the action or the map under which the two projections correspond, or by a counterexample within a regular minimal family.

Chapter G2

Gaussian Realization at C_*

This chapter asks what the directed law of Chapter 4 becomes when every kernel is linear Gaussian, and what the blocks of the resulting precision mean. The central tool is one Schur-complement lemma (Lemma G2.6): the marginal precision of a set of directions is a Schur complement, dominated in the Loewner order by the precision restricted to those directions, with equality exactly under Λ-orthogonality. With it the chapter computes the assembled precision of a directed forest (Proposition G2.16), proves frame covariance of every kernel under the common frame (Proposition G2.17), and states the Gaussian forms of the M-coordinate, the comparison lift, and the linear record of selective communication. The associated-bundle morphism $\widetilde{\Phi}$ of Chapter 2 is realized locally by the law map (G2.37) of a root kernel, of which the bridge matrix B_i is one parameter. The interaction family of Chapter G3, the recognition costs of Chapter G4, the Fisher charts of Chapter G5, and the coarse maps of Chapter G7 all start here.

The Gaussian is a tractable realization, not the ambient theory, and no matrix identity of this chapter is asserted for a non-Gaussian belief fiber. The common frame ties the gauge choices of the two channels but does not identify the dimensions K and d_m, make B_i invertible, turn Φ and $\widetilde{\Phi}$ into inverses, or add a reverse generative factor. [NOT-CLAIMED]

G2.1 Frames and sectors

Hypothesis G2.1 (Represented frame group of the Gaussian realization). Write $\Gamma_b := \rho_b(G) \subseteq \mathrm{GL}(K)$ and $\Gamma_m := \rho_m(G) \subseteq \mathrm{GL}(d_m)$ for the represented frame groups of the two channels. Where a result names this hypothesis, it adopts the specialization $\Gamma_b = \mathrm{GL}^+(K)$, realized for example by $G = \mathrm{GL}^+(K)$ with ρ_b its defining representation. Identities that use only the frame-change laws, such as the kernel covariance of Proposition G2.17 and the congruence covariance of precisions and pencils, hold for every Γ_b. Three families of statements use the specialization: the gain-absorption and orientation statements of Section G3.2, which need exactly the orientation-preserving frames; the frame dependence of ordinary eigenvalues in Section G5.3, which needs $\Gamma_b \supseteq \mathbb{R}_{>0} I_K$; and the rank-only classification of Proposition G5.10, which needs $\Gamma_b \supseteq \mathrm{GL}^+(K)$. The hypothesis excludes orientation-reversing represented frames and compact structure groups, whose represented frames lie in $\mathrm{O}(K)$ after a choice of invariant inner product.

Proposition G2.2 (Compact represented frames). Let $\Gamma_b \subseteq \mathrm{O}(K)$. A block reframing $\mathbf{P} = \mathrm{diag}(\mathbf{P}_1, \dots, \mathbf{P}_N)$ with $\mathbf{P}_i \in \Gamma_b$ acts on a precision by the orthogonal similarity $\mathbf{P}^{-\top} \Lambda \mathbf{P}^{-1} = \mathbf{P} \Lambda \mathbf{P}^{\top}$,

so ordinary spectra are frame invariant, as are log determinants of Λ restricted to a subspace and computed in orthonormal bases. For $M \succ 0$ and $\mathbf{P}_0 \in \Gamma_b$ the congruence $\mathbf{P}_0^\top M \mathbf{P}_0$ is an orthogonal similarity, so the eigenvalues of M are invariants; when Γ_b is $\mathrm{SO}(K)$ or $\mathrm{O}(K)$, the orbit of M is exactly its orthogonal-similarity class, labeled by its K eigenvalues. [ESTABLISHED]

Proof. For orthogonal $\mathbf{P}$, $\mathbf{P}^{-\top} = \mathbf{P}$. A restricted log determinant in an orthonormal basis E becomes one in the orthonormal basis $\mathbf{P}E$, because $(\mathbf{P}E)^\top(\mathbf{P}\Lambda\mathbf{P}^\top)(\mathbf{P}E) = E^\top \Lambda E$. A spectral decomposition $M = ODO^\top$ with $O \in \mathrm{O}(K)$ can be taken with $\det O = 1$ by reversing one eigenvector, so $\mathrm{SO}(K)$ already reaches every orthogonal similarity. □

G2.1.1 The common frame

Throughout this supplement the belief and model channels of agent i use one local frame u_i, the gauge choice of the main manuscript (Section 2.2.2); one passive re-expression $g_i \in G$ then acts on samples by

$$k_i \longmapsto \rho_b(g_i) k_i, \qquad m_i \longmapsto \rho_m(g_i) m_i. \tag{G2.1}$$

The two channels may still carry inequivalent representations and independent link families.

The belief-sector precision analyzed in Chapters G3 and G4 is that of the belief conditional $p(k \mid m, o, X)$, the fast belief at frozen model latents in the sense of Section 8.2: bridge terms, offsets, and the model part of each observation mean shift only the information vector at fixed m, as Proposition G3.10 computes. The model sector $p(m \mid k, o, X)$ has the same structure with K replaced by d_m.

G2.2 Information form

Let ν be Lebesgue measure on $\mathbb{R}^n$. The Gaussian family in information form has the information vector $\mathsf{h} \in \mathbb{R}^n$ and the precision $\Lambda \in \mathbb{S}^n_{++}$, with density

$$q(y \mid \mathsf{h}, \Lambda) = \exp\left(\mathsf{h}^\top y - \tfrac{1}{2} y^\top \Lambda y - \mathsf{A}(\mathsf{h}, \Lambda)\right), \qquad y \in \mathbb{R}^n. \tag{G2.2}$$

The sans-serif A is the log normalizer; the self terms of Chapter G3 are the $K \times K$ matrices A_i.

G2.2.1 Coordinate charts

Definition G2.3 (Gaussian coordinate charts). Pair two symmetric matrices by the trace form $\langle M, M' \rangle = \mathrm{Tr}(M^\top M')$ and use the chart of $\mathbb{S}^n$ whose coordinates are the diagonal entries and the off-diagonal entries scaled by $\sqrt{2}$, in which the Euclidean inner product of coordinate vectors is this pairing. With the statistic $\mathsf{T}(y) = (y, yy^\top)$, the natural parameter is

$$\lambda = (\lambda_1, \lambda_2) = \left(\mathsf{h}, -\tfrac{1}{2}\Lambda\right), \tag{G2.3}$$

the information-form chart is (h, Λ), the expectation chart is $\boldsymbol{\mu} = (\mu, M_2)$ with $M_2 = \mathbb{E}_q[YY^\top]$, and the moment chart is (μ, Σ). The moment chart is not Legendre dual to the natural chart, because

$\Sigma = M_2 - \mu\mu^\top$. With the unscaled half-vectorization every off-diagonal derivative would acquire a factor two; all matrix blocks in this supplement are written in the scaled chart.

Proposition G2.4 (Log normalizer, moments, and dual charts). (i) For every $(\mathsf{h}, \Lambda) \in \mathbb{R}^n \times \mathbb{S}^n_{++}$ the density (G2.2) is normalizable, with

$$\mathsf{A}(\mathsf{h}, \Lambda) = \tfrac{1}{2}\mathsf{h}^\top \Lambda^{-1}\mathsf{h} - \tfrac{1}{2}\log\det\Lambda + \tfrac{n}{2}\log 2\pi, \tag{G2.4}$$

and

$$d\mathsf{A} = \mu^\top d\mathsf{h} - \tfrac{1}{2}\operatorname{Tr}\left[\left(\Sigma + \mu\mu^\top\right) d\Lambda\right], \qquad \mu = \Lambda^{-1}\mathsf{h}, \qquad \Sigma = \Lambda^{-1}, \tag{G2.5}$$

so $q(\cdot \mid \mathsf{h}, \Lambda)$ has mean μ and covariance Σ; the Hessian of A in h at fixed Λ, which is the Fisher information in h, is Σ, and the Fisher information in μ at fixed Σ is Λ. (ii) In the natural chart the family is the regular minimal canonical family

$$q(y) = \exp\left(\langle \lambda, \mathsf{T}(y)\rangle - \mathsf{A}(\lambda)\right), \qquad \lambda \in \mathbb{R}^n \times \{\lambda_2 \prec 0\}, \tag{G2.6}$$

with

$$\mathsf{A}(\lambda) = -\tfrac{1}{4}\lambda_1^\top \lambda_2^{-1}\lambda_1 - \tfrac{1}{2}\log\det(-2\lambda_2) + \tfrac{n}{2}\log 2\pi, \tag{G2.7}$$

strictly convex on its open convex natural domain, and with mean parameter

$$\boldsymbol{\mu} = \nabla_\lambda \mathsf{A} = (\mu, M_2), \qquad M_2 = \Sigma + \mu\mu^\top. \tag{G2.8}$$

(iii) The Fisher metric in the natural chart is

$$g_\lambda = \nabla^2_\lambda \mathsf{A}(\lambda) = \operatorname{Cov}_q\left(\mathsf{T}(Y)\right), \tag{G2.9}$$

and in the expectation chart it is

$$g_{\boldsymbol{\mu}} = \left(\frac{\partial\lambda}{\partial\boldsymbol{\mu}}\right)^\top g_\lambda \left(\frac{\partial\lambda}{\partial\boldsymbol{\mu}}\right) = g_\lambda^{-1}, \tag{G2.10}$$

the Hessian of A^*, whose value at $\boldsymbol{\mu}$ is the negative differential entropy of q. [ESTABLISHED]

Proof. (i) Completing the square,

$$\mathsf{h}^\top y - \tfrac{1}{2}y^\top \Lambda y = -\tfrac{1}{2}\left(y - \Lambda^{-1}\mathsf{h}\right)^\top \Lambda \left(y - \Lambda^{-1}\mathsf{h}\right) + \tfrac{1}{2}\mathsf{h}^\top\Lambda^{-1}\mathsf{h}, \tag{G2.11}$$

and the Gaussian integral $(2\pi)^{n/2}(\det\Lambda)^{-1/2}$ give (G2.4); differentiating termwise with $d\Lambda^{-1} = -\Lambda^{-1}d\Lambda\Lambda^{-1}$ and $d\log\det\Lambda = \operatorname{Tr}(\Lambda^{-1}d\Lambda)$ gives (G2.5). At fixed Λ the family is canonical in h with statistic y, so its Fisher information is the covariance Σ; at fixed Σ the score in μ is $\Lambda(y-\mu)$, with covariance $\Lambda\Sigma\Lambda = \Lambda$. (ii) In the scaled chart $\langle\lambda, \mathsf{T}(y)\rangle = \mathsf{h}^\top y - \frac{1}{2}y^\top\Lambda y$, and substituting $\mathsf{h} = \lambda_1$, $\Lambda = -2\lambda_2$ into (G2.4) gives (G2.7). The natural domain is open, and the family is minimal, because a nonzero affine function of $(y, yy^\top)$ is a nonzero polynomial of degree at most two and vanishes only on a Lebesgue-null set. Proposition G1.3 then gives strict convexity, $\nabla_\lambda\mathsf{A} = \mathbb{E}\mathsf{T}$,

which is (G2.8), and (G2.9). Since $\boldsymbol{\mu} = \nabla \mathsf{A}(\lambda)$, the Jacobian $\partial\boldsymbol{\mu}/\partial\lambda$ is g_λ, which is positive definite, so the chain rule gives (G2.10); by Legendre duality $\nabla \mathsf{A}^*(\boldsymbol{\mu}) = \lambda$, so $g_\lambda^{-1} = \nabla^2 \mathsf{A}^*$, and $\mathsf{A}^*(\boldsymbol{\mu}) = \langle \lambda, \boldsymbol{\mu} \rangle - \mathsf{A}(\lambda) = \mathbb{E}_q \log q$. □

By (i) the precision is at once a coupling matrix and the Fisher metric of the mean, and the chapters below use both readings.

G2.3 Conditional reading and the Schur lemma

Let $\mathcal{P}$ be a partition of the n coordinates into blocks of dimensions n_b, and write Λ_{bc} for the corresponding blocks.

Proposition G2.5 (Conditional reading of the precision). Let $Y \sim \mathcal{N}(\mu, \Lambda^{-1})$ with $\Lambda \succ 0$. (i) For any block b,

$$\mathbb{E}\left[Y_b \mid Y_{-b} = y_{-b}\right] = \mu_b - \Lambda_{bb}^{-1} \sum_{c \neq b} \Lambda_{bc} \left(y_c - \mu_c\right), \qquad \operatorname{Cov}\left(Y_b \mid Y_{-b}\right) = \Lambda_{bb}^{-1}. \tag{G2.12}$$

(ii) For distinct blocks b and c, Y_b and Y_c are conditionally independent given all remaining coordinates if and only if $\Lambda_{bc} = 0$. (iii) If b and c have equal dimension and $\Lambda_{bc} = -W_{bc}$ with $W_{bc} \succeq 0$, the displacement $\Lambda_{bb}^{-1} W_{bc} \delta_c$ of the conditional mean of Y_b caused by $\delta_c = y_c - \mu_c$ points toward δ_c in the Λ_{bb} inner product, since $\delta_c^\top \Lambda_{bb} (\Lambda_{bb}^{-1} W_{bc} \delta_c) = \delta_c^\top W_{bc} \delta_c \geq 0$; in the Euclidean inner product it can point away: for $\Lambda_{bb} = \operatorname{diag}(1, 10)$, $W_{bc} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $\Lambda_{cc} = 10 I_2$, whose joint matrix has leading minors $1, 10, 89, 780$, and $\delta_c = (1, -2)$, the displacement is $(-1, -\frac{1}{10})$ and its Euclidean pairing with δ_c is $-\frac{4}{5}$. [ESTABLISHED]

Proof. At fixed y_{-b} the exponent of (G2.2) is $-\frac{1}{2} y_b^\top \Lambda_{bb} y_b$ plus a linear term, so the conditional law is Gaussian with precision Λ_{bb}, and completing the square around μ gives the conditional mean. Conditioning on the coordinates outside $b \cup c$ leaves a Gaussian in (y_b, y_c) with precision $\left(\begin{smallmatrix} \Lambda_{bb} & \Lambda_{bc} \\ \Lambda_{cb} & \Lambda_{cc} \end{smallmatrix}\right)$, which factorizes exactly when the quadratic form has no cross term. The numbers in (iii) are direct evaluation. This is the Gaussian case of the pairwise Markov property [21, Ch. 5]; see also Rue and Held [27, Ch. 2]. □

Lemma G2.6 (Schur complement). Let $\Lambda \in \mathbb{S}^n_{++}$ and let $[E\ F]$ be an orthogonal matrix with $E \in \mathbb{R}^{n \times k}$. (i) The marginal precision of $E^\top Y$ under $\mathcal{N}(\mu, \Lambda^{-1})$ is the Schur complement

$$\left(E^\top \Lambda^{-1} E\right)^{-1} = E^\top \Lambda E - E^\top \Lambda F \left(F^\top \Lambda F\right)^{-1} F^\top \Lambda E; \tag{G2.13}$$

for the coordinate block b of a partition this reads

$$\left(\Lambda^{-1}\right)_{bb} = \left(\Lambda_{bb} - \Lambda_{b,-b} \Lambda_{-b,-b}^{-1} \Lambda_{-b,b}\right)^{-1}. \tag{G2.14}$$

(ii) The difference $E^\top \Lambda E - (E^\top \Lambda^{-1} E)^{-1}$ is positive semidefinite of rank $\operatorname{rank}(E^\top \Lambda F)$, and it vanishes exactly when $E^\top \Lambda F = 0$, that is when $\operatorname{range} E$ and $\operatorname{range} F$ are Λ-orthogonal. (iii)

$(E^\top \Lambda E)^{-1} \preceq E^\top \Lambda^{-1} E$, with equality exactly when $E^\top \Lambda F = 0$. (iv) The determinant factorizes as

$$\det \Lambda = \det(E^\top \Lambda E) \det\big(F^\top \Lambda F - F^\top \Lambda E (E^\top \Lambda E)^{-1} E^\top \Lambda F\big) = \frac{\det(E^\top \Lambda E)}{\det(F^\top \Lambda^{-1} F)}.$$

(v) A zero precision block says nothing about the corresponding covariance block:

$$\Lambda = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad \Lambda^{-1} = \frac{1}{4} \begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \tag{G2.15}$$

has $\Lambda_{13} = 0$ and $(\Lambda^{-1})_{13} = \frac{1}{4}$. [ESTABLISHED]

Proof. In the basis $[E\ F]$, $\widetilde{\Lambda} = [E\ F]^\top \Lambda [E\ F]$ has blocks $\widetilde{\Lambda}_{EE} = E^\top \Lambda E$, $\widetilde{\Lambda}_{EF} = E^\top \Lambda F$, $\widetilde{\Lambda}_{FF} = F^\top \Lambda F$, and orthogonality gives $\widetilde{\Lambda}^{-1} = [E\ F]^\top \Lambda^{-1} [E\ F]$. Multiplying out the factorization

$$\widetilde{\Lambda} = \begin{pmatrix} I & \widetilde{\Lambda}_{EF} \widetilde{\Lambda}_{FF}^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \widetilde{\Lambda}_{EE} - \widetilde{\Lambda}_{EF} \widetilde{\Lambda}_{FF}^{-1} \widetilde{\Lambda}_{FE} & 0 \\ 0 & \widetilde{\Lambda}_{FF} \end{pmatrix} \begin{pmatrix} I & 0 \\ \widetilde{\Lambda}_{FF}^{-1} \widetilde{\Lambda}_{FE} & I \end{pmatrix}$$

and inverting shows that the EE block of $\widetilde{\Lambda}^{-1}$ is the inverse of the Schur complement, which is (i); the coordinate case takes for E the columns of the identity in block b. The difference in (ii) is $M^\top \widetilde{\Lambda}_{FF}^{-1} M$ with $M = \widetilde{\Lambda}_{FE}$ and $\widetilde{\Lambda}_{FF}^{-1} \succ 0$, hence positive semidefinite with the rank of M. For (iii), if $0 \prec S \preceq T$ then $T^{-1/2} S T^{-1/2} \preceq I$, so $S^{-1} \succeq T^{-1}$, with equality only when $S = T$ [17, §7.7]; apply it to (ii). The factorization with the roles of E and F exchanged gives the first formula in (iv), and (i) with the roles exchanged identifies that Schur complement with $(F^\top \Lambda^{-1} F)^{-1}$. Part (v) is direct multiplication. □

The lemma is the reason this supplement works in information form: a restriction on the recognition family constrains conditional dependence, which is a coordinate of Λ, while its effect on marginal covariances runs through every path in the remaining coordinates. Its corollaries below are the four forms in which it is used.

Corollary G2.7 (The block conditional covariance is dominated by the marginal). For every block b,

$$\Lambda_{bb}^{-1} \preceq \big(\Lambda^{-1}\big)_{bb}, \tag{G2.16}$$

with equality if and only if $\Lambda_{b,-b} = 0$. The left side is the block covariance of the optimal factorized recognition law of Theorem G4.2, so that law understates every marginal block covariance of the target except in the decoupled case. [ESTABLISHED]

Proof. Lemma G2.6(iii) with E the coordinate columns of block b. □

Corollary G2.8 (Marginal precision against restricted precision). Let $[B\ B_\perp]$ be orthogonal with $B \in \mathbb{R}^{n \times r}$, and write $\widetilde{\Lambda} = [B\ B_\perp]^\top \Lambda [B\ B_\perp]$ with blocks indexed by $\parallel$ and $\perp$. The marginal

precision of the directions $B_\perp$ is the Schur complement

$$\left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} = \widetilde{\Lambda}_{\perp\perp} - \widetilde{\Lambda}_{\perp\|} \widetilde{\Lambda}_{\|\|}^{-1} \widetilde{\Lambda}_{\|\perp}, \tag{G2.17}$$

and

$$B_\perp^\top \Lambda B_\perp - \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} = B_\perp^\top \Lambda B \left(B^\top \Lambda B\right)^{-1} B^\top \Lambda B_\perp \succeq 0, \tag{G2.18}$$

a difference of rank at most r that vanishes exactly when $B^\top \Lambda B_\perp = 0$. [ESTABLISHED]

Proof. Lemma G2.6(i) and (ii) with $E = B_\perp$ and $F = B$; the rank of $B^\top \Lambda B_\perp$ is at most r. □

Corollary G2.9 (Fisher monotonicity of the mean submodel). Let $B \in \mathbb{R}^{n\times k}$ have orthonormal columns and let $B_\perp$ span the orthogonal complement. The mean submodel $\{\mathcal{N}(B\bar{m}, \Lambda^{-1}) : \bar{m} \in \mathbb{R}^k\}$ has Fisher metric $B^\top \Lambda B$, the pullback of the mean metric Λ along $\bar{m} \mapsto B\bar{m}$. The deterministic channel $y \mapsto B^\top y$ carries it to $\{\mathcal{N}(\bar{m}, B^\top \Lambda^{-1} B)\}$, whose Fisher metric is the marginal precision $(B^\top \Lambda^{-1} B)^{-1}$, and

$$B^\top \Lambda B - \left(B^\top \Lambda^{-1} B\right)^{-1} = \left(B^\top \Lambda B_\perp\right) \left(B_\perp^\top \Lambda B_\perp\right)^{-1} \left(B_\perp^\top \Lambda B\right) \succeq 0. \tag{G2.19}$$

The right side is the Fisher loss of the channel, and it vanishes exactly when $B^\top y$ is sufficient for $\bar{m}$, that is when $B^\top \Lambda B_\perp = 0$. [ESTABLISHED]

Proof. The score of the submodel is $B^\top \Lambda(y - B\bar{m})$, with covariance $B^\top \Lambda B$; the pushed family has covariance $B^\top \Lambda^{-1} B$, whose inverse is its Fisher metric. Lemma G2.6(i) and (ii) with $E = B$, $F = B_\perp$ give (G2.19). Writing $y = BB^\top y + B_\perp B_\perp^\top y$, the conditional covariance of the score given $B^\top Y$ is $B^\top \Lambda B_\perp \operatorname{Cov}(B_\perp^\top Y \mid B^\top Y) B_\perp^\top \Lambda B$, and $\operatorname{Cov}(B_\perp^\top Y \mid B^\top Y) = (B_\perp^\top \Lambda B_\perp)^{-1}$ by Proposition G2.5(i) in the basis $[B\ B_\perp]$; so the right side of (G2.19) is the expected conditional score covariance of Theorem 7.8. It vanishes exactly when $B^\top \Lambda B_\perp = 0$, which is when the score, and hence the likelihood ratio, depends on y only through $B^\top y$. □

Corollary G2.10 (Full-rank inclusion and averaging). Let $S \in \mathbb{R}^{n\times k}$ have full column rank, set $M_S = S^\top S$, and let $S^\dagger = M_S^{-1} S^\top$. The mean submodel $\mu = S\bar{\mu}$ has Fisher matrix $S^\top \Lambda S$, while the mean of the pushforward under $y \mapsto S^\dagger y$ has Fisher matrix

$$\Lambda_{\mathrm{av}} = M_S \left(S^\top \Lambda^{-1} S\right)^{-1} M_S. \tag{G2.20}$$

With $B = S M_S^{-1/2}$ completed to an orthogonal matrix $[B\ B_\perp]$,

$$S^\top \Lambda S - \Lambda_{\mathrm{av}} = M_S^{1/2} \left[B^\top \Lambda B - \left(B^\top \Lambda^{-1} B\right)^{-1}\right] M_S^{1/2} \succeq 0, \tag{G2.21}$$

with equality exactly when $\operatorname{range} S$ and its Euclidean orthogonal complement are Λ-orthogonal. [ESTABLISHED]

Proof. The pushforward covariance is $S^\dagger \Lambda^{-1} (S^\dagger)^\top = M_S^{-1} S^\top \Lambda^{-1} S M_S^{-1}$, whose inverse is (G2.20). Substituting $S = B M_S^{1/2}$ into both matrices gives (G2.21), and Corollary G2.9 supplies its positivity and equality condition. □

G2.4 Moments and vector residuals

Proposition G2.11 (Finite-second-moment congruence). Let a belief law q on $\mathbb{R}^K$ have finite mean μ and covariance Σ, and let $g \in G$ act by the linear map $\rho_b(g)$. The pushed law $\rho_b(g)_\# q$ has

$$\mu' = \rho_b(g)\mu, \qquad \Sigma' = \rho_b(g)\Sigma\rho_b(g)^\top, \tag{G2.22}$$

and if $\Sigma \succ 0$ its precision is

$$\Lambda' = \rho_b(g)^{-\top} \Lambda \rho_b(g)^{-1}, \qquad \Lambda = \Sigma^{-1}. \tag{G2.23}$$

The model channel has the same statement in dimension d_m. [ESTABLISHED]

Proof. Linearity of expectation gives the mean, and the centered pushed variable $\rho_b(g)(Y - \mu)$ has covariance $\rho_b(g)\Sigma\rho_b(g)^\top$; inversion gives (G2.23). □

G2.4.1 Residuals of aligned and twisted links

Let μ_i be a belief-channel vector coordinate of agent i, let U_i be the frame of u_i relative to a reference point of the fiber, as in Proposition 2.12, and let $e : j \to i$ be an edge copy with belief link Θ_e^b and per-edge twist $\mathring{\Theta}_e^b = U_i^{-1} \Theta_e^b U_j$ ((2.32)).

Proposition G2.12 (Vector residuals). With $z_i = \rho_b(U_i)^{-1}\mu_i$,

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i)\big(z_i - \rho_b(\mathring{\Theta}_e^b) z_j\big), \tag{G2.24}$$

and for frame-aligned links (Definition 2.14), whose twists equal e_G,

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i)(z_i - z_j). \tag{G2.25}$$

More generally, if $\Theta_e^b = U_i'(U_j')^{-1}$ on a belief-trivializing subfamily of links and $z_i' = \rho_b(U_i')^{-1}\mu_i$, then

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i')(z_i' - z_j'). \tag{G2.26}$$

[ESTABLISHED]

Proof. Substitute $\mu_i = \rho_b(U_i) z_i$ and $\rho_b(\Theta_e^b) = \rho_b(U_i)\rho_b(\mathring{\Theta}_e^b)\rho_b(U_j)^{-1}$; (G2.25) is the case $\mathring{\Theta}_e^b = e_G$, and (G2.26) is the same substitution with the potentials U_i'. □

Belief-link trivialization says nothing about model links or either cross morphism.

G2.5 The directed linear-Gaussian model

Definition G2.13 (Directed linear-Gaussian forest). Take

$$k_i \in \mathbb{R}^K, \qquad m_i \in \mathbb{R}^{d_m}, \qquad o_i \in \mathbb{R}^{d_{o,i}}, \tag{G2.27}$$

and a directed forest on the agents, so that each nonroot i has one parent $j = \mathrm{par}(i)$, joined by an edge copy $j \to i$ with links $\Theta^b_{ij}, \Theta^m_{ij} \in G$ (Definition 2.11). For a nonroot,

$$m_i \mid m_j, X \sim \mathcal{N}\left(F^m_i \rho_m(\Theta^m_{ij}) m_j + \xi^m_i, \Sigma^m_i\right), \tag{G2.28}$$

$$k_i \mid k_j, m_i, X \sim \mathcal{N}\left(F^b_i \rho_b(\Theta^b_{ij}) k_j + \mathsf{B}_i m_i + \xi^b_i, \Sigma^b_i\right), \tag{G2.29}$$

$$o_i \mid k_i, m_i, X \sim \mathcal{N}\left(\mathsf{C}^b_i k_i + \mathsf{C}^m_i m_i + \xi^o_i, \Sigma^o_i\right), \tag{G2.30}$$

with parameter shapes

$$\rho_b(\Theta^b_{ij}), F^b_i \in \mathbb{R}^{K\times K}, \quad \rho_m(\Theta^m_{ij}), F^m_i \in \mathbb{R}^{d_m\times d_m}, \quad \mathsf{B}_i \in \mathbb{R}^{K\times d_m}, \tag{G2.31}$$

$$\mathsf{C}^b_i \in \mathbb{R}^{d_{o,i}\times K}, \quad \mathsf{C}^m_i \in \mathbb{R}^{d_{o,i}\times d_m}, \quad \xi^b_i \in \mathbb{R}^K, \quad \xi^m_i \in \mathbb{R}^{d_m}, \quad \xi^o_i \in \mathbb{R}^{d_{o,i}}. \tag{G2.32}$$

Each root r has the laws

$$m_r \mid X \sim \mathcal{N}(\bar{m}_r, \Sigma^m_{r,0}), \tag{G2.33}$$

$$k_r \mid m_r, X \sim \mathcal{N}(\mathsf{B}_r m_r + \xi^b_{r,0}, \Sigma^b_{r,0}), \tag{G2.34}$$

followed by (G2.30). A general directed acyclic graph replaces each parent term by a finite sum. The represented links are invertible; the gains F^b_i, F^m_i need not be.

Hypothesis G2.14 (Positive-definite receiving covariances). Every receiving covariance is positive definite:

$$\Sigma^m_i, \Sigma^m_{r,0} \in \mathbb{S}^{d_m}_{++}, \qquad \Sigma^b_i, \Sigma^b_{r,0} \in \mathbb{S}^K_{++}, \qquad \Sigma^o_i \in \mathbb{S}^{d_{o,i}}_{++}. \tag{G2.35}$$

The hypothesis excludes degenerate Gaussian kernels, whose laws have no Lebesgue density on the receiving space.

Proposition G2.15 (Normalization and root moments). Under Hypothesis G2.14, every kernel of Definition G2.13 is a nondegenerate normalized density and the assembled directed law is a probability measure. The root belief coordinate has

$$\bar{k}_r = \mathsf{B}_r \bar{m}_r + \xi^b_{r,0}, \qquad \Sigma^{b,\mathrm{marg}}_{r,0} = \Sigma^b_{r,0} + \mathsf{B}_r \Sigma^m_{r,0} \mathsf{B}_r^\top \succeq \Sigma^b_{r,0}, \tag{G2.36}$$

and, for a root model law $s = \mathcal{N}(\mu_s, \Sigma_s)$, the law map induced by the root kernel is

$$\widetilde{\phi}_r(s) = \mathcal{N}\left(\mathsf{B}_r \mu_s + \xi^b_{r,0},\ \Sigma^b_{r,0} + \mathsf{B}_r \Sigma_s \mathsf{B}_r^\top\right). \tag{G2.37}$$

This law map, of which B_r is one parameter, is the local realization of $\widetilde{\Phi}$. [ESTABLISHED]

Proof. A Gaussian density with positive-definite covariance integrates to one for every mean, so Proposition 4.4 applies. The tower property gives the means and the law of total covariance gives the covariances; the added term $\mathsf{B}_r\Sigma_s\mathsf{B}_r^\top$ is positive semidefinite. □

At fixed model latents, write $F_i := F_i^b$, $\Theta_{ij} := \rho_b(\Theta_{ij}^b) \in \Gamma_b$, and $\Lambda_i := (\Sigma_i^b)^{-1}$; the belief sector is the forest of receiving kernels

$$k_i \mid k_j \sim \mathcal{N}\left(F_i\Theta_{ij}k_j, \Lambda_i^{-1}\right), \qquad j = \mathrm{par}(i), \tag{G2.38}$$

with a proper prior $\mathcal{N}(0, \Lambda_r^{-1})$ at each root; the bridge term, the offsets, and a nonzero prior mean shift only the information vector. A single subscript on Λ denotes the innovation precision of one receiving kernel, a double subscript a block of the assembled precision.

Proposition G2.16 (Assembled precision blocks). Write $\Omega_i := F_i\Theta_{i\,\mathrm{par}(i)}$ for the transition of a nonroot i and $\mathrm{ch}(i) = \{i' : \mathrm{par}(i') = i\}$. The joint precision of the belief sector has blocks

$$\Lambda_{i\,\mathrm{par}(i)} = -\Lambda_i\Omega_i, \qquad \Lambda_{ii} = \Lambda_i + \sum_{i'\in\mathrm{ch}(i)} \Omega_{i'}^\top\Lambda_{i'}\Omega_{i'}, \tag{G2.39}$$

and $\Lambda_{ij} = 0$ whenever i and j are not adjacent in the forest. In the joint precision of (k, m), the block between k_i and m_i is $-\Lambda_i\mathsf{B}_i + \mathsf{C}_i^{b\top}(\Sigma_i^o)^{-1}\mathsf{C}_i^m \in \mathbb{R}^{K\times d_m}$, whose first term is the prior contribution; a family built from square $K \times K$ agent blocks cannot host it when $K \neq d_m$. [ESTABLISHED]

Proof. The negative log joint density of the belief sector is $\frac{1}{2}\sum_i(k_i - \Omega_i k_{\mathrm{par}(i)})^\top\Lambda_i(k_i - \Omega_i k_{\mathrm{par}(i)})$ plus terms free of k, with no parent term at a root. One summand contributes Λ_i to the ii block, $-\Lambda_i\Omega_i$ to the $i\,\mathrm{par}(i)$ block, and $\Omega_i^\top\Lambda_i\Omega_i$ to the parent's diagonal block; collecting them gives (G2.39). In the joint energy, the factors (G2.29) and (G2.30) are the only ones containing both k_i and m_i, and their mixed second derivatives are $-\Lambda_i\mathsf{B}_i$ and $\mathsf{C}_i^{b\top}(\Sigma_i^o)^{-1}\mathsf{C}_i^m$. □

Proposition G2.17 (Gaussian kernel covariance). Let $g_i \in G$ act by (G2.1), write $\rho_i^b := \rho_b(g_i)$, $\rho_i^m := \rho_m(g_i)$, let the links transform by (4.21), so that $\rho_x(\Theta_{ij}^{x\prime}) = \rho_i^x\rho_x(\Theta_{ij}^x)(\rho_j^x)^{-1}$, and let the parameters transform by

$$F_i^{b\prime} = \rho_i^b F_i^b(\rho_i^b)^{-1}, \qquad F_i^{m\prime} = \rho_i^m F_i^m(\rho_i^m)^{-1}, \tag{G2.40}$$

$$\mathsf{B}_i' = \rho_i^b\mathsf{B}_i(\rho_i^m)^{-1}, \qquad \xi_i^{b\prime} = \rho_i^b\xi_i^b, \qquad \xi_i^{m\prime} = \rho_i^m\xi_i^m, \tag{G2.41}$$

$$\Sigma_i^{b\prime} = \rho_i^b\Sigma_i^b(\rho_i^b)^\top, \qquad \Sigma_i^{m\prime} = \rho_i^m\Sigma_i^m(\rho_i^m)^\top, \tag{G2.42}$$

so that for either channel

$$\Lambda_i = \Sigma_i^{-1}, \qquad \Sigma_i' = \rho_i\Sigma_i\rho_i^\top, \qquad \Lambda_i' = \rho_i^{-\top}\Lambda_i\rho_i^{-1}, \tag{G2.43}$$

with the root laws

$$\bar{m}'_r = \rho^m_r \bar{m}_r, \qquad \Sigma^{m\prime}_{r,0} = \rho^m_r \Sigma^m_{r,0} (\rho^m_r)^\top, \tag{G2.44}$$

$$\mathsf{B}'_r = \rho^b_r \mathsf{B}_r (\rho^m_r)^{-1}, \qquad \xi^{b\prime}_{r,0} = \rho^b_r \xi^b_{r,0}, \qquad \Sigma^{b\prime}_{r,0} = \rho^b_r \Sigma^b_{r,0} (\rho^b_r)^\top, \tag{G2.45}$$

and gauge-inert observations

$$\mathsf{C}^{b\prime}_i = \mathsf{C}^b_i (\rho^b_i)^{-1}, \qquad \mathsf{C}^{m\prime}_i = \mathsf{C}^m_i (\rho^m_i)^{-1}, \qquad \xi^{o\prime}_i = \xi^o_i, \qquad \Sigma^{o\prime}_i = \Sigma^o_i. \tag{G2.46}$$

Then every transformed kernel is the receiving-frame pushforward of the original, so the kernel identities (4.23), (4.24), and (4.25) hold and the joint law and its evidence obey Proposition 4.9. The induced root law map (G2.37) obeys the frame law (2.23) of $\widetilde{\Phi}$, the root marginal (G2.36) transforms by belief-channel congruence, and positive definiteness is preserved in each channel. [ESTABLISHED]

Proof. For the belief kernel, the transformed conditional mean at transformed inputs is

$$F^{b\prime}_i \rho_b(\Theta^{b\prime}_{ij})(\rho^b_j k_j) + \mathsf{B}'_i(\rho^m_i m_i) + \xi^{b\prime}_i = \rho^b_i \left(F^b_i \rho_b(\Theta^b_{ij}) k_j + \mathsf{B}_i m_i + \xi^b_i \right), \tag{G2.47}$$

and the covariance transforms by (G2.42), so the transformed Gaussian is the pushforward of the original under ρ^b_i. The model kernel is the same computation without a bridge. At the observation kernel,

$$\mathsf{C}^{b\prime}_i(\rho^b_i k_i) + \mathsf{C}^{m\prime}_i(\rho^m_i m_i) + \xi^{o\prime}_i = \mathsf{C}^b_i k_i + \mathsf{C}^m_i m_i + \xi^o_i, \tag{G2.48}$$

with fixed covariance. The root calculation is identical, and the law map (G2.37) of a pushed model law $\rho^m_{r\#} s$ is $\rho^b_{r\#} \widetilde{\phi}_r(s)$ by the same substitution. Composition is Proposition 4.9. Both summands of (G2.36) transform by belief-channel congruence, and congruence by an invertible matrix preserves positive definiteness. □

G2.6 The Gaussian M-coordinate

Proposition G2.18 (Gaussian M-coordinate and its existence proviso). Suppose the factors whose parameters lie in θ carry disjoint parameter blocks, that each such factor is a law $\mathcal{N}(Fw, \Sigma)$ of its target coordinate t given a regressor vector w listing its parent coordinates and the constant 1, with F free over all matrices of its shape and Σ in the open cone of (G2.35), and that $\mathbb{Q}_{V,o,X}$ has finite second moments. The expected complete log-likelihood of (5.58) is then a sum of separately maximized factor terms. For one factor, with expectations under $\mathbb{Q}_{V,o,X}$, put $M_{ww} = \mathbb{E}[ww^\top]$, let F_* solve $F_* M_{ww} = \mathbb{E}[tw^\top]$, and set $\Sigma_* = \mathbb{E}[(t - F_* w)(t - F_* w)^\top]$. Then

$$\mathbb{E}\big[(t - Fw)(t - Fw)^\top\big] = \Sigma_* + (F - F_*) M_{ww} (F - F_*)^\top, \qquad \mathbb{E} \log \mathcal{N}(t; Fw, \Sigma) \le -\tfrac{1}{2} \log\det(2\pi e \Sigma_*), \tag{G2.49}$$

with equality at (F_*, Σ_*) when $\Sigma_* \succ 0$. If Σ_* is singular the term is unbounded above and has no maximizer. The matrix Σ_* is positive definite exactly when no nonzero linear combination of t is

almost surely an affine function of the parents, which holds for a latent target whenever $\mathbb{Q}_{V,o,X}$ has a Lebesgue density on the latent coordinates. Two cases violate the proviso: a record factor at its single observed value, whose intercept can equal that value with zero coefficients on the parents, giving $\Sigma_* = 0$; and a link learned inside the invertible group with the rest of its mean map fixed, which for $K = 1$, link map $\mathsf{t} = \rho_b(\Theta^b_{ij}) \neq 0$, and k_i, k_j independent and centered, has covariance-optimized term $-\frac{1}{2}\log\big(2\pi e(\mathbb{E}k_i^2 + \mathsf{t}^2\mathbb{E}k_j^2)\big)$, which approaches its supremum $-\frac{1}{2}\log(2\pi e\mathbb{E}k_i^2)$ as $\mathsf{t} \to 0$ without attaining it. [ESTABLISHED]

Proof. Expanding $t - Fw = (t - F_*w) - (F - F_*)w$ and using the normal equations to cancel the cross term gives the first identity. For $\Sigma_* \succ 0$ and ϱ_k the eigenvalues of $\Sigma^{-1/2}\Sigma_*\Sigma^{-1/2}$, $-\log\det\Sigma - \mathrm{Tr}(\Sigma^{-1}\Sigma_*) = -\log\det\Sigma_* + \sum_k(\log\varrho_k - \varrho_k)$, which is maximal exactly at $\Sigma = \Sigma_*$; the first identity orders every residual second moment above Σ_*, and $\mathrm{Tr}(\Sigma^{-1}\cdot)$ is monotone, which gives the inequality. If Σ_* is singular, $\Sigma = \Sigma_* + \varepsilon I$ with $\varepsilon \downarrow 0$ drives the term to $+\infty$. A nonzero a with $a^\top\Sigma_* a = 0$ makes $a^\top t$ almost surely affine in the parents, and such a relation confines a latent vector to a Lebesgue-null hyperplane. The two witnesses are direct evaluation. □

G2.7 Gaussian selective communication

The comparison maps S_e^Q, S_e^K, the value map $\mathsf{L}_e^{\mathrm{val}}$, and the encoder, transmission, and decoder maps $\mathsf{L}_e^{\mathrm{enc}}, \mathsf{L}_e^{\mathrm{tr}}, \mathsf{L}_e^{\mathrm{dec}}$ are those of Sections 6.4 and 6.5.

Proposition G2.19 (A comparison lift requires receiver geometry). Fix $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ with $\Sigma_i \succ 0$ and a full-row-rank S_e^Q. (i) The reduced covariance $S_e^Q\Sigma_i S_e^{Q\top}$ is positive definite. (ii) The lift

$$R_e = \Sigma_i S_e^{Q\top}(S_e^Q\Sigma_i S_e^{Q\top})^{-1}, \qquad S_e^Q R_e = I, \qquad \mathsf{L}_e^{\mathrm{cmp}} = R_e S_e^K, \tag{G2.50}$$

makes $R_e y$ the unique minimizer of $\delta^\top\Sigma_i^{-1}\delta$ subject to $S_e^Q\delta = y$, and $\mathbb{E}[k_i \mid S_e^Q k_i = y] = \mu_i + R_e(y - S_e^Q\mu_i)$. (iii) If comparison and content share a Gaussian source $z \sim \mathcal{N}(\mu_z, \Sigma_z)$, $\Sigma_z \succ 0$, with content map L and full-row-rank key map S_e^K, then

$$\mathbb{E}[\mathsf{L}z \mid S_e^K z = y] = \mathsf{L}\mu_z + \mathsf{L}\Sigma_z S_e^{K\top}(S_e^K\Sigma_z S_e^{K\top})^{-1}(y - S_e^K\mu_z), \tag{G2.51}$$

and the correction vanishes for all y exactly when $\mathsf{L}\Sigma_z S_e^{K\top} = 0$. (iv) Under $S_e^{Q\prime} = \phi_e^{\mathrm{cmp}} S_e^Q \rho_b(g_i)^{-1}$ with ϕ_e^{cmp} invertible on the comparison space, and $\Sigma_i' = \rho_b(g_i)\Sigma_i\rho_b(g_i)^\top$, the lift transforms by $R_e' = \rho_b(g_i)R_e(\phi_e^{\mathrm{cmp}})^{-1}$. (v) In the comparison-record model of Proposition 6.19, with S_i^Q of full row rank, R_e formed from S_i^Q, and $\widetilde{W}_i^{-1} \to 0$, $\mathsf{L}_e^{\mathrm{val}} \to R_e S_e^K$ and $\Sigma_i^+ \to \Sigma_i - R_e S_i^Q\Sigma_i$. (vi) For comparison and content copies $Y = V = z \sim \mathcal{N}(0,1)$, every product law $q(Y)q(V)$ with continuous marginals has infinite relative entropy to the diagonal-supported joint, even with identical comparison marginals. [ESTABLISHED]

Proof. (i) For $u \neq 0$, $S_e^{Q\top}u \neq 0$ by full row rank, so $u^\top S_e^Q\Sigma_i S_e^{Q\top}u > 0$. (ii) Write a feasible displacement as $R_e y + w$ with $S_e^Q w = 0$; its squared norm has cross term $y^\top(S_e^Q\Sigma_i S_e^{Q\top})^{-1}S_e^Q w = 0$ and the additional term $w^\top\Sigma_i^{-1}w$, positive unless $w = 0$. The residual $k_i - \mu_i - R_e(S_e^Q k_i - S_e^Q\mu_i)$

has zero covariance with $S_e^Q k_i$, hence is independent of it, which gives the conditional mean. (iii) The same argument applied to z and followed by L. (iv) Substitution. (v) The Woodbury identity $\Sigma_i^+ S_i^{Q\top} \widetilde{W}_i = \Sigma_i S_i^{Q\top} (\Sigma_i^{\mathrm{pred}})^{-1}$, with $\Sigma_i^{\mathrm{pred}} = \widetilde{W}_i^{-1} + S_i^Q \Sigma_i S_i^{Q\top}$ of (6.42), gives $\mathsf{L}_e^{\mathrm{val}} = \Sigma_i S_i^{Q\top} (\Sigma_i^{\mathrm{pred}})^{-1} S_e^K$ and $\Sigma_i^+ = \Sigma_i - \Sigma_i S_i^{Q\top} (\Sigma_i^{\mathrm{pred}})^{-1} S_i^Q \Sigma_i$; as $\widetilde{W}_i^{-1} \to 0$, $(\Sigma_i^{\mathrm{pred}})^{-1} \to (S_i^Q \Sigma_i S_i^{Q\top})^{-1}$ by (i). (vi) The joint of (Y, V) is concentrated on the diagonal, which a product of continuous laws assigns mass zero, so absolute continuity fails. □

The content map L_e of a communication layer is a separate parameter; the lift $\mathsf{L}_e^{\mathrm{cmp}}$ is what receiver geometry supplies when content is read back from a comparison.

Proposition G2.20 (The linear Gaussian record). Let the three stages of (6.26) be $h^- = \mathsf{L}_e^{\mathrm{enc}} k_j + \epsilon_-$, $h^+ = \mathsf{L}_e^{\mathrm{tr}} h^- + \epsilon_+$, and $o_e = \mathsf{L}_e^{\mathrm{dec}} h^+ + \epsilon_o$, with mutually independent centered Gaussian noises, independent of k_j, of covariances $\Sigma_e^-, \Sigma_e^+, \Sigma_e^o \succeq 0$. (i) Then

$$o_e \mid k_j \sim \mathcal{N}(\mathsf{L}_e k_j, \Sigma_e^{\mathrm{rec}}), \qquad \Sigma_e^{\mathrm{rec}} = \mathsf{L}_e^{\mathrm{dec}} \mathsf{L}_e^{\mathrm{tr}} \Sigma_e^- \mathsf{L}_e^{\mathrm{tr}\top} \mathsf{L}_e^{\mathrm{dec}\top} + \mathsf{L}_e^{\mathrm{dec}} \Sigma_e^+ \mathsf{L}_e^{\mathrm{dec}\top} + \Sigma_e^o, \tag{G2.52}$$

with $\mathsf{L}_e = \mathsf{L}_e^{\mathrm{dec}} \mathsf{L}_e^{\mathrm{tr}} \mathsf{L}_e^{\mathrm{enc}}$; the kernel is normalized at every rank, has a Lebesgue density when $\Sigma_e^{\mathrm{rec}} \succ 0$, for instance when $\Sigma_e^o \succ 0$, and for input $\mathcal{N}(\mu_j, \Sigma_j)$ gives $o_e \sim \mathcal{N}(\mathsf{L}_e \mu_j, \Sigma_e^{\mathrm{rec}} + \mathsf{L}_e \Sigma_j \mathsf{L}_e^\top)$; if the receiver frame changes by g_i, with $\mathsf{L}_e^{\mathrm{dec}\prime} = \rho_b(g_i) \mathsf{L}_e^{\mathrm{dec}}$ and $\Sigma_e^{o\prime} = \rho_b(g_i) \Sigma_e^o \rho_b(g_i)^\top$, then $\Sigma_e^{\mathrm{rec}\prime} = \rho_b(g_i) \Sigma_e^{\mathrm{rec}} \rho_b(g_i)^\top$. Deterministic placement is normalized as a measure but need not have a Lebesgue density. (ii) For $\Sigma_e^{\mathrm{rec}} \succ 0$ and $k_j \sim \mathcal{N}(0, \Sigma_j)$,

$$I(k_j; o_e) = \tfrac{1}{2} \log \frac{\det(\Sigma_e^{\mathrm{rec}} + \mathsf{L}_e \Sigma_j \mathsf{L}_e^\top)}{\det \Sigma_e^{\mathrm{rec}}}, \tag{G2.53}$$

which is unchanged by a congruent transformation of signal and noise; for $\mathsf{L}_e = (1, 0)$, $\Sigma_j = I_2$, and scalar noise variance $\sigma^2 > 0$ it equals $\frac{1}{2} \log(1 + 1/\sigma^2)$, so at the fixed rank one it takes every value in $(0, \infty)$. (iii) The comparison energy has the normalized realization $o_e \mid k_i, k_j \sim \mathcal{N}(S_e^Q k_i - S_e^K k_j, W_e^{-1})$, $W_e \succ 0$, whose negative log density at $o_e = 0$ is V_e of (6.37) plus the Gaussian normalizer; with a proper normalized baseline and finitely many such records, the evidence at this slice is finite and positive, for arbitrary rectangular S_e^Q, S_e^K. [ESTABLISHED]

Proof. (i) Substituting the stages gives $o_e = \mathsf{L}_e k_j + \mathsf{L}_e^{\mathrm{dec}} \mathsf{L}_e^{\mathrm{tr}} \epsilon_- + \mathsf{L}_e^{\mathrm{dec}} \epsilon_+ + \epsilon_o$, a sum of independent Gaussians, so the conditional covariance is Σ_e^{rec} and the marginal adds $\mathsf{L}_e \Sigma_j \mathsf{L}_e^\top$. A Gaussian law is a probability measure for every positive-semidefinite covariance. (ii) The difference of the output entropy and the conditional entropy $\frac{1}{2} \log \det(2\pi e \Sigma_e^{\mathrm{rec}})$ is the displayed ratio, and a common congruence multiplies both determinants by the same factor; the example is direct evaluation. (iii) Each likelihood is bounded by its normalizer and strictly positive, so integrating it against a normalized baseline gives finite positive evidence. □

Chapter G3

The Interaction Family

The aggregation, Kron, and operator results of this supplement act on precisions of one shape: a matrix-weighted graph Laplacian plus positive semidefinite self terms. This chapter asks when the belief-sector precision of Chapter G2 has that shape. The answer is that the shape is a restriction, not a consequence of linear-Gaussian directedness: an orientation-reversing gain, and for odd K a v-structure, leave it in every chart reached by orientation-preserving frames (Proposition G3.9), while on forests every orientation-preserving invertible gain is absorbed (Proposition G3.8); the condition for membership is Proposition G3.10, and the restriction is adopted as Hypothesis G3.13. It is used by hard aggregation (Section G7.2), by the operator arrow (Section G9.1), and by the cone dynamics of Supplement B.

Membership in the family is a statement in a chosen chart, and nothing here selects a chart other than the link-trivializing one of Hypothesis G3.4. The family is stated for the belief sector at fixed model latents and not for the joint belief–model precision, and no conditioning bound for the family is asserted. [NOT-CLAIMED]

G3.1 The interaction family

Throughout, agents are $i \in V = \{1, \dots, N\}$, belief latents are $k_i \in \mathbb{R}^K$, and the stacked latent has dimension $n = NK$.

Definition G3.1 (Interaction family). A matrix $\Lambda \in \mathbb{S}^{NK}$, partitioned into $K \times K$ agent blocks, belongs to the interaction family $\mathcal{I}(V)$ when there are self terms $A_i \in \mathbb{S}^K_+$ and edge weights $W_{ij} = W_{ji} \in \mathbb{S}^K_+$, $i \neq j$, with

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j). \tag{G3.1}$$

Its Laplacian part is $L := \Lambda - \mathrm{blockdiag}(A_1, \dots, A_N)$.

The edge weights are matrices, so L is a matrix-weighted graph Laplacian, the object of the matrix-weighted consensus literature [30].

Proposition G3.2 (Energy and kernel). For $\Lambda \in \mathcal{I}(V)$ and $y = (y_1, \dots, y_N)$,

$$y^\top \Lambda y = \sum_{i \in V} y_i^\top A_i y_i + \sum_{i<j} (y_i - y_j)^\top W_{ij} (y_i - y_j) . \tag{G3.2}$$

Consequently $\Lambda \succeq 0$, the consensus subspace $\mathbf{1} \otimes \mathbb{R}^K$ lies in $\ker L$, and $\Lambda \succ 0$ if and only if the only y with $A_i y_i = 0$ for all i and $W_{ij}(y_i - y_j) = 0$ for all $i < j$ is $y = 0$. A necessary condition for $\Lambda \succ 0$ is $\sum_i A_i \succ 0$. [ESTABLISHED]

Proof. Each unordered pair contributes $y_i^\top W_{ij} y_i + y_j^\top W_{ij} y_j - 2 y_i^\top W_{ij} y_j$, placing W_{ij} on the two diagonal blocks and $-W_{ij}$ on the two off-diagonal blocks; adding the self terms reproduces (G3.1). Every term is nonnegative, and for $M \succeq 0$, $v^\top M v = 0$ exactly when $Mv = 0$, which gives the kernel criterion. For $y = \mathbf{1} \otimes v$ every difference vanishes and the energy is $v^\top (\sum_i A_i) v$. □

With scalar weights a connected graph forces $\ker L$ to be the consensus subspace. A singular matrix weight leaves $y_i - y_j$ free along $\ker W_{ij}$, so $\ker L$ can be larger on a connected graph, and the criterion of Proposition G3.2 replaces connectivity.

Proposition G3.3 (The interaction family is thin). For $K \geq 2$ and $N \geq 2$, $\mathcal{I}(V)$ lies in a linear subspace of $\mathbb{S}^{NK}$ of codimension at least $\binom{N}{2} \frac{K(K-1)}{2}$, hence has Lebesgue measure zero there. Membership is not invariant under block-diagonal changes of coordinates: the congruence of the member with $N = K = 2$, $A_1 = A_2 = I_2$, $W_{12} = I_2$ by $\mathbf{P} = \mathrm{diag}(I_2, S)$, $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, has off-diagonal block $-S$, which is not symmetric. [ESTABLISHED]

Proof. Membership requires each off-diagonal block Λ_{ij} to be symmetric as a $K \times K$ matrix, since $W_{ij} = W_{ji}$ and $W_{ji} = W_{ij}^\top$; these are $\frac{K(K-1)}{2}$ independent linear equations per unordered pair, and the conic conditions only shrink the set. For the witness, $(\mathbf{P}^\top \Lambda \mathbf{P})_{12} = I_2(-I_2)S = -S$. □

G3.2 When the interaction form holds

G3.2.1 Trivialized coordinates and flat comparison

Write $\Theta_{ij} := \rho_b(\Theta_{ij}^b) \in \Gamma_b \subseteq \mathrm{GL}(K)$ for the represented belief link of the edge copy $j \to i$, with Γ_b of Hypothesis G2.1. The residual between two agents is $k_i - \Theta_{ij} k_j$, not $k_i - k_j$, so the plain differences of (G3.2) are a statement in coordinates in which the links have been removed. Proposition 2.13 decides when that is possible: frames with $U_i^{-1} \Theta_{ij} U_j = I$ exist exactly when every cycle product of the links is the identity, and they are then unique up to one common right factor on each connected component.

Hypothesis G3.4 (Flat comparison). The represented belief links form a graph coboundary: there are frames $U_i \in \Gamma_b$ with $\Theta_{ij} = U_i U_j^{-1}$ on every edge. In the coordinates $z_i = U_i^{-1} k_i$, applied equally to latents and to means, every residual is a plain difference,

$$k_i - \Theta_{ij} k_j = U_i (z_i - z_j) . \tag{G3.3}$$

The hypothesis excludes exactly the link assignments with a nontrivial cycle product, which the graph links of Definition 2.11 permit. It states Definition G3.1 in one chart and is the sufficient domain of Proposition G7.3.

Lemma G3.5 (Internal-edge annihilation). Let $\check{\Theta}_{ij}$ be the represented link in a chart common to i and j, and let $W_{ij} \succeq 0$. Identifying $z_i = z_j = z$ leaves the edge energy $z^\top (I - \check{\Theta}_{ij})^\top W_{ij} (I - \check{\Theta}_{ij}) z$, which vanishes for every z if and only if $\operatorname{ran}(I - \check{\Theta}_{ij}) \subseteq \ker W_{ij}$. For $W_{ij} \succ 0$ this means $\check{\Theta}_{ij} = I$; a singular weight can hide a nontrivial link in its kernel. [ESTABLISHED]

Proof. The energy is $\|W_{ij}^{1/2}(I - \check{\Theta}_{ij})z\|^2$, and $\ker W_{ij}^{1/2} = \ker W_{ij}$. □

Remark G3.6 (Membership depends on the frame). The change of variables $y = Uz$, $U = \operatorname{blockdiag}(U_1, \dots, U_N)$, acts on precisions by the congruence $\Lambda^{(z)} = U^\top \Lambda^{(y)} U$, which preserves inertia by Sylvester's law [17, Ch. 4] but not membership in $\mathcal{I}(V)$, since the block $U_i^\top \Lambda_{ij}^{(y)} U_j$ need not be symmetric when $U_i \neq U_j$ (Proposition G3.3). A frame-independent criterion is therefore invariant under block reframings with blocks in Γ_b. Two families of such invariants exist: generalized eigenvalues of the pair (L, Λ), which transform by one congruence (Proposition G5.5), and law functionals of unions of agent blocks, such as determinant gaps and mutual informations, which a reframing leaves unchanged because it changes no law (Section G5.3). Eigenvalues of either matrix alone are invariant when $\Gamma_b \subseteq \mathrm{O}(K)$ (Proposition G2.2) and frame dependent under $\Gamma_b = \mathrm{GL}^+(K)$ (Proposition G5.9). [ESTABLISHED]

G3.2.2 What a directed linear-Gaussian model contributes

For the belief sector (G2.38) with transitions $\Omega_i = F_i \Theta_{i\,\mathrm{par}(i)}$, the blocks (G2.39) show that membership requires the candidate weight

$$W_{i\,\mathrm{par}(i)} := \Lambda_i \Omega_i = \Lambda_i F_i \Theta_{i\,\mathrm{par}(i)} \tag{G3.4}$$

to be symmetric and positive semidefinite. Under Hypothesis G3.4, with $j = \mathrm{par}(i)$, set

$$\widetilde{\Omega}_i := U_i^{-1} \Omega_i U_j = U_i^{-1} F_i U_i, \qquad \widetilde{\Lambda}_i := U_i^\top \Lambda_i U_i, \tag{G3.5}$$

the second form of $\widetilde{\Omega}_i$ using $\Theta_{ij} = U_i U_j^{-1}$. The links are then removed, and any remaining obstruction is attributable to the gain.

Proposition G3.7 (Two global-condition witnesses in the link-trivializing chart). Under Hypothesis G3.4, the assembled precision in the chart $z = U^{-1}k$ is $U^\top \Lambda U$, whose blocks are (G2.39) with (Λ_i, Ω_i) replaced by $(\widetilde{\Lambda}_i, \widetilde{\Omega}_i)$. The chart is unique up to $U_i \mapsto U_i g$, $g \in \Gamma_b$, independently on each connected component, which acts on its blocks by $M \mapsto g^\top M g$, so symmetry of an off-diagonal block and the sign of a symmetric block are the same in every such chart. In this chart there are admissible parameters outside $\mathcal{I}(V)$:

$$\widetilde{\Lambda}_i = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right) \succ 0, \qquad \widetilde{\Omega}_i = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{GL}^+(2) \qquad \Longrightarrow \qquad -\widetilde{\Lambda}_i \widetilde{\Omega}_i = \left(\begin{smallmatrix} 1 & -2 \\ 3 & -1 \end{smallmatrix}\right), \tag{G3.6}$$

which differs from its transpose by a matrix of Frobenius norm $5\sqrt{2}$, so no symmetric W exists; and $K = 1$, $F_i = -1$, $\Theta_{ij} = \Lambda_i = 1$ gives the symmetric block with $W = -1 < 0$, as does $F_i = -I$, $\Theta_{ij} = \Lambda_i = I$ with $W = -I$ at every K. [ESTABLISHED]

Proof. The residual of each factor is $U_i(z_i - \widetilde{\Omega}_i z_j)$, so the energy in z is that of (G2.39) with the tilded parameters. Uniqueness up to a common right factor on each connected component is part of Proposition 2.13; under it $\widetilde{\Omega}_i \mapsto g^{-1}\widetilde{\Omega}_i g$ and $\widetilde{\Lambda}_i \mapsto g^\top \widetilde{\Lambda}_i g$, so $-\widetilde{\Lambda}_i\widetilde{\Omega}_i \mapsto g^\top(-\widetilde{\Lambda}_i\widetilde{\Omega}_i)g$; a matrix M is symmetric exactly when $g^\top(M - M^\top)g = 0$, and a symmetric M satisfies $M \preceq 0$ exactly when $g^\top M g \preceq 0$. The witnesses are direct computation: $\det \widetilde{\Lambda}_i = 5$, $\det \widetilde{\Omega}_i = 1$, and the antisymmetric part of the displayed block is $\left(\begin{smallmatrix} 0 & -5 \\ 5 & 0 \end{smallmatrix}\right)$. □

Proposition G3.8 (Gain absorption and the orientation criterion). (o) The directed law depends on gain and link only through Ω_i, and a block reframing $k_i = \mathbf{P}_i z_i$ sends Ω_i to $\mathbf{P}_i^{-1}\Omega_i\mathbf{P}_{\mathrm{par}(i)}$, the transformation law of a link. Assume Hypothesis G2.1, so that $\mathbf{P}_i \in \mathrm{GL}^+(K)$. (i) On a directed forest whose transitions are invertible with $\det \Omega_i > 0$, the blocks $\mathbf{P}_r = I$ at every root and $\mathbf{P}_i = \Omega_i\mathbf{P}_{\mathrm{par}(i)}$ otherwise lie in $\mathrm{GL}^+(K)$, and in the coordinates $z_i = \mathbf{P}_i^{-1}k_i$ the assembled precision lies in $\mathcal{I}(V)$ with

$$W_{i\,\mathrm{par}(i)} = \mathbf{P}_i^\top \Lambda_i \mathbf{P}_i, \qquad A_r = \Lambda_r \quad \text{at roots}, \qquad A_i = 0 \quad \text{otherwise.} \tag{G3.7}$$

Observation terms $\mathrm{blockdiag}(O_1, \dots, O_N)$ with $O_i \succeq 0$ keep it in $\mathcal{I}(V)$, with self terms $A_i + \mathbf{P}_i^\top O_i \mathbf{P}_i$. (ii) Let $\Lambda \in \mathbb{S}^{2K}_{++}$ be a two-agent precision with invertible off-diagonal block Λ_{12}. There are $\mathbf{P}_1, \mathbf{P}_2 \in \mathrm{GL}^+(K)$ with $\mathrm{diag}(\mathbf{P}_1, \mathbf{P}_2)^\top \Lambda\, \mathrm{diag}(\mathbf{P}_1, \mathbf{P}_2) \in \mathcal{I}(\{1, 2\})$ if and only if

$$(-1)^K \det \Lambda_{12} > 0. \tag{G3.8}$$

(iii) A directed forest with invertible transitions has a chart reached by frames in $\mathrm{GL}^+(K)$ in which its assembled precision lies in $\mathcal{I}(V)$ if and only if $\det \Omega_i > 0$ for every nonroot i. (iv) If orientation-reversing frames are admitted, $\mathbf{P}_i \in \mathrm{GL}(K)$, every two-agent precision with invertible Λ_{12} is reached. (v) The witness (G3.6) has $\det \widetilde{\Omega}_i = 1$ and is absorbed: with j its root, the blocks $\mathbf{P}_i = I$, $\mathbf{P}_j = \widetilde{\Omega}_i^{-1}$ give the pure agreement factor with $W = \widetilde{\Lambda}_i$, $A_i = 0$, and $A_j = \widetilde{\Omega}_i^{-\top}\widetilde{\Lambda}_j\widetilde{\Omega}_i^{-1}$ for the root precision $\widetilde{\Lambda}_j$. For even K the gain $-I$ has determinant one and is absorbed in the same way. [ESTABLISHED]

Proof. (o) In the new coordinates the residual is

$$\mathbf{P}_i z_i - \Omega_i \mathbf{P}_{\mathrm{par}(i)} z_{\mathrm{par}(i)} = \mathbf{P}_i(z_i - \mathbf{P}_i^{-1}\Omega_i\mathbf{P}_{\mathrm{par}(i)} z_{\mathrm{par}(i)}).$$

(i) Each $\mathbf{P}_i$ is a product of matrices of positive determinant, and the residual becomes $\mathbf{P}_i(z_i - z_{\mathrm{par}(i)})$, so the energy is $\frac{1}{2}\sum_i (z_i - z_{\mathrm{par}(i)})^\top \mathbf{P}_i^\top \Lambda_i \mathbf{P}_i (z_i - z_{\mathrm{par}(i)}) + \frac{1}{2}\sum_r z_r^\top \Lambda_r z_r$, which is (G3.2) with the parameters (G3.7); an observation term adds $z_i^\top \mathbf{P}_i^\top O_i \mathbf{P}_i z_i$. (ii) If the reframed off-diagonal block is $-W$ with $W \succeq 0$ symmetric, invertibility of Λ_{12} makes $W \succ 0$, and $\det \mathbf{P}_1 \det \Lambda_{12} \det \mathbf{P}_2 = (-1)^K \det W$ forces (G3.8). Conversely, put $\mathbf{P}_1 = I$ and $\mathbf{P}_2 = -\Lambda_{12}^{-1}\Lambda_{11}$, whose determinant $(-1)^K \det \Lambda_{11}/\det \Lambda_{12}$ is positive under (G3.8). The reframed off-diagonal block is $-\Lambda_{11}$, so $W = \Lambda_{11} \succ 0$ and $A_1 = 0$, while $A_2 = \Lambda_{11}\Lambda_{12}^{-\top}\Lambda_{22}\Lambda_{12}^{-1}\Lambda_{11} - \Lambda_{11}$ is the congruence by $\Lambda_{12}^{-1}\Lambda_{11}$ of the Schur complement $\Lambda_{22} - \Lambda_{12}^\top\Lambda_{11}^{-1}\Lambda_{12} \succ 0$. (iii) The block of the edge to the parent is

$-\Lambda_i\Omega_i$, and $(-1)^K \det(-\Lambda_i\Omega_i) = \det\Lambda_i \det\Omega_i$; a block reframing multiplies that determinant by $\det\mathbf{P}_i \det\mathbf{P}_{\mathrm{par}(i)} > 0$, so the necessity argument of (ii) applies edge by edge, and (i) gives sufficiency. (iv) The construction of (ii) with $\mathbf{P}_2 = -\Lambda_{12}^{-1}\Lambda_{11}$ uses no sign condition once $\mathbf{P}_2 \in \mathrm{GL}(K)$ is allowed. (v) With $z_j = \widetilde{\Omega}_i k_j$ the residual is $z_i - z_j$ and the root prior becomes $\frac{1}{2} z_j^\top \widetilde{\Omega}_i^{-\top} \widetilde{\Lambda}_j \widetilde{\Omega}_i^{-1} z_j$; these blocks differ from those of (i) by the common right factor $\widetilde{\Omega}_i^{-1}$. □

Proposition G3.9 (Chart-independent witnesses). (i) Under Hypothesis G2.1, a single receiving factor with invertible transition of negative determinant, for instance $\Omega_i = \mathrm{diag}(-1, 1, \ldots, 1)$, has an assembled precision outside $\mathcal{I}(\{i, j\})$ in every chart reached by block reframings in $\mathrm{GL}^+(K)$. (ii) For every cycle $i_0, i_1, \ldots, i_{n_\circ} = i_0$ of $n_\circ \geq 3$ distinct agents whose blocks $\Lambda_{i_t i_{t+1}}$ are invertible, the sign of $\prod_t \det\Lambda_{i_t i_{t+1}}$ is invariant under every block reframing with $\mathbf{P}_i \in \mathrm{GL}(K)$, and membership in $\mathcal{I}(V)$ forces it to equal $(-1)^{Kn_\circ}$. (iii) Let a child i have two parents j, j' with invertible transitions $\Omega_j, \Omega_{j'}$, receiving precision Λ_i, and proper root priors, the directed acyclic generalization of (G2.38) in which parent terms add. For odd K its assembled precision lies outside $\mathcal{I}(\{i, j, j'\})$ in every chart reached by block reframings in $\mathrm{GL}(K)$, orientation reversing or not. [ESTABLISHED]

Proof. (i) By (G3.8), since $(-1)^K \det(-\Lambda_i\Omega_i) = \det\Lambda_i \det\Omega_i < 0$. (ii) A block reframing multiplies $\det\Lambda_{i_t i_{t+1}}$ by $\det\mathbf{P}_{i_t} \det\mathbf{P}_{i_{t+1}}$, and each agent of the cycle occurs in two factors, so the product is multiplied by $\prod_t (\det\mathbf{P}_{i_t})^2 > 0$. In $\mathcal{I}(V)$ an invertible off-diagonal block is $-W_e$ with $W_e \succ 0$, so the product equals $(-1)^{Kn_\circ} \prod_e \det W_e$. (iii) The factor $\frac{1}{2}(k_i - \Omega_j k_j - \Omega_{j'} k_{j'})^\top \Lambda_i (k_i - \Omega_j k_j - \Omega_{j'} k_{j'})$ gives $\Lambda_{ij} = -\Lambda_i\Omega_j$, $\Lambda_{ij'} = -\Lambda_i\Omega_{j'}$, and $\Lambda_{jj'} = \Omega_j^\top \Lambda_i \Omega_{j'}$, all invertible, while the root priors add only to diagonal blocks. The product of the three determinants around the triangle is $(\det\Lambda_i)^3 (\det\Omega_j)^2 (\det\Omega_{j'})^2 > 0$, whereas (ii) requires the sign $(-1)^{3K} = -1$ for odd K. □

The gain F_i is a free generative parameter, not required to be symmetric, positive, or close to the identity. On forests the witnesses of Proposition G3.7 are properties of the link-trivializing chart, and the law-level obstructions are an orientation-reversing gain and, for odd K, a v-structure (Proposition G3.9). The interaction family is therefore not a consequence of the general directed model, and Hypothesis G3.13 imposes it in the link-trivializing coordinates.

G3.2.3 Exactly when the form holds

Proposition G3.10 (Exact global interaction condition). (i) The assembled prior precision (G2.39) lies in $\mathcal{I}(V)$, necessarily with the edge weights (G3.4), if and only if, for every nonroot i,

$$W_i := \Lambda_i\Omega_i \in \mathbb{S}_+^K, \tag{G3.9}$$

and, for every agent i,

$$A_i := \Lambda_i - W_i + \sum_{i' \in \mathrm{ch}(i)} \left(W_{i'} \Lambda_{i'}^{-1} W_{i'} - W_{i'} \right) \succeq 0, \tag{G3.10}$$

with the term $-W_i$ omitted at a root. (ii) With the observation kernel (G2.30), the belief conditional $p(k \mid m, o, X)$ has precision

$$\Lambda^{\mathrm{post}} = \Lambda^{\mathrm{prior}} + \mathrm{blockdiag}(O_1, \dots, O_N), \qquad O_i := \mathsf{C}_i^{b\top} (\Sigma_i^o)^{-1} \mathsf{C}_i^b \succeq 0, \tag{G3.11}$$

and $\Lambda^{\mathrm{post}} \in \mathcal{I}(V)$ if and only if (G3.9) holds and $A_i + O_i \succeq 0$ at every agent, while bridge terms, offsets, and the model part of each observation mean enter only its information vector. The prior condition of (i) is therefore sufficient for membership of the belief conditional. (iii) It is not necessary: for one scalar receiving factor $j \to i$ with $\Lambda_i = 1$, $\Omega_i = \frac{1}{2}$, root precision $\Lambda_j = \frac{1}{10}$, and an observation on j with $O_j = \frac{1}{2}$, the prior precision $\left(\begin{smallmatrix} 1 & -1/2 \\ -1/2 & 7/20 \end{smallmatrix}\right)$ is positive definite, with determinant $\frac{1}{10}$, and has $A_j = -\frac{3}{20}$, so it lies outside the family, while the belief conditional has $A_j + O_j = \frac{7}{20}$ and $A_i = \frac{1}{2}$ and lies inside it. (iv) Factorwise membership is sufficient but not necessary for global membership: for $\Lambda_i = 1$, $\Omega_i = \frac{1}{2}$, and $\Lambda_j = 1$, in the block order (i, j),

$$\Lambda = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{4} \end{pmatrix} = \begin{pmatrix} A_i + W & -W \\ -W & A_j + W \end{pmatrix}, \qquad W = \tfrac{1}{2}, \quad A_i = \tfrac{1}{2}, \quad A_j = \tfrac{3}{4}, \tag{G3.12}$$

is positive definite and in $\mathcal{I}(\{i, j\})$, although the receiving factor alone is not: its parent-side residual is $\Omega_i^2 \Lambda_i - W = -\frac{1}{4}$, offset by the root prior. [ESTABLISHED]

Proof. (i) Condition (G3.9) is exactly that each off-diagonal block has the form $-W$ with W symmetric positive semidefinite, and the off-diagonal blocks of a member determine its edge weights. Given it, $\Omega_{i'} = \Lambda_{i'}^{-1} W_{i'}$ and $\Omega_{i'}^\top \Lambda_{i'} \Omega_{i'} = W_{i'} \Lambda_{i'}^{-1} W_{i'}$ by symmetry of $W_{i'}$. The self term is the residual $A_i = \Lambda_{ii} - \sum_{j \sim i} W_{ij}$ over the edge to the parent, of weight W_i, and the edges to the children, of weights $W_{i'}$; substituting (G2.39) gives (G3.10). (ii) At fixed m and o, the k-dependent terms of the negative log joint density are the belief factors, in which m and the offsets enter only the means, and the observation factors, each contributing $\frac{1}{2} k_i^\top O_i k_i$ plus terms at most linear in k_i. The Hessian in k is therefore (G3.11), with the off-diagonal blocks of the prior and self-term residuals $A_i + O_i$. (iii) and (iv) are direct evaluation of (G3.10): in (iii) $A_j = \frac{1}{10} + \frac{1}{4} - \frac{1}{2}$ and $A_i = 1 - \frac{1}{2}$; in (iv) the leading principal minors are 1 and 1, and $A_j = 1 - \frac{1}{4}$. □

Proposition G3.11 (Edge-local characterization with singular gains allowed). Consider a single receiving factor $j \to i$ in isolation, contributing Λ_i to the ii block, $\Omega_i^\top \Lambda_i \Omega_i$ to the jj block, and $-\Lambda_i \Omega_i$ to the ij block. It lies in $\mathcal{I}(\{i, j\})$ if and only if

$$\Pi_i := \Lambda_i^{-1/2} \, (\Lambda_i \Omega_i) \, \Lambda_i^{-1/2} \text{ is an orthogonal projection,} \tag{G3.13}$$

equivalently if and only if $\Lambda_i \Omega_i$ is symmetric and Ω_i is idempotent, that is Ω_i is a Λ_i-orthogonal projection. Its self terms are then $A_i = \Lambda_i^{1/2} (I - \Pi_i) \Lambda_i^{1/2}$ and $A_j = 0$, and the full-rank case $\Pi_i = I$ is $\Omega_i = I$, the pure agreement factor with $W = \Lambda_i$. The gains satisfying the condition form a Lebesgue-null subset of $\mathbb{R}^{K \times K}$. [ESTABLISHED]

Proof. Membership requires $W := \Lambda_i \Omega_i$ symmetric and positive semidefinite, $\Lambda_i - W \succeq 0$, and $\Omega_i^\top \Lambda_i \Omega_i - W \succeq 0$. With W symmetric, Π_i is symmetric; the first two conditions read $0 \preceq \Pi_i \preceq I$, and

since $\Omega_i^\top \Lambda_i \Omega_i = W\Lambda_i^{-1}W = \Lambda_i^{1/2}\Pi_i^2\Lambda_i^{1/2}$, the third reads $\Pi_i^2 \succeq \Pi_i$. As $0 \preceq \Pi_i \preceq I$ implies $\Pi_i^2 \preceq \Pi_i$, the conditions hold exactly when $\Pi_i^2 = \Pi_i$. Conversely every orthogonal projection satisfies them, with $\Omega_i = \Lambda_i^{-1/2}\Pi_i\Lambda_i^{1/2}$ idempotent; the self terms follow by substitution. Idempotents of rank r form the conjugation orbit of $\mathrm{diag}(I_r, 0)$, whose stabilizer is $\mathrm{GL}(r) \times \mathrm{GL}(K-r)$, a smooth submanifold of dimension $K^2 - r^2 - (K-r)^2 = 2r(K-r) < K^2$; the finite union over r is null. □

Corollary G3.12 (Invertible-gain factor). In the link-trivializing chart, an isolated factor with invertible transition lies in $\mathcal{I}(\{i,j\})$ if and only if $\Omega_i = I$. When every chart reached by frames in $\mathrm{GL}^+(K)$ is allowed, an invertible factor with $\det \Omega_i > 0$ is instead absorbed by Proposition G3.8. [ESTABLISHED]

Proof. Proposition G3.11 makes Ω_i idempotent, and multiplying $\Omega_i^2 = \Omega_i$ by Ω_i^{-1} gives $\Omega_i = I$; the converse is its full-rank case. □

With singular gains a proper projection constrains only the components of the residual inside $\mathrm{range}(\Pi_i)$ and leaves the complement to the receiving-side self term.

Hypothesis G3.13 (Global interaction form). Where a result names this hypothesis, the precision it acts on belongs to $\mathcal{I}(V)$ in the link-trivializing coordinates. For the directed model (G2.38) the precision is that of the belief conditional (G3.11), and by Proposition G3.10(ii) the hypothesis holds exactly when (G3.9) holds and $A_i + O_i \succeq 0$ at every agent; the prior balance (G3.10) is sufficient but not necessary, and the receiving factors need not satisfy Proposition G3.11 ((G3.12)). The hypothesis licenses the aggregation closure of Section G7.2 and the operator arrow of Section G9.1, which are stated for the family, on a precision assembled from the directed model. It excludes precisions whose off-diagonal blocks are asymmetric or sign-incorrect in the link-trivializing chart, which the directed model does not prevent (Proposition G3.7) and which form all but a null set of precisions when $K \geq 2$ and $N \geq 2$ (Proposition G3.3).

Remark G3.14 (Factorwise-agreement realization). Requiring every receiving factor to satisfy Proposition G3.11, so that the interaction parameters are read from the generative model one factor at a time, is strictly stronger than Hypothesis G3.13: it admits Λ_i-orthogonal projections with singular gains and only $\Omega_i = I$ with invertible gains, and it excludes the global member (G3.12). [ESTABLISHED]

With latent links in the exponential chart, the matter precision of Section G11.4 is a function $\Lambda(\mathring{\omega})$ of the generators that belongs to this family at zero generators, and at zero information vector one half its log-determinant is, up to a constant, the induced action of Proposition G11.5.

G3.3 Nonemptiness

Proposition G3.15 (Constructive nonemptiness and the kernel). Under Hypothesis G3.4, let each edge of a graph (V, E) carry an agreement factor with residual $\mu_i - \Theta_{ij}\mu_j$ and link covariance $\Sigma_{ij}^{\mathrm{link}} \in \mathbb{S}_{++}^K$, and let each vertex carry a quadratic self term $A_i \succeq 0$ in the trivialized coordinates;

a self term is the precision of a proper full-dimensional prior only when $A_i \succ 0$. The assembled trivialized energy is

$$\tfrac{1}{2} z^\top \Lambda z = \tfrac{1}{2} \sum_i z_i^\top A_i z_i + \tfrac{1}{2} \sum_{(i,j)\in E} (z_i - z_j)^\top W_{ij} (z_i - z_j), \qquad W_{ij} = U_i^\top (\Sigma_{ij}^{\mathrm{link}})^{-1} U_i, \tag{G3.14}$$

so $\Lambda \in \mathcal{I}(V) \cap \mathbb{S}_+^{NK}$, every edge weight is positive definite, and

$$\ker \Lambda = \{ z : A_i z_i = 0 \text{ for every } i, \quad z_i = z_j \text{ for every } (i,j) \in E \}. \tag{G3.15}$$

If V_α are the connected components of (V, E), equivalently

$$\ker \Lambda = \bigoplus_\alpha \left\{ \mathbf{1}_{V_\alpha} \otimes v : v \in \bigcap_{i \in V_\alpha} \ker A_i \right\}. \tag{G3.16}$$

Consequently $\Lambda \succ 0$ exactly when $\bigcap_{i\in V_\alpha} \ker A_i = \{0\}$ for every component; one positive-definite self term per component suffices, and taking every $A_i \succ 0$ constructs an element of $\mathcal{I}(V) \cap \mathbb{S}_{++}^{NK}$ whose self terms are proper priors. [ESTABLISHED]

Proof. By (G3.3) the residual is $U_i(z_i - z_j)$, so the edge quadratic form has the matrix $U_i^\top (\Sigma_{ij}^{\mathrm{link}})^{-1} U_i$ between two copies of $z_i - z_j$, and that matrix is symmetric positive definite because U_i is invertible. Comparison with (G3.2) identifies the parameters. The energy vanishes exactly when $A_i z_i = 0$ at every vertex and, since every $W_{ij} \succ 0$, $z_i = z_j$ on every edge; the latter makes z constant on each component. A positive-semidefinite matrix is positive definite exactly when its kernel is trivial. □

Without trivialization the same agreement factor contributes the off-diagonal block $-(\Sigma_{ij}^{\mathrm{link}})^{-1}\Theta_{ij}$, which is symmetric only in special position; the trivialization is what places the construction in the family.

Chapter G4

Recognition Restrictions

This chapter specializes the restriction principle of Proposition 5.9 to a nondegenerate Gaussian target and asks two questions: what a restriction of the recognition law costs at a fixed model, and whether that cost can select a recognition structure. The cost of factorizing the recognition covariance across a partition is a determinant gap (Theorem G4.3) equal to the dual total correlation of the target (Corollary G4.5) and independent of the realized observation (Corollary G4.7); the cost of restricting the recognition mean is a quadratic form in the marginal precision of the restricted directions and carries the data (Proposition G4.8). Neither cost selects a structure: the gap is monotone under refinement and vanishes at the one-block partition (Proposition G4.16), a Gaussian instance of Open Problem 5.14. The costs of identifying agents under aggregation (Section G4.3) and the coarse maps of Chapter G7 use these results.

Every cost here is computed at one fixed model and target; nothing orders recognition structures across different models, which Proposition 5.13 shows is impossible without a common evidence-preserving channel. The block restriction constrains only the recognition law and does not assert that any covariance block of the target vanishes. [NOT-CLAIMED] The selection of a nondegenerate structure is Open Problem 5.14, and the selection of a partition for coarse-graining is Open Problem 10.20.

G4.1 The block restriction and its optimum

Throughout this chapter the target is Gaussian and nondegenerate,

$$p^\star = \mathcal{N}\left(\mu, \Lambda^{-1}\right), \qquad \Lambda \in \mathbb{S}^n_{++}, \qquad \mu = \Lambda^{-1}\mathsf{h}, \tag{G4.1}$$

with $n = NK$ for the belief sector of Chapter G3; nothing here uses the interaction family, so a joint belief–model target is equally admitted. For a partition $\mathcal{P}$ of the n coordinates into blocks of dimensions n_b, the restricted family is

$$\mathcal{Q}_{\mathcal{P}} = \left\{\mathcal{N}\left(\mu_Q, \Sigma_Q\right) : \mu_Q \in \mathbb{R}^n, \quad \Sigma_Q = \operatorname{blockdiag}_{b\in\mathcal{P}}\left(\Sigma_b\right), \quad \Sigma_b \in \mathbb{S}^{n_b}_{++}\right\}. \tag{G4.2}$$

A block may collect one agent's coordinates, split one agent's belief and model channels, or collect several agents; every statement is relative to $\mathcal{P}$.

Proposition G4.1 (Gaussian reverse relative entropy). For $Q = \mathcal{N}(\mu_Q, \Sigma_Q)$ with $\Sigma_Q \succ 0$,

$$D_{\mathrm{KL}}\left(Q \| p^\star\right) = \frac{1}{2}\left[\mathrm{Tr}\left(\Lambda\Sigma_Q\right) + \left(\mu_Q - \mu\right)^\top \Lambda \left(\mu_Q - \mu\right) - n - \log\det\Lambda - \log\det\Sigma_Q\right]. \tag{G4.3}$$

The mean enters only the quadratic term and the covariance only the trace and log determinant. [ESTABLISHED]

Proof. $\mathbb{E}_Q \log q = -\frac{1}{2}\log\det(2\pi e\Sigma_Q)$, and $-\log p^\star(y) = \frac{1}{2}(y-\mu)^\top\Lambda(y-\mu) + \frac{n}{2}\log 2\pi - \frac{1}{2}\log\det\Lambda$ with $\mathbb{E}_Q[(Y-\mu)^\top\Lambda(Y-\mu)] = \mathrm{Tr}(\Lambda\Sigma_Q) + (\mu_Q-\mu)^\top\Lambda(\mu_Q-\mu)$; adding the two cancels the $\frac{n}{2}\log 2\pi$ terms. □

Theorem G4.2 (Block optimum). Over $\mathcal{Q}_\mathcal{P}$ the objective (G4.3) has the unique minimizer

$$\mu_Q^\star = \mu, \qquad \Sigma_b^\star = \Lambda_{bb}^{-1} \quad \text{for every } b \in \mathcal{P}, \tag{G4.4}$$

so the optimal recognition mean equals the target mean although the covariance is restricted. Holding every factor but b fixed, the coordinate maximizer of the bound is Gaussian with covariance Λ_{bb}^{-1} and mean

$$\mu_b^{\mathrm{new}} = \Lambda_{bb}^{-1}\left(\mathsf{h}_b - \sum_{c\neq b}\Lambda_{bc}\mu_c\right), \tag{G4.5}$$

which is (G2.12) read as an update, and every fixed point of the full sweep satisfies $\Lambda\mu_Q = \mathsf{h}$, hence $\mu_Q = \mu$. [ESTABLISHED]

Proof. On the block-diagonal set $\mathrm{Tr}(\Lambda\Sigma_Q) = \sum_b \mathrm{Tr}(\Lambda_{bb}\Sigma_b)$ and $\log\det\Sigma_Q = \sum_b \log\det\Sigma_b$, so

$$D_{\mathrm{KL}} = \frac{1}{2}\left(\mu_Q - \mu\right)^\top\Lambda\left(\mu_Q-\mu\right) + \frac{1}{2}\sum_{b\in\mathcal{P}}\left[\mathrm{Tr}\left(\Lambda_{bb}\Sigma_b\right) - \log\det\Sigma_b\right] + \text{const}, \tag{G4.6}$$

a strictly convex quadratic in μ_Q plus one strictly convex term per block, since $-\log\det$ is strictly convex on the positive-definite cone [7, §3.1.5]. The parameter set is convex, so a minimizer is unique, and differentiating on the open cone gives

$$\nabla_{\mu_Q} D_{\mathrm{KL}} = \Lambda\left(\mu_Q - \mu\right), \qquad \nabla_{\Sigma_b} D_{\mathrm{KL}} = \tfrac{1}{2}\left(\Lambda_{bb} - \Sigma_b^{-1}\right), \tag{G4.7}$$

whose common zero is (G4.4); $\Lambda_{bb} \succ 0$ as a principal submatrix. For the coordinate update, the terms of (G4.6) involving block b at fixed other means are minimized at (G4.5), and the fixed-point equations stacked over blocks read $\Lambda\mu_Q = \mathsf{h}$ [5, 6]. □

The covariance is fixed in one step and the mean only at a fixed point of the sweep.

Theorem G4.3 (Determinant gap). Let $\mathcal{L}_\mathcal{P}^\star$ be the bound at the optimum of Theorem G4.2. Then

$$\log p_\theta(o \mid X) - \mathcal{L}_\mathcal{P}^\star = \frac{1}{2}\left[\sum_{b\in\mathcal{P}}\log\det\Lambda_{bb} - \log\det\Lambda\right] \geq 0, \tag{G4.8}$$

with equality if and only if $\Lambda_{bc} = 0$ for every pair of distinct blocks. [ESTABLISHED]

Proof. At the optimum $\mathrm{Tr}(\Lambda\Sigma_Q^\star) = \sum_b n_b = n$, $\log\det\Sigma_Q^\star = -\sum_b \log\det\Lambda_{bb}$, and the mean term vanishes, so

$$\inf_{Q\in\mathcal{Q}_\mathcal{P}} D_{\mathrm{KL}}\left(Q\|p^\star\right) = \frac{1}{2}\left[\sum_{b\in\mathcal{P}} \log\det\Lambda_{bb} - \log\det\Lambda\right], \tag{G4.9}$$

and (5.23) converts this into (G4.8). For a first block and its complement, Lemma G2.6(iv) gives

$$\det\Lambda = \det\Lambda_{11}\cdot\det\left(\Lambda_{22} - \Lambda_{21}\Lambda_{11}^{-1}\Lambda_{12}\right), \tag{G4.10}$$

with the Schur complement $\preceq \Lambda_{22}$. If $0 \prec S \preceq T$, the eigenvalues of $T^{-1/2}ST^{-1/2}$ lie in $(0,1]$, so $\det S \le \det T$ with equality only when $S = T$; hence $\det\Lambda \le \det\Lambda_{11}\det\Lambda_{22}$, with equality exactly when $\Lambda_{21}\Lambda_{11}^{-1}\Lambda_{12} = 0$, that is $\Lambda_{12} = 0$. Induction on the number of blocks gives the claim, which is Fischer's inequality [17, §7.8]. □

Proposition G4.4 (Canonical-correlation form and telescoping). For a two-block precision with $\Lambda_{11}\in\mathbb{S}^{n_1}_{++}$, $\Lambda_{22}\in\mathbb{S}^{n_2}_{++}$, and cross block Λ_{12}, write

$$\Lambda = \begin{pmatrix}\Lambda_{11} & \Lambda_{12}\\ \Lambda_{12}^\top & \Lambda_{22}\end{pmatrix}, \qquad \mathcal{K} := \Lambda_{11}^{-1/2}\Lambda_{12}\Lambda_{22}^{-1/2}. \tag{G4.11}$$

Then

$$\Lambda_{22} - \Lambda_{12}^\top\Lambda_{11}^{-1}\Lambda_{12} = \Lambda_{22}^{1/2}\left(I - \mathcal{K}^\top\mathcal{K}\right)\Lambda_{22}^{1/2}, \qquad \det\Lambda = \det\Lambda_{11}\det\Lambda_{22}\det\left(I - \mathcal{K}^\top\mathcal{K}\right). \tag{G4.12}$$

The singular values $\check{\rho}_a$ of $\mathcal{K}$ are the canonical correlations between the two blocks under the target, $0 \le \check{\rho}_a < 1$, and the two-block gap is the mutual information of the blocks,

$$\Delta(\Lambda_{11},\Lambda_{22};\Lambda_{12}) = -\frac{1}{2}\log\det\left(I - \mathcal{K}^\top\mathcal{K}\right) = -\frac{1}{2}\sum_{a=1}^{\min(n_1,n_2)}\log\left(1-\check{\rho}_a^2\right). \tag{G4.13}$$

For ordered blocks $b_1,\dots,b_m$ of $\mathcal{P}$ with cumulative sets $\bar{b}_k := b_1\cup\cdots\cup b_k$, the increment

$$\delta_k = \frac{1}{2}\left[\log\det\Lambda_{\bar{b}_{k-1}\bar{b}_{k-1}} + \log\det\Lambda_{b_kb_k} - \log\det\Lambda_{\bar{b}_k\bar{b}_k}\right], \qquad k \ge 2, \tag{G4.14}$$

is the conditional mutual information of $Y_{\bar{b}_{k-1}}$ and Y_{b_k} given the coordinates outside $\bar{b}_k$, and the intermediate determinants cancel:

$$\sum_{k=2}^{m}\delta_k = \frac{1}{2}\left[\sum_{k=1}^{m}\log\det\Lambda_{b_kb_k} - \log\det\Lambda\right]. \tag{G4.15}$$

The total does not depend on the order; the increments can. [ESTABLISHED]

Proof. Substituting $\Lambda_{12} = \Lambda_{11}^{1/2} \mathcal{K} \Lambda_{22}^{1/2}$ into the Schur complement gives the first identity of (G4.12), and (G4.10) the second; positive definiteness of the Schur complement gives $\check{\rho}_a < 1$. The block-diagonal change of variables that whitens Λ_{11} and Λ_{22} carries Λ to $\left(\begin{smallmatrix} I & \mathcal{K} \\ \mathcal{K}^\top & I \end{smallmatrix}\right)$, and rotations within the blocks by the singular vectors of $\mathcal{K}$ split it into two-by-two precisions $\left(\begin{smallmatrix} 1 & \check{\rho}_a \\ \check{\rho}_a & 1 \end{smallmatrix}\right)$ and identities; each pair has correlation $-\check{\rho}_a$, and canonical correlations are unchanged by invertible changes of variables within each block. The mutual information of a Gaussian pair with correlation $\pm\check{\rho}_a$ is $-\frac{1}{2}\log(1-\check{\rho}_a^2)$. The precision of the conditional law of $Y_{\bar{b}_k}$ given the coordinates outside $\bar{b}_k$ is $\Lambda_{\bar{b}_k \bar{b}_k}$, so δ_k is the two-block gap of that conditional law, which is its mutual information; summing (G4.14) telescopes. □

Corollary G4.5 (The determinant gap is the dual total correlation). With h the differential entropy under $p^\star$,

$$\frac{1}{2}\left[\sum_{b\in\mathcal{P}} \log\det \Lambda_{bb} - \log\det \Lambda\right] = h(Y) - \sum_{b\in\mathcal{P}} h\left(Y_b \mid Y_{-b}\right), \tag{G4.16}$$

the dual total correlation of the blocks [15]. For two blocks it is their mutual information. For three or more blocks it generally differs from the total correlation, which for a Gaussian law with covariance Σ is

$$\sum_b h(Y_b) - h(Y) = \tfrac{1}{2}\left(\sum_b \log\det \Sigma_{bb} - \log\det \Sigma\right); \tag{G4.17}$$

for three scalar blocks with unit variances and all correlations $\frac{1}{2}$, the total correlation is $\frac{1}{2}\log 2$ and the dual total correlation is $\frac{1}{2}\log\frac{27}{16}$. [ESTABLISHED]

Proof. $h(Y) = \frac{1}{2}\log\det(2\pi e\Lambda^{-1})$ and, by (G2.12), $h(Y_b \mid Y_{-b}) = \frac{1}{2}\log\det(2\pi e\Lambda_{bb}^{-1})$; the factors $2\pi e$ cancel because $\sum_b n_b = n$, which gives (G4.16), and the same substitution with marginal entropies gives (G4.17). The two-block case is Proposition G4.4. In the example $\det\Sigma = \frac{1}{2}$ and each diagonal entry of $\Lambda = \Sigma^{-1}$ is $\frac{3}{2}$, with $\det\Lambda = 2$. □

The block restriction changes only the candidate Q: it sets $\Sigma^\star_{bc} = 0$ for $b \neq c$ and $\Sigma^\star_{bb} = \Lambda_{bb}^{-1} \preceq (\Lambda^{-1})_{bb}$ (Corollary G2.7), and the gap measures a dependence functional of the fixed target, not a covariance of the candidate.

G4.2 Observation independence and its scope

Proposition G4.6 (Conditional precision is observation free). Suppose the joint law of (o, Y) given X is Gaussian with precision $\widehat{\Lambda}$ and information vector $\widehat{\mathsf{h}}$, both functions of (θ, X) alone. Then the conditional law of Y given o has

$$\Lambda = \widehat{\Lambda}_{YY}, \qquad \mu = \Lambda^{-1}\left(\widehat{\mathsf{h}}_Y - \widehat{\Lambda}_{Yo} o\right), \tag{G4.18}$$

so Λ does not depend on the realized o while μ is affine in it. [ESTABLISHED]

Proof. At fixed o the exponent of the joint density has quadratic part $-\frac{1}{2}y^\top \widehat{\Lambda}_{YY} y$ and linear part $(\widehat{\mathrm{h}}_Y - \widehat{\Lambda}_{Yo} o)^\top y$, which by (G2.2) are the natural parameters of the conditional. □

Corollary G4.7 (Observation independence of the determinant gap). Under the hypothesis of Proposition G4.6, the gap of Theorem G4.3 is a function of (θ, X) and $\mathcal{P}$ alone. [ESTABLISHED]

Proof. The gap depends on o only through Λ, because the mean term vanishes at the optimum (Theorem G4.2), and Λ is observation free. □

Both hypotheses enter the proof: without an observation-independent joint precision, (G4.18) is unavailable, and a restriction that constrains the mean leaves the quadratic term nonzero at the optimum, as the next proposition computes.

Proposition G4.8 (Cost of a mean restriction). Let $[B\ B_\perp]$ be orthogonal with $B \in \mathbb{R}^{n\times r}$, and restrict the recognition family to

$$\mathcal{Q}_B = \left\{ \mathcal{N}(\mu_Q, \Sigma_Q) : \mu_Q \in \mathrm{range}(B), \quad \Sigma_Q \in \mathbb{S}^n_{++} \right\}. \tag{G4.19}$$

With $\mu_\perp := B_\perp^\top \mu$,

$$\inf_{Q\in\mathcal{Q}_B} D_{\mathrm{KL}}(Q\|p^\star) = \frac{1}{2}\mu_\perp^\top \left(B_\perp^\top \Lambda^{-1} B_\perp \right)^{-1} \mu_\perp, \tag{G4.20}$$

attained at $\Sigma_Q^\star = \Lambda^{-1}$ and $\mu_Q^\star = \mu - \Lambda^{-1}B_\perp(B_\perp^\top\Lambda^{-1}B_\perp)^{-1}\mu_\perp$. The matrix $(B_\perp^\top\Lambda^{-1}B_\perp)^{-1}$ is the marginal precision of the restricted directions, which differs from the restricted precision $B_\perp^\top\Lambda B_\perp$ by the correction of Corollary G2.8 except under Λ-orthogonality. [ESTABLISHED]

Proof. By the separation in Proposition G4.1, the covariance term is minimized at $\Sigma_Q^\star = \Lambda^{-1}$, where it vanishes. It remains to minimize $\frac{1}{2}v^\top\Lambda v$, $v = \mu_Q - \mu$, subject to $B_\perp^\top v = -\mu_\perp$; stationarity gives $v = \Lambda^{-1}B_\perp\omega$ and the constraint gives $\omega = -(B_\perp^\top\Lambda^{-1}B_\perp)^{-1}\mu_\perp$, so the value is $\frac{1}{2}\omega^\top(B_\perp^\top\Lambda^{-1}B_\perp)\omega$, which is (G4.20). The identification uses $\mathrm{Cov}(B_\perp^\top Y) = B_\perp^\top\Lambda^{-1}B_\perp$. □

Corollary G4.9 (A mean restriction carries the data). Under the hypothesis of Proposition G4.6, the cost (G4.20) is quadratic in the realized observation, and it is observation free if and only if $B_\perp^\top\Lambda^{-1}\widehat{\Lambda}_{Yo} = 0$. [ESTABLISHED]

Proof. $\mu_\perp = B_\perp^\top\Lambda^{-1}(\widehat{\mathrm{h}}_Y - \widehat{\Lambda}_{Yo}o)$ is affine in o by (G4.18), and the quadratic form is positive definite. □

Corollary G4.10 (Costs of combined restrictions add). Restricting the mean to $\mathrm{range}(B)$ and the covariance to (G4.2) together has cost

$$\frac{1}{2}\left[\sum_{b\in\mathcal{P}} \log\det\Lambda_{bb} - \log\det\Lambda\right] + \frac{1}{2}\mu_\perp^\top\left(B_\perp^\top\Lambda^{-1}B_\perp\right)^{-1}\mu_\perp, \tag{G4.21}$$

the first term structural and the second data dependent. [ESTABLISHED]

Proof. (G4.3) has no term coupling μ_Q and Σ_Q, and the two constraints act on disjoint parameter blocks, so the minima add; Theorem G4.3 and proposition G4.8 supply them. □

G4.3 Costs of identifying agents

Let $S \in \mathbb{R}^{n \times n_c K}$ be the matrix of a hard aggregation of the N agents into n_c clusters (Section G7.2), so that $\operatorname{range} S$ consists of the configurations constant on each cluster; let B have orthonormal columns spanning $\operatorname{range} S$, so $r = n_c K$, complete it to an orthogonal $[B\ B_\perp]$, and write $p = \mathcal{N}(\mu, \Lambda^{-1})$ for the target.

Corollary G4.11 (Mean-tie cost and factorization gap of an aggregation). The mean-tie family, with recognition mean in $\operatorname{range} S$ and unrestricted covariance, has optimized cost

$$\mathbb{D}_{\text{tie}} = \inf_{\mu_Q \in \operatorname{range} S,\ \Sigma_Q \succ 0} D_{\mathrm{KL}}\big(\mathcal{N}(\mu_Q, \Sigma_Q) \| p\big) = \frac{1}{2} \mu_\perp^\top \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} \mu_\perp, \qquad \mu_\perp = B_\perp^\top \mu, \tag{G4.22}$$

whose marginal precision obeys

$$\left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} \preceq B_\perp^\top \Lambda B_\perp, \tag{G4.23}$$

with equality exactly when $B^\top \Lambda B_\perp = 0$, so the restricted precision overstates the cost otherwise. Covariance factorization across the cluster blocks I has gap

$$\mathbb{D}_{\text{fact}} = \frac{1}{2} \left[\sum_I \log\det \Lambda_{II} - \log\det \Lambda \right]. \tag{G4.24}$$

The mean-tie cost vanishes at the finest partition and the factorization gap at the coarsest; the gap decreases under merging. [ESTABLISHED]

Proof. Proposition G4.8 with B spanning $\operatorname{range} S$, Corollary G2.8, and Theorem G4.3 with the cluster blocks give the three displays. At the finest partition $\operatorname{range} S = \mathbb{R}^n$ and $B_\perp$ is empty; at the coarsest the one block is everything; monotonicity is Proposition G4.14. □

Proposition G4.12 (Regularized identification cost). Assume $r < n$, so at least one cluster contains two agents. A recognition law supported on $\operatorname{range} S$ is singular with respect to p, so its relative entropy to p is $+\infty$. With $\mu_Q^\star$ the mean optimizer of (G4.22) and the full-rank regularization

$$\Sigma_\varepsilon = B(B^\top \Lambda B)^{-1} B^\top + \varepsilon B_\perp B_\perp^\top, \qquad q_\varepsilon = \mathcal{N}(\mu_Q^\star, \Sigma_\varepsilon), \qquad \varepsilon > 0, \tag{G4.25}$$

one has

$$\begin{aligned} \mathbb{D}_\varepsilon := D_{\mathrm{KL}}(q_\varepsilon \| p) = \mathbb{D}_{\text{tie}} + \frac{n-r}{2} \log \frac{1}{\varepsilon} - \frac{n-r}{2} + \frac{1}{2} \log\det(B_\perp^\top \Lambda^{-1} B_\perp) \\ + \frac{\varepsilon}{2} \operatorname{Tr}(B_\perp^\top \Lambda B_\perp), \end{aligned} \tag{G4.26}$$

so $\mathbb{D}_\varepsilon \to +\infty$ as $\varepsilon \downarrow 0$. Identification of agents is therefore a generative construction and not a finite recognition restriction; the identified law is the conditional of p on $\operatorname{range} S$ (Proposition G7.4). [ESTABLISHED]

Proof. In the basis $[B\ B_\perp]$, $\det \Sigma_\varepsilon = \varepsilon^{n-r}/\det(B^\top \Lambda B)$ and $\operatorname{Tr}(\Lambda\Sigma_\varepsilon) = r + \varepsilon \operatorname{Tr}(B_\perp^\top \Lambda B_\perp)$. Substituting into (G4.3), with the optimized mean term $\mathbb{D}_{\text{tie}}$, and using $\det \Lambda = \det(B^\top \Lambda B)/\det(B_\perp^\top \Lambda^{-1} B_\perp)$ from Lemma G2.6(iv) gives (G4.26). □

Proposition G4.13 (The hard trace changes the evidence). (i) Precomposing the joint energy of latents and observations with $y = Sc$ generally changes the observation marginal: for independent standard Gaussian y_1, y_2 and $o = y_1 + \xi$ with independent standard Gaussian ξ, $\operatorname{Var}(o) = 2$, whereas the normalized joint precomposed with $y_1 = y_2 = c$ has precision $\left(\begin{smallmatrix} 3 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ in (c, o) and $\operatorname{Var}(o) = \frac{3}{2}$. (ii) Under the hypothesis of Proposition G4.6, with $\Lambda = \widehat{\Lambda}_{YY}$ at fixed (θ, X), the deterministic linear map $P_\Lambda = (S^\top \Lambda S)^{-1} S^\top \Lambda$ carries the posterior $\mathcal{N}(\mu, \Lambda^{-1})$ at every observation to the posterior of the precomposed joint, and $\mathrm{id}_O \otimes P_\Lambda$ is a recognition-independent channel that preserves the evidence, so Corollary 10.8 applies to it. [ESTABLISHED]

Proof. (i) The fine joint has $\operatorname{Var}(o) = \operatorname{Var}(y_1) + 1 = 2$. The energy $\frac{1}{2}(y_1^2 + y_2^2 + (o - y_1)^2)$ at $y_1 = y_2 = c$ is $\frac{1}{2}(3c^2 - 2co + o^2)$, whose normalized law has the displayed precision and $\operatorname{Var}(o) = 3/(3 \cdot 1 - 1) = \frac{3}{2}$. (ii) With $\Lambda_c = S^\top \Lambda S$, the pushed posterior is $\mathcal{N}(\Lambda_c^{-1} S^\top \Lambda \mu, \Lambda_c^{-1})$. Precomposing the energy with $y = Sc$ at fixed o leaves the quadratic part $\frac{1}{2} c^\top \Lambda_c c$ and the linear part $(S^\top \mathsf{h})^\top c$, with $\mathsf{h} = \Lambda\mu$, so the precomposed posterior is the same law. The map P_Λ depends on (θ, X) only, so the channel is fixed across observations and recognition laws, and it acts on latents only, leaving the observation marginal unchanged. □

A single kernel independent of the precision does not exist when $\operatorname{range}(\Lambda S)$ varies; the Galerkin coarse law is the precision-dependent pushforward of Proposition G7.5.

G4.4 Comparing recognition structures at fixed model

Proposition G4.14 (Refinement monotonicity for every comparable pair). Let $\mathcal{P}'$ refine $\mathcal{P}$, so that every block of $\mathcal{P}'$ lies in a block of $\mathcal{P}$. Then $\mathcal{Q}_{\mathcal{P}'} \subseteq \mathcal{Q}_{\mathcal{P}}$ and

$$\mathcal{L}^\star_{\mathcal{P}} - \mathcal{L}^\star_{\mathcal{P}'} = \sum_{b \in \mathcal{P}} \frac{1}{2} \left[\sum_{b' \in \mathcal{P}' :\, b' \subseteq b} \log\det \Lambda_{b'b'} - \log\det \Lambda_{bb} \right] \geq 0, \tag{G4.27}$$

each summand being the gap of Λ_{bb} under the partition induced on b, and vanishing exactly when Λ_{bb} is block diagonal under it. [ESTABLISHED]

Proof. A matrix block diagonal under $\mathcal{P}'$ is block diagonal under $\mathcal{P}$. Subtracting two instances of (G4.8) cancels $\log\det \Lambda$ and regroups by coarse block; each bracket is Theorem G4.3 applied to $\Lambda_{bb} \succ 0$. □

Proposition G4.15 (Non-nested structures are not ordered). Outside the refinement order the bounds are ordered by no structural feature such as the number or sizes of blocks. For $n = 3$ take

$$\Lambda(a,b) = \begin{pmatrix} 1 & a & 0 \\ a & 1 & b \\ 0 & b & 1 \end{pmatrix}, \qquad a^2 + b^2 < 1, \tag{G4.28}$$

and the partitions $\mathcal{P}_1 = \{\{1,2\},\{3\}\}$ and $\mathcal{P}_2 = \{\{1\},\{2,3\}\}$, which have equal block sizes and neither of which refines the other. Their gaps are $\frac{1}{2}\log\frac{1-a^2}{1-a^2-b^2}$ and $\frac{1}{2}\log\frac{1-b^2}{1-a^2-b^2}$, so the order is decided by whether $|a|$ exceeds $|b|$ and reverses when they are exchanged; at $(a,b) = (0.8, 0.2)$ the values are 0.0589 and 0.5493. [ESTABLISHED]

Proof. The leading minors are 1, $1 - a^2$, and $1 - a^2 - b^2$. The gaps follow from (G4.8) with $\det\Lambda = 1 - a^2 - b^2$, $\det\Lambda_{\{1,2\}\{1,2\}} = 1 - a^2$, $\det\Lambda_{\{2,3\}\{2,3\}} = 1 - b^2$, and unit diagonal entries; the values are $\frac{1}{2}\log(0.36/0.32)$ and $\frac{1}{2}\log(0.96/0.32)$. □

A recognition structure is therefore well chosen relative to a given coupling matrix, and any partition-selection rule is an optimization over structures rather than a reading of a scalar such as the block count.

Proposition G4.16 (The gap does not select a structure). Along any nested refinement chain of partitions, each refining the next, that ends at the one-block partition, the gap of (G4.8) is nonincreasing under coarsening and vanishes at the one-block partition for every $\Lambda \succ 0$; it supplies no model-independent order of nonnested partitions. Minimizing the gap therefore selects the unrestricted family and returns no information about structure. [ESTABLISHED]

Proof. At the one-block partition $\sum_b \log\det\Lambda_{bb} = \log\det\Lambda$. Monotonicity along the chain is Proposition G4.14, and Proposition G4.15 supplies the nonnested boundary. □

This is the Gaussian instance of Open Problem 5.14: the determinant gap is minimized at the trivial partition, and the mean-restriction cost (G4.20) is a cost at a fixed structure.

Chapter G5

Fisher Geometry and Invariants

Which geometric statements about the Gaussian recognition family are properties of its laws, and which are properties of a chart or of a frame? This chapter gives two answers. The first concerns the Fisher metric of the family. Its mean sector equals the precision Λ in the moment chart, and in the expectation chart it equals Λ on the quotient by the second-moment directions but not on the coordinate mean directions, where it exceeds Λ by a term governed by the squared Mahalanobis norm of the mean (Proposition G5.2). The second concerns the interaction precision. The generalized spectrum of the pencil (L, Λ) is invariant under every block reframing whenever the pencil is regular (Proposition G5.5), whereas ordinary eigenvalues and single subspace determinants are frame dependent as soon as the block reframing group contains the positive scalars (Proposition G5.9). The pencil results type the singular boundary of the interaction family in Chapter G9, and the invariance of agent-aligned law functionals is what makes the determinant gap of Theorem G4.3 a quantity about the model. The Fisher metric is the tensor adopted in Chapter 7; its Hessian and dual-inverse forms on the exponential charts are (G2.9) and (G2.10), and the four charts and the Frobenius pairing of symmetric-matrix coordinates are fixed in Section G2.2.1.

The chapter works with the Gaussian recognition family on the assembled latent space $\mathbb{R}^n$, $n = NK$, in information parameters (h, Λ) with $\Lambda \succ 0$, moments $\mu = \Lambda^{-1}\mathsf{h}$ and $\Sigma = \Lambda^{-1}$, and second moment $M_2 = \Sigma + \mu\mu^\top$. It does not assert that a rescaled coarse-graining step contracts an information-geometric quantity along a history on which the precision changes, that the dissipation identity of the fixed-covariance relaxation of Section G8.2 is derived from the fixed-precision pushforward of Proposition G7.5 or conversely, or that an update other than that fixed-covariance mean flow is a natural-gradient flow. [NOT-CLAIMED] The monotonicity question beyond a fixed precision is Open Problem G9.4.

G5.1 The chart correction

It is often said that the mean sector of the Gaussian Fisher metric is the precision. The statement is true in the moment chart (μ, Σ). In the expectation chart (μ, M_2) it is true for the metric induced on the quotient by the second-moment directions and false for the restriction to the coordinate mean directions. The two propositions below compute both readings.

Proposition G5.1 (Fisher metric in the moment chart). In the moment chart (μ, Σ), with $\Lambda = \Sigma^{-1}$, the score of the Gaussian family at y in the direction of a tangent vector $(\dot\mu, \dot\Sigma)$, with $\dot\Sigma$

symmetric, is

$$\partial_{(\dot\mu,\dot\Sigma)} \log q(y) = \dot\mu^\top \Lambda(y-\mu) + \tfrac{1}{2}(y-\mu)^\top \Lambda\dot\Sigma\Lambda(y-\mu) - \tfrac{1}{2}\operatorname{Tr}(\Lambda\dot\Sigma), \tag{G5.1}$$

and the Fisher metric is block diagonal,

$$g^F_{(\mu,\Sigma)}\big((\dot\mu_1,\dot\Sigma_1),(\dot\mu_2,\dot\Sigma_2)\big) = \dot\mu_1^\top\Lambda\dot\mu_2 + \tfrac{1}{2}\operatorname{Tr}\left(\Lambda\dot\Sigma_1\Lambda\dot\Sigma_2\right). \tag{G5.2}$$

Every affine map $(\mu,\Sigma)\mapsto(Q\mu+\mathrm{v}, Q\Sigma Q^\top)$ with $Q\in\mathrm{GL}(n)$ and $\mathrm{v}\in\mathbb{R}^n$ is an isometry of (G5.2). [ESTABLISHED]

Proof. Write $e=y-\mu$, so that $\log q(y) = -\frac{1}{2}e^\top\Lambda e + \frac{1}{2}\log\det\Lambda - \frac{n}{2}\log 2\pi$. A displacement $(\dot\mu,\dot\Sigma)$ moves e by $-\dot\mu$ and Λ by $-\Lambda\dot\Sigma\Lambda$, and differentiating gives (G5.1). Under $e\sim\mathcal{N}(0,\Sigma)$ the linear term has second moment $\dot\mu^\top\Lambda\Sigma\Lambda\dot\mu = \dot\mu^\top\Lambda\dot\mu$, the cross moment of the linear and quadratic terms is a third central moment of a centered Gaussian and vanishes, and the Gaussian variance of a quadratic form gives the quadratic term the variance $\frac{1}{4}\operatorname{Var}(e^\top\Lambda\dot\Sigma\Lambda e) = \frac{1}{2}\operatorname{Tr}(\Lambda\dot\Sigma\Lambda\Sigma\Lambda\dot\Sigma\Lambda\Sigma) = \frac{1}{2}\operatorname{Tr}(\Lambda\dot\Sigma\Lambda\dot\Sigma)$. Polarization gives (G5.2). Under the affine map the tangent vector becomes $(Q\dot\mu, Q\dot\Sigma Q^\top)$ and the precision becomes $Q^{-\top}\Lambda Q^{-1}$, and both terms of (G5.2) are unchanged. □

This is the Fisher metric of the multivariate normal model whose Riemannian geometry is developed by Skovgaard [28]; the form for a general parameterization and the affine isometry are Eqs. (7) and (8) of Pinele et al. [24].

Proposition G5.2 (Fisher metric in the expectation chart). In the expectation chart (μ, M_2) the Fisher metric has the following properties. (i) Its mean block, cross block, and second block are

$$g_{[\mu\mu]} = \left(1+\mu^\top\Lambda\mu\right)\Lambda + \Lambda\mu\mu^\top\Lambda, \tag{G5.3}$$

$$g_{[\mu M_2]}\big(\dot\mu,\dot M_2\big) = -\dot\mu^\top\Lambda\dot M_2\Lambda\mu, \tag{G5.4}$$

and $g_{[M_2M_2]}(\dot M_2,\dot M_2') = \frac{1}{2}\operatorname{Tr}(\Lambda\dot M_2\Lambda\dot M_2')$, so the metric is block diagonal in this chart exactly when $\mu=0$. (ii) The metric induced on the quotient of the tangent space by the second-moment directions, equivalently the restriction of g^F to the Fisher-orthogonal complement of those directions read in the coordinate μ, is the Schur complement

$$g^{\mathrm{quot}}_\mu = g_{[\mu\mu]} - g_{[\mu M_2]}g^{-1}_{[M_2M_2]}g_{[M_2\mu]} = \Lambda. \tag{G5.5}$$

(iii) With $\hat\mu=\Lambda^{1/2}\mu$,

$$g_{[\mu\mu]} = \Lambda^{1/2}\left[\left(1+\|\hat\mu\|^2\right)I_n + \hat\mu\hat\mu^\top\right]\Lambda^{1/2}, \tag{G5.6}$$

so the eigenvalues of $\Lambda^{-1}g_{[\mu\mu]}$ are $1+\mu^\top\Lambda\mu$ with multiplicity $n-1$ and $1+2\mu^\top\Lambda\mu$ with multiplicity one; hence $g_{[\mu\mu]}\succeq\Lambda$, with equality if and only if $\mu=0$. [ESTABLISHED]

Proof. (i) The chart change is $\Sigma = M_2 - \mu\mu^\top$, so a displacement $(\dot\mu, \dot M_2)$ induces

$$\delta\mu = \dot\mu, \qquad \delta\Sigma = \dot M_2 - \left(\dot\mu\mu^\top + \mu\dot\mu^\top\right), \tag{G5.7}$$

and the metric in this chart is the pullback of (G5.2) along (G5.7). For the mean block put $\dot M_2 = 0$; cycling each of the four traces gives

$$\mathrm{Tr}\left(\Lambda\delta\Sigma\Lambda\delta\Sigma\right) = 2\left(\dot\mu^\top\Lambda\mu\right)^2 + 2\left(\mu^\top\Lambda\mu\right)\left(\dot\mu^\top\Lambda\dot\mu\right), \tag{G5.8}$$

and adding the mean term of (G5.2) to one half of (G5.8) gives

$$\dot\mu^\top\Lambda\dot\mu + \left(\mu^\top\Lambda\mu\right)\left(\dot\mu^\top\Lambda\dot\mu\right) + \left(\dot\mu^\top\Lambda\mu\right)^2 = \dot\mu^\top\left[\left(1+\mu^\top\Lambda\mu\right)\Lambda + \Lambda\mu\mu^\top\Lambda\right]\dot\mu, \tag{G5.9}$$

which is (G5.3). Pairing $(\dot\mu, 0)$ with $(0, \dot M_2)$, only the covariance parts meet, and they give $\frac{1}{2}\,\mathrm{Tr}\big(\Lambda[-(\dot\mu\mu^\top + \mu\dot\mu^\top)]\Lambda\dot M_2\big) = -\dot\mu^\top\Lambda\dot M_2\Lambda\mu$. For $\dot\mu = 0$ nothing changes. The cross block vanishes for all arguments exactly when $\Lambda^2\mu = 0$, that is when $\mu = 0$. (ii) The metric is positive definite on this minimal family, so the second block and its Schur complement are invertible, and the block-inverse identity gives $(g_\mu^{\mathrm{quot}})^{-1} = (g_{\boldsymbol{\mu}}^{-1})_{[\mathsf{hh}]}$, where $g_{\boldsymbol{\mu}}$ is the metric in the expectation chart. By (G2.10), $g_{\boldsymbol{\mu}}^{-1}$ is the metric in the natural chart, whose block in the coordinate h paired with μ is $\mathrm{Cov}_q(Y) = \Sigma$ by (G2.9); inverting gives (G5.5). The Schur complement is the metric on the orthogonal quotient. (iii) Equation (G5.6) is (G5.3) rewritten, and $\Lambda^{-1}g_{[\mu\mu]}$ is similar to the bracket, whose eigenvalues are $1+2\|\hat\mu\|^2$ along $\hat\mu$ and $1+\|\hat\mu\|^2$ on its orthogonal complement. Finally $g_{[\mu\mu]} - \Lambda = (\mu^\top\Lambda\mu)\Lambda + \Lambda\mu\mu^\top\Lambda$ is a sum of two positive semidefinite matrices, which vanishes only if $\mu^\top\Lambda\mu = 0$, that is $\mu = 0$. □

The two restriction mean blocks, Λ in (G5.2) and (G5.3), differ because the coordinate vector $\partial/\partial\mu$ at fixed Σ and the one at fixed M_2 are different tangent vectors at the same law: the first moves the mean and changes M_2, the second moves the mean and compensates in Σ. Their discrepancy is governed by $\mu^\top\Lambda\mu$ and is unbounded, while the quotient metric equals Λ at every point. A coordinate identity that mixes the mean and covariance sectors of a Gaussian family must therefore name its chart and say whether it restricts or quotients.

G5.2 The mean sector and the interaction Laplacian

The recognition family is a location family in its mean, so the mean parameter takes values in the latent space $\mathbb{R}^n$ itself, and the mean block of (G5.2) has the same components as the sample-space quadratic form $y \mapsto \frac{1}{2}y^\top\Lambda y$. The coincidence is one of components in matched charts and not of objects: one side is a metric on a manifold of laws, the other a bilinear form on the latent space, and for a family that is not a location family the two do not even have the same dimension. The next proposition records the scoped relation to the Laplacian part L of the interaction family of Definition G3.1.

Proposition G5.3 (Edge relative entropy at fixed covariance). Let the edge (i,j) carry a positive-definite weight $W_{ij} \in \mathbb{S}^K_{++}$. For endpoint laws with means $\mu_i, \mu_j \in \mathbb{R}^K$ and the common covariance W_{ij}^{-1},

$$D_{\mathrm{KL}}\big(\mathcal{N}(\mu_i, W_{ij}^{-1}) \| \mathcal{N}(\mu_j, W_{ij}^{-1})\big) = \tfrac{1}{2}(\mu_i - \mu_j)^\top W_{ij}(\mu_i - \mu_j) = \tfrac{1}{2} g^F_{(\mu, W_{ij}^{-1})}\big((\mu_i - \mu_j, 0), (\mu_i - \mu_j, 0)\big). \tag{G5.10}$$

If every nonzero edge weight is positive definite, the sum of these relative entropies over the edges equals $\frac{1}{2}\mu^\top L \mu$, the Laplacian part of the energy (G3.2) evaluated at the stacked mean μ. [ESTABLISHED]

Proof. The relative entropy of two Gaussians with a common covariance Σ_0 is $\frac{1}{2}(\mu_i - \mu_j)^\top \Sigma_0^{-1}(\mu_i - \mu_j)$, and the mean block of (G5.2) at covariance W_{ij}^{-1} is W_{ij}. The sum over edges is the difference form $\sum_{i<j}(\mu_i - \mu_j)^\top W_{ij}(\mu_i - \mu_j) = \mu^\top L \mu$ of the Laplacian part. □

The assembled $NK \times NK$ matrix L enters only through this sum over typed edge quadratics, and a singular weight has no fixed-covariance reading along its kernel. For a regular non-Gaussian family the analogous quadratic is only the second-order term of the relative entropy at a frozen base law (Proposition 7.6), so outside these Gaussian or frozen-local hypotheses a sample-space Laplacian is neither a recognition Fisher tensor nor a natural-gradient operator.

G5.3 Pencils and gauge invariants

A change of local trivialization acts on every fiber and changes the components of every object while changing no law. On the assembled latent coordinate it acts agent by agent.

Definition G5.4 (Block reframing and the interaction pencil). Let $\Gamma_b = \rho_b(G)$ be the represented frame group of Hypothesis G2.1. A block reframing is the invertible new-to-old coordinate map

$$\mathbf{P} = \mathrm{diag}(\mathbf{P}_1, \ldots, \mathbf{P}_N), \qquad \mathbf{P}_i \in \Gamma_b, \qquad y_{\mathrm{old}} = P y_{\mathrm{new}}. \tag{G5.11}$$

Split the interaction precision once, in the chart in which Definition G3.1 is imposed, as $\Lambda = L + A$, with L its Laplacian part and $A = \mathrm{diag}(A_1, \ldots, A_N)$ its self terms. A block reframing acts on these quadratic forms by congruence,

$$\Lambda \longmapsto \mathbf{P}^\top \Lambda \mathbf{P}, \qquad L \longmapsto \mathbf{P}^\top L \mathbf{P}, \qquad A \longmapsto \mathbf{P}^\top A \mathbf{P}. \tag{G5.12}$$

The interaction pencil is the pair (L, Λ); its generalized eigenvalues are the roots d of $\det(L - \mathrm{d}\Lambda)$, with eigenvectors

$$Lx = \mathrm{d}\Lambda x. \tag{G5.13}$$

Because the reframing is block diagonal, $\mathbf{P}^\top A \mathbf{P}$ is again block diagonal, while $\mathbf{P}^\top L \mathbf{P}$ need not be a matrix-weighted Laplacian when neighboring blocks are reframed differently, since the row sums that define that normal form are not preserved. The invariants below compare forms and are unaffected

by that change of normal form. The law $\mathcal{N}(\mu, \Sigma)$ in old coordinates is $\mathcal{N}(\mathbf{P}^{-1}\mu, \mathbf{P}^{-1}\Sigma\mathbf{P}^{-\top})$ in new coordinates, so by the affine isometry of Proposition G5.1 with $Q = \mathbf{P}^{-1}$ every block reframing is an isometry of the Fisher metric.

Proposition G5.5 (Regular-pencil invariance). Let L and Λ be symmetric with $\det(L - \mathrm{d}\Lambda)$ not the zero polynomial, and let $\mathbf{P}$ be invertible. Then

$$\det\left(\mathbf{P}^{\top} L\mathbf{P} - \mathrm{d}\mathbf{P}^{\top}\Lambda\mathbf{P}\right) = \det(\mathbf{P})^2 \det(L - \mathrm{d}\Lambda), \tag{G5.14}$$

so (L, Λ) and $(\mathbf{P}^{\top} L\mathbf{P}, \mathbf{P}^{\top}\Lambda\mathbf{P})$ have the same generalized eigenvalues with the same multiplicities, and x solves (G5.13) exactly when $\mathbf{P}^{-1}x$ solves the reframed problem with the same d. If Λ is invertible, there are exactly $n = NK$ finite roots. [ESTABLISHED]

Proof. Equation (G5.14) is multiplicativity of the determinant, and the nonzero factor $\det(\mathbf{P})^2$ changes neither the roots nor their multiplicities. If $Lx = \mathrm{d}\Lambda x$, then $\mathbf{P}^{\top} L\mathbf{P}(\mathbf{P}^{-1}x) = \mathbf{P}^{\top} Lx = \mathrm{d}\mathbf{P}^{\top}\Lambda\mathbf{P}(\mathbf{P}^{-1}x)$, and the argument with $\mathbf{P}^{-1}$ gives the converse. When Λ is invertible the coefficient of d^n in $\det(L - \mathrm{d}\Lambda)$ is $(-1)^n \det\Lambda \neq 0$. □

Proposition G5.6 (Regularity criterion). For $L, A \succeq 0$ and $\Lambda = L + A$, the pencil (L, Λ) is regular if and only if

$$\ker L \cap \ker A = \{0\}. \tag{G5.15}$$

In that case $\Lambda \succ 0$. [ESTABLISHED]

Proof. Positive semidefiniteness gives $\ker(L + A) = \ker L \cap \ker A$. A nonzero vector in the intersection annihilates $L - \mathrm{d}\Lambda$ for every d, so the determinant vanishes identically. Conversely, if the intersection is trivial then $L + \Lambda = 2L + A \succ 0$, so the determinant is nonzero at $\mathrm{d} = -1$, and $\ker\Lambda = \{0\}$. □

Regularity is load bearing. Congruence covariance holds for every pencil, but for an irregular one the determinant vanishes identically. At the boundary operator $A = 0$ of the interaction family one has $\Lambda = L$, and $\det(L - \mathrm{d}L) \equiv 0$ whenever $\ker L \neq \{0\}$; restricting to a complement of $\ker L$ leaves the constant spectrum $\{1\}$, which carries no information about the graph. A quotient, pin, or mass repair must be declared and its pencil proved regular before the statements below apply to such a point.

Proposition G5.7 (Localization and endpoints). Let $L \succeq 0$ and $A \succeq 0$ with $\Lambda = L + A \succ 0$. Every generalized eigenvalue of (G5.13) is real and lies in $[0, 1]$; the eigenspace of 0 is $\ker L$, and the eigenspace of 1 is $\ker A$. If L is the Laplacian part of Definition G3.1, every configuration that is constant across agents lies in $\ker L$, so $\dim\ker L \geq K$. [ESTABLISHED]

Proof. Because $\Lambda \succ 0$, the pencil is equivalent to the symmetric matrix $\Lambda^{-1/2} L\Lambda^{-1/2}$, so its n generalized eigenvalues are real and are the stationary values of the Rayleigh quotient $x^{\top} Lx / x^{\top}\Lambda x$. Since $x^{\top}\Lambda x = x^{\top} Lx + x^{\top} Ax$ with both terms nonnegative and the sum positive, the quotient lies in $[0, 1]$. $Lx = 0$ is (G5.13) with $\mathrm{d} = 0$, and $Lx = \Lambda x$ is $Ax = 0$. For the Laplacian part,

$x^\top L x = \sum_{i<j}(x_i - x_j)^\top W_{ij}(x_i - x_j)$ vanishes on every $x = \mathbf{1} \otimes w$ with $w \in \mathbb{R}^K$, and these configurations form a K-dimensional space. □

The generalized eigenvalue has a frame-independent reading: d_k is the fraction of the precision along the k-th pencil direction that interaction supplies rather than self terms. Consensus directions sit at $\mathsf{d} = 0$, because interaction supplies no precision there, and a direction that no self term constrains sits at $\mathsf{d} = 1$.

Proposition G5.8 (Mass-pencil transfer). Let $L \succeq 0$, $\Lambda_{\mathrm{ref}} \succ 0$, $\mathsf{a} > 0$, and $\Lambda = L + \mathsf{a}\Lambda_{\mathrm{ref}}$. The generalized eigenvalues $\mathsf{d}_{\mathrm{ref}}$ of $(L, \Lambda_{\mathrm{ref}})$ and d of (L, Λ) correspond by

$$\mathsf{d} = \frac{\mathsf{d}_{\mathrm{ref}}}{\mathsf{d}_{\mathrm{ref}} + \mathsf{a}}, \qquad \mathsf{d}_{\mathrm{ref}} = \frac{\mathsf{a}\mathsf{d}}{1 - \mathsf{d}}, \tag{G5.16}$$

with the same eigenvectors and multiplicities, and the correspondence preserves order. Consequently the counting functions $\mathsf{n}(t) = \#\{k : \mathsf{d}_k \le t\}$ and $\mathsf{n}_{\mathrm{ref}}(s) = \#\{k : \mathsf{d}_{\mathrm{ref},k} \le s\}$ satisfy

$$\mathsf{n}(t) = \mathsf{n}_{\mathrm{ref}}\left(\frac{\mathsf{a}t}{1-t}\right), \qquad 0 \le t < 1, \tag{G5.17}$$

and if $\mathsf{n}_{\mathrm{ref}}(s) - \mathsf{n}_{\mathrm{ref}}(0) \sim \mathrm{c}_1 s^\alpha$ as $s \downarrow 0$, then $\mathsf{n}(t) - \mathsf{n}(0) \sim \mathrm{c}_1 \mathsf{a}^\alpha t^\alpha$ as $t \downarrow 0$. [ESTABLISHED]

Proof. For $\mathsf{d} \neq 1$, the equation $Lv = \mathsf{d}(L + \mathsf{a}\Lambda_{\mathrm{ref}})v$ is equivalent to $Lv = \big(\mathsf{a}\mathsf{d}/(1-\mathsf{d})\big)\Lambda_{\mathrm{ref}}v$, and $\mathsf{d} = 1$ would require $\Lambda_{\mathrm{ref}}v = 0$, which is impossible. Since $L \succeq 0$, every $\mathsf{d}_{\mathrm{ref}}$ is nonnegative, and the map $\mathsf{d}_{\mathrm{ref}} \mapsto \mathsf{d}_{\mathrm{ref}}/(\mathsf{d}_{\mathrm{ref}} + \mathsf{a})$ is increasing on $[0, \infty)$; this gives the order, the counting identity, and, because $\mathsf{a}t/(1-t) = \mathsf{a}t(1 + O(t))$, the asymptotic consequence. Without $L \succeq 0$ the order can fail: for $L = \mathrm{diag}(-2, 1, 3)$, $\Lambda_{\mathrm{ref}} = I$, and $\mathsf{a} = 1$, the roots $-2, 1, 3$ map to $2, \frac{1}{2}, \frac{3}{4}$. □

Mass removal and the small-eigenvalue limit do not commute,

$$\lim_{\mathsf{a}\downarrow 0}\lim_{\mathsf{d}_{\mathrm{ref}}\downarrow 0}\frac{\mathsf{d}_{\mathrm{ref}}}{\mathsf{d}_{\mathrm{ref}} + \mathsf{a}} = 0, \qquad \lim_{\mathsf{d}_{\mathrm{ref}}\downarrow 0}\lim_{\mathsf{a}\downarrow 0}\frac{\mathsf{d}_{\mathrm{ref}}}{\mathsf{d}_{\mathrm{ref}} + \mathsf{a}} = 1, \tag{G5.18}$$

so a positive mass regularizes each finite problem but supplies no mass-independent spectrum at the boundary operator.

Proposition G5.9 (Frame dependence of absolute spectra and subspace determinants). (i) If $\Gamma_b \supseteq \mathbb{R}_{>0} I_K$, in particular under the specialization $\Gamma_b = \mathrm{GL}^+(K)$ of Hypothesis G2.1, then $\mathbf{P} = \mathrm{c} I_n$ with $\mathrm{c} > 0$ is a block reframing and $\mathrm{spec}(\mathbf{P}^\top \Lambda \mathbf{P}) = \mathrm{c}^2\,\mathrm{spec}(\Lambda)$. (ii) Let $\mathcal{U}'$ be a subspace of the new coordinate space with orthonormal basis B', let $\mathcal{U} = \mathbf{P}\mathcal{U}'$ have orthonormal basis B, and let $\Lambda' = \mathbf{P}^\top \Lambda \mathbf{P}$. Then

$$\log\det\left(B'^\top \Lambda' B'\right) = \log\det\left(B^\top \Lambda B\right) + \log\det\left(B'^\top \mathbf{P}^\top \mathbf{P} B'\right), \tag{G5.19}$$

so a subspace log-determinant is frame independent if and only if the restriction of $\mathbf{P}$ to $\mathcal{U}'$ has unit absolute determinant; every orthogonal $\mathbf{P}$ satisfies this condition. (iii) For a partition of the agents

into blocks I, the determinant gap $\frac{1}{2}\big[\sum_I \log\det \Lambda_{II} - \log\det \Lambda\big]$ of Theorem G4.3 is invariant under every block reframing. A partition of the coordinates that splits an agent does not have this property: under the specialization $\Gamma_b = \mathrm{GL}^+(2)$, for $N = 1$, $K = 2$, $\Lambda = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and the blocks $\{1\}$, $\{2\}$, the reframing $\mathbf{P} = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ changes the gap from $\frac{1}{2}\log\frac{4}{3}$ to $\frac{1}{2}\log 4$. [ESTABLISHED]

Proof. (i) $\mathbf{P}^\top \Lambda \mathbf{P} = \mathrm{c}^2 \Lambda$. (ii) Since $\mathbf{P}\mathcal{U}' = \mathcal{U}$, write $\mathbf{P}B' = BS$ with S invertible. Then $B'^\top \Lambda' B' = S^\top (B^\top \Lambda B) S$ and $B'^\top \mathbf{P}^\top \mathbf{P} B' = S^\top S$, so $\log\det(B'^\top \Lambda' B') = \log\det(B^\top \Lambda B) + \log\det(S)^2$ and $\log\det(B'^\top \mathbf{P}^\top \mathbf{P} B') = \log\det(S)^2$, which gives the identity. (iii) For blocks that are unions of agents, the (I, I) block of $\mathbf{P}^\top \Lambda \mathbf{P}$ is $\mathbf{P}_I^\top \Lambda_{II} \mathbf{P}_I$ with $\mathbf{P}_I = \mathrm{diag}(\mathbf{P}_i : i \in I)$, so $\sum_I \log\det \Lambda_{II}$ shifts by $\sum_I \log\det(\mathbf{P}_I)^2 = \log\det(\mathbf{P})^2$, which is the shift of $\log\det \Lambda$. In the witness, $\mathbf{P}^\top \Lambda \mathbf{P} = \left(\begin{smallmatrix} 6 & 3 \\ 3 & 2 \end{smallmatrix}\right)$ and $\det \mathbf{P} = 1$, and the two gaps are $\frac{1}{2}\log(2 \cdot 2/3)$ and $\frac{1}{2}\log(6 \cdot 2/3)$. □

A reframing changes no law, so a quantity computed from components that is not invariant under (G5.12) for every $\mathbf{P}$ with blocks in Γ_b is a property of the chart. When Γ_b contains the positive scalars, part (i) lets a threshold on absolute eigenvalues of Λ accept or reject any given direction by the choice of c, so such a threshold states nothing about the model. Two invariant families remain available: generalized spectra of pencils whose two members transform by the same congruence (Proposition G5.5), and law functionals of agent-aligned blocks, such as determinant gaps, mutual informations, canonical correlations, and relative entropies between laws, which are invariant because a reframing changes no law and which part (iii) exhibits in coordinates for the gap. A construction that uses one subspace determinant must declare its frame, carry the Jacobian term of (G5.19), or combine determinants so that the Jacobians cancel, as the gap does. When $\Gamma_b \subseteq \mathrm{O}(K)$, as for a compact structure group, every block reframing is orthogonal and ordinary spectra are invariant (Section G2.1).

Proposition G5.10 (Residual fiber label). Assume $\Gamma_b \supseteq \mathrm{GL}^+(K)$, as under the specialization of Hypothesis G2.1. For $M \in \mathbb{S}_+^K$, the orbit of M under global reframing $M \mapsto \bar{\mathbf{P}}^\top M \bar{\mathbf{P}}$, $\bar{\mathbf{P}} \in \Gamma_b$, is the set of positive semidefinite matrices of rank $\mathrm{rank}\, M$. Hence the matrix factor M of the biadditive family (G9.6) has $K + 1$ congruence classes and no continuous modulus. [ESTABLISHED]

Proof. Congruence by an invertible matrix preserves rank and positive semidefiniteness. For the converse, let $r = \mathrm{rank}\, M$ and write $M = QDQ^\top$ with Q orthogonal and D diagonal, with positive entries $d_1, \ldots, d_r$ followed by zeros. Then

$$\bar{\mathbf{P}} = Q\,\mathrm{diag}\left(d_1^{-1/2}, \ldots, d_r^{-1/2}, 1, \ldots, 1\right) \quad \text{gives} \quad \bar{\mathbf{P}}^\top M \bar{\mathbf{P}} = \mathrm{diag}(I_r, 0),$$

and if $\det \bar{\mathbf{P}} < 0$, changing the sign of the first column of $\bar{\mathbf{P}}$ keeps this normal form and makes the determinant positive. Two positive semidefinite matrices of equal rank are therefore congruent to the same normal form by elements of $\mathrm{GL}^+(K) \subseteq \Gamma_b$. This is the positive semidefinite case of Sylvester's law of inertia [17, §4.5]. □

The hypothesis on Γ_b is load bearing: when Γ_b is $\mathrm{SO}(K)$ or $\mathrm{O}(K)$, the orbit of $M \succ 0$ is its orthogonal-similarity class, labeled by K continuous eigenvalues (Section G2.1). The proposition also does

not extend unchanged to a nonflat pair (W, Θ), whose residual action combines congruence on W with conjugation on Θ.

Chapter G6

Ambient Score Tangents and the Hermite Spectrum

At a fixed law, three different objects are called tangent directions: the tangent space of a finite-dimensional family, the Hilbert space of centered square-integrable scores, and the bounded action quotient on which the action calculus of Theorem 12.4 is analytic. This chapter asks how the three are nested at a Gaussian law, and what a parameter-independent block channel does to scores once a block of independent copies has been formed. The main result, Theorem G6.8, computes the score operator of the normalized sum of b independent standard Gaussian coordinates: in the normalized Hermite basis it is diagonal with eigenvalues $b^{1-k/2}$, compact, and injective with dense but proper range. It rests on the exact extensive Fisher budget of an arbitrary parameter-independent block channel (Theorem G6.4) and on the realization of every centered square-integrable score by a two-sided path (Lemma G6.2), and Proposition G6.7 decides which Hermite directions also admit two-sided exponential action charts. The chapter is an explicit instance of a distinction drawn in general in the main manuscript. Exact arrows create no growth of a tangent direction: the reverse conditional expectation is a contraction (Theorem 12.5), the bounded action sector is nonexpansive (12.11), and the exact interaction cocycle is bounded uniformly in depth (Proposition 12.21). The eigenvalue $b^{1/2} > 1$ of the degree-one direction therefore comes entirely from the extensive replication factor $\sqrt{b}$, a declared normalization between two reference tangent spaces.

The statements of the chapter are linear statements about score tangents at one law, and its spectral statements concern the scalar independent normalized sum. The chapter does not assert a convergence statement for the nonlinear map that the normalized sum induces on laws, or a statement about the nonlinear bounded action map beyond its derivative. [NOT-CLAIMED]

G6.1 Quadratic-mean scores

Definition G6.1 (Quadratic-mean differentiability and score). Let $(\mathsf{W}, \mathscr{W}, \nu)$ be a probability space and write

$$L_0^2(\nu) = \left\{ h \in L^2(\nu) : \int h d\nu = 0 \right\}. \tag{G6.1}$$

A family $(\nu_t)_{|t|<t_0}$ of probability laws with $\nu_t \ll \nu$ and $\nu_0 = \nu$ is differentiable in quadratic mean at $t = 0$ with score $h \in L^2(\nu)$ when, for $t \to 0$ from both sides,

$$\left\| \sqrt{\frac{d\nu_t}{d\nu}} - 1 - \frac{t}{2}h \right\|_{L^2(\nu)} = o(|t|). \tag{G6.2}$$

The Fisher information of the path at $t = 0$ is $\|h\|^2_{L^2(\nu)}$.

Lemma G6.2 (Every centered square-integrable score is realized two-sided). The score of every path that is differentiable in quadratic mean at ν lies in $L^2_0(\nu)$. Conversely, let $h \in L^2_0(\nu)$ and, for every real t, put $Z_t = 1 + t^2\|h\|_2^2/4$ and

$$\frac{d\nu_t}{d\nu} = \frac{(1 + th/2)^2}{Z_t}. \tag{G6.3}$$

Then every ν_t is a probability law, $\nu_0 = \nu$, and the path is differentiable in quadratic mean at $t = 0$ with score h, for $t \to 0^+$ and for $t \to 0^-$. No exponential moment and no boundedness of h is used. The quadratic-mean tangent set at ν is therefore exactly $L^2_0(\nu)$. [ESTABLISHED]

Proof. For the first statement write $p_t = d\nu_t/d\nu$. Since $\int p_t d\nu = 1$, $\int(\sqrt{p_t}-1)d\nu = -\frac{1}{2}\int(\sqrt{p_t}-1)^2 d\nu$. By (G6.2) the left side is $\frac{t}{2}\int h d\nu + o(|t|)$ and the right side is $O(t^2)$, so $\int h d\nu = 0$.

For the converse, centering gives $\int(1 + th/2)^2 d\nu = 1 + t\nu(h) + \frac{t^2}{4}\nu(h^2) = Z_t$, so $p_t \geq 0$ integrates to one. Put $w_t = 1 + th/2$, so that $\sqrt{p_t} = |w_t|/\sqrt{Z_t}$ and

$$\sqrt{p_t} - w_t = \frac{|w_t| - w_t}{\sqrt{Z_t}} + w_t\left(\frac{1}{\sqrt{Z_t}} - 1\right). \tag{G6.4}$$

On $\{w_t < 0\}$ one has $th/2 < -1$, hence $|h| > 2/|t|$ and $|w_t| - w_t = 2|w_t| \leq |th|$; elsewhere the difference vanishes. Since $Z_t \geq 1$,

$$\frac{1}{t^2}\left\|\frac{|w_t| - w_t}{\sqrt{Z_t}}\right\|_2^2 \leq \int h^2 \mathbf{1}_{\{|h|>2/|t|\}} d\nu \longrightarrow 0 \tag{G6.5}$$

by dominated convergence, because h^2 is integrable. For the second term of (G6.4), $\|w_t\|_2 = \sqrt{Z_t}$, so its norm is exactly $|1 - \sqrt{Z_t}| = O(t^2)$. Adding the two bounds and using $w_t = 1 + \frac{t}{2}h$ gives (G6.2). □

G6.2 The extensive block lift and the pushed score

Proposition G6.3 (Independent block replication of a differentiable path). Fix an integer $b \geq 1$. (i) Let (ν_t) satisfy (G6.2) with score h. Then the product path $\nu_t^{\otimes b}$ is differentiable in quadratic mean at $\nu^{\otimes b}$ with score

$$(\mathscr{I}_b h)(x_1, \ldots, x_b) = \sum_{i=1}^{b} h(x_i), \tag{G6.6}$$

and the replication lift is a bounded linear injection

$$\mathscr{I}_b : L_0^2(\nu) \longrightarrow L_0^2(\nu^{\otimes b}), \qquad \|\mathscr{I}_b h\|^2_{L^2(\nu^{\otimes b})} = b\|h\|^2_{L^2(\nu)}, \tag{G6.7}$$

so $\|\mathscr{I}_b\| \le \sqrt{b}$, with equality when $L_0^2(\nu) \neq \{0\}$, and $b^{-1/2}\mathscr{I}_b$ is an isometry onto its range. The lift is not the Hoeffding assembly E_ℓ of (12.24). (ii) For $b \ge 2$, no parameter-independent Markov kernel K satisfies $\mathcal{N}(x,1)\mathsf{K} = \mathcal{N}(x\mathbf{1}_b, I_b)$ for every $x \in \mathbb{R}$. [ESTABLISHED]

Proof. (i) Write $r_t = \sqrt{d\nu_t/d\nu} = 1 + \frac{t}{2}h + t\varepsilon_t$ with $\|\varepsilon_t\|_{L^2(\nu)} \to 0$. The product amplitude is $\prod_{i=1}^b (1+\delta_i)$ with $\delta_i(x) = \frac{t}{2}h(x_i) + t\varepsilon_t(x_i)$, and

$$\prod_{i=1}^b (1+\delta_i) = 1 + \sum_{i=1}^b \delta_i + \sum_{|S|\ge 2} \prod_{i\in S} \delta_i. \tag{G6.8}$$

Each factor depends on one coordinate, so $L^2(\nu^{\otimes b})$ norms of products factor, $\|\prod_{i\in S}\delta_i\|_2 = \prod_{i\in S}\|\delta_i\|_2 = O(|t|^{|S|})$, and every term with $|S| \ge 2$ is $O(t^2)$. The linear part is $\frac{t}{2}\mathscr{I}_b h + t\sum_i \varepsilon_t(x_i)$, whose remainder has norm at most $b|t|\|\varepsilon_t\|_2 = o(|t|)$. Independence and centering give $\mathbb{E}(\mathscr{I}_b h)^2 = \sum_i \mathbb{E}h(x_i)^2 = b\|h\|_2^2$. (ii) The family $x \mapsto \mathcal{N}(x,1)$ has Fisher information one and the family $x \mapsto \mathcal{N}(x\mathbf{1}_b, I_b)$ has Fisher information b. A parameter-independent Markov kernel cannot increase Fisher information (Theorem 7.8), so such a K would force $b \le 1$. □

The replication is extensive. Replacing the sum (G6.6) by the arithmetic mean without rescaling the path parameter would change the Fisher information from $b\|h\|_2^2$ to $\|h\|_2^2/b$, which is a different construction. Part (ii) says that the replication pair is a lift between two reference tangent spaces and not a pushforward along any arrow, so it can serve neither as a counterexample to Markov contraction nor as evidence that Fisher information increased along a coarse-graining arrow. The factor $\sqrt{b}$ belongs to the declared normalization that identifies the two reference spaces.

Theorem G6.4 (Pushed score and the exact extensive Fisher defect). Let $C_b : \mathsf{W}^b \rightsquigarrow \mathsf{Z}$ be a parameter-independent Markov kernel between standard Borel spaces, put $\nu^{(b)} = \nu^{\otimes b}$, $\mathbb{J}_b(dx,dz) = \nu^{(b)}(dx)C_b(x,dz)$, and $\nu^{(b),c} = \nu^{(b)}C_b$, and let $\mathsf{U}_b : L^2(\nu^{(b)}) \to L^2(\nu^{(b),c})$ be the reverse conditional expectation of Theorem 12.5 for this arrow. For every $h \in L_0^2(\nu)$ and every path with score h, the pushed path $\nu_t^{\otimes b}C_b$ is differentiable in quadratic mean at $\nu^{(b),c}$ with score

$$\mathscr{L}_b h := \mathsf{U}_b \mathscr{I}_b h = \mathbb{E}_{\mathbb{J}_b}\left[\sum_{i=1}^b h(x_i)\middle| z\right], \tag{G6.9}$$

so $\mathscr{L}_b : L_0^2(\nu) \to L_0^2(\nu^{(b),c})$ is linear with $\|\mathscr{L}_b\| \le \sqrt{b}$. The information budget is exact:

$$b\|h\|^2_{L^2(\nu)} - \|\mathscr{L}_b h\|^2_{L^2(\nu^{(b),c})} = \int \mathrm{Var}_{\mathbb{J}_b}(\mathscr{I}_b h|z)\,\nu^{(b),c}(dz) \ge 0, \tag{G6.10}$$

with equality exactly when $\mathscr{I}_b h$ agrees $\mathbb{J}_b$-almost surely with a measurable function of z. Equivalently, $b^{-1/2}\mathscr{L}_b$ contracts L_0^2 norms. [ESTABLISHED]

Proof. Suppose first that h is bounded and that the path is (G6.3). For $|t|$ small, $w_t \geq 1-|t|\|h\|_\infty/2 > 0$, so $d\nu_t/d\nu = w_t^2/Z_t = 1 + th + O(t^2)$ uniformly, the product density is $1 + t\mathscr{I}_b h + O(t^2)$ uniformly, and, exactly as in (12.4),

$$\frac{d(\nu_t^{\otimes b} C_b)}{d\nu^{(b),c}}(z) = \mathbb{E}_{\mathbb{J}_b}\left[\frac{d\nu_t^{\otimes b}}{d\nu^{(b)}}(x)\middle| z\right] = 1 + t\mathscr{L}_b h(z) + O(t^2) \tag{G6.11}$$

uniformly. Since $\sqrt{1+v} = 1 + v/2 + O(v^2)$ uniformly for $|v| \leq 1/2$, the square root of (G6.11) is $1 + \frac{t}{2}\mathscr{L}_b h + O(t^2)$ uniformly, hence in $L^2(\nu^{(b),c})$.

For a general path ν_t with score $h \in L_0^2(\nu)$, choose bounded centered $h_n \to h$ in $L^2(\nu)$ and let $\nu_t^{(n)}$ be their paths (G6.3). Squared Hellinger distance is the f-divergence of $f(v) = (\sqrt{v}-1)^2$, so conditional Jensen applied to the disintegration of the pushed joint law gives

$$\left\| \sqrt{\frac{d(\nu_t^{\otimes b} C_b)}{d\nu^{(b),c}}} - \sqrt{\frac{d((\nu_t^{(n)})^{\otimes b} C_b)}{d\nu^{(b),c}}} \right\|_2 \leq \left\| \sqrt{\frac{d\nu_t^{\otimes b}}{d\nu^{(b)}}} - \sqrt{\frac{d(\nu_t^{(n)})^{\otimes b}}{d\nu^{(b)}}} \right\|_2 . \tag{G6.12}$$

Both input paths have scores $\mathscr{I}_b h$ and $\mathscr{I}_b h_n$ by Proposition G6.3, so the right side divided by $|t|$ has limit superior $\frac{1}{2}\|\mathscr{I}_b(h-h_n)\|_2 = \frac{\sqrt{b}}{2}\|h-h_n\|_2$. Adding and subtracting the bounded-case expansion and using $\|\mathscr{L}_b(h-h_n)\|_2 \leq \sqrt{b}\|h-h_n\|_2$,

$$\limsup_{t\to 0} \frac{1}{|t|}\left\| \sqrt{\frac{d(\nu_t^{\otimes b} C_b)}{d\nu^{(b),c}}} - 1 - \frac{t}{2}\mathscr{L}_b h \right\|_2 \leq \sqrt{b}\|h-h_n\|_2. \tag{G6.13}$$

The left side does not depend on n, so it is zero. Conditional Jensen gives $\|\mathsf{U}_b\| \leq 1$ and preservation of the mean, so $\mathscr{L}_b$ has the stated type and $\|\mathscr{L}_b\| \leq \|\mathscr{I}_b\| \leq \sqrt{b}$. Finally the conditional variance identity (B.10) applied to $\mathscr{I}_b h$, together with (G6.7), is (G6.10), and a nonnegative conditional variance integrates to zero exactly when it vanishes almost everywhere. □

Both ends of the budget are attained in simple cases. A channel that maps every block to one point sends every score to zero, so a fixed channel can annihilate the whole tangent. In the other direction, equality in (G6.10) at one parameter is local score sufficiency and not recovery of the experiment, as the witness following Theorem 7.8 shows.

G6.3 Bounded actions and ambient score tangents

The bounded action calculus of the main manuscript perturbs a law ν by bounded actions $\varphi \in \mathfrak{B} = L^\infty(\nu;\mathbb{R})$ and works on the quotient $\overline{\mathfrak{B}} = \mathfrak{B}/\mathbb{R}\mathbf{1}$ with the oscillation norm (12.15). Its normalized exponential path $t \mapsto \widehat{\nu}^{t\varphi}$ is (12.12). For a coarse channel with pushed law ν^c and reverse conditional expectation U, $\overline{\mathsf{U}}$ denotes the descent of U to the quotient.

Proposition G6.5 (Bounded action classes embed isometrically in the Fisher tangent). For $\varphi \in \mathfrak{B}$, the path $t \mapsto \widehat{\nu}^{t\varphi}$ is differentiable in quadratic mean with score

$$\mathscr{S}_\nu[\varphi] = -\varphi + \nu(\varphi), \tag{G6.14}$$

which depends only on the class $[\varphi] \in \overline{\mathfrak{B}}$. With the Fisher norm

$$\|[\varphi]\|_F = \inf_{c\in\mathbb{R}} \|\varphi - c\|_{L^2(\nu)} = \|\varphi - \nu(\varphi)\|_{L^2(\nu)}, \tag{G6.15}$$

the map $\mathscr{S}_\nu : (\overline{\mathfrak{B}}, \|\cdot\|_F) \to L^2_0(\nu)$ is a linear isometry onto the bounded centered subspace; this range is dense and, whenever $L^2_0(\nu)$ contains an unbounded element, proper, so the Fisher-norm completion of $\overline{\mathfrak{B}}$ is canonically $L^2_0(\nu)$. The two quotient norms satisfy $\|[\varphi]\|_F \le \|[\varphi]\|_{\mathrm{osc}}$. For a coarse channel the centering square commutes,

$$\mathscr{S}_{\nu^c}\big(\overline{\mathsf{U}}[\varphi]\big) = \mathsf{U}\mathscr{S}_\nu[\varphi], \tag{G6.16}$$

and the replicated block action $(x_1, \dots, x_b) \mapsto \sum_i \varphi(x_i)$ has score $\mathscr{I}_b\mathscr{S}_\nu[\varphi]$ at $\nu^{\otimes b}$ and, after the block channel of Theorem G6.4, pushed score $\mathscr{L}_b\mathscr{S}_\nu[\varphi]$. [ESTABLISHED]

Proof. Boundedness gives the uniform expansions $e^{-t\varphi} = 1 - t\varphi + O(t^2)$ and $\nu(e^{-t\varphi}) = 1 - t\nu(\varphi) + O(t^2)$, so the density ratio of $\widehat{\nu}^{t\varphi}$ is $1 - t[\varphi - \nu(\varphi)] + O(t^2)$ uniformly and its square root is $1 + \frac{t}{2}\mathscr{S}_\nu[\varphi] + O(t^2)$, which is (G6.2). Adding a constant to φ leaves (G6.14) unchanged. The L^2 minimizer over constants is the mean, which gives (G6.15) and the isometry, and $\inf_c\|\varphi - c\|_2 \le \inf_c\|\varphi - c\|_\infty$ gives the norm order. Bounded functions are dense in $L^2(\nu)$ and centering is continuous, so the range is dense; it misses every unbounded element of $L^2_0(\nu)$, for example $h(x) = x$ when $\nu = \mathcal{N}(0,1)$. Identity (G6.16) is mean preservation of U together with linearity, and the replication statement is Proposition G6.3 with (G6.9). □

The two norms on $\overline{\mathfrak{B}}$ do different work. The oscillation norm controls the nonlinear map of Theorem 12.4; the Fisher norm controls the quadratic-mean tangent. Completing in the Fisher norm introduces unbounded scores on which the nonlinear bounded action map is not defined, so a spectral statement proved on L^2_0 is not a statement about the nonlinear bounded action chart, and conversely. At a Gaussian law the nesting of the three tangent objects is exact.

Remark G6.6 (Three tangent objects at a Gaussian law). At a nondegenerate Gaussian law ν on $\mathbb{R}^n$, (a) the tangent space of the Gaussian family, spanned by the scores (G5.1), is a subspace of $L^2_0(\nu)$ of dimension $n + n(n+1)/2$; (b) $L^2_0(\nu)$ is infinite dimensional and, by Lemma G6.2, consists exactly of the scores of paths differentiable in quadratic mean; and (c) the bounded action quotient embeds isometrically in $L^2_0(\nu)$ with dense proper range by Proposition G6.5. The tangent space (a) meets the range of (c) only in 0. [ESTABLISHED]

Proof. The score (G5.1) is a polynomial of degree at most two whose linear part is $\dot{\mu}^\top \Lambda y$ and whose quadratic part is $\frac{1}{2} y^\top \Lambda \dot{\Sigma} \Lambda y$, so it vanishes only for $\dot{\mu} = 0$ and $\dot{\Sigma} = 0$, and the dimension is that of the moment chart. The monomials are linearly independent in $L^2(\nu)$, so $L^2_0(\nu)$ is infinite

dimensional. A nonzero element of (a) is a nonconstant polynomial and hence unbounded, while the range of (c) consists of bounded functions. □

The gap between (b) and (c) is visible along the Hermite directions of the standard Gaussian.

Proposition G6.7 (Exact two-sided domain of a Hermite exponential action). Let $\nu = \mathcal{N}(0,1)$, let He_k be the probabilists' Hermite polynomial of degree $k \geq 1$, and set

$$Z_k(t) = \int e^{-t\mathrm{He}_k(x)}\nu(dx) \in (0, +\infty]. \tag{G6.17}$$

Then $Z_1(t) < \infty$ for every real t; $Z_2(t) < \infty$ exactly for $t > -\frac{1}{2}$, where $Z_2(t) = e^t(1+2t)^{-1/2}$; for odd $k \geq 3$, $Z_k(t) = +\infty$ for every $t \neq 0$; and for even $k \geq 4$, $Z_k(t) < \infty$ exactly for $t \geq 0$. Consequently every He_k with $k \geq 1$ lies in $L^2_0(\nu)$ and is realized as a two-sided quadratic-mean score, while for $k \geq 3$ no two-sided exponential action neighborhood of the origin contains it; the lowest-degree direction whose exponential normalizer diverges on both sides is $\mathrm{He}_3(x) = x^3 - 3x$, with $\mathbb{E}_\nu \mathrm{He}_3^2 = 6$. [ESTABLISHED]

Proof. For $k = 1$ the integral is the moment-generating function $e^{t^2/2}$. For $k = 2$, $\mathrm{He}_2(x) = x^2 - 1$, so $Z_2(t) = e^t \int e^{-tx^2}\nu(dx)$, which converges exactly when $t + \frac{1}{2} > 0$, with value $e^t(1+2t)^{-1/2}$. For $k \geq 3$ write $\mathrm{He}_k(x) = x^k + O(x^{k-2})$, so the exponent $-t\mathrm{He}_k(x) - x^2/2$ of the integrand against Lebesgue measure has leading term $-tx^k$. For odd k and $t \neq 0$, exactly one tail has $-tx^k \to +\infty$, and the integral diverges; for He_3 and $t > 0$, for instance, $-t\mathrm{He}_3(-r) - r^2/2 = tr^3 - r^2/2 - 3tr \to +\infty$ as $r \to +\infty$. For even k and $t < 0$ both tails diverge; for $t > 0$ the integrand is eventually bounded by $e^{-\mathrm{c}_1 x^k}$ with $\mathrm{c}_1 > 0$, and $Z_k(0) = 1$. Hermite polynomials are square integrable against ν and orthogonal to the constants, the realization is Lemma G6.2, and $\mathbb{E}_\nu \mathrm{He}_3^2 = 3!$. □

The two score directions of the Gaussian family lie in the benign part of this classification. At $n = 1$, $\mu = 0$, and $\Lambda = 1$, the score (G5.1) reads $\dot{\mu}x + \frac{\dot{\Sigma}}{2}(x^2 - 1) = \dot{\mu}\mathrm{He}_1 + \frac{\dot{\Sigma}}{2}\mathrm{He}_2$, so the mean and variance directions occupy Hermite degrees one and two and both admit two-sided exponential neighborhoods; the constraint $t > -\frac{1}{2}$ in the variance direction is the natural-domain boundary $\Lambda \succ 0$ seen through the exponential chart. A complete Hermite basis of $L^2_0(\nu)$ is therefore neither a chart of the Gaussian family nor a set of directions of the bounded action space, and the spectral statement below concerns the L^2_0 tangent.

G6.4 The Gaussian block spectrum

Theorem G6.8 (Gaussian block spectrum in the normalized Hermite basis). Let $\nu = \mathcal{N}(0,1)$, let $b \geq 2$ be an integer, let $\xi_1, \ldots, \xi_b$ be independent with law ν, and let the block channel be the deterministic normalized sum

$$\bar{\xi} = b^{-1/2} \sum_{i=1}^{b} \xi_i, \tag{G6.18}$$

whose law is again ν. That normalization, and no general principle, makes source and target the same space and hence makes

$$\mathscr{L}_b = \mathsf{U}_b \mathscr{I}_b : L^2_0(\nu) \longrightarrow L^2_0(\nu) \tag{G6.19}$$

an endomorphism. With $\widehat{\mathrm{He}}_k = \mathrm{He}_k/\sqrt{k!}$, the family $\{\widehat{\mathrm{He}}_k\}_{k\geq 1}$ is a complete orthonormal basis of $L^2_0(\nu)$ and

$$\mathscr{L}_b \widehat{\mathrm{He}}_k = b^{1-k/2} \widehat{\mathrm{He}}_k, \qquad k \geq 1. \tag{G6.20}$$

Consequently $\mathscr{L}_b$ is self-adjoint, positive, and diagonal in that basis; it is Hilbert–Schmidt with $\sum_{k\geq 1} b^{2-k} = b^2/(b-1)$, hence compact; its norm and spectral radius are $\sqrt{b}$; and

$$\sigma(\mathscr{L}_b) = \left\{ b^{1-k/2} : k \geq 1 \right\} \cup \{0\}. \tag{G6.21}$$

Every displayed nonzero value is a simple eigenvalue. The value 0 is not an eigenvalue; it is the unique accumulation point of the spectrum and belongs to the continuous spectrum. [ESTABLISHED]

Proof. Completeness of the normalized Hermite system in $L^2(\nu)$ is standard, and deleting $\widehat{\mathrm{He}}_0 = 1$ leaves a complete orthonormal basis of the centered subspace. For each i, the pair $(\xi_i, \bar{\xi})$ is jointly centered Gaussian with unit variances and covariance $b^{-1/2}$, so ξ_i given $\bar{\xi} = z$ has law $\mathcal{N}(z/\sqrt{b}, 1 - 1/b)$. Taking the conditional expectation of the generating function $\exp(t\xi_i - t^2/2)$ and completing the square,

$$\mathbb{E}\left[e^{t\xi_i - t^2/2} \middle| \bar{\xi} = z \right] = \exp\left(\frac{tz}{\sqrt{b}} + \frac{t^2}{2}\left(1 - \frac{1}{b}\right) - \frac{t^2}{2} \right) = \exp\left(sz - \frac{s^2}{2} \right), \qquad s = \frac{t}{\sqrt{b}}. \tag{G6.22}$$

The generating identity $\exp(sz - s^2/2) = \sum_{k\geq 0} \mathrm{He}_k(z) s^k/k!$ converges in $L^2(\nu)$ locally uniformly in s, and conditional expectation is L^2 continuous, so the two power series in t may be compared coefficient by coefficient. This gives the Mehler regression identity $\mathbb{E}[\mathrm{He}_k(\xi_i) \mid \bar{\xi}] = b^{-k/2}\mathrm{He}_k(\bar{\xi})$, and summing over the b replicated coordinates and dividing by $\sqrt{k!}$ proves (G6.20), uniformly in b.

Since $\mathscr{L}_b$ is bounded and acts diagonally on a complete orthonormal basis, it is that diagonal operator, hence self-adjoint and positive with eigenvalues $b^{1-k/2} \downarrow 0$. Square summability of the eigenvalues is the stated geometric series, so $\mathscr{L}_b$ is Hilbert–Schmidt and compact, and its norm is the largest eigenvalue $b^{1/2}$. The spectrum of a diagonal operator is the closure of its eigenvalue set, which is (G6.21), and the eigenvalues are pairwise distinct, so each is simple. Every eigenvalue is strictly positive, so $\ker \mathscr{L}_b = \{0\}$. The range contains every finite Hermite expansion and is dense, but it is proper: $\sum_{k\geq 1} b^{1-k/2}\widehat{\mathrm{He}}_k$ lies in $L^2_0(\nu)$, while any preimage would have every Hermite coefficient equal to one, which is not square summable. An injective operator with dense, nonclosed range has 0 in its continuous spectrum. □

The case $b = 2$ of (G6.20) is the linearization of the normalized-sum map on densities at the standard Gaussian computed by Jona-Lasinio [18, Sec. 2, Eq. (2.14)], who notes there that completing a limit theorem requires control of the nonlinear terms, which he does not pursue. The

theorem above is a statement about the derivative at the Gaussian law, a linear operator on the L^2_0 tangent; the eigenvalue $b^{1/2}$ of the degree-one direction exceeds one only because the replication factor of (G6.7) is extensive while the norm is held fixed, since the one-arrow operator U_b is a contraction.

Chapter G7

Coarse Maps I: Aggregation and Kron Closure

Which coarse maps of the Gaussian interaction family are exact, and when does the family close under them? The core distinguishes Markov pushforward, energy precomposition, and restriction of a recognition family (Section 10.4); this chapter realizes the first two on the interaction family of Definition G3.1. Hard aggregation precomposes the energy with the identification of each cluster to one coordinate. For graph-exponential energies it closes with a linear parameter map (Theorem G7.2); on the Gaussian family it sums cut weights and self terms and annihilates internal edges (Proposition G7.3), and the coarse law is the fine law conditioned on zero within-cluster disagreement (Proposition G7.4), which is also a precision-dependent linear pushforward (Proposition G7.5). When a linear-Gaussian exterior record reads the latent coordinates, the pencil of the record and reference precisions selects the range of that pushforward: at each rank it gives the linear statistic with the largest exterior-predictive information, and on its range the coarse posterior of a surrogate that reads the record at the projected latent coincides with the coarse posterior (Proposition G7.6). Kron reduction marginalizes hidden agents. It is exact on the complete Gaussian action (Proposition G7.11) but leaves the matrix-weighted interaction family in general (Proposition G7.7), and Table G7.1 collects the regimes in which it closes, among them the graph criterion of Theorem G7.15, its prior-carrying form Corollary G7.16, and the nonflat series closure of Theorem G7.21. These results are the Gaussian instance of the coarse channels of Chapter 10 and of the network coarse-graining of Chapter 12. Nonflat internal links, relaxation, and quotient volumes continue in Chapter G8, the operator-level aggregation semigroup is Proposition G9.2, and reciprocal pairs are treated in Chapter G10.

Symbols without a channel index refer to the belief channel, of fiber dimension K; a model-channel realization, of dimension d_m, uses its own self terms, weights, and links, and no equality of the two channels is implied. Coarse families admit newly generated positive-semidefinite self terms; the family that forbids them is the no-new-anchor pair subfamily of Definition 11.23. A law has density proportional to $e^{-\mathsf{E}}$ with respect to its stated reference, as in (10.8), and every quadratic energy carries the factor one half. The chapter uses these local symbols: $\mathfrak{h}$ for a Gaussian source (information vector), $\mathfrak{c}$ for a scalar offset, subscripts r and h for retained and hidden (eliminated) coordinates, V_{r} and V_{h} for retained and hidden vertices, M for the pivot of an elimination, H for the frame of a congruence-diagonal cone $\mathcal{W}_H$, and t_1, t_2 for positive test scales. The symbols of the exterior interface (Λ_{rec}, $\mathsf{d}_{\mathrm{ref},k}$, x_k, c_k, $\tilde{o}_k$, $\widetilde{P}_r$, $\mathbf{W}$, and o_0) are declared in Proposition G7.6.

This chapter does not claim that a separately fitted coarse coupling, or a learned or soft pooling, realizes an exact coarse map; those are different approximate models. Nor does it claim a canonical decomposition of a coarse operator into links and self terms, monotonicity of an information functional along aggregation when the precision varies, or optimality of the exterior-informed interface among nonlinear statistics. Unrestricted Kron closure is refuted, and balanced cut links admit no continuous extension to all singular cuts (Proposition 11.28). The open items are monotonicity along aggregation with varying precision (Open Problem G9.4) and Kron classification beyond finite descriptions (Open Problem GA.14). [NOT-CLAIMED]

G7.1 Graph-exponential energy closure

Hard identification of a cluster is precomposition of the energy with the map that copies one coarse coordinate to every member. For energies built from node and pair statistics this precomposition stays in the same form.

Definition G7.1 (Graph-exponential energy). Let V be finite, let Z_0 be standard Borel with a σ-finite reference ν_0, and let $\mathsf{T}_0 : \mathsf{Z}_0 \to \mathsf{V}_0$ and $\mathsf{T}_1 : \mathsf{Z}_0^2 \to \mathsf{V}_1$ be measurable statistics with values in real vector spaces, T_1 symmetric. For node parameters θ_i and pair parameters $\theta_{ij} = \theta_{ji}$, paired with V_0 and V_1, the graph-exponential energy on Z_0^V is

$$\mathsf{E}_\theta(z) = -\sum_{i\in V}\langle\theta_i, \mathsf{T}_0(z_i)\rangle - \sum_{\{i,j\}}\langle\theta_{ij}, \mathsf{T}_1(z_i, z_j)\rangle. \tag{G7.1}$$

It defines a law, with density $e^{-\mathsf{E}_\theta}/Z_\theta$ relative to $\nu_0^{\otimes V}$, exactly when $Z_\theta = \int e^{-\mathsf{E}_\theta} d\nu_0^{\otimes V}$ is finite. For a partition $\mathcal{P}$ of V, with $I(i)$ the cluster of i, the identification map is $\iota_\mathcal{P} : \mathsf{Z}_0^\mathcal{P} \to \mathsf{Z}_0^V$, $(\iota_\mathcal{P}\bar{z})_i = \bar{z}_{I(i)}$.

Theorem G7.2 (Graph-exponential trace closure). Suppose the diagonal restriction of the pair statistic is affine in the node statistic,

$$\mathsf{T}_1(z, z) = \mathsf{D}\mathsf{T}_0(z) + \mathsf{T}_1^\circ, \tag{G7.2}$$

for a linear map $\mathsf{D} : \mathsf{V}_0 \to \mathsf{V}_1$ and a constant T_1°; this excludes pair statistics whose diagonal the node statistic cannot represent. Then for every partition $\mathcal{P}$,

$$\mathsf{E}_\theta(\iota_\mathcal{P}\bar{z}) = -\sum_I \langle\theta_I, \mathsf{T}_0(\bar{z}_I)\rangle - \sum_{I<J}\langle\theta_{IJ}, \mathsf{T}_1(\bar{z}_I, \bar{z}_J)\rangle + \mathfrak{c}_\mathcal{P}(\theta), \tag{G7.3}$$

where

$$\theta_I = \sum_{i\in I}\theta_i + \mathsf{D}^* \sum_{\{i,j\}\subseteq I}\theta_{ij}, \qquad \theta_{IJ} = \sum_{\substack{i\in I\\ j\in J}}\theta_{ij}, \qquad \mathfrak{c}_\mathcal{P}(\theta) = -\Big\langle \sum_I \sum_{\{i,j\}\subseteq I}\theta_{ij}, \mathsf{T}_1^\circ\Big\rangle. \tag{G7.4}$$

The parameter map $\theta \mapsto (\theta_I, \theta_{IJ})$ is linear. A coarse law, with density proportional to $e^{-\mathsf{E}_\theta \circ \iota_{\mathcal{P}}}$ relative to $\nu_0^{\otimes \mathcal{P}}$, exists if and only if

$$\bar{Z}_{\mathcal{P}}(\theta) = \int e^{-\mathsf{E}_\theta(\iota_{\mathcal{P}} \bar{z})} \nu_0^{\otimes \mathcal{P}}(d\bar{z}) < \infty. \tag{G7.5}$$

If $\mathsf{T}_1(z, z)$ is constant, $\mathsf{D} = 0$ is admissible, internal edges contribute only to $\mathfrak{c}_{\mathcal{P}}$, and node parameters add. When Z_0 is a group and T_1 is translation invariant, the diagonal is constant; the converse fails, as $\mathsf{T}_1(z, z') = (z - z')^2 zz'$ on $\mathbb{R}$ shows. If no nonzero functional of T_0 is ν_0-almost everywhere constant, the map D in (G7.2) is unique. [ESTABLISHED]

Proof. Substitute the identified variables into (G7.1). A cut edge $\{i, j\}$ with $i \in I$, $j \in J$ contributes $-\langle \theta_{ij}, \mathsf{T}_1(\bar{z}_I, \bar{z}_J) \rangle$, and these sum to the θ_{IJ} term. An internal edge contributes, by (G7.2), $-\langle \mathsf{D}^* \theta_{ij}, \mathsf{T}_0(\bar{z}_I) \rangle - \langle \theta_{ij}, \mathsf{T}_1^\circ \rangle$; the first part joins θ_I and the second joins $\mathfrak{c}_{\mathcal{P}}$. This proves (G7.3) and (G7.4), and the parameter map is a sum of linear maps. The normalizer clause is the definition of a law on the stated reference; the algebra does not decide finiteness. A translation-invariant T_1 has $\mathsf{T}_1(z, z) = \mathsf{T}_1(0, 0)$. The displayed T_1 vanishes on the diagonal but changes under $z \mapsto z + a$. If two maps D, D' satisfy (G7.2), then $(\mathsf{D} - \mathsf{D}')\mathsf{T}_0$ is almost everywhere constant, so $\langle (\mathsf{D} - \mathsf{D}')^* \varphi, \mathsf{T}_0 \rangle$ is almost everywhere constant for every functional φ, and the hypothesis forces $\mathsf{D} = \mathsf{D}'$. □

The theorem is energy closure, not closure of normalized laws. The reference $\nu_0^{\otimes \mathcal{P}}$, the normalizer, and any gauge action are further data: a compatible normalized family would satisfy $\mathbb{P}_\theta C_{\mathcal{P}} = \bar{\mathbb{P}}_{\theta_{\mathcal{P}}}$ for a declared channel $C_{\mathcal{P}}$ and the parameter map above, and energy precomposition does not supply that channel (Section 10.4). For the Gaussian family the channel exists and is identified in Proposition G7.5.

G7.2 Hard aggregation of the interaction precision

In trivialized coordinates z_i the interaction precision of Definition G3.1 and its energy are

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j), \qquad A_i \succeq 0, \quad W_{ij} = W_{ji} \succeq 0, \tag{G7.6}$$

$$\mathsf{E}(z) = \frac{1}{2} z^\top \Lambda z = \frac{1}{2} \sum_i z_i^\top A_i z_i + \frac{1}{2} \sum_{i<j} (z_i - z_j)^\top W_{ij} (z_i - z_j), \tag{G7.7}$$

the second identity being Proposition G3.2 with the factor one half. The form is a declared subfamily of the directed linear-Gaussian models (Propositions G3.7 and G3.9).

Proposition G7.3 (Exact Gaussian aggregation). For a hard partition $\mathcal{P}$ of the N agents, let $\widehat{S} \in \{0, 1\}^{N \times |\mathcal{P}|}$ with $\widehat{S}_{iI} = \mathbf{1}_{\{i \in I\}}$, put $S = \widehat{S} \otimes I_K$ and $\Lambda_{\mathrm{c}} = S^\top \Lambda S$, and let $n_I = |I|$. The columns of S are orthogonal but not normalized,

$$S^\top S = \mathrm{diag}(n_I) \otimes I_K. \tag{G7.8}$$

For every Λ of the form (G7.6),

$$(\Lambda_{\mathrm{c}})_{IJ} = -\sum_{\substack{i\in I\\ j\in J}} W_{ij} \quad (I \neq J), \qquad (\Lambda_{\mathrm{c}})_{II} = \sum_{i\in I} A_i + \sum_{\substack{i\in I\\ j\notin I}} W_{ij}, \tag{G7.9}$$

so $\Lambda_{\mathrm{c}} \succeq 0$ is again an interaction precision, with $A_I = \sum_{i\in I} A_i$ and $W_{IJ} = \sum_{i\in I, j\in J} W_{ij}$; every internal edge is annihilated, and a source h becomes $S^\top \mathsf{h}$. [ESTABLISHED]

Proof. Equation (G7.8) holds because each agent lies in exactly one cluster. The Gaussian energy with source, $\mathsf{E}(z) - \mathsf{h}^\top z$, is the graph-exponential energy (G7.1) with $\mathsf{T}_0(z) = (z, -\frac{1}{2}zz^\top)$, $\theta_i = (\mathsf{h}_i, A_i)$, $\mathsf{T}_1(z, z') = -\frac{1}{2}(z-z')(z-z')^\top$, and $\theta_{ij} = W_{ij}$, paired by $\langle(\mathsf{h}, A), (z, Z)\rangle = \mathsf{h}^\top z + \mathrm{Tr}(AZ)$. The pair statistic vanishes on the diagonal, so Theorem G7.2 applies with $\mathsf{D} = 0$ and $\mathsf{T}_1^\circ = 0$ and gives $A_I = \sum_{i\in I} A_i$, $\mathsf{h}_I = \sum_{i\in I} \mathsf{h}_i = (S^\top \mathsf{h})_I$, and $W_{IJ} = \sum W_{ij}$, with no internal-edge term. Reading the blocks of the coarse energy through Proposition G3.2 gives (G7.9), and $\Lambda_{\mathrm{c}} \succeq 0$ by congruence. □

This is the Galerkin coarse operator of unsmoothed aggregation algebraic multigrid, whose prolongator is the Boolean aggregate-membership matrix [23, Eqs. (2.1)–(2.2)]; for graph Laplacians the same piecewise-constant interpolation returns a graph Laplacian [22, Sec. 2]. Smoothed aggregation smooths this tentative prolongator and builds it from declared zero-energy modes, which covers systems with several unknowns per node [33]; $S = \widehat{S} \otimes I_K$ is the piecewise-constant prolongator for K unknowns per node. The general Galerkin construction is Section 10.4.1.

Proposition G7.4 (Aggregation is conditioning on zero internal disagreement). Let $\Lambda \succ 0$ on $\mathbb{R}^n$, let $\mathsf{h} \in \mathbb{R}^n$, and let $p = \mathcal{N}(\Lambda^{-1}\mathsf{h}, \Lambda^{-1})$. Let $B \in \mathbb{R}^{n\times r}$ have full column rank, let the columns of $B_\perp$ form an orthonormal basis of $\mathrm{range}(B)^\perp$, and put $w = B_\perp^\top z$. The conditional law of $z \sim p$ given w has a unique version that is weakly continuous in w. At $w = 0$ it is supported on $\mathrm{range}(B)$, and in the coordinate y with $z = By$ it is

$$p(dy \mid B_\perp^\top z = 0) = \mathcal{N}\big((B^\top \Lambda B)^{-1} B^\top \mathsf{h}, (B^\top \Lambda B)^{-1}\big), \tag{G7.10}$$

the Gaussian with natural parameters $(B^\top \mathsf{h}, B^\top \Lambda B)$. The same law results from conditioning, in the same continuous sense, on $\psi z = 0$ for any linear map ψ with $\ker \psi = \mathrm{range}(B)$. The linearity is material: when $r < n$, for a smooth χ on $\mathbb{R}^r$ with $\chi(y) \geq c > 0$ for every y, conditioning on the statistic $\chi(B^+ z) B_\perp^\top z$, which has the same zero set, multiplies the density of (G7.10) by $\chi(y)^{-(n-r)}$ before normalization, and so changes the law whenever χ is not constant. For $B = S$ the event $B_\perp^\top z = 0$ says $z_i = z_j$ whenever i and j share a cluster, so $(S^\top \mathsf{h}, \Lambda_{\mathrm{c}})$ are the natural parameters of the fine law conditioned on zero within-cluster disagreement. When $r < n$ this conditional law is singular with respect to p. [ESTABLISHED]

Proof. The map $(y, w) \mapsto By + B_\perp w$ is a linear bijection, so in the coordinates (y, w) the law p has a density proportional to $\exp[-\frac{1}{2}(By + B_\perp w)^\top \Lambda (By + B_\perp w) + \mathsf{h}^\top (By + B_\perp w)]$, with a constant Jacobian. Since $B_\perp^\top B = 0$ and $B_\perp^\top B_\perp = I$, the statistic is $B_\perp^\top z = w$. The joint density is continuous and positive, so dividing by the w-marginal density gives a conditional density jointly continuous in

(y, w); it is Gaussian with mean affine in w and constant covariance, hence weakly continuous, and two versions agree for almost every w, hence for every w by continuity. At $w = 0$ the exponent is $-\frac{1}{2}y^\top B^\top \Lambda B y + (B^\top \mathsf{h})^\top y$, which gives (G7.10). If $\ker \psi = \mathrm{range}(B)$, then $\psi B = 0$ gives $\psi = \psi B_\perp B_\perp^\top$ with $\psi B_\perp$ injective, so $\psi z = (\psi B_\perp) w$ and $w = (\psi B_\perp)^+ \psi z$; the two statistics generate the same σ-algebra, and their continuous versions correspond under $w \mapsto \psi B_\perp w$, which fixes 0. For the nonlinear statistic, $B^+ z = y$, and in the coordinates (y, u) with $u = \chi(y) w$ the density of p acquires the Jacobian factor $\chi(y)^{-(n-r)}$. The lower bound on χ and positive definiteness of Λ bound these joint densities near $u = 0$ by an integrable Gaussian function of y. Dominated convergence therefore gives a positive finite marginal density and a weakly continuous conditional version at zero, with the stated density. For $r < n$ this is a different law exactly when χ is not constant. Without an integrability condition the nonlinear statement fails: for independent standard normal (Y, W) and $\chi(y) = e^{-y^2}$, the candidate density at $u = 0$ is proportional to $e^{y^2/2}$, which is not integrable. For $B = S$, $\mathrm{range}(S)$ is the set of configurations constant on each cluster. Finally $\mathrm{range}(B)$ is Lebesgue null when $r < n$, while p has a positive density. □

Proposition G7.5 (Galerkin aggregation is a precision-dependent pushforward). Let $\Lambda \succ 0$ on $\mathbb{R}^n$, let $S \in \mathbb{R}^{n \times r}$ be injective, and put

$$\Lambda_{\mathrm{c}} = S^\top \Lambda S, \qquad P_\Lambda = \Lambda_{\mathrm{c}}^{-1} S^\top \Lambda. \tag{G7.11}$$

(i) The map $y \mapsto P_\Lambda y$ pushes $\mathcal{N}(\Lambda^{-1}\mathsf{h}, \Lambda^{-1})$ to $\mathcal{N}(\Lambda_{\mathrm{c}}^{-1} S^\top \mathsf{h}, \Lambda_{\mathrm{c}}^{-1})$, the Gaussian with natural parameters $(S^\top \mathsf{h}, S^\top \Lambda S)$ obtained by precomposing the energy with $y = Sc$. (ii) On the fixed-precision mean fiber $\{\mathcal{N}(\mu, \Lambda^{-1}) : \mu \in \mathbb{R}^n\}$, P_Λ is one parameter-independent deterministic kernel, and the Fisher metric of the pushed family satisfies $P_\Lambda^\top \Lambda_{\mathrm{c}} P_\Lambda = \Lambda S \Lambda_{\mathrm{c}}^{-1} S^\top \Lambda \preceq \Lambda$, with equality exactly on tangent directions in $\mathrm{range}(S)$. (iii) For nested aggregations $S = S_1 S_2$ with both factors injective, $P^{(S_2)}_{S_1^\top \Lambda S_1} P^{(S_1)}_\Lambda = P^{(S_1 S_2)}_\Lambda$, where the superscript names the aggregation matrix. (iv) Let the joint law of (o, Y) given the structural data be Gaussian with precision $\widehat{\Lambda}$ and information vector $\widehat{\mathsf{h}}$, both functions of (θ, X) alone, so that by Proposition G4.6 the conditional precision of Y given o is $\widehat{\Lambda}_{YY}$. Then $\mathrm{id}_O \otimes P_{\widehat{\Lambda}_{YY}}$ is a recognition-independent channel determined by (θ, X). It leaves the observation marginal, hence the evidence, unchanged, and it carries the conditional law of Y given o to the Gaussian with natural parameters $(S^\top(\widehat{\mathsf{h}}_Y - \widehat{\Lambda}_{Yo} o), S^\top \widehat{\Lambda}_{YY} S)$, which is the conditional law of c given o under the joint precomposed with $y = Sc$; that precomposed joint generally has a different observation marginal. (v) Let $\mathfrak{L}$ be a set of positive-definite $n \times n$ matrices. A single Markov kernel κ with $\mathcal{N}(\mu, \Lambda^{-1})\kappa = \mathcal{N}(P_\Lambda \mu, \Lambda_{\mathrm{c}}^{-1})$ for every $\mu \in \mathbb{R}^n$ and every $\Lambda \in \mathfrak{L}$ exists if and only if the subspace $\mathrm{range}(\Lambda S)$ is the same for every $\Lambda \in \mathfrak{L}$; then P_Λ is constant on $\mathfrak{L}$ and $\kappa(y, \cdot) = \delta_{P_\Lambda y}$ for Lebesgue-almost every y. For a hard partition with a cluster of at least two agents, the interaction precision with every $A_i = I_K$ and no edges, and the one obtained from it by replacing one self term in that cluster by $3I_K$, give different subspaces, so no parameter-independent kernel realizes aggregation on the interior of the interaction cone. [ESTABLISHED]

Proof. (i) $P_\Lambda \Lambda^{-1} P_\Lambda^\top = \Lambda_{\mathrm{c}}^{-1} S^\top \Lambda \Lambda^{-1} \Lambda S \Lambda_{\mathrm{c}}^{-1} = \Lambda_{\mathrm{c}}^{-1}$ and $P_\Lambda \Lambda^{-1} \mathsf{h} = \Lambda_{\mathrm{c}}^{-1} S^\top \mathsf{h}$, and a linear image of a Gaussian is Gaussian. (ii) The pushed family $\mu \mapsto \mathcal{N}(P_\Lambda \mu, \Lambda_{\mathrm{c}}^{-1})$ has Fisher matrix $P_\Lambda^\top \Lambda_{\mathrm{c}} P_\Lambda$, and

$\Lambda - \Lambda S \Lambda_c^{-1} S^\top \Lambda = \Lambda^{1/2}(I - \Pi)\Lambda^{1/2}$ with Π the orthogonal projection onto $\mathrm{range}(\Lambda^{1/2}S)$, whose null directions are exactly $\mathrm{range}(S)$. (iii) Substitute $\Lambda_1 = S_1^\top \Lambda S_1$ into $(S_2^\top \Lambda_1 S_2)^{-1} S_2^\top \Lambda_1 \Lambda_1^{-1} S_1^\top \Lambda$. (iv) The channel is the identity on o and depends only on $\widehat{\Lambda}_{YY}$, a function of (θ, X). The conditional law is $\mathcal{N}(\widehat{\Lambda}_{YY}^{-1}(\widehat{\mathsf{h}}_Y - \widehat{\Lambda}_{Yo}o), \widehat{\Lambda}_{YY}^{-1})$, so (i) with source $\widehat{\mathsf{h}}_Y - \widehat{\Lambda}_{Yo}o$ gives the stated image, and reading the precomposed joint density at fixed o as a function of c gives the same natural parameters. (v) Suppose κ exists and fix $\Lambda_0 \in \mathfrak{L}$. For a bounded continuous f on $\mathbb{R}^r$ put $F(y) = \int f(u)\kappa(y, du)$ and $F_0 = f \circ P_{\Lambda_0}$. By the hypothesis on κ and by (i), the Gaussian averages of F and of F_0 against $\mathcal{N}(\mu, \Lambda_0^{-1})$ agree for every μ, so the bounded function $F - F_0$ has zero convolution with the $\mathcal{N}(0, \Lambda_0^{-1})$ density. Its Fourier transform is a tempered distribution annihilated by multiplication with the nowhere-vanishing Gaussian transform, hence zero, and $F = F_0$ almost everywhere. Running f over $u \mapsto e^{i\omega^\top u}$ with rational ω and using continuity of characteristic functions gives $\kappa(y, \cdot) = \delta_{P_{\Lambda_0} y}$ for almost every y. Gaussian laws are absolutely continuous, so the hypothesis at any $\Lambda \in \mathfrak{L}$ requires $P_{\Lambda_0}\mu = P_\Lambda \mu$ for every μ. Both maps are left inverses of S, with kernels $\mathrm{range}(\Lambda_0 S)^\perp$ and $\mathrm{range}(\Lambda S)^\perp$, and a left inverse of S is determined by its kernel, so they agree exactly when $\mathrm{range}(\Lambda S) = \mathrm{range}(\Lambda_0 S)$. Conversely, if $\Lambda S = \Lambda_0 S G$ with G invertible, then $P_\Lambda = P_{\Lambda_0}$ and $\Lambda^{-1}\Lambda_0 S = S G^{-1}$ give $P_{\Lambda_0}\Lambda^{-1}P_{\Lambda_0}^\top = G^{-1}(S^\top \Lambda_0 S)^{-1} = \Lambda_c^{-1}$, so $\delta_{P_{\Lambda_0} y}$ serves every $\Lambda \in \mathfrak{L}$. For the final clause, the first precision is the identity, with $\mathrm{range}(\Lambda S) = \mathrm{range}(S)$, the vectors constant on every cluster. For the second, apply ΛS to the coarse vector equal to $v \neq 0$ on the cluster and zero elsewhere: the result equals $3v$ at one agent of the cluster and v at another, so it is not constant on the cluster and $\mathrm{range}(\Lambda S) \neq \mathrm{range}(S)$. □

By part (i), the conditional law of Proposition G7.4 is the image of p under P_Λ, which depends on Λ. With $B = \iota_{\mathcal{P}}$, the injection of Section 11.5, Proposition G7.4 identifies $\iota_{\mathcal{P}}^\top \Lambda \iota_{\mathcal{P}}$ as the precision of the fine law conditioned on exact internal parallel transport, the law-level reading of (11.26). The Schur term of (G2.19) is the data-processing defect of the mean submodel under $y \mapsto B^\top y$, and part (ii) exhibits the corresponding defect $\Lambda - \Lambda S \Lambda_c^{-1} S^\top \Lambda$ for P_Λ. When an inference history moves within one fixed-precision mean fiber, the Galerkin map acts pointwise as the pushforward of the kernel P_Λ, so the pointwise-Markov hypothesis of Corollary 7.51 is met; part (v) records what fails when the precision moves.

Part (ii) of Proposition G7.5 gives Fisher contraction on each fixed-precision fiber, and part (iii) gives it at every finite depth of a nested aggregation. Families on which $\mathrm{range}(\Lambda S)$ varies, the singular boundary member of the biadditive family, and unbounded depth are Open Problem G9.4.

G7.3 The exterior-informed interface

Proposition G7.5 takes the aggregation matrix S as given. When the latent coordinates are read by a linear-Gaussian exterior record, the precision that the record supplies and the reference precision form a pencil, and its leading generalized eigenvectors select a range. The next proposition shows that the linear statistic with this range keeps the most information about the record among linear statistics of its rank, computes the information it discards, identifies its projector with the Galerkin

projector of Proposition G7.5(iv), and records the metric conventions under which the discarded information is bracketed by Hilbert–Schmidt tails. Its untruncated two-block case is (G4.13).

Proposition G7.6 (Exterior-informed linear interface). Let the latent $Y \in \mathbb{R}^n$ have the reference law $p_\Lambda = \mathcal{N}(\Lambda^{-1}\mathsf{h}, \Lambda^{-1})$ with $\Lambda \succ 0$, and let the exterior record $o \in \mathbb{R}^m$ have the conditional law $p(\cdot \mid y) = \mathcal{N}(\mathsf{C}y + o_0, \Sigma_o)$ given $Y = y$, with $\mathsf{C} \in \mathbb{R}^{m\times n}$, $o_0 \in \mathbb{R}^m$, and $\Sigma_o \succ 0$; every jointly Gaussian pair (Y, o) with positive-definite joint covariance has this form. Put $\Lambda_{\mathrm{rec}} = \mathsf{C}^\top \Sigma_o^{-1} \mathsf{C}$, choose a generalized eigenbasis of the pencil $(\Lambda_{\mathrm{rec}}, \Lambda)$,

$$\Lambda_{\mathrm{rec}} x_k = \mathsf{d}_{\mathrm{ref},k} \Lambda x_k, \qquad x_j^\top \Lambda x_k = \delta_{jk}, \qquad \mathsf{d}_{\mathrm{ref},1} \geq \cdots \geq \mathsf{d}_{\mathrm{ref},n} \geq 0, \tag{G7.12}$$

and for $0 \leq r \leq n$ let $S_r = [x_1, \ldots, x_r]$ and

$$\widetilde{P}_r = S_r S_r^\top \Lambda, \tag{G7.13}$$

the Λ-orthogonal projector onto $\mathrm{span}\{x_1, \ldots, x_r\}$; the statistic $\widetilde{P}_r Y$ is the interface of rank r. Write $c_k = x_k^\top \Lambda Y$ for the Λ-dual coordinates, and $\tilde{o}_k = \mathsf{d}_{\mathrm{ref},k}^{-1/2} x_k^\top \mathsf{C}^\top \Sigma_o^{-1}(o - o_0)$ whenever $\mathsf{d}_{\mathrm{ref},k} > 0$. For a linear statistic $\mathbf{W}Y$ with $\mathbf{W} \in \mathbb{R}^{n'\times n}$, call $I(o; \mathbf{W}Y)$ its exterior-predictive information and $I(o; Y \mid \mathbf{W}Y) = I(o; Y) - I(o; \mathbf{W}Y)$ its discarded information, and let the reference-conditioned lift $\bar{p}_o^{\mathbf{W}}$ be the law of Y obtained by drawing $\mathbf{W}Y$ from its conditional law given o and then Y from the conditional law of p_Λ given $\mathbf{W}Y$. For $\mathbf{W} \in \mathbb{R}^{n\times n}$ the surrogate posterior and surrogate evidence

$$\widehat{p}_o^{\mathbf{W}}(dy) = \frac{p_\Lambda(y) p(o \mid \mathbf{W}y)}{\widehat{Z}^{\mathbf{W}}(o)} dy, \qquad \widehat{Z}^{\mathbf{W}}(o) = \int_{\mathbb{R}^n} p_\Lambda(y) p(o \mid \mathbf{W}y) dy, \tag{G7.14}$$

keep the full reference law and read the record at $\mathbf{W}y$, that is, with mean $\mathsf{C}\mathbf{W}y + o_0$. Write $p(\cdot \mid o)$ for the posterior law of Y and $p(o)$ for the evidence. Then the following hold for every h and every o_0. (i) The conditional precision of Y given o is $\Lambda + \Lambda_{\mathrm{rec}}$, and the x_k are generalized eigenvectors of $(\Lambda_{\mathrm{rec}}, \Lambda + \Lambda_{\mathrm{rec}})$ with eigenvalues $\mathsf{d}_k = \mathsf{d}_{\mathrm{ref},k}/(1 + \mathsf{d}_{\mathrm{ref},k})$. At most $\min(n, m)$ of the $\mathsf{d}_{\mathrm{ref},k}$ are positive, and $\check{\rho}_k = \mathsf{d}_k^{1/2}$, $k \leq \min(n, m)$, are the canonical correlations of (Y, o) in the sense of Proposition G4.4, so that $\mathsf{d}_{\mathrm{ref},k} = \check{\rho}_k^2/(1 - \check{\rho}_k^2)$; put $\check{\rho}_k = 0$ for $k > \min(n, m)$. The coordinates are canonical variates: $\mathrm{Cov}(c_j, c_k) = \delta_{jk}$, $\mathrm{Cov}(\tilde{o}_j, \tilde{o}_k) = (1 + \mathsf{d}_{\mathrm{ref},k})\delta_{jk}$, the correlation of c_j and $\tilde{o}_k$ is $\check{\rho}_k \delta_{jk}$, and a linear functional of o uncorrelated with every $\tilde{o}_k$ is uncorrelated with Y. The interface $\widetilde{P}_r Y = S_r(c_1, \ldots, c_r)^\top$ generates the same σ-algebra as $(c_1, \ldots, c_r)$. (ii) For every $\mathbf{W} \in \mathbb{R}^{n'\times n}$ of rank at most r,

$$I(o; \mathbf{W}Y) \leq \frac{1}{2} \sum_{k=1}^{r} \log(1 + \mathsf{d}_{\mathrm{ref},k}) = I(o; \widetilde{P}_r Y), \qquad I(o; Y \mid \mathbf{W}Y) = \int D_{\mathrm{KL}}\left(p(\cdot \mid o) \| \bar{p}_o^{\mathbf{W}}\right) p(o) do, \tag{G7.15}$$

so among linear statistics of rank at most r the interface maximizes the exterior-predictive information and minimizes the discarded information, which is the evidence-averaged relative entropy of the posterior from the reference-conditioned lift. If $r < n$ and $\mathsf{d}_{\mathrm{ref},r} > \mathsf{d}_{\mathrm{ref},r+1}$, equality in the inequality holds only when $\mathbf{W}Y$ and $\widetilde{P}_r Y$ generate the same σ-algebra. For every $\mathbf{W} \in \mathbb{R}^{n\times n}$ of

rank at most r, $\int D_{\mathrm{KL}}(p(\cdot \mid o)\|\widehat{p}_o^{\mathbf{W}})p(o)do \geq I(o;Y \mid \mathbf{W}Y)$. (iii) The minimal discarded information is

$$I(o;Y \mid \widetilde{P}_r Y) = \frac{1}{2}\sum_{k>r}\log\left(1+\mathsf{d}_{\mathrm{ref},k}\right) = -\frac{1}{2}\sum_{k>r}\log\left(1-\check{\rho}_k^2\right) = \int D_{\mathrm{KL}}\left(p(\cdot \mid o)\|\widehat{p}_o^{\widetilde{P}_r}\right)p(o)do \tag{G7.16}$$

nats. At $r=0$ it is $I(o;Y)$, the two-block gap (G4.13) of the joint precision of (Y,o). By (ii), $\widetilde{P}_r$ minimizes the evidence-averaged surrogate relative entropy over all $\mathbf{W}\in\mathbb{R}^{n\times n}$ of rank at most r. (iv) Under p_Λ the components $\widetilde{P}_r Y$ and $(I-\widetilde{P}_r)Y$ are independent, and with $w=(I-\widetilde{P}_r)y$,

$$p(o \mid y) = p(o \mid \widetilde{P}_r y)\exp\left[(o-o_0)^\top \Sigma_o^{-1}\mathsf{C}w - \frac{1}{2}w^\top \Lambda_{\mathrm{rec}} w\right]. \tag{G7.17}$$

For every o the law of $\widetilde{P}_r Y$ is the same under $p(\cdot \mid o)$ and under $\widehat{p}_o^{\widetilde{P}_r}$, so the coarse relative entropy vanishes, and $\widehat{p}_o^{\widetilde{P}_r} = \bar{p}_o^{\widetilde{P}_r}$. In the notation of (G7.11) with $S=S_r$, $P_{\Lambda+\Lambda_{\mathrm{rec}}} = P_\Lambda = S_r^\top \Lambda$, so $\widetilde{P}_r = S_r P_{\Lambda+\Lambda_{\mathrm{rec}}}$ and the posterior law of $\widetilde{P}_r Y$ is the image under S_r of the Galerkin pushforward of Proposition G7.5(iv). The identities (G7.16) and (G7.17), the independence, the vanishing of the coarse relative entropy, $\widehat{p}_o^{\widetilde{P}_r} = \bar{p}_o^{\widetilde{P}_r}$, the Galerkin identification, and (G7.18) hold verbatim when $x_1,\dots,x_n$ is replaced by any reordering of any Λ-orthonormal generalized eigenbasis of the pencil, the sums over $k>r$ running over its last $n-r$ vectors, while the minimality statements of (ii) and (iii) use the ordering; when $r<n$ and $\mathsf{d}_{\mathrm{ref},r} > \mathsf{d}_{\mathrm{ref},r+1}$, the range of $\widetilde{P}_r$ is the sum of the complete eigenspaces with eigenvalue at least $\mathsf{d}_{\mathrm{ref},r}$, and $\widetilde{P}_r$ is determined by $(\Lambda,\Lambda_{\mathrm{rec}})$. Neither the spectral range nor Λ-orthogonality can be dropped: for $n=2$, $m=1$, $\mathsf{h}=0$, $\Lambda=I_2$, $\mathsf{C}=e_1^\top$, $\Sigma_o=1$, $o_0=0$, and $r=1$, the Λ-orthogonal projector onto the span of $(\cos\theta,\sin\theta)^\top$ with $0<\theta<\pi/2$ gives the coarse coordinate the posterior variance $1-\frac{1}{2}\cos^2\theta$ under $p(\cdot \mid o)$ and $(1+\cos^2\theta)^{-1}$ under the surrogate, and the projector $y\mapsto(y_1+\kappa y_2,0)^\top$ with $\kappa\neq 0$, whose range $\mathrm{span}\{x_1\}$ is spectral, gives $\frac{1}{2}+\kappa^2$ and $(1+\kappa^2)/(2+\kappa^2)$; in both cases the coarse relative entropy is positive for every o. The surrogate evidence is a probability density on $\mathbb{R}^m$, and

$$\int \log\frac{\widehat{Z}^{\widetilde{P}_r}(o)}{p(o)}p(o)do = -\frac{1}{2}\sum_{k>r}\left[\mathsf{d}_{\mathrm{ref},k} - \log\left(1+\mathsf{d}_{\mathrm{ref},k}\right) + \mathsf{d}_{\mathrm{ref},k}\left(x_k^\top \mathsf{h}\right)^2\right], \tag{G7.18}$$

which equals $-D_{\mathrm{KL}}(p(o)do\|\widehat{Z}^{\widetilde{P}_r}(o)do)$; $\log(\widehat{Z}^{\widetilde{P}_r}(o)/p(o))$ is unbounded below in o when $\mathsf{d}_{\mathrm{ref},k}>0$ for some $k>r$. (v) Under a passive change of coordinates $y'=R_Y y+a_Y$, $o'=R_o o+a_o$ with $R_Y\in\mathrm{GL}(n,\mathbb{R})$, $R_o\in\mathrm{GL}(m,\mathbb{R})$, $a_Y\in\mathbb{R}^n$, and $a_o\in\mathbb{R}^m$, the transformed model has $\Lambda'=R_Y^{-\top}\Lambda R_Y^{-1}$ and $\Lambda'_{\mathrm{rec}}=R_Y^{-\top}\Lambda_{\mathrm{rec}}R_Y^{-1}$. The eigenvalues $\mathsf{d}_{\mathrm{ref},k}$, d_k, and $\check{\rho}_k$ and the discarded information (G7.16) are invariant, the vectors $R_Y x_k$ form a Λ'-orthonormal generalized eigenbasis, and the projector and interface are covariant: $\widetilde{P}'_r = R_Y\widetilde{P}_r R_Y^{-1}$, and $\widetilde{P}'_r Y' = R_Y\widetilde{P}_r Y + \widetilde{P}'_r a_Y$ generates the same σ-algebra as $\widetilde{P}_r Y$. (vi) For $\mathsf{G}\succ 0$ on $\mathbb{R}^m$ define the Hilbert–Schmidt tail, without a factor one half,

$$\mathrm{HS}_r(\mathsf{G}) = \mathrm{Tr}\left(\Lambda^{-1}(I-\widetilde{P}_r)^\top \mathsf{C}^\top \mathsf{G}\mathsf{C}(I-\widetilde{P}_r)\right). \tag{G7.19}$$

Then $\mathrm{HS}_r(\Sigma_o^{-1}) = \sum_{k>r} \mathsf{d}_{\mathrm{ref},k}$, which equals the expected realized exterior error $\int |\Sigma_o^{-1/2}\mathsf{C}(I - \widetilde{P}_r)(y - \Lambda^{-1}\mathsf{h})|^2 p_\Lambda(dy)$; for the marginal covariance $\Sigma_{\mathrm{ext}} = \mathsf{C}\Lambda^{-1}\mathsf{C}^\top + \Sigma_o$ of o, $\mathrm{HS}_r(\Sigma_{\mathrm{ext}}^{-1}) = \sum_{k>r} \check{\rho}_k^2$; and

$$\frac{1}{2}\mathrm{HS}_r\left(\Sigma_{\mathrm{ext}}^{-1}\right) \le I(o; Y \mid \widetilde{P}_r Y) \le \frac{1}{2}\mathrm{HS}_r\left(\Sigma_o^{-1}\right), \tag{G7.20}$$

with equality in either place only when $\mathsf{d}_{\mathrm{ref},k} = 0$ for every $k > r$. [ESTABLISHED]

Proof. Coordinates. The matrix $\Lambda^{-1/2}\Lambda_{\mathrm{rec}}\Lambda^{-1/2} \succeq 0$ has an orthonormal eigenbasis, and its image under $\Lambda^{-1/2}$ is a basis as in (G7.12), with every $\mathsf{d}_{\mathrm{ref},k} \ge 0$. With $S_n = [x_1, \dots, x_n]$ one has $S_n^\top \Lambda S_n = I_n$ and $S_n^\top \Lambda_{\mathrm{rec}} S_n = \mathrm{diag}(\mathsf{d}_{\mathrm{ref},k})$, hence $\Lambda^{-1} = S_n S_n^\top$ and $y = S_n c$ with $c = S_n^\top \Lambda y$; under p_Λ the vector c is $\mathcal{N}(\bar{c}, I_n)$ with $\bar{c} = S_n^\top \mathsf{h}$, so $\bar{c}_k = x_k^\top \mathsf{h}$. By congruence the number q of positive $\mathsf{d}_{\mathrm{ref},k}$ is $\mathrm{rank}\,\Lambda_{\mathrm{rec}} \le \min(n, m)$, and these are $k = 1, \dots, q$. The vectors $f_k = \mathsf{d}_{\mathrm{ref},k}^{-1/2}\Sigma_o^{-1/2}\mathsf{C}x_k$, $k \le q$, are orthonormal in $\mathbb{R}^m$, because $f_j^\top f_k = (\mathsf{d}_{\mathrm{ref},j}\mathsf{d}_{\mathrm{ref},k})^{-1/2} x_j^\top \Lambda_{\mathrm{rec}} x_k$, and $\Sigma_o^{-1/2}\mathsf{C}x_k = 0$ for $k > q$, because its squared norm is $\mathsf{d}_{\mathrm{ref},k}$. Hence $\Sigma_o^{-1/2}\mathsf{C}y = \sum_{k \le q} \mathsf{d}_{\mathrm{ref},k}^{1/2} c_k f_k$ and $\tilde{o}_k = f_k^\top \Sigma_o^{-1/2}(o - o_0)$. Let $o^\perp$ be the orthogonal projection of $\Sigma_o^{-1/2}(o - o_0)$ onto the orthogonal complement of $\mathrm{span}\{f_k\}$. The map $o \mapsto (\tilde{o}_1, \dots, \tilde{o}_q, o^\perp)$ is an affine bijection with constant Jacobian, and

$$\log p(o \mid y) = -\frac{m}{2}\log(2\pi) - \frac{1}{2}\log\det\Sigma_o - \frac{1}{2}\sum_{k \le q}\left(\tilde{o}_k - \mathsf{d}_{\mathrm{ref},k}^{1/2} c_k\right)^2 - \frac{1}{2}|o^\perp|^2. \tag{G7.21}$$

Consequently the pairs $(c_k, \tilde{o}_k)$, $k \le q$, the coordinates c_k, $k > q$, and $o^\perp$ are independent; $\tilde{o}_k = \mathsf{d}_{\mathrm{ref},k}^{1/2} c_k + \tilde{\xi}_k$ with $\tilde{\xi}_k$ standard normal; and given o the c_k are independent with laws $\mathcal{N}\big((\bar{c}_k + \mathsf{d}_{\mathrm{ref},k}^{1/2}\tilde{o}_k)/(1+\mathsf{d}_{\mathrm{ref},k}), (1+\mathsf{d}_{\mathrm{ref},k})^{-1}\big)$ for $k \le q$ and $\mathcal{N}(\bar{c}_k, 1)$ for $k > q$. The joint law of (Y, o) is the image of the nondegenerate Gaussian law of $(Y, o - \mathsf{C}Y - o_0)$ under a linear bijection; conversely, for a jointly Gaussian pair with positive-definite covariance, $\mathsf{C} = \Sigma_{oY}\Sigma_{YY}^{-1}$ and $\Sigma_o = \Sigma_{oo} - \Sigma_{oY}\Sigma_{YY}^{-1}\Sigma_{Yo} \succ 0$ give the stated form.

(i) The conditional covariance of c is $\mathrm{diag}((1 + \mathsf{d}_{\mathrm{ref},k})^{-1})$, so that of Y is $S_n\,\mathrm{diag}((1 + \mathsf{d}_{\mathrm{ref},k})^{-1})S_n^\top = (\Lambda + \Lambda_{\mathrm{rec}})^{-1}$, since $S_n^{-\top} S_n^{-1} = \Lambda$ and $S_n^{-\top}\,\mathrm{diag}(\mathsf{d}_{\mathrm{ref},k})S_n^{-1} = \Lambda_{\mathrm{rec}}$. The eigenvalue correspondence is Proposition G5.8 with $L = \Lambda_{\mathrm{rec}}$, $\Lambda_{\mathrm{ref}} = \Lambda$, and $\mathsf{a} = 1$. Expanding $\log p_\Lambda(y) + \log p(o \mid y)$ shows that the joint precision of (Y, o) has blocks $\widehat{\Lambda}_{YY} = \Lambda + \Lambda_{\mathrm{rec}}$, $\widehat{\Lambda}_{Yo} = -\mathsf{C}^\top\Sigma_o^{-1}$, and $\widehat{\Lambda}_{oo} = \Sigma_o^{-1}$. For the factor $\mathcal{K}$ of (G4.11) built from these blocks, $\mathcal{K}\mathcal{K}^\top = \widehat{\Lambda}_{YY}^{-1/2}\Lambda_{\mathrm{rec}}\widehat{\Lambda}_{YY}^{-1/2}$ is similar to $\widehat{\Lambda}_{YY}^{-1}\Lambda_{\mathrm{rec}}$, whose eigenvalues are the d_k. The eigenvalues of $\mathcal{K}\mathcal{K}^\top$ are the squared singular values of the $n \times m$ matrix $\mathcal{K}$ together with $n - \min(n, m)$ zeros; since $\mathsf{d}_k = 0$ exactly for $k > q$ and $q \le \min(n, m)$, the ordered squared singular values are $\mathsf{d}_1, \dots, \mathsf{d}_{\min(n,m)}$, which Proposition G4.4 identifies with the squared canonical correlations; $1 + \mathsf{d}_{\mathrm{ref},k} = (1 - \mathsf{d}_k)^{-1}$ gives the inverse relation. The covariances of the canonical variates follow from the independence structure above and $\tilde{o}_k = \mathsf{d}_{\mathrm{ref},k}^{1/2} c_k + \tilde{\xi}_k$, and the correlation is $\mathsf{d}_{\mathrm{ref},k}^{1/2}(1 + \mathsf{d}_{\mathrm{ref},k})^{-1/2} = \check{\rho}_k$. For $a \in \mathbb{R}^m$, $\mathrm{Cov}(o) = \Sigma_{\mathrm{ext}}$ and $\Sigma_{\mathrm{ext}}\Sigma_o^{-1}\mathsf{C}x_k = \mathsf{C}(\Lambda^{-1}\Lambda_{\mathrm{rec}} + I)x_k = (1 + \mathsf{d}_{\mathrm{ref},k})\mathsf{C}x_k$ give $\mathrm{Cov}(a^\top o, \tilde{o}_k) = \mathsf{d}_{\mathrm{ref},k}^{-1/2}(1 + \mathsf{d}_{\mathrm{ref},k})a^\top \mathsf{C}x_k$. If this vanishes for every $k \le q$, then $a^\top \mathsf{C}S_n = 0$, because $\mathsf{C}x_k = 0$ for $k > q$, so $a^\top \mathsf{C} = 0$ and $\mathrm{Cov}(Y, a^\top o) = \Lambda^{-1}\mathsf{C}^\top a = 0$. Finally $\widetilde{P}_r Y = S_r S_r^\top \Lambda Y = S_r(c_1, \dots, c_r)^\top$ with S_r injective.

(ii) Let $r' \le r$ be the rank of $\mathbf{W}$, let the columns of $\mathbf{O}_1 \in \mathbb{R}^{n \times r'}$ be an orthonormal basis of the row space of $\mathbf{W}S_n$, and complete them to an orthogonal matrix $\mathbf{O} = [\mathbf{O}_1, \mathbf{O}_2]$. Put $z_1 = \mathbf{O}_1^\top c$

and $z_2 = \mathbf{O}_2^\top c$. Since $\mathbf{W}S_n\mathbf{O}_2 = 0$, $\mathbf{W}Y = \mathbf{W}S_n c = (\mathbf{W}S_n\mathbf{O}_1)z_1$ with $\mathbf{W}S_n\mathbf{O}_1$ injective, so $\mathbf{W}Y$ and z_1 are linear functions of each other, generate the same σ-algebra, and have the same mutual information with o. Under p_Λ, $(z_1, z_2) \sim \mathcal{N}(\mathbf{O}^\top \bar{c}, I_n)$, so z_1 and z_2 are independent with unit covariances; given o, z_1 has the constant covariance $\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1$ with $\mathsf{M} = \operatorname{diag}((1 + \mathsf{d}_{\mathrm{ref},k})^{-1})$. The expected logarithm of the ratio of the conditional and marginal Gaussian densities of z_1 is therefore

$$I(o; \mathbf{W}Y) = I(o; z_1) = -\frac{1}{2} \log\det\left(\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1\right). \tag{G7.22}$$

Order eigenvalues increasingly. By the Courant–Fischer characterization, $\lambda_i(\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1)$ is the minimum over i-dimensional subspaces $U \subseteq \mathbb{R}^{r'}$ of the maximum of $u^\top \mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1 u$ over unit $u \in U$; since $\mathbf{O}_1$ is an isometry, $\mathbf{O}_1 U$ runs over some of the i-dimensional subspaces of $\mathbb{R}^n$, so $\lambda_i(\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1) \geq \lambda_i(\mathsf{M}) = (1 + \mathsf{d}_{\mathrm{ref},i})^{-1}$ for $i \leq r'$. Hence $\det(\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1) \geq \prod_{i \leq r'}(1 + \mathsf{d}_{\mathrm{ref},i})^{-1} \geq \prod_{i \leq r}(1 + \mathsf{d}_{\mathrm{ref},i})^{-1}$, the last step because every factor is at most one, and (G7.22) gives the inequality in (G7.15). For $\mathbf{W} = \widetilde{P}_r$, by (i), $\mathbf{O}_1$ may be taken to be the first r coordinate vectors, and $\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1 = \operatorname{diag}_{k \leq r}((1 + \mathsf{d}_{\mathrm{ref},k})^{-1})$ attains the bound. Suppose now $r < n$, $\mathsf{d}_{\mathrm{ref},r} > \mathsf{d}_{\mathrm{ref},r+1}$, and equality. Then $\mathsf{d}_{\mathrm{ref},i} \geq \mathsf{d}_{\mathrm{ref},r} > 0$ for $i \leq r$, so the second step is strict unless $r' = r$; and since each of the r factor ratios is at least one, $\lambda_i(\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1) = \mathsf{M}_{ii}$ for every $i \leq r$. With weights $\omega_k = |\mathbf{O}_1^\top e_k|^2 \in [0, 1]$, $\sum_k \omega_k = r$, one has $\operatorname{Tr}(\mathbf{O}_1^\top \mathsf{M}\mathbf{O}_1) = \sum_k \mathsf{M}_{kk}\omega_k$, and

$$\sum_k \mathsf{M}_{kk}\omega_k - \sum_{k \leq r} \mathsf{M}_{kk} = \sum_{k \leq r} \mathsf{M}_{kk}(\omega_k - 1) + \sum_{k > r} \mathsf{M}_{kk}\omega_k \geq (\mathsf{M}_{r+1,r+1} - \mathsf{M}_{rr}) \sum_{k > r} \omega_k, \tag{G7.23}$$

using $\mathsf{M}_{kk} \leq \mathsf{M}_{rr}$ and $\omega_k \leq 1$ for $k \leq r$, $\mathsf{M}_{kk} \geq \mathsf{M}_{r+1,r+1}$ for $k > r$, and $\sum_{k \leq r}(\omega_k - 1) = -\sum_{k > r} \omega_k$. The left side vanishes by the equality of eigenvalues and $\mathsf{M}_{r+1,r+1} > \mathsf{M}_{rr}$, so $\omega_k = 0$ for $k > r$, the range of $\mathbf{O}_1$ is spanned by the first r coordinate vectors, and z_1 is an invertible linear function of $(c_1, \ldots, c_r)$, whose σ-algebra is that of $\widetilde{P}_r Y$ by (i). For the lift, pass to the coordinates (z_1, z_2) by a linear bijection, which preserves relative entropy (Lemma 3.12(e)). The posterior density is $p(z_1 \mid o)p(z_2 \mid z_1, o)$ and the lift density is $p(z_1 \mid o)p_\Lambda(z_2)$, because under the reference z_2 is independent of z_1. By Bayes' rule $p(z_2 \mid z_1, o) = p(o \mid z)p_\Lambda(z_2)/p(o \mid z_1)$ with $p(o \mid z_1) = \int p(o \mid z)p_\Lambda(dz_2)$, so the logarithm of the ratio of the two densities is $\log p(o \mid Y) - \log p(o \mid z_1)$, and its expectation under the joint law is $\mathbb{E}\log(p(o \mid Y)/p(o)) - \mathbb{E}\log(p(o \mid z_1)/p(o)) = I(o; Y) - I(o; z_1)$, all terms being finite Gaussian second moments. This is the identity in (G7.15). For $\mathbf{W} \in \mathbb{R}^{n \times n}$, $\mathbf{W}y = \mathbf{W}S_n\mathbf{O}_1 z_1$, so $p(o \mid \mathbf{W}y)$ depends on z_1 alone and the density of $\widehat{p}_o^{\mathbf{W}}$ is proportional to $p_\Lambda(z_1)p(o \mid \mathbf{W}S_n\mathbf{O}_1 z_1)p_\Lambda(z_2)$; its conditional law of z_2 given z_1 is $p_\Lambda(z_2)$, that of the lift. Lemma 3.12(a), with z_1 as the first coordinate, writes $D_{\mathrm{KL}}(p(\cdot \mid o) \| \widehat{p}_o^{\mathbf{W}})$ as the relative entropy of the z_1-marginals plus $\int D_{\mathrm{KL}}(p(z_2 \mid z_1, o) \| p_\Lambda(z_2))p(dz_1 \mid o)$, and the same lemma writes $D_{\mathrm{KL}}(p(\cdot \mid o) \| \bar{p}_o^{\mathbf{W}})$ as this second term alone, since the lift has the posterior z_1-marginal. Integrating over o gives the last inequality of (ii).

(iii) With $\mathbf{O}_1 = \mathbf{O} = I_n$, (G7.22) gives $I(o; Y) = \frac{1}{2}\sum_{k \leq n} \log(1 + \mathsf{d}_{\mathrm{ref},k})$, and (ii) gives $I(o; \widetilde{P}_r Y) = \frac{1}{2}\sum_{k \leq r} \log(1 + \mathsf{d}_{\mathrm{ref},k})$; their difference is the first expression in (G7.16), and $1 + \mathsf{d}_{\mathrm{ref},k} = (1 - \check{\rho}_k^2)^{-1}$ with (i) gives the second. At $r = 0$, Proposition G4.4 identifies $-\frac{1}{2}\sum_{k \leq \min(n,m)} \log(1 - \check{\rho}_k^2)$ with

the two-block gap and with the mutual information of the blocks. The last equality in (G7.16) is $\widehat{p}_o^{\widetilde{P}_r} = \bar{p}_o^{\widetilde{P}_r}$ from (iv) with the identity in (G7.15). None of these steps uses h or o_0.

(iv) $\widetilde{P}_r Y = \sum_{k\le r} c_k x_k$ and $(I-\widetilde{P}_r)Y = \sum_{k>r} c_k x_k$ are functions of disjoint sets of the independent coordinates c_k. The point $\widetilde{P}_r y$ has coordinates $(c_1,\dots,c_r,0,\dots,0)$, so $p(o \mid \widetilde{P}_r y)$ is (G7.21) with $c_k = 0$ for $k > r$, and the ratio $p(o \mid y)/p(o \mid \widetilde{P}_r y)$ is $\exp\sum_{r<k\le q}[\mathsf{d}_{\mathrm{ref},k}^{1/2}\tilde{o}_k c_k - \frac{1}{2}\mathsf{d}_{\mathrm{ref},k}c_k^2]$. Since $(o-o_0)^\top \Sigma_o^{-1}\mathsf{C}x_k = \mathsf{d}_{\mathrm{ref},k}^{1/2}\tilde{o}_k$ for $k \le q$, $\mathsf{C}x_k = 0$ for $k > q$, and $w^\top \Lambda_{\mathrm{rec}} w = \sum_{k>r}\mathsf{d}_{\mathrm{ref},k}c_k^2$, this is (G7.17). In the coordinates c, the density of $\widehat{p}_o^{\widetilde{P}_r}$ is proportional to $\prod_k \mathcal{N}(c_k;\bar{c}_k,1)\exp[-\frac{1}{2}\sum_{k\le\min(r,q)}(\tilde{o}_k - \mathsf{d}_{\mathrm{ref},k}^{1/2}c_k)^2]$. Under it the c_k are independent, with the posterior laws above for $k \le r$ and the reference laws $\mathcal{N}(\bar{c}_k,1)$ for $k > r$. The first statement gives equality of the laws of $(c_1,\dots,c_r)$, hence of $\widetilde{P}_r Y$; the second says that the surrogate is the posterior law of $(c_1,\dots,c_r)$ followed by the reference conditional law of the remaining coordinates, which by independence is their reference law, and this is $\bar{p}_o^{\widetilde{P}_r}$. For the Galerkin projector, $x_k^\top \Lambda_{\mathrm{rec}} = \mathsf{d}_{\mathrm{ref},k}x_k^\top\Lambda$ gives $S_r^\top(\Lambda+\Lambda_{\mathrm{rec}}) = \mathrm{diag}_{k\le r}(1+\mathsf{d}_{\mathrm{ref},k})S_r^\top\Lambda$ and $S_r^\top(\Lambda+\Lambda_{\mathrm{rec}})S_r = \mathrm{diag}_{k\le r}(1+\mathsf{d}_{\mathrm{ref},k})$, so $P_{\Lambda+\Lambda_{\mathrm{rec}}} = S_r^\top\Lambda = P_\Lambda$. By (i), $\Lambda+\Lambda_{\mathrm{rec}} = \widehat{\Lambda}_{YY}$, so the posterior law of $P_{\widehat{\Lambda}_{YY}}Y$ is the pushforward of Proposition G7.5(iv), and $\widetilde{P}_r Y = S_r P_{\widehat{\Lambda}_{YY}}Y$. The arguments of (iii) and of this part use only that $x_1,\dots,x_n$ is a Λ-orthonormal generalized eigenbasis; the order enters only the inequality of (ii). When $\mathsf{d}_{\mathrm{ref},r} > \mathsf{d}_{\mathrm{ref},r+1}$, the vectors $x_1,\dots,x_r$ span the sum of the eigenspaces with eigenvalue at least $\mathsf{d}_{\mathrm{ref},r}$, a subspace fixed by the pencil, and a Λ-orthogonal projector is determined by its range. In the example, $\Lambda_{\mathrm{rec}} = \mathrm{diag}(1,0)$, $x_k = e_k$, and $o = Y_1 + \xi$ with $\xi \sim \mathcal{N}(0,1)$. For $v = (\cos\theta,\sin\theta)^\top$ the coarse coordinate $t = v^\top Y$ has variance 1 and covariance $\cos\theta$ with o, whose variance is 2, so its posterior variance is $1-\frac{1}{2}\cos^2\theta$; the surrogate reads the record at $\mathsf{C}vv^\top y = t\cos\theta$, and since t and $v_\perp^\top Y$ are independent standard normal under the reference, its coarse law is proportional to $\mathcal{N}(t;0,1)\mathcal{N}(o;t\cos\theta,1)$, with variance $(1+\cos^2\theta)^{-1}$. Since $0<\cos^2\theta<1$, $(1-\frac{1}{2}\cos^2\theta)(1+\cos^2\theta) = 1+\frac{1}{2}\cos^2\theta\sin^2\theta > 1$, so the variances differ. For the second projector, which is idempotent with range $\mathrm{span}\{e_1\}$ and not symmetric, the coarse coordinate $t = Y_1 + \kappa Y_2$ has variance $1+\kappa^2$ and covariance 1 with o, so its posterior variance is $\frac{1}{2}+\kappa^2$; the surrogate reads the record at t, and a surrogate density of the form $p_\Lambda(y)g(t)$ has t-marginal proportional to the reference law of t times g, here $\mathcal{N}(t;0,1+\kappa^2)\mathcal{N}(o;t,1)$, with variance $(1+\kappa^2)/(2+\kappa^2)$; and $(\frac{1}{2}+\kappa^2)(2+\kappa^2) - (1+\kappa^2) = \frac{3}{2}\kappa^2+\kappa^4 > 0$. Two Gaussian laws with different variances have positive relative entropy. For the evidence, $\int \widehat{Z}^{\mathbf{W}}(o)do = 1$ by Tonelli's theorem. By the independence structure, under $p(o)$ the coordinates $\tilde{o}_k$, $k\le q$, are independent $\mathcal{N}(\mathsf{d}_{\mathrm{ref},k}^{1/2}\bar{c}_k, 1+\mathsf{d}_{\mathrm{ref},k})$ and independent of the standard normal $o^\perp$; under $\widehat{Z}^{\widetilde{P}_r}$, the law of o when $Y \sim p_\Lambda$ and $o \sim p(\cdot \mid \widetilde{P}_r Y)$, the same holds except that $\tilde{o}_k \sim \mathcal{N}(0,1)$ for $r < k \le q$. The two densities share the Jacobian of the affine bijection, so

$$\log\frac{\widehat{Z}^{\widetilde{P}_r}(o)}{p(o)} = \sum_{r<k\le q}\left[\frac{(\tilde{o}_k - \mathsf{d}_{\mathrm{ref},k}^{1/2}\bar{c}_k)^2}{2(1+\mathsf{d}_{\mathrm{ref},k})} - \frac{\tilde{o}_k^2}{2} + \frac{1}{2}\log(1+\mathsf{d}_{\mathrm{ref},k})\right]. \tag{G7.24}$$

Under $p(o)$, $\mathbb{E}(\tilde{o}_k - \mathsf{d}_{\mathrm{ref},k}^{1/2}\bar{c}_k)^2 = 1+\mathsf{d}_{\mathrm{ref},k}$ and $\mathbb{E}\tilde{o}_k^2 = 1+\mathsf{d}_{\mathrm{ref},k}+\mathsf{d}_{\mathrm{ref},k}\bar{c}_k^2$, which gives (G7.18) with $\bar{c}_k = x_k^\top \mathsf{h}$, the terms with $k > q$ being zero; its left side is $-D_{\mathrm{KL}}(p(o)do\|\widehat{Z}^{\widetilde{P}_r}(o)do)$ because both

evidences are probability densities. The coefficient of $\tilde{o}_k^2$ is $-\frac{1}{2}\mathsf{d}_{\mathrm{ref},k}/(1+\mathsf{d}_{\mathrm{ref},k}) < 0$ for $r < k \le q$, and $o \mapsto \tilde{o}_k$ is onto $\mathbb{R}$, so the logarithm is unbounded below.

(v) The reference law of Y' is $\mathcal{N}(R_Y\Lambda^{-1}\mathsf{h} + a_Y, R_Y\Lambda^{-1}R_Y^\top)$, and $o' \mid y' \sim \mathcal{N}(\mathsf{C}'y' + o'_0, R_o\Sigma_o R_o^\top)$ with $\mathsf{C}' = R_o\mathsf{C}R_Y^{-1}$ and $o'_0 = R_o o_0 + a_o - \mathsf{C}'a_Y$, which gives Λ' and

$$\Lambda'_{\mathrm{rec}} = R_Y^{-\top}\mathsf{C}^\top R_o^\top R_o^{-\top}\Sigma_o^{-1}R_o^{-1}R_o\mathsf{C}R_Y^{-1} = R_Y^{-\top}\Lambda_{\mathrm{rec}}R_Y^{-1}.$$

Proposition G5.5 with $\mathbf{P} = R_Y^{-1}$ gives the invariance of the $\mathsf{d}_{\mathrm{ref},k}$ with multiplicities and shows that x is a generalized eigenvector exactly when $R_Y x$ is one for the transformed pencil; $(R_Y x_j)^\top\Lambda'(R_Y x_k) = x_j^\top\Lambda x_k$. The d_k, $\check{\rho}_k$, and (G7.16) are functions of the $\mathsf{d}_{\mathrm{ref},k}$. Substitution gives

$$\widetilde{P}'_r = R_Y S_r S_r^\top R_Y^\top R_Y^{-\top}\Lambda R_Y^{-1} = R_Y\widetilde{P}_r R_Y^{-1},$$

hence $\widetilde{P}'_r Y' = R_Y\widetilde{P}_r Y + \widetilde{P}'_r a_Y$, an injective affine function of $\widetilde{P}_r Y$.

(vi) Since $\Lambda^{-1} = S_n S_n^\top$ and $\widetilde{P}_r S_n = S_n\,\mathrm{diag}(I_r, 0)$, $(I-\widetilde{P}_r)S_n = S_n\,\mathrm{diag}(0, I_{n-r})$, and cyclicity of the trace gives $\mathrm{HS}_r(\mathsf{G}) = \sum_{k>r} x_k^\top\mathsf{C}^\top\mathsf{G}\mathsf{C}x_k$. For $\mathsf{G} = \Sigma_o^{-1}$ each term is $x_k^\top\Lambda_{\mathrm{rec}}x_k = \mathsf{d}_{\mathrm{ref},k}$, and the realized error equals $\mathrm{Tr}(\Sigma_o^{-1/2}\mathsf{C}(I-\widetilde{P}_r)\Lambda^{-1}(I-\widetilde{P}_r)^\top\mathsf{C}^\top\Sigma_o^{-1/2}) = \mathrm{HS}_r(\Sigma_o^{-1})$ because Λ^{-1} is the reference covariance. For the marginal metric put $\mathsf{C}_{\mathrm{w}} = \Sigma_o^{-1/2}\mathsf{C}S_n$, so that $\mathsf{C}_{\mathrm{w}}^\top\mathsf{C}_{\mathrm{w}} = \mathrm{diag}(\mathsf{d}_{\mathrm{ref},k})$ and $\Sigma_{\mathrm{ext}} = \Sigma_o^{1/2}(\mathsf{C}_{\mathrm{w}}\mathsf{C}_{\mathrm{w}}^\top + I_m)\Sigma_o^{1/2}$; then $S_n^\top\mathsf{C}^\top\Sigma_{\mathrm{ext}}^{-1}\mathsf{C}S_n = \mathsf{C}_{\mathrm{w}}^\top(\mathsf{C}_{\mathrm{w}}\mathsf{C}_{\mathrm{w}}^\top + I_m)^{-1}\mathsf{C}_{\mathrm{w}} = (\mathsf{C}_{\mathrm{w}}^\top\mathsf{C}_{\mathrm{w}} + I_n)^{-1}\mathsf{C}_{\mathrm{w}}^\top\mathsf{C}_{\mathrm{w}} = \mathrm{diag}(\mathsf{d}_k)$, whose diagonal entries are the $\check{\rho}_k^2$. The bounds follow termwise from (G7.16), $\log(1+x) \le x$ for $x \ge 0$, and $-\log(1-x) \ge x$ for $0 \le x < 1$, each with equality only at $x = 0$. □

The interface coordinates read Y along the projection directions of the Gaussian information bottleneck. By $\Lambda_{\mathrm{rec}}x_k = \mathsf{d}_{\mathrm{ref},k}\Lambda x_k$, the vector Λx_k is a left eigenvector of $\mathrm{Cov}(Y \mid o)\,\mathrm{Cov}(Y)^{-1} = (\Lambda + \Lambda_{\mathrm{rec}})^{-1}\Lambda$ with eigenvalue $1 - \mathsf{d}_k$, so $c_k = x_k^\top\Lambda Y$ reads Y along a direction that the optimal projections of Chechik et al. [10, Sec. 3, Theorem 3.1] switch on in ascending order of these eigenvalues, and their Sec. 7.1 identifies these directions with those of canonical correlation analysis. That theorem solves the Lagrangian tradeoff between compression and prediction, not the fixed-rank maximization of part (ii), which the Courant–Fischer argument above supplies.

G7.4 Kron reduction: coefficient families and graph boundaries

Kron reduction marginalizes hidden agents, and it is the dual of aggregation. For a coordinate split $z = (z_{\mathrm{r}}, z_{\mathrm{h}})$, Proposition G7.4 with $\mathrm{range}(B) = \{z_{\mathrm{h}} = 0\}$ gives the conditional precision Λ_{rr}, the restriction of the precision. Integrating out z_{h} keeps instead the restriction $(\Lambda^{-1})_{\mathrm{rr}}$ of the covariance, whose inverse is the Schur complement $\Lambda_{\mathrm{rr}} - \Lambda_{\mathrm{rh}}\Lambda_{\mathrm{hh}}^{-1}\Lambda_{\mathrm{hr}}$ (Lemma G2.6), and Corollary G2.8 orders the two precisions. Aggregation stays in the interaction family; marginalization does not in general (Proposition G7.7). The regimes in which it does close carry different quantifiers: which coefficient families pass tests at one fixed hidden anchor, which fixed graphs close for every independent positive-definite assignment, and which nonflat graphs close while keeping their link marks. In every regime a Schur reduction is integration over the hidden coordinates with the source and scalar offset carried along (Proposition G7.11), and a component whose elimination generates a

positive-semidefinite self term stays in the coarse family. Table G7.1 collects the verdicts. For scalar weights the closure of loopy Laplacians under Kron reduction is Dörfler and Bullo [13, Lemma II.1], and Hansen and Ghrist [16, Sec. 4.2, Thm. 4.3] treat Kron reduction for cellular sheaves; the positive-pair obstruction below is compared with their results where it is stated.

Table G7.1: Closure regimes of Kron reduction in the Gaussian interaction family. "Generated self terms" records whether elimination produces new positive-semidefinite self terms, which the coarse family admits.

Admissible family and quantifier	Verdict	Generated self terms	Statement
Independent matrix weights and anchors on an arbitrary graph	not closed; rational three-agent witness	not applicable	Proposition G7.7
All self terms and weights in one fixed cone $\mathcal{W}_H$	closed on every graph	yes, inside $\mathcal{W}_H$	Theorem G7.10
Pair tests of an arbitrary family at one fixed SPD hidden anchor Q_0	closed iff the family lies in a cone $\mathcal{W}_H$ with $HH^\top = Q_0$; maximal families are these cones	yes, inside $\mathcal{W}_H$	Theorem G7.12
Coordinates of $\mathcal{W}_H$ confined to prescribed domains	closed iff every star update stays in the domains	yes	Remark G7.13
Positive-semidefinite hidden anchor sharing its kernel with the family	quotient form of the fixed-anchor classification	yes, on the quotient	Remark G7.14
Fixed flat graph, independent SPD weights, $K \geq 2$	closed iff every hidden component has at most two terminals, ground counted	anchor increments from components touching ground	Theorem G7.15
Fixed flat graph, SPD anchor on every hidden agent, $K \geq 2$	closed iff every hidden component touches at most one retained agent; failure generic otherwise	SPD increment at the touched agent	Corollary G7.16
Nonflat links, hidden vertices of degree two with zero anchor	closed in every order, loop holonomy retained	none	Theorem G7.21
Nonflat links, hidden vertices of degree two with PSD anchors	one-link-plus-anchor presentation exists, not unique	yes	Proposition G7.24
Branching (three terminals), proper SPD law	marginal outside every positive two-node presentation; rational dual certificate	not applicable	Proposition G7.17
Finite semialgebraic description of fine and coarse families	decidable by quantifier elimination	encoded in the description	Theorem GA.4
Declared quotient metric with variable support	decidable by quantifier elimination	encoded in the description	Theorem GA.12

Proposition G7.7 (Unrestricted Kron reduction leaves the family). Take $N = 3$, $K = 2$, $A_1 = A_2 = A_3 = I_2$, and

$$W_{12} = 0, \qquad W_{13} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad W_{23} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{G7.25}$$

Then the assembled Λ is positive definite, and eliminating agent 3 manufactures

$$W_{12}^{\mathrm{Kron}} = -\left(\Lambda^{\mathrm{eff}}\right)_{12} = W_{13}\Lambda_{33}^{-1}W_{23} = \frac{1}{19}\begin{pmatrix} 9 & 3 \\ 4 & 14 \end{pmatrix}, \tag{G7.26}$$

which is not symmetric and hence is not a positive-semidefinite interaction weight. [ESTABLISHED]

Proof. $\Lambda_{33} = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ has determinant 19, and direct multiplication gives (G7.26). The leading principal minors of Λ are $2, 6, 18, 48, 102, 233$, so $\Lambda \succ 0$ by Sylvester's criterion. The full Schur complement is symmetric, its two mirrored off-diagonal blocks being transposes; membership in (G7.6) requires each block to be symmetric, which fails. □

Lemma G7.8 (Scalar Kron closure). For $K = 1$, call a symmetric matrix with nonpositive off-diagonal entries and nonnegative row sums a loopy Laplacian. The Schur complement of a loopy Laplacian at any nonsingular principal block is a loopy Laplacian; one-node elimination with pivot $m = \Lambda_{kk} > 0$ adds $w_{ik}w_{kj}/m \geq 0$ to the weight between distinct retained neighbors i, j of k. [ESTABLISHED]

Proof. Dörfler and Bullo [13, Lemma II.1] prove the statement for an irreducible loopy Laplacian and a proper retained set of at least two nodes; the following argument covers every case. A loopy Laplacian is diagonally dominant with nonnegative diagonal, hence positive semidefinite, so a nonsingular principal block is positive definite; its successive pivots are therefore positive, and the Schur complement at the block equals the composition of one-node eliminations. Eliminating k replaces Λ_{ij} by $\Lambda_{ij} - \Lambda_{ik}\Lambda_{kj}/m$. For $i \neq j$ the subtracted term is nonnegative because $\Lambda_{ik}, \Lambda_{kj} \leq 0$, so off-diagonal entries stay nonpositive and w_{ij} increases by $w_{ik}w_{kj}/m$. With row sums $\varrho_i = \sum_j \Lambda_{ij}$, the new row sum of a retained i is $\varrho_i - \Lambda_{ik}\varrho_k/m \geq \varrho_i \geq 0$. Symmetry is preserved. □

The matrix analog is closed on a larger subfamily than common orthogonal diagonalization.

Definition G7.9 (Congruence-diagonal cone). For $H \in \mathrm{GL}(K)$, the congruence-diagonal cone is

$$\mathcal{W}_H = \{H\,\mathrm{diag}(d)H^\top : d \in [0,\infty)^K\}. \tag{G7.27}$$

Theorem G7.10 (Congruence-diagonal Kron closure). If every A_i and W_{ij} lies in one cone $\mathcal{W}_H$, then every Schur complement at a nonsingular block of hidden agents is again an interaction precision with all reduced self terms and weights in $\mathcal{W}_H$. The cones $\mathcal{W}_H$ contain pairs that no common orthogonal eigenbasis diagonalizes: for $H = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $W = H\,\mathrm{diag}(1,2)H^\top$ and $W' = H\,\mathrm{diag}(3,1)H^\top$ satisfy $WW' - W'W = \left(\begin{smallmatrix} 0 & -5 \\ 5 & 0 \end{smallmatrix}\right)$. [ESTABLISHED]

Proof. Factor

$$\Lambda = (I_N \otimes H)\Lambda_{\mathrm{d}}(I_N \otimes H^\top), \tag{G7.28}$$

where, after grouping coordinates by channel, Λ_{d} is a direct sum of K scalar loopy Laplacians. For retained agents V_{r} and hidden agents V_{h},

$$\Lambda^{\mathrm{eff}} = (I_{V_{\mathrm{r}}} \otimes H)\Big[(\Lambda_{\mathrm{d}})_{\mathrm{rr}} - (\Lambda_{\mathrm{d}})_{\mathrm{rh}}(\Lambda_{\mathrm{d}})_{\mathrm{hh}}^{-1}(\Lambda_{\mathrm{d}})_{\mathrm{hr}}\Big](I_{V_{\mathrm{r}}} \otimes H^\top), \tag{G7.29}$$

and $(\Lambda_{\mathrm{d}})_{\mathrm{hh}}$ is nonsingular because Λ_{hh} is. The bracket eliminates channel by channel, and each channel's Schur complement is a loopy Laplacian by Lemma G7.8. Reassembling the channels gives reduced self terms and weights in $\mathcal{W}_H$. The commutator is direct multiplication, and commuting symmetric matrices are exactly those with a common orthogonal eigenbasis. □

G7.4.1 The complete Gaussian marginal action

Fix a coordinate split $z = (z_{\mathrm{r}}, z_{\mathrm{h}})$, with $n = \dim z$ and $n_{\mathrm{h}} = \dim z_{\mathrm{h}}$, an SPD precision Λ, a source h, and a real scalar offset $\mathfrak{c}$, with coordinate Lebesgue reference measures.

Proposition G7.11 (Normalized Schur marginal and reverse kernel). The action, its law, and its raw mass are

$$\begin{aligned} \mathsf{E}(z) &= \frac{1}{2}z^\top \Lambda z - \mathsf{h}^\top z + \mathfrak{c}, \\ p_\Lambda(z) &= Z_\Lambda^{-1} e^{-\mathsf{E}(z)}, \qquad \mathbb{P}_\Lambda = \mathcal{N}(\Lambda^{-1}\mathsf{h}, \Lambda^{-1}), \\ Z_\Lambda &= e^{-\mathfrak{c}}(2\pi)^{n/2}(\det \Lambda)^{-1/2}\exp\Big(\frac{1}{2}\mathsf{h}^\top\Lambda^{-1}\mathsf{h}\Big). \end{aligned} \tag{G7.30}$$

With $M = \Lambda_{\mathrm{hh}}$, the action decomposes as

$$\begin{aligned} \mathsf{E}(x, z_{\mathrm{h}}) = \frac{1}{2}\big(z_{\mathrm{h}} - M^{-1}(\mathsf{h}_{\mathrm{h}} - \Lambda_{\mathrm{hr}}x)\big)^\top M\big(z_{\mathrm{h}} - M^{-1}(\mathsf{h}_{\mathrm{h}} - \Lambda_{\mathrm{hr}}x)\big) \\ + \frac{1}{2}x^\top\Lambda_{\mathrm{rr}}x - \mathsf{h}_{\mathrm{r}}^\top x + \mathfrak{c} - \frac{1}{2}(\mathsf{h}_{\mathrm{h}} - \Lambda_{\mathrm{hr}}x)^\top M^{-1}(\mathsf{h}_{\mathrm{h}} - \Lambda_{\mathrm{hr}}x), \end{aligned} \tag{G7.31}$$

and the exact marginal action and reverse kernel are

$$\begin{aligned} \Lambda^{\mathrm{eff}} &= \Lambda_{\mathrm{rr}} - \Lambda_{\mathrm{rh}}M^{-1}\Lambda_{\mathrm{hr}}, \\ \mathsf{h}^{\mathrm{eff}} &= \mathsf{h}_{\mathrm{r}} - \Lambda_{\mathrm{rh}}M^{-1}\mathsf{h}_{\mathrm{h}}, \\ \mathfrak{c}^{\mathrm{eff}} &= \mathfrak{c} + \frac{1}{2}\log\det M - \frac{n_{\mathrm{h}}}{2}\log(2\pi) - \frac{1}{2}\mathsf{h}_{\mathrm{h}}^\top M^{-1}\mathsf{h}_{\mathrm{h}}, \\ \mathbb{P}_\Lambda(dz_{\mathrm{h}} \mid z_{\mathrm{r}} = x) &= \mathcal{N}\big(M^{-1}(\mathsf{h}_{\mathrm{h}} - \Lambda_{\mathrm{hr}}x), M^{-1}\big)(dz_{\mathrm{h}}). \end{aligned} \tag{G7.32}$$

Both M and Λ^{eff} are SPD. Integration preserves the raw mass Z_Λ, and the normalized marginal is the coordinate pushforward of $\mathbb{P}_\Lambda$. Successive eliminations agree with direct elimination, sources and scalar offsets included, and the conditional reconstruction kernels compose. For an observation $o = \mathsf{C}_{\mathrm{r}}z_{\mathrm{r}} + \mathsf{C}_{\mathrm{h}}z_{\mathrm{h}} + \xi$ with $\xi \sim \mathcal{N}(0, \Sigma_o)$ independent of z and $\Sigma_o \succ 0$,

$$o \mid z_{\mathrm{r}} = x \sim \mathcal{N}\Big((\mathsf{C}_{\mathrm{r}} - \mathsf{C}_{\mathrm{h}}M^{-1}\Lambda_{\mathrm{hr}})x + \mathsf{C}_{\mathrm{h}}M^{-1}\mathsf{h}_{\mathrm{h}},\ \Sigma_o + \mathsf{C}_{\mathrm{h}}M^{-1}\mathsf{C}_{\mathrm{h}}^\top\Big), \tag{G7.33}$$

with SPD covariance, and mixing this likelihood with the marginal recovers the evidence at every record, so the coordinate projection pushes the whole observed joint. The posterior hidden conditional differs in general from the prior conditional in (G7.32), and the two agree when $\mathsf{C}_{\mathrm{h}} = 0$. [ESTABLISHED]

Proof. The moment formulas are the standard Gaussian marginal and conditional [5, Secs. 2.3.1–2.3.2, Eqs. (2.73), (2.75), (2.91)–(2.93)]; the raw mass and scalar offset are recorded here. Translating z by $\Lambda^{-1}\mathsf{h}$ and diagonalizing Λ orthogonally reduce $\int e^{-\mathsf{E}}$ to one-dimensional integrals $\int_{\mathbb{R}} e^{-\lambda u^2/2} du = (2\pi/\lambda)^{1/2}$, which gives Z_Λ. Expanding the square proves (G7.31); integrating its first line over z_{h} gives $(2\pi)^{n_{\mathrm{h}}/2}(\det M)^{-1/2}$, and expanding its last term proves every line of (G7.32). A principal block of an SPD matrix is SPD, and for nonzero x, minimizing $(x, z_{\mathrm{h}})^\top \Lambda (x, z_{\mathrm{h}})$ over z_{h} gives $x^\top \Lambda^{\mathrm{eff}} x > 0$. For a nonnegative Borel function on a finite product of Euclidean spaces the product integral equals its iterated integrals in any order, with infinite values allowed; the Gaussian integrals here are finite, which proves equality of elimination orders and the composition of reconstruction kernels formed from the corresponding marginals. For the sensor, write z_{h} given $z_{\mathrm{r}} = x$ as its conditional mean plus independent centered Gaussian noise of covariance M^{-1}, which gives (G7.33) [5, Sec. 2.3.3, Eq. (2.115)]; integrating the likelihood of the full coordinates against the prior in the order hidden first shows that the evidence is unchanged. When $\mathsf{C}_{\mathrm{h}} \neq 0$ the likelihood depends on z_{h}, which changes the hidden conditional given o. □

Because the whole observed joint is pushed, the finite-KL chain identity of Section 10.2 gives the exact free-energy difference with the same evidence on both sides.

When Λ is the matter precision of Section G11.4 at given links and every coordinate is hidden, the scalar $\mathfrak{c}^{\mathrm{eff}} - \mathfrak{c}$ of (G7.32) is the induced action of Proposition G11.5 at those links, and for positive-definite self terms and zero source it has the hopping expansion over weighted Wilson loops of Proposition G11.8.

G7.4.2 Arbitrary independent families at a fixed anchor

Theorem G7.12 (Fixed-anchor classification of arbitrary families). Fix $Q_0 = Q_0^\top \succ 0$ and an arbitrary subset $\mathscr{F} \subseteq \mathbb{S}_+^K$; no closedness, convexity, scaling closure, or membership of Q_0 in $\mathscr{F}$ is assumed. For $W, W' \in \mathscr{F}$ and $t_1, t_2 > 0$, the two-edge test eliminates a hidden agent with anchor Q_0 and incident weights $t_1 W$, $t_2 W'$, with pivot and manufactured coefficient

$$M = Q_0 + t_1 W + t_2 W', \qquad W_{W,W'}(t_1, t_2) = t_1 t_2 W M^{-1} W'. \tag{G7.34}$$

Giving both retained agents the anchor Q_0 makes each test a proper three-agent Gaussian, with energy at least $\lambda_{\min}(Q_0)\|z\|^2/2$. Symmetry of $W_{W,W'}(t_1, t_2)$ for at least one pair of scales, for every $W, W' \in \mathscr{F}$, is equivalent to

$$W Q_0^{-1} W' = W' Q_0^{-1} W \quad \text{for all } W, W' \in \mathscr{F}, \tag{G7.35}$$

and to the existence of one $O \in O(K)$ with

$$H = Q_0^{1/2} O, \qquad Q_0 = HH^\top, \qquad \mathscr{F} \subseteq \mathcal{W}_H. \tag{G7.36}$$

One H serves the whole family, even an uncountable one. Under these conditions every pair of positive scales gives $W_{W,W'}(t_1, t_2) \in \mathcal{W}_H$, and the inclusion-maximal families with the test property are exactly the cones $\mathcal{W}_H$. In particular, if a closed convex cone $\mathcal{W} \subseteq \mathbb{S}_+^K$ has an SPD order unit M_0 in its relative interior, its anchors and edge coefficients range independently over $\mathcal{W}$ with arbitrary positive scaling, and every defined one-agent elimination must manufacture minus a symmetric member of $\mathcal{W}$, then $\mathcal{W} \subseteq \mathcal{W}_H$ for some H with $HH^\top = M_0$, and every maximal such cone is a cone $\mathcal{W}_H$. Neither premise can be dropped. If only equal pairs $W = W'$ are tested, every test output is symmetric for the family $\{I, \operatorname{diag}(1,2), \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)\}$, which lies in no common cone $\mathcal{W}_H$. If the weights lie in a common cone but the anchor does not, the test can fail: for $W = \operatorname{diag}(1,0)$, $W' = \operatorname{diag}(0,1)$, $Q_0 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and $t_1 = t_2 = 1$, the manufactured coefficient is $\left(\begin{smallmatrix} 0 & -1/8 \\ 0 & 0 \end{smallmatrix}\right)$. [ESTABLISHED]

Proof. Whiten by Q_0, writing $\widehat{W} = Q_0^{-1/2} W Q_0^{-1/2}$, $\widehat{W}' = Q_0^{-1/2} W' Q_0^{-1/2}$, and $\widehat{M} = I + t_1 \widehat{W} + t_2 \widehat{W}'$. Substituting $\widehat{W}' = (\widehat{M} - I - t_1 \widehat{W})/t_2$ gives

$$\begin{aligned} \widehat{W}\widehat{M}^{-1}\widehat{W}' - \widehat{W}'\widehat{M}^{-1}\widehat{W} &= t_2^{-1}(\widehat{M}^{-1}\widehat{W} - \widehat{W}\widehat{M}^{-1}) = \widehat{M}^{-1}(\widehat{W}\widehat{W}' - \widehat{W}'\widehat{W})\widehat{M}^{-1}, \\ W_{W,W'} - W_{W,W'}^\top &= t_1 t_2 Q_0 M^{-1}(WQ_0^{-1}W' - W'Q_0^{-1}W)M^{-1}Q_0. \end{aligned} \tag{G7.37}$$

All outside factors are invertible or positive scalars, so one tested pair of positive scales already forces (G7.35). The whitened family is real symmetric and pairwise commuting, and simultaneous orthogonal diagonalization of the whole family follows by induction on dimension: if every member is scalar any orthonormal basis works; otherwise every member preserves the eigenspaces of one nonscalar member, which have smaller dimension, and the induction applies to the restricted families. Only finitely many reductions are used, so no countability premise is needed, and the empty family is trivial. Positive semidefiniteness makes the diagonal entries nonnegative, which proves (G7.36). Conversely, if $W = H \operatorname{diag}(d) H^\top$ and $W' = H \operatorname{diag}(d') H^\top$ with $d, d' \in [0, \infty)^K$, then

$$W_{W,W'}(t_1, t_2) = H \operatorname{diag}(v) H^\top, \qquad v = \frac{t_1 t_2 d \odot d'}{\mathbf{1} + t_1 d + t_2 d'} \text{ coordinatewise}, \tag{G7.38}$$

which proves sufficiency and positivity. A tested family therefore lies in a full cone $\mathcal{W}_H$, and a maximal family equals it. Conversely, a PSD matrix compatible with every member of $\mathcal{W}_H$ becomes, after whitening and the same orthogonal change, a matrix commuting with every coordinate rank-one projector, hence nonnegative diagonal, so it belongs to the cone; this proves maximality.

For the cone statement, $M_0 \in \mathcal{W}$ and scaling closure admit the pair test with $Q_0 = M_0$, incident weights $t_1 W, t_2 W'$ for $W, W' \in \mathcal{W}$, and retained anchors M_0; the required symmetry gives (G7.35) with $Q_0 = M_0$, hence $\mathcal{W} \subseteq \mathcal{W}_H$ with $HH^\top = M_0$. The cone $\mathcal{W}_H$ is closed and convex, contains $M_0 = H I_K H^\top$ in its relative interior, and is closed under elimination by Theorem G7.10, so it satisfies the same hypotheses and every maximal cone is of this form. For the omitted premises, each $F(I + tF)^{-1}F$ is symmetric, while $\mathcal{W}_H \ni I$ forces the common orthogonal basis $HD_0^{1/2}$ for

the family, and $\operatorname{diag}(1,2)$ and $\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$ do not commute; the last coefficient is direct multiplication with $M=\left(\begin{smallmatrix}3&1\\1&3\end{smallmatrix}\right)$. □

Anchors that vary with a context over the base are treated in Supplement B.

Remark G7.13 (Stability inside a prescribed diagonal domain). Fix H and coordinate domains $\mathscr{D}_A, \mathscr{D}_W \subseteq [0,\infty)^K$ with $0 \in \mathscr{D}_W$, and write $A_i = H\operatorname{diag}(a_i)H^\top$ and $W_{ij} = H\operatorname{diag}(w_{ij})H^\top$. A defined elimination of agent k has coordinatewise pivot $m = a_k + \sum_{i\neq k} w_{ik} > 0$ and updates

$$w'_{ij} = w_{ij} + \frac{w_{ik}\odot w_{kj}}{m}, \qquad a'_i = a_i + \frac{w_{ik}\odot a_k}{m}, \tag{G7.39}$$

with products and quotients taken coordinatewise. Closure inside the domains for every finite PSD graph and every defined one-agent elimination is equivalent to

$$a_i + \frac{u_i\odot a_0}{m} \in \mathscr{D}_A, \qquad w_{ij} + \frac{u_i\odot u_j}{m} \in \mathscr{D}_W, \qquad m = a_0 + \sum_i u_i > 0, \tag{G7.40}$$

for every finite star inventory of a center anchor a_0 and retained anchors a_i in $\mathscr{D}_A$, and center weights u_i and retained weights w_{ij} in $\mathscr{D}_W$. If only full-space SPD graphs are admitted, necessity is restricted to inventories producing such graphs. Containment in $\mathcal{W}_H$ does not imply closure inside a smaller family: with the coefficient set $\{1\}$ in one dimension, anchor one at three agents and two unit edges through the middle one, elimination produces the weight $1/3$ and retained anchors $4/3$. [ESTABLISHED]

Proof. The off-diagonal Schur blocks give the first update. In each coordinate, subtracting the new incident-weight sum from the new diagonal leaves

$$a_i + w_{ik} - \frac{w_{ik}(w_{ik} + \sum_{j\neq i,k} w_{kj})}{m} = a_i + \frac{w_{ik}a_k}{m}, \tag{G7.41}$$

which is the anchor update. Applying the updates at every star proves sufficiency of (G7.40). Conversely, realize an inventory as a star with the given retained anchors and weights and eliminate its center; its pivot is positive, so the outputs must satisfy exactly the displayed conditions. For several hidden agents, positivity of the hidden principal block gives positive successive pivots, as in Proposition G7.11. The one-dimensional example is the update with $m = 3$. □

Remark G7.14 (The shared-kernel version). If $Q_0 \succeq 0$ and $\ker Q_0 \subseteq \ker W$ for every $W \in \mathscr{F}$, all forms descend to $\mathbb{R}^K/\ker Q_0$, the descended anchor is SPD, and Theorem G7.12 applies there with an ordinary inverse. If $B_0 : \mathbb{R}^r \to (\ker Q_0)^\perp$ is an orthonormal inclusion with $r = \operatorname{rank} Q_0$, the pair-test property is equivalent to

$$H_0 = B_0(B_0^\top Q_0 B_0)^{1/2}O, \qquad Q_0 = H_0H_0^\top, \qquad W = H_0\operatorname{diag}(d_W)H_0^\top, \quad d_W \geq 0, \tag{G7.42}$$

for one $O \in O(r)$ and all members W. This is a quotient statement, not an inverse or a normalized density on the null directions. Without the shared-kernel hypothesis, $Q_0 = 0$ and SPD W, W' give

$$\left[W(t_1W + t_2W')^{-1}W'\right]^{-1} = t_1W'^{-1} + t_2W^{-1} \succ 0, \tag{G7.43}$$

a symmetric output even when W and W' do not commute, so the three-matrix family of Theorem G7.12 passes every test with no common cone. [ESTABLISHED]

Proof. Symmetry and $Wv = 0$ on $\ker Q_0$ make each form independent of the representative of a quotient class. Positivity gives $\ker(Q_0 + t_1W + t_2W') = \ker Q_0$, since a zero of the quadratic annihilates every nonnegative summand, so the quotient pivot is positive, and whitening in the coordinates B_0 gives (G7.42); when $r = 0$ all forms vanish and the quotient is a point. Expanding the inverse of $W(t_1W+t_2W')^{-1}W'$ proves (G7.43). A positive-dimensional null direction has infinite Lebesgue integral, with a constant or an exponential linear factor along it, so no full-space Gaussian normalizer is supplied. □

G7.4.3 A graph criterion for unrestricted matrix weights

Now let every present edge weight vary independently over all SPD $K \times K$ matrices and restrict the graph instead. Let $\mathfrak{I}$ be a finite undirected multigraph with edge set E and no self-loops. If anchors occur, add a ground vertex 0, represent each present SPD anchor by an edge to 0, and keep 0 among the retained vertices throughout elimination; its coordinate is set to zero only when the anchored law is formed. In a common frame,

$$\mathsf{E}_{\mathrm{aug}}(z) = \frac{1}{2}\sum_{e\in E}(z_{i_e} - z_{j_e})^{\top}W_e(z_{i_e} - z_{j_e}), \qquad W_e \succ 0. \tag{G7.44}$$

Let V_{h} be the hidden ordinary vertices and V_{r} the retained vertices, ground included. For each connected component $\mathcal{Z}$ of the graph induced on V_{h}, its terminal boundary is

$$\partial\mathcal{Z} = \{r \in V_{\mathrm{r}} : \text{some edge joins } r \text{ to a vertex of } \mathcal{Z}\}. \tag{G7.45}$$

Theorem G7.15 (Universal closure at a fixed retained boundary). Suppose $\partial\mathcal{Z} \neq \varnothing$ for every hidden component; this is exactly the condition that makes the hidden Dirichlet block positive definite for every SPD assignment, and a component without boundary has a nonzero constant hidden null vector. For $K \geq 2$, the Schur complement of (G7.44) is an augmented matrix-weighted Laplacian, with symmetric nonpositive off-diagonal blocks and diagonal blocks equal to minus the sum of the off-diagonal blocks, for every independent SPD edge assignment if and only if

$$|\partial\mathcal{Z}| \leq 2 \quad \text{for every hidden component } \mathcal{Z}, \tag{G7.46}$$

with ground counted as one terminal. A two-terminal component contributes one SPD edge and a one-terminal component contributes zero. For $K = 1$ every graph with nonempty boundaries closes, so the bound is not necessary there. Sufficiency extends to PSD weights whenever the hidden

Dirichlet block remains SPD, the two-terminal output then being only semidefinite. For a declared nested collection of hidden sets, universal closure at every declared level is equivalent to (G7.46) at every such set; closure at every intermediate one-vertex prefix requires the criterion at every prefix, because a component with two final terminals can expose three during an ordering. An excluded graph has some failing SPD assignments, not failure for every assignment. [ESTABLISHED]

Proof. With retained coordinates zero, zero edge energy forces equality throughout a hidden component and then zero at its boundary, which proves the boundary-positivity statement. The hidden Dirichlet block is block diagonal over the components, and each component's Schur form is the minimum of its energy with boundary values fixed, hence PSD. Equal boundary values admit the same constant at every hidden vertex and give zero energy, so a one-terminal component contributes zero. With two terminals write its symmetric block matrix as $M_{\mathcal{Z}}$; the equations $M_{11} + M_{12} = 0$, $M_{21} + M_{22} = 0$, and $M_{21} = M_{12}^{\top}$ force

$$M_{\mathcal{Z}} = \begin{pmatrix} W_{\mathcal{Z}} & -W_{\mathcal{Z}} \\ -W_{\mathcal{Z}} & W_{\mathcal{Z}} \end{pmatrix}, \qquad W_{\mathcal{Z}} = W_{\mathcal{Z}}^{\top} \succeq 0. \tag{G7.47}$$

A nonzero terminal difference with zero energy would have a minimizing hidden extension with every edge difference zero, and connectedness would force equal terminal values; hence $W_{\mathcal{Z}} \succ 0$. Summing the component contributions and the direct retained edges proves sufficiency, an anchor increment arising when one terminal is ground.

For necessity, take a hidden center 4 with three retained terminals $1, 2, 3$ and legs

$$W_{14} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad W_{24} = I_2, \qquad W_{34} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{G7.48}$$

Its pivot and manufactured coefficient between terminals 1 and 2 are

$$M = W_{14} + W_{24} + W_{34} = \begin{pmatrix} 4 & 1 \\ 1 & 5 \end{pmatrix}, \qquad W_{12}^{\mathrm{Kron}} = W_{14} M^{-1} W_{24} = \frac{1}{19} \begin{pmatrix} 5 & -1 \\ -2 & 8 \end{pmatrix}, \tag{G7.49}$$

which is not symmetric. If one terminal is ground, give it the W_{34} leg, leaving the defect between two ordinary terminals, and direct sums with I_{K-2} extend the obstruction to every $K > 2$. Now take any hidden component with at least three terminals. Select one incident edge from each of three terminals, join their hidden endpoints by a minimal tree inside the component, and extend it to a tree containing every hidden vertex of the component. The three-terminal core has a unique branching vertex, allowing coincident hidden endpoints and legs of length one, and extra hidden branches have no terminal, so they minimize to their attachment value and contribute nothing. On a core leg, minimizing $\sum_j d_j^{\top} W_j d_j / 2$ subject to $\sum_j d_j = d$ gives $W_j d_j = \lambda$ and

$$W_{\mathrm{leg}} = \Big(\sum_j W_j^{-1} \Big)^{-1}, \tag{G7.50}$$

the unique minimum by positivity of the Hessian. Assigning lW_{14} to each edge of a leg of length l realizes W_{14}, and likewise for the other legs; give the extra tree edges I_K and temporarily the unused edges zero. The tree still joins every hidden vertex to a terminal, so its Dirichlet block is SPD and its reduced core has the defect (G7.49). Restoring every unused present edge with weight δI_K, the Dirichlet inverse and Schur complement are continuous at $\delta = 0$, where the determinant is nonzero, so the nonzero antisymmetric block persists for small $\delta > 0$, with every final weight SPD. Other hidden components can use scalar identity weights and direct retained edges are symmetric, so neither cancels the antisymmetric block. For $K = 1$, closure is Lemma G7.8.

For the PSD extension, add δI_K to every present weight and let $\delta \downarrow 0$; continuity of the inverse at the positive limiting Dirichlet block preserves the Schur matrix, and symmetry, zero row sums, and PSD weights are closed conditions. Direct and successive elimination agree by Proposition G7.11, which gives the nested statement, and the prefix statement is the same criterion applied at each prefix. □

Flatness here means plain differences in a common frame. A graph coboundary can first be trivialized; genuinely nonflat links are the subject of Section G7.4.5.

Corollary G7.16 (Marginalizing prior-carrying agents). Let $K \geq 2$, and suppose every hidden agent carries a present SPD anchor, for instance a proper agent-level prior self term. Then the Schur complement of (G7.44) is an augmented matrix-weighted Laplacian for every independent SPD assignment of the edge weights and anchors if and only if every hidden component is adjacent to at most one retained ordinary agent. Such a component adds an SPD increment to the anchor of that agent, or nothing when it has no retained ordinary neighbor. On every other graph, in particular when a hidden agent relays between two retained agents, the failing assignments contain an open dense subset of full Lebesgue measure in the SPD parameter domain. [ESTABLISHED]

Proof. Each present anchor is an edge to ground, so ground lies in $\partial\mathcal{Z}$ for every hidden component, and (G7.46) holds exactly when $\mathcal{Z}$ has at most one retained ordinary neighbor. By Theorem G7.15, a component whose terminals are one retained agent r and ground contributes one SPD edge from r to ground, an anchor increment at r, and a component whose only terminal is ground contributes zero. On an excluded graph the theorem supplies an SPD assignment at which some retained off-diagonal block is nonsymmetric. By the adjugate formula every entry of the antisymmetric part of that block is a polynomial in the weight entries divided by $\det \Lambda_{\mathrm{hh}} > 0$; the SPD parameter domain is an open connected subset of a Euclidean space, and a polynomial that is nonzero at one of its points vanishes there only on a closed nowhere-dense set of Lebesgue measure zero. For a single relay with hidden anchor Q_0 and incident weights W, W', (G7.37) with $t_1 = t_2 = 1$ shows that the manufactured block is symmetric exactly when $WQ_0^{-1}W' = W'Q_0^{-1}W$. Proposition G7.7 is such a relay with $Q_0 = I_2$: it is the star (G7.48) with its $W_{24} = I_2$ leg grounded, and its manufactured coefficient is $W_{14}M^{-1}W_{34}$. □

The interaction family only requires $A_i \succeq 0$, and a directed forest places proper priors at its roots alone (Proposition G3.8), so the hypothesis is a declared restriction. When every agent does carry an SPD self term, marginalizing an agent that relays between two retained agents generically

leaves the positive matrix-Laplacian family, while marginalizing a hidden component attached to a single retained agent stays in it and only raises that agent's anchor.

G7.4.4 A proper Gaussian beyond every positive two-node decomposition

The three-terminal star also separates failure of the difference-weight form from failure of every positive pair representation on the same retained nodes. Hansen and Ghrist [16, Sec. 4.2] show that Kron reduction is impossible for general cellular sheaves: their witness is a star with three boundary vertices, one-dimensional boundary stalks, and a two-dimensional internal stalk, whose boundary section space no sheaf on the boundary vertices alone reproduces. For one-dimensional vertex stalks, Hansen and Ghrist [16, Thm. 4.3] prove the opposite: matrices of factor width at most two, which are exactly the Laplacians of such sheaves [16, Sec. 3.2.2], are closed under Schur complements. The positive two-node class below is the block analog of factor width two, with K-dimensional node blocks, and its dual cone consists of the symmetric matrices whose principal submatrices on at most two nodes are positive semidefinite. Relative to that work, the proposition supplies a proper SPD Gaussian law rather than the section space of a singular sheaf Laplacian, with two-dimensional node blocks and an exact rational certificate in the dual cone.

Proposition G7.17 (Rational obstruction to positive pair decomposition). There is a proper four-agent Gaussian with SPD edge weights and retained anchors whose marginal on three two-dimensional nodes is not a sum of PSD quadratic terms each supported on at most two of those nodes. Hence new invertible links, rectangular pair residuals, parallel edges, self-loops, and freely redistributed PSD anchors cannot represent that marginal within the positive two-node class, and neither can block-diagonal invertible changes of the retained frames, which preserve the class by congruence. The marginal action is of Möbius order two: every Möbius potential (12.46) of order three or more of a Gaussian marginal action vanishes, so the obstruction concerns positivity of the pair terms, not interaction order. [ESTABLISHED]

Proof. Let L_{st} be the six-dimensional retained Schur Laplacian of the unanchored star (G7.48), and put

$$\mathsf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad t = \frac{1}{104}, \qquad \delta = \frac{1}{20000}. \tag{G7.51}$$

Define a symmetric matrix Y in three two-dimensional blocks by

$$\begin{aligned} Y_{ii} &= (1+t^2)I_2, \\ Y_{12} &= I_2 + t\mathsf{J}, \qquad Y_{21} = I_2 - t\mathsf{J}, \\ Y_{ij} &= I_2 \quad \text{for the other distinct pairs.} \end{aligned} \tag{G7.52}$$

Every principal submatrix of Y on one or two nodes is PSD: for diagonal blocks αI and a cross block C the quadratic is at least $\alpha(\|u\|^2+\|v\|^2)-2\|C\|\|u\|\|v\| \geq 0$ when $\|C\| \leq \alpha$, and $(I_2+t\mathsf{J})^{\top}(I_2+t\mathsf{J}) = (1+t^2)I_2$ gives the special cross norm $\sqrt{1+t^2} \leq 1+t^2$, the others being one. Consequently $\mathrm{Tr}(YP) \geq 0$ for every PSD matrix P supported on at most two nodes, since $\mathrm{Tr}(YP) = \mathrm{Tr}(Y_{TT}P_{TT})$

for P supported on a node set T: this is the dual-cone membership of Y, and every sum of positive one-node or two-node terms has nonnegative pairing with Y.

Substitution into $W_{i4} - W_{i4}M^{-1}W_{i4}$ gives diagonal traces $36/19$, $29/19$, $39/19$, hence $\operatorname{Tr} L_{\mathrm{st}} = 104/19$. Write $Y = (\mathbf{1}\mathbf{1}^\top) \otimes I_2 + tY_1 + t^2 I_6$, with $(Y_1)_{12} = \mathsf{J}$, $(Y_1)_{21} = -\mathsf{J}$, and other blocks zero. The constant-vector kernel of L_{st} makes the first pairing vanish, and since $W_{12}^{\mathrm{Kron}} - (W_{12}^{\mathrm{Kron}})^\top = \mathsf{J}/19$,

$$\operatorname{Tr}(Y_1 L_{\mathrm{st}}) = \operatorname{Tr}\big(\mathsf{J}(W_{12}^{\mathrm{Kron}} - (W_{12}^{\mathrm{Kron}})^\top)\big) = -\frac{2}{19}, \qquad \operatorname{Tr}(Y L_{\mathrm{st}}) = \frac{-2t + 104t^2}{19} = -\frac{1}{1976}. \tag{G7.53}$$

Put the anchor δI_2 at each of the three retained vertices of the star, keeping its hidden anchor zero. Zero total energy forces the anchored terminal coordinates and then the connected hidden coordinate to vanish, so the full precision is SPD, and its retained precision is $\Lambda_{\mathrm{r}} = L_{\mathrm{st}} + \delta I_6 \succ 0$, but

$$\operatorname{Tr}(Y\Lambda_{\mathrm{r}}) < -\frac{1}{1976} + \frac{7}{20000} < 0, \qquad \operatorname{Tr} Y = 6(1 + t^2) < 7, \tag{G7.54}$$

the last step by $7 \cdot 1976 = 13832 < 20000$. This contradicts the nonnegative pairing of every positive two-node decomposition. All constants are rational, and (G7.30) supplies the finite normalizer. A block-diagonal invertible change of retained frames maps positive one-node and two-node terms to terms of the same kind. For the Möbius statement, a Gaussian action is a sum of functions of at most two node blocks; for a function of the blocks in a set D and a block $a \notin D$, the Möbius potential of any set containing a vanishes, because pairing each subset B without a with $B \cup \{a\}$ gives equal terms of opposite sign. □

Mixing distinct retained nodes, adding latent vertices, using signed pair terms, or allowing positive terms on three or more nodes changes the representation problem, and the SPD marginal remains a valid complete quadratic Gaussian action. The hyperedges that exact closure generically requires for general laws (Section 12.5) do not arise for Gaussian marginals.

G7.4.5 Nonflat series and cycle marginalization

We now retain link data instead of requiring a flat difference presentation. Loop holonomy at c_* is in scope here, and the content is Gaussian.

Definition G7.18 (Marked nonflat series class). Fix a finite multigraph on V, one orientation of each edge copy $e = (i, j)$, invertible links $\Theta_e \in \mathrm{GL}(K)$ mapping the fiber at j to the fiber at i, SPD weights W_e, PSD anchors A_i, sources h_i, and a scalar offset $\mathfrak{c}$, with action

$$\mathsf{E}(z) = \mathfrak{c} + \frac{1}{2}\sum_{e=(i,j)} (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j) + \frac{1}{2}\sum_i z_i^\top A_i z_i - \sum_i \mathsf{h}_i^\top z_i. \tag{G7.55}$$

A split $V = V_{\mathrm{r}} \sqcup V_{\mathrm{h}}$ belongs to the marked nonflat series class when every hidden vertex has total degree two, a self-loop counted twice, and zero anchor, the retained anchors are kept throughout elimination, and the assembled precision Λ is SPD. Hidden sources need not vanish.

Lemma G7.19 (Properness of a marked nonflat graph). In each connected component choose a root r and a spanning tree, with tree transports $\Theta_{i\leftarrow r}$ and $\Theta_{r\leftarrow r} = I$. Then

$$\Lambda \succ 0 \quad\Longleftrightarrow\quad \bigcap_{\substack{e=(i,j)\\ \text{non-tree}}} \ker(\Theta_{i\leftarrow r} - \Theta_e \Theta_{j\leftarrow r}) \;\cap\; \bigcap_i \ker(A_i \Theta_{i\leftarrow r}) = \{0\} \quad \text{in each component.} \tag{G7.56}$$

One retained SPD anchor in every component suffices. An all-hidden unanchored cycle with holonomy Hol is admitted exactly when $I - \mathrm{Hol}$ is invertible, and $\mathrm{Hol} \neq I$ alone does not suffice. [ESTABLISHED]

Proof. The quadratic form of Λ is a sum of nonnegative residual and anchor terms, so its kernel consists of the vectors with $z_i = \Theta_e z_j$ on every edge and $A_i z_i = 0$ at every vertex. The tree edges force $z_i = \Theta_{i\leftarrow r} u$ with $u = z_r$, and the remaining conditions are the displayed intersection. An SPD anchor at a vertex i forces $\Theta_{i\leftarrow r} u = 0$, hence $u = 0$. On a cycle with one non-tree edge the condition reads $(I - \mathrm{Hol})u = 0$ after conjugation by an invertible tree transport, so it has only the zero solution exactly when $I - \mathrm{Hol}$ is invertible; $\mathrm{Hol} = \mathrm{diag}(1, 2)$ is a nonidentity holonomy with a nonzero kernel. □

Proposition G7.20 (Exact nonflat series pivot). Let i_0 be a hidden vertex of the class of Definition G7.18, and orient its two non-loop edges so that their residuals are $x - \Theta_1 z_{i_0}$ and $z_{i_0} - \Theta_2 y$, with SPD weights W_1, W_2, where $x = z_{i_1}$ and $y = z_{i_2}$ may be coordinates of the same vertex. Set

$$M = \Theta_1^\top W_1 \Theta_1 + W_2, \qquad W_{\mathrm{new}} = (W_1^{-1} + \Theta_1 W_2^{-1} \Theta_1^\top)^{-1}, \qquad \Theta_{\mathrm{new}} = \Theta_1 \Theta_2. \tag{G7.57}$$

Eliminating z_{i_0} replaces the two edge terms by $\frac{1}{2}(x - \Theta_{\mathrm{new}} y)^\top W_{\mathrm{new}} (x - \Theta_{\mathrm{new}} y)$, and the source and scalar updates are

$$\begin{aligned} \mathsf{h}_{i_1}^{\mathrm{new}} &= \mathsf{h}_{i_1} + W_1 \Theta_1 M^{-1} \mathsf{h}_{i_0}, \\ \mathsf{h}_{i_2}^{\mathrm{new}} &= \mathsf{h}_{i_2} + \Theta_2^\top W_2 M^{-1} \mathsf{h}_{i_0}, \\ \mathfrak{c}^{\mathrm{new}} &= \mathfrak{c} + \frac{1}{2} \log\det M - \frac{K}{2} \log(2\pi) - \frac{1}{2} \mathsf{h}_{i_0}^\top M^{-1} \mathsf{h}_{i_0}. \end{aligned} \tag{G7.58}$$

If $i_1 = i_2$, both source increments add at that vertex and the new edge is the marked self-loop $(\Theta_{\mathrm{new}}, W_{\mathrm{new}})$. No new anchor is generated. [ESTABLISHED]

Proof. The hidden terms are

$$\frac{1}{2} z_{i_0}^\top M z_{i_0} - z_{i_0}^\top q + \frac{1}{2} x^\top W_1 x + \frac{1}{2} y^\top \Theta_2^\top W_2 \Theta_2 y, \qquad q = \Theta_1^\top W_1 x + W_2 \Theta_2 y + \mathsf{h}_{i_0}. \tag{G7.59}$$

Since $M \succ 0$, completing the square multiplies the remaining density by

$$(2\pi)^{K/2} (\det M)^{-1/2} \exp\!\big(q^\top M^{-1} q / 2\big).$$

Let $\bar{W} = W_1 - W_1\Theta_1 M^{-1}\Theta_1^\top W_1$. From $M = \Theta_1^\top W_1 \Theta_1 + W_2$ one has $\bar{W}\Theta_1 = W_1\Theta_1 M^{-1} W_2$, and consequently

$$\begin{aligned} \bar{W}(W_1^{-1} + \Theta_1 W_2^{-1}\Theta_1^\top) &= I, \\ W_2 - W_2 M^{-1} W_2 &= \Theta_1^\top \bar{W}\Theta_1. \end{aligned} \tag{G7.60}$$

The first identity gives $\bar{W} = W_{\mathrm{new}}$; the second and $\bar{W}\Theta_1 = W_1\Theta_1 M^{-1} W_2$ identify the endpoint diagonal and cross terms as the new edge energy, and the linear and constant terms of $q^\top M^{-1} q/2$ give (G7.58). Substituting $x = y$ gives the loop. Reversing an edge uses $(\Theta, W) \mapsto (\Theta^{-1}, \Theta^\top W \Theta)$, because the old residual is minus Θ times the reversed one, so the displayed orientation costs nothing. □

Theorem G7.21 (Arbitrary-order nonflat series closure). In the class of Definition G7.18, eliminating any subset of hidden vertices in any order gives the exact normalized Gaussian marginal within the same class: every surviving hidden vertex still has degree two and zero anchor, the remaining precision is SPD, and distinct parallel paths remain distinct edges. Along an oriented path $v_0, \dots, v_m$, with Θ_i mapping from v_i to v_{i-1}, put $\Theta_{\le 0} = I$ and $\Theta_{\le j} = \Theta_1 \cdots \Theta_j$; the fused data are

$$\Theta_{\mathrm{path}} = \Theta_{\le m}, \qquad W_{\mathrm{path}}^{-1} = \sum_{i=1}^{m} \Theta_{\le i-1} W_i^{-1} \Theta_{\le i-1}^\top. \tag{G7.61}$$

A returning path has retained loop holonomy $\mathrm{Hol} = \Theta_{\le m}$ and quadratic

$$Q_{\mathrm{loop}} = (I - \mathrm{Hol})^\top W_{\mathrm{path}} (I - \mathrm{Hol}). \tag{G7.62}$$

A component meeting V_{r} decomposes into maximal hidden paths between retained endpoints, which may coincide; a cycle with one retained vertex becomes one loop, and with several retained vertices its ordered coarse link product equals its original cycle product, so each cycle meeting the retained set keeps its holonomy mark. An all-hidden cycle integrates away exactly under the condition of Lemma G7.19; its raw partition factor survives, but its normalized marginal on the empty coordinate set does not identify its holonomy, and a retained Q_{loop} need not identify the separate marks Hol and W_{path}. These marks are structural data that the Gaussian law does not determine. [ESTABLISHED]

Proof. A degree-two hidden vertex without a loop has two incident edges counted with multiplicity, so Proposition G7.20 applies. If its neighbors differ, one edge replaces two and degrees are preserved; if they coincide, a loop carries the same two incidences. No hidden anchor is created. A degree-two vertex already carrying a loop is an isolated component, arising initially or as the last vertex of an all-hidden cycle; its loop block is SPD by global positivity and integrates as one Gaussian block, contributing only the scalar of (G7.32). Positivity of every remaining Schur complement holds by Proposition G7.11, which proves closure by induction for every order. Write a path segment as (Θ, C) with $C = W^{-1}$; the series operation is

$$(\Theta, C) * (\Theta', C') = (\Theta\Theta',\ C + \Theta C' \Theta^\top). \tag{G7.63}$$

For a third pair (Θ'', C'') either parenthesization gives transport $\Theta\Theta'\Theta''$ and covariance $C+\Theta C'\Theta^\top+\Theta\Theta' C''\Theta'^\top\Theta^\top$, which proves associativity and, by induction, (G7.61); each summand is SPD. Equal endpoints give (G7.62). The scalar and source terms agree for all orders by the Gaussian integration argument of Proposition G7.11. The coarse link product of a cycle is the ordered product of the fused links, which is the original cycle product. Different pairs $(\mathrm{Hol}, W_{\mathrm{path}})$ can give the same Q_{loop}, for instance $(2I, W)$ and $(3I, W/4)$, both with $Q_{\mathrm{loop}} = W$. □

Proposition G7.22 (Gauge and rescaling covariance with scalar factors). For nodewise passive coordinates $z_i' = g_i z_i$, $g_i \in \mathrm{GL}(K)$, the data transform as

$$\begin{aligned} \Theta'_e &= g_i \Theta_e g_j^{-1}, \qquad W'_e = g_i^{-\top} W_e g_i^{-1}, \\ A'_i &= g_i^{-\top} A_i g_i^{-1}, \qquad \mathsf{h}'_i = g_i^{-\top} \mathsf{h}_i, \end{aligned} \tag{G7.64}$$

and the series rule commutes with the change: $M' = g_{i_0}^{-\top} M g_{i_0}^{-1}$, $W'_{\mathrm{new}} = g_{i_1}^{-\top} W_{\mathrm{new}} g_{i_1}^{-1}$, and $\Theta'_{\mathrm{new}} = g_{i_1} \Theta_{\mathrm{new}} g_{i_2}^{-1}$. With the raw scalar $\mathfrak{c}$ unchanged,

$$\begin{aligned} \mathfrak{c}'_{\mathrm{r}} &= \mathfrak{c}_{\mathrm{r}} - \sum_{i \in V_{\mathrm{h}}} \log |\det g_i|, \qquad Z' = \Big(\prod_{i \in V} |\det g_i| \Big) Z, \\ p'_{\mathrm{r}}(x') &= |\det g_{\mathrm{r}}|^{-1} p_{\mathrm{r}}(g_{\mathrm{r}}^{-1} x'), \qquad g_{\mathrm{r}} = \mathrm{blkdiag}_{i \in V_{\mathrm{r}}} g_i; \end{aligned} \tag{G7.65}$$

if instead the raw measure itself is transformed, $\mathfrak{c}' = \mathfrak{c} + \sum_{i \in V} \log |\det g_i|$, then $Z' = Z$ and $\mathfrak{c}'_{\mathrm{r}} = \mathfrak{c}_{\mathrm{r}} + \sum_{i \in V_{\mathrm{r}}} \log |\det g_i|$, and both conventions give the same normalized law. A retained loop transforms by $\mathrm{Hol}' = g_r \mathrm{Hol} g_r^{-1}$. An invertible retained rescaling $x' = gx$ has the normalized pushforward

$$\mu'_{\mathrm{r}} = g\mu_{\mathrm{r}}, \qquad \Sigma'_{\mathrm{r}} = g\Sigma_{\mathrm{r}} g^\top, \qquad p'_{\mathrm{r}}(x') = |\det g|^{-1} p_{\mathrm{r}}(g^{-1} x'), \tag{G7.66}$$

which preserves the marked graph presentation when g is block diagonal and may mix vertices otherwise. [ESTABLISHED]

Proof. Substitution into each residual proves (G7.64). The inverse covariance of (G7.57) transforms by

$$W_1'^{-1} + \Theta'_1 W_2'^{-1} \Theta_1'^\top = g_{i_1} (W_1^{-1} + \Theta_1 W_2^{-1} \Theta_1^\top) g_{i_1}^\top. \tag{G7.67}$$

The source increments transform as endpoint covectors, $\mathsf{h}_{i_0}^\top M^{-1} \mathsf{h}_{i_0}$ is invariant, and $\det M' = \det M / |\det g_{i_0}|^2$, which gives the hidden scalar shift. Changing variables in the full integral gives the total Jacobian and then the retained one; adding the total log-Jacobian to the initial action gives the raw-measure convention. The loop and rescaling laws are the same change of variables. □

A restriction of the links to a matrix subgroup must be closed under products and inverses, and gauges in that subgroup preserve the restricted domain. No preferred physical scale transformation is selected by this finite pushforward.

Proposition G7.23 (A noncommuting nonflat proper example). Take one retained vertex 1, hidden vertices $2, 3$, and two distinct two-edge cycles $1-2-1$ and $1-3-1$, whose edge copies

are distinct interactions. Give vertex 1 the anchor I_2, the hidden vertices zero anchors, and put

$$\Theta_{\mathrm{a}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \Theta_{\mathrm{b}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad W_{\mathrm{a}} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad W_{\mathrm{b}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{G7.68}$$

The first cycle has links Θ_{a}, I_2 and weights $W_{\mathrm{a}}, W_{\mathrm{b}}$; the second has links Θ_{b}, I_2 and weights I_2, I_2. With $\mathsf{h}_2 = (1,0)^\top$, all other sources zero, and $\mathfrak{c} = 0$, this is a proper Gaussian with noncommuting loop holonomies and coefficients outside every cone $\mathcal{W}_H$ containing its anchor, and its exact marginal on vertex 1 is given by (G7.70). [ESTABLISHED]

Proof. Every weight is SPD and the retained SPD anchor gives (G7.56). The holonomies satisfy $\Theta_{\mathrm{a}}\Theta_{\mathrm{b}} - \Theta_{\mathrm{b}}\Theta_{\mathrm{a}} = \mathrm{diag}(1,-1) \neq 0$, so no based conjugation makes both flat or diagonal. The triple $I_2, W_{\mathrm{a}}, W_{\mathrm{b}}$ lies in no common cone: containing I_2 forces a common orthogonal basis, but $W_{\mathrm{a}}W_{\mathrm{b}} - W_{\mathrm{b}}W_{\mathrm{a}} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. This fixed-chart coefficient statement and the gauge-invariant holonomy statement are distinct. The hidden pivots and loop weights are

$$\begin{aligned} M_2 &= \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix}, \quad \widehat{W}_2 = \frac{1}{11}\begin{pmatrix} 7 & -2 \\ -2 & 10 \end{pmatrix}, \\ M_3 &= \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \widehat{W}_3 = \frac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}, \end{aligned} \tag{G7.69}$$

with $\det M_2 = 11$, $\det M_3 = 5$, and loop quadratics $\mathrm{diag}(0, 7/11)$ and $\mathrm{diag}(2/5, 0)$. The formulas above give

$$\begin{aligned} \Lambda^{\mathrm{eff}} &= \mathrm{diag}(7/5, 18/11), & \mathsf{h}^{\mathrm{eff}} &= (1, -3/11)^\top, \\ \mathfrak{c}^{\mathrm{eff}} &= \frac{1}{2}\log 55 - 2\log(2\pi) - \frac{5}{22}, & \mu_1 &= (5/7, -1/6)^\top, \\ \Sigma_1 &= \mathrm{diag}(5/7, 11/18), & Z &= (2\pi)^3 \exp(17/28)/\sqrt{126}. \end{aligned} \tag{G7.70}$$

Indeed $\det \Lambda = 11 \cdot 5 \cdot (126/55) = 126$, $M_2^{-1}\mathsf{h}_2 = (5,-2)^\top/11$, and the completed-square value is $5/11 + 5/7 + 1/22 = 17/14$, which verifies source, covariance, scalar, and normalization in the same proper law. □

The zero hidden anchor is what makes the series rule produce no new anchor. Under admitted positive-semidefinite self terms, anchored hidden vertices still close, with a presentation that is no longer unique.

Proposition G7.24 (Anchored hidden vertices). Keep the setting of Proposition G7.20 with distinct endpoints, but give the hidden vertex a PSD anchor A_{i_0}, and put $M = \Theta_1^\top W_1 \Theta_1 + W_2 + A_{i_0}$. Eliminating z_{i_0} replaces the two edge terms and the anchor by $\frac{1}{2}x^\top \bar{W} x - x^\top R y + \frac{1}{2} y^\top Q y$, with

$$\bar{W} = W_1 - W_1\Theta_1 M^{-1}\Theta_1^\top W_1 \succ 0, \qquad R = W_1\Theta_1 M^{-1} W_2 \Theta_2, \qquad Q = \Theta_2^\top (W_2 - W_2 M^{-1} W_2)\Theta_2,$$

where R is invertible and $Q - R^\top \bar{W}^{-1} R \succeq 0$, so that

$$\tfrac{1}{2}(x - \bar{W}^{-1}Ry)^\top \bar{W}(x - \bar{W}^{-1}Ry) + \tfrac{1}{2} y^\top (Q - R^\top \bar{W}^{-1} R) y$$

is one invertible link with SPD weight plus a generated PSD anchor at i_2; the source and scalar updates are those of (G7.58) with this M. For $A_{i_0} = 0$ this is the series rule, with $\bar{W} = W_{\mathrm{new}}$, $\bar{W}^{-1}R = \Theta_{\mathrm{new}}$, and zero anchor. For $A_{i_0} \neq 0$ new anchors arise: in the scalar chain with unit edges, identity links, and hidden anchor $\alpha > 0$, the pivot is $M = 2 + \alpha$ and the retained quadratic matrix is

$$\begin{pmatrix} 1 - 1/M & -1/M \\ -1/M & 1 - 1/M \end{pmatrix} = \frac{1}{M}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \frac{\alpha}{M} I_2. \tag{G7.71}$$

The presentation is not unique: the flat link with weight $1/M$ and anchors α/M at both endpoints, and the link $1/(1+\alpha)$ with weight $(1+\alpha)/M$ and anchor $\alpha/(1+\alpha)$ at i_2, both reproduce (G7.71). [ESTABLISHED]

Proof. The hidden terms are those of (G7.59) with the anchor added to M, and completing the square leaves $\frac{1}{2}x^\top W_1 x + \frac{1}{2}y^\top \Theta_2^\top W_2 \Theta_2 y - \frac{1}{2}q^\top M^{-1} q$, whose blocks are $\bar{W}$, $-R$, and Q; the source and scalar terms are as before. The matrix $\left(\begin{smallmatrix} W_1 & W_1\Theta_1 \\ \Theta_1^\top W_1 & M \end{smallmatrix}\right)$ is SPD because its Schur complement at W_1 is $W_2 + A_{i_0} \succ 0$, so its Schur complement $\bar{W}$ at M is SPD. R is a product of invertible matrices. The eliminated quadratic is the minimum over z_{i_0} of a nonnegative form, hence PSD, and with $\bar{W} \succ 0$ its Schur complement $Q - R^\top \bar{W}^{-1} R$ is PSD; completing the square in x gives the displayed presentation. For $A_{i_0} = 0$, $\bar{W} = W_{\mathrm{new}}$ by (G7.60), $R = W_{\mathrm{new}}\Theta_1\Theta_2$ by $\bar{W}\Theta_1 = W_1\Theta_1 M^{-1} W_2$, and $Q = \Theta_2^\top \Theta_1^\top W_{\mathrm{new}} \Theta_1 \Theta_2 = R^\top \bar{W}^{-1} R$. The scalar chain is direct substitution: $\bar{W} = Q = 1 - 1/M$ and $R = 1/M$, and both presentations expand to (G7.71). □

In the flat graph theorem a hidden anchor is an edge to ground, which can create a third terminal, and in the fixed-anchor theorem the positive hidden anchor is the premise that forces the commutation constraint. Degree two in the nonflat theorem guarantees every intermediate order, whereas the graph criterion concerns each selected final hidden set; these quantifiers are distinct. The nonflat series closure assumes zero hidden anchors and does not cover prior-carrying hidden agents, for which the presentation of Proposition G7.24 carries a link that is not determined by the marginal.

Chapter G8

Coarse Maps II: Nonflat Links, Relaxation, and Quotient Volume

What does hard aggregation do to nonflat internal links, how does a Gaussian mean relax onto the fixed sector, what is the Gaussian form of the holonomy-constrained parent, and how does the volume of hard membership enter a partition function? A twisted internal link adds a positive-semidefinite self term to its cluster, which only a singular weight can hide, and one effective quotient for all links exists exactly under the condition of Proposition G8.1. The constant-metric flow relaxes exponentially onto the fixed sector and is the Fisher natural-gradient flow on the fixed-covariance mean family (Proposition G8.2). The Gaussian forward-KL barycenter is moment matching with a log-determinant score (Proposition G8.3), and under compact holonomy the constrained parent is the Haar-projected moment match (Corollary G8.4), while no parameter-independent Markov kernel realizes the joint barycenter of two normal laws for all parameters (Proposition G8.5). Hard membership carries the quotient metric of Proposition G8.6, whose Jacobian changes raw normalizers and not normalized laws. These results continue Chapter G7 and specialize the structural fixed sectors (Section 11.5) and the holonomy-constrained marginal parents (Theorem 11.10) of the main manuscript; the relaxation is the identified natural-gradient example of Chapter G5. Notation follows Chapter G7; in addition G denotes a constant metric, g a passive linear change of coordinates or an element of a holonomy group, and w_i mixture weights.

This chapter does not identify a generic, state-dependent Fisher flow with the constant projector and rate of the constant-metric relaxation, and it does not claim that a partition is selected by quotient volume or by transported relative entropy. A natural partition selector remains open (Open Problem 10.20). [NOT-CLAIMED] Balanced cut links admit no continuous extension to all singular cuts (Proposition 11.28).

G8.1 Nonflat links and effective support

We return to hard identification with twisted internal links. For an edge $e = (i, j)$ with link Θ_e and weight $W_e \succeq 0$, the twisted pair energy is

$$\frac{1}{2}(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \tag{G8.1}$$

With a latent link, $\Theta_e = e^{\mathring{\omega}_e}$ in reference coordinates; at the means of Gaussian laws whose transported source has the receiver's covariance, and with W_e the receiver's precision, this energy is the law-level transported divergence of Proposition G11.1.

Proposition G8.1 (Nonflat internal links and effective support). Let the assembled precision Λ, with twisted links, be SPD, and let S be the aggregation matrix of a hard partition. (i) The hard trace of Λ, with natural parameters $(S^\top \mathsf{h}, S^\top \Lambda S)$, is the conditional law of Proposition G7.4 with $B = S$; it is in general not the coordinate marginal of Theorem G7.21. (ii) Under hard identification an internal edge contributes to its cluster the PSD self term

$$(I - \Theta_e)^\top W_e (I - \Theta_e), \tag{G8.2}$$

which vanishes exactly when

$$W_e^{1/2}(I - \Theta_e) = 0; \tag{G8.3}$$

a positive-definite W_e forces $\Theta_e = I$, while a singular weight can hide a nonidentity link. (iii) With $\mathrm{pr}_e : \mathbb{R}^K \to \mathbb{R}^K / \ker W_e$ the quotient map,

$$W_e^{1/2}(I - \Theta_e) = 0 \quad \Longleftrightarrow \quad \mathrm{pr}_e\, \Theta_e = \mathrm{pr}_e\,. \tag{G8.4}$$

(iv) For links $\Theta_e \in \mathrm{GL}(K)$, a single subspace $\mathscr{N} \subseteq \mathbb{R}^K$ such that every pair form descends to $\mathbb{R}^K/\mathscr{N}$ with positive-definite induced weight exists exactly when

$$\ker W_e = \mathscr{N}, \qquad \Theta_e \mathscr{N} = \mathscr{N} \quad \text{for every relevant edge,} \tag{G8.5}$$

and on that quotient group-valued holonomy is defined; when the kernels differ there is no canonical product of the edge-local quotients. (v) On the triangle with $W_{12} = \mathrm{diag}(1,0)$, $\Theta_{12} = \mathrm{diag}(1,2)$, $W_{23} = \mathrm{diag}(0,1)$, $\Theta_{23} = \mathrm{diag}(2,1)$, and $W_{31} = \Theta_{31} = I$, every edge satisfies (G8.3) and the loop product is $2I$, which fixes no nonzero vector; the kernels differ, so no common quotient exists. Nevertheless the kernel of the cluster's connection energy, cut out by the section constraints (11.25), is the two-dimensional space of constant sections, so hard fixed-section aggregation of this cluster keeps the full fiber. [ESTABLISHED]

Proof. (i) Precomposing the energy with $z = Sy$ gives the natural parameters $(S^\top \mathsf{h}, S^\top \Lambda S)$, which are those of the conditional law by Proposition G7.4; the marginal precision is a Schur complement, which is dominated by the restricted precision with equality only in the Λ-orthogonal case (Corollary G2.8). (ii) Setting $z_i = z_j = y$ turns (G8.1) into $\frac{1}{2} y^\top (I - \Theta_e)^\top W_e (I - \Theta_e) y = \frac{1}{2} \| W_e^{1/2} (I - \Theta_e) y \|^2$ (Lemma G3.5), which vanishes for every y exactly under (G8.3). If $W_e \succ 0$ this forces $I - \Theta_e = 0$, while $W_e = \mathrm{diag}(1,0)$ and $\Theta_e = \mathrm{diag}(1,2)$ satisfy it. (iii) $\mathrm{pr}_e\, \Theta_e = \mathrm{pr}_e$ means $\mathrm{range}(I - \Theta_e) \subseteq \ker W_e = \ker W_e^{1/2}$. (iv) The pair form descends in both endpoint variables exactly when $W_e^{1/2} n = 0$ and $W_e^{1/2} \Theta_e n' = 0$ for all $n, n' \in \mathscr{N}$, that is, $\mathscr{N} \subseteq \ker W_e$ and $\Theta_e \mathscr{N} \subseteq \ker W_e$. Positive definiteness of the induced weight forces $\ker W_e = \mathscr{N}$, and invertibility of Θ_e turns $\Theta_e \mathscr{N} \subseteq \mathscr{N}$ into $\Theta_e \mathscr{N} = \mathscr{N}$ and makes the induced link invertible. (v) Each weighted residual vanishes by direct multiplication, and $\Theta_{12}\Theta_{23}\Theta_{31} = 2I$. The kernels are $\mathrm{span}(e_2)$, $\mathrm{span}(e_1)$, and

$\{0\}$. The section constraints $W_e^{1/2}(z_i - \Theta_e z_j) = 0$ give $z_1^{(1)} = z_2^{(1)}$, $z_2^{(2)} = z_3^{(2)}$, and $z_3 = z_1$, whose solutions are exactly the constant sections. □

G8.2 Constant-metric Gaussian relaxation

Proposition G8.2 (Relaxation onto the fixed sector). Let $\mathsf{G} = \mathsf{G}^\top \succ 0$ and $L = L^\top \succeq 0$ be constant matrices on $\mathbb{R}^d$, let the columns of B_0 form a basis of $\ker L$, and define the G-orthogonal projector

$$\Pi_0 = B_0(B_0^\top \mathsf{G} B_0)^{-1} B_0^\top \mathsf{G}, \tag{G8.6}$$

with $\Pi_0 = 0$ when $\ker L = \{0\}$. The flow

$$\mathsf{G}\dot{z} = -Lz, \qquad z(0) = z_0, \tag{G8.7}$$

converges to $\Pi_0 z_0$. If $L \neq 0$ and

$$r_+ = \lambda_{\min}^+\big(\mathsf{G}^{-1/2} L \mathsf{G}^{-1/2}\big) \tag{G8.8}$$

is the smallest positive eigenvalue, then

$$\|z(t) - \Pi_0 z_0\|_{\mathsf{G}} \le e^{-r_+ t}\|z_0 - \Pi_0 z_0\|_{\mathsf{G}}, \tag{G8.9}$$

and for $\mathsf{E}(z) = \frac{1}{2} z^\top L z$,

$$\frac{d}{dt}\mathsf{E}(z(t)) = -\dot{z}^\top \mathsf{G}\dot{z} = -z^\top L \mathsf{G}^{-1} L z \le 0. \tag{G8.10}$$

A constant passive reframing $z' = gz$, $g \in \mathrm{GL}(d)$, preserves the equation exactly under

$$\mathsf{G}' = g^{-\top}\mathsf{G}g^{-1}, \qquad L' = g^{-\top} L g^{-1}, \qquad \Pi_0' = g\Pi_0 g^{-1}, \tag{G8.11}$$

so the choice $\mathsf{G} = I$ is equivariant only under orthogonal reframings. For a time-dependent $g = g(t)$ with vanishing original temporal connection, the covariant derivative

$$D_t' z' = \dot{z}' - \dot{g}g^{-1}z' = g\dot{z}, \qquad \mathsf{G}' D_t' z' = -L' z', \tag{G8.12}$$

restores the equation. On the fixed-covariance Gaussian mean family $q_z = \mathcal{N}(z, \mathsf{G}^{-1})$ the Fisher metric of the mean is G and

$$D_{\mathrm{KL}}(q_z \| q_w) = \frac{1}{2}(z - w)^\top \mathsf{G}(z - w), \tag{G8.13}$$

so (G8.7) is exactly the Fisher natural-gradient flow of E on that family. For a family (q_λ) with Fisher metric g^F whose log-density is twice continuously differentiable in λ, with the domination of Hypothesis 7.1 permitting differentiation under the integral through second order, only the local expansion

$$D_{\mathrm{KL}}(q_\lambda \| q_{\lambda + d\lambda}) = \frac{1}{2} d\lambda^\top g^F(\lambda) d\lambda + o(\|d\lambda\|^2) \tag{G8.14}$$

holds. [ESTABLISHED]

Proof. Set $y = \mathsf{G}^{1/2}z$ and $\widetilde{L} = \mathsf{G}^{-1/2}L\mathsf{G}^{-1/2} \succeq 0$. Then $\dot{y} = -\widetilde{L}y$, and the spectral theorem gives convergence to the Euclidean projection of y_0 onto $\ker \widetilde{L} = \mathsf{G}^{1/2} \ker L$, with transverse norm decaying at the smallest positive eigenvalue of $\widetilde{L}$. Transforming back gives (G8.6) and (G8.9), and differentiating E along the flow gives (G8.10). Substituting $z = g^{-1}z'$ proves (G8.11), and $\dot{z}' = \dot{g}z + g\dot{z}$ proves (G8.12). The Gaussian relative entropy with equal covariances gives (G8.13), whose Hessian is the Fisher metric G, and the natural gradient of E in that metric is $\mathsf{G}^{-1}Lz$. For the local expansion write $\mathbf{s}_j = \partial_j \log q_\lambda$. At $\lambda' = \lambda$ the first derivative of $D_{\mathrm{KL}}(q_\lambda \| q_{\lambda'})$ in λ' is $-\mathbb{E}_\lambda[\mathbf{s}_j] = 0$, and the second is $-\mathbb{E}_\lambda[\partial_i \mathbf{s}_j] = \mathbb{E}_\lambda[\mathbf{s}_i \mathbf{s}_j] = g^F_{ij}(\lambda)$, by differentiating $\int q_\lambda \mathbf{s}_j = 0$; Taylor's theorem in λ' gives (G8.14). □

Freezing $g^F(\lambda_0) = \mathsf{G}$ yields only a local linearization of a general family. The rate r_+ and the dissipation per unit t belong to the flow parameter, not to the unparameterized orbit.

G8.3 Gaussian marginal-law specialization

Specialize Theorem 11.10 to Gaussian laws after transporting every constituent marginal to one root, with the full nondegenerate Gaussian family as admissible parent family. With trivial represented holonomy the holonomy-fixed subfamily is that whole family; the parent family itself is declared, not selected by the holonomy.

Proposition G8.3 (Full-Gaussian forward-KL barycenter). Let $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ with $\Sigma_i \succ 0$, and let $w_i > 0$ with $\sum_i w_i = 1$. The unique minimizer over nondegenerate Gaussians of

$$\mathcal{D}_I(\mu, \Sigma) = \sum_i w_i D_{\mathrm{KL}}\big(\mathcal{N}(\mu_i, \Sigma_i) \| \mathcal{N}(\mu, \Sigma)\big) \tag{G8.15}$$

matches the first two moments,

$$\bar{\mu} = \sum_i w_i \mu_i, \qquad \bar{\Sigma} = \sum_i w_i \big[\Sigma_i + (\mu_i - \bar{\mu})(\mu_i - \bar{\mu})^\top\big], \tag{G8.16}$$

and its minimized score is

$$\mathcal{D}_I^\star = \frac{1}{2}\Big[\log\det \bar{\Sigma} - \sum_i w_i \log\det \Sigma_i\Big]. \tag{G8.17}$$

If every constituent and the parent are restricted to one covariance $\Sigma_0 \succ 0$, the minimizer is $\mathcal{N}(\bar{\mu}, \Sigma_0)$, and with the symmetrized convention

$$D_{\mathrm{SKL}}(p, q) := \frac{1}{2}\big[D_{\mathrm{KL}}(p\|q) + D_{\mathrm{KL}}(q\|p)\big] \tag{G8.18}$$

its score satisfies

$$\sum_i w_i D_{\mathrm{KL}}\big(q_i\|\mathcal{N}(\bar\mu,\Sigma_0)\big)=\sum_{i<j} w_i w_j D_{\mathrm{SKL}}(q_i,q_j)=\frac12\sum_{i,j} w_i w_j D_{\mathrm{SKL}}(q_i,q_j), \tag{G8.19}$$

which is not the free-covariance optimum, whose covariance includes the between-mean term of (G8.16). [ESTABLISHED]

Proof. Minimizing $\sum_i w_i D_{\mathrm{KL}}(q_i\|q)$ over an exponential family q is minimizing $D_{\mathrm{KL}}(\sum_i w_i q_i\|q)$ up to a constant, which is moment matching [5, Sec. 10.7, Eqs. (10.184)–(10.187)]; we record the score. The Gaussian relative entropy gives

$$\begin{aligned} 2\mathcal{D}_I(\mu,\Sigma)&=\mathrm{Tr}(\Sigma^{-1}\Xi(\mu))-K+\log\det\Sigma-\sum_i w_i\log\det\Sigma_i,\\ \Xi(\mu)&=\sum_i w_i\big[\Sigma_i+(\mu_i-\mu)(\mu_i-\mu)^\top\big]. \end{aligned} \tag{G8.20}$$

Completing the weighted square gives $\Xi(\mu)=\bar\Sigma+(\mu-\bar\mu)(\mu-\bar\mu)^\top$, so $\mu=\bar\mu$ is the unique minimizer in μ. At that value $\mathrm{Tr}(\Sigma^{-1}\bar\Sigma)-\log\det(\Sigma^{-1}\bar\Sigma)\ge K$, with equality exactly when $\Sigma=\bar\Sigma$, and substitution gives (G8.17). With one covariance Σ_0, $D_{\mathrm{KL}}(q_i\|\mathcal{N}(\mu,\Sigma_0))=\frac12\|\mu_i-\mu\|^2_{\Sigma_0^{-1}}$ is minimized in the weighted sum at $\bar\mu$, and $D_{\mathrm{SKL}}(q_i,q_j)=\frac12\|\mu_i-\mu_j\|^2_{\Sigma_0^{-1}}$; the weighted variance identity $\sum_i w_i\|\mu_i-\bar\mu\|^2=\frac12\sum_{i,j}w_iw_j\|\mu_i-\mu_j\|^2$ in the Σ_0^{-1} inner product gives (G8.19). □

Corollary G8.4 (Compact-closure holonomy Gaussian barycenter). In the setting of Proposition G8.3, let $\mathrm{Hol}_I\subset\mathrm{GL}(K)$ be a represented holonomy group whose closure $\overline{\mathrm{Hol}}_I$ is compact, with normalized Haar probability dg. A nondegenerate Gaussian is Hol_I-invariant exactly when it is $\overline{\mathrm{Hol}}_I$-invariant. Define

$$\bar\mu=\sum_i w_i\mu_i,\qquad \Xi=\sum_i w_i(\Sigma_i+\mu_i\mu_i^\top), \tag{G8.21}$$

and their Haar projections

$$\mu_{\mathrm{Hol}}=\int_{\overline{\mathrm{Hol}}_I} g\bar\mu\,dg,\qquad \Xi_{\mathrm{Hol}}=\int_{\overline{\mathrm{Hol}}_I} g\Xi g^\top dg,\qquad \Sigma_{\mathrm{Hol}}=\Xi_{\mathrm{Hol}}-\mu_{\mathrm{Hol}}\mu_{\mathrm{Hol}}^\top. \tag{G8.22}$$

Then $\Sigma_{\mathrm{Hol}}\succ 0$, and $\mathcal{N}(\mu_{\mathrm{Hol}},\Sigma_{\mathrm{Hol}})$ is the unique minimizer of (G8.15) over Hol_I-invariant nondegenerate Gaussians, with score

$$\mathcal{D}^\star_{I,\mathrm{Hol}}=\frac12\Big[\log\det\Sigma_{\mathrm{Hol}}-\sum_i w_i\log\det\Sigma_i\Big]. \tag{G8.23}$$

The constraint can be active without any disagreement among the constituents: for $K=2$, $\mathrm{Hol}_I=\{I_2,-I_2\}$, and the one-law ensemble $\mathcal{N}(u,I_2)$ with $u\neq 0$, the unconstrained optimizer $\mathcal{N}(u,I_2)$ is not invariant, while the constrained optimizer is $\mathcal{N}(0,I_2+uu^\top)$ with score $\frac12\log(1+\|u\|^2)>0$. [ESTABLISHED]

Proof. The stabilizer of a Gaussian under the continuous pushforward action of $\mathrm{GL}(K)$ is closed, so invariance under Hol_I and under its closure agree. For every invariant candidate q, invariance of relative entropy under a common invertible pushforward gives

$$\sum_i w_i D_{\mathrm{KL}}(q_i\|q) = \int_{\overline{\mathrm{Hol}}_I} \sum_i w_i D_{\mathrm{KL}}(g_{\#}q_i\|q)dg, \tag{G8.24}$$

so the objective is that of the orbit-averaged ensemble, the Haar-symmetrized mixture (11.21), whose first and second moments are μ_{Hol} and Ξ_{Hol}. Haar invariance makes both moments invariant, so the moment-matched Gaussian of that ensemble is an admissible candidate, and Proposition G8.3, applied to the averaged ensemble, makes it the unique minimizer among all nondegenerate Gaussians and hence among invariant ones; this is the Gaussian clause of Proposition 11.11 with the moments made explicit. The covariance Σ_{Hol} is the covariance of the averaged mixture, which dominates the Haar average of the positive-definite matrices $g\Sigma_i g^\top$, so it is positive definite. Every element of a compact matrix group has $|\det g| = 1$, so $\log\det(g\Sigma_i g^\top) = \log\det\Sigma_i$, which gives (G8.23). In the witness the Haar average over $\{\pm I_2\}$ sends $\bar\mu = u$ to 0 and leaves $\Xi = I_2 + uu^\top$ unchanged, and $\det(I_2 + uu^\top) = 1 + \|u\|^2$. □

The trivial-holonomy case of the corollary is Proposition G8.3. Compactness cannot be dropped: a holonomy element with an eigenvalue off the unit circle, such as $2I_K$, fixes no nondegenerate Gaussian (Corollary 11.12).

The Markov property of a configuration arrow (Definition 7.44) can fail even for natural meta-agent maps, as the barycenter of two normal laws shows.

Proposition G8.5 (Normal barycenters and parameter-independent kernels). Let $\Sigma_0 \succ 0$ be an $n \times n$ covariance, $\mathsf{H} \in \mathbb{R}^{n'\times n}$, $\mathsf{b} \in \mathbb{R}^{n'}$, and $\Sigma_1 \succeq 0$ an $n' \times n'$ covariance, and let κ be a Markov kernel from $\mathbb{R}^n$ to $\mathbb{R}^{n'}$ with

$$\mathcal{N}(\mu, \Sigma_0)\kappa = \mathcal{N}(\mathsf{H}\mu + \mathsf{b}, \Sigma_1) \quad \text{for every } \mu \in \mathbb{R}^n. \tag{G8.25}$$

Then $\Sigma_1 \succeq \mathsf{H}\Sigma_0\mathsf{H}^\top$ and $\kappa(y,\cdot) = \mathcal{N}(\mathsf{H}y+\mathsf{b}, \Sigma_1 - \mathsf{H}\Sigma_0\mathsf{H}^\top)$ for Lebesgue-almost every y. Consequently no Markov kernel from $\mathbb{R}^2$ to $\mathbb{R}$ satisfies

$$\big(\mathcal{N}(\mu_1,\varsigma^2)\otimes\mathcal{N}(\mu_2,\varsigma^2)\big)\kappa = \mathcal{N}\left(\tfrac{\mu_1+\mu_2}{2}, \varsigma^2\right) \quad \text{for all } \mu_1,\mu_2 \in \mathbb{R} \text{ and all } \varsigma > 0, \tag{G8.26}$$

and none sends $\mathcal{N}(\mu_1,\varsigma_1^2)\otimes\mathcal{N}(\mu_2,\varsigma_2^2)$ to the joint barycenter $\mathcal{N}\big(\frac{\mu_1+\mu_2}{2}, \frac{\varsigma_1^2+\varsigma_2^2}{2}\big)$ for all parameters. For each fixed ς, the kernel $y \mapsto \mathcal{N}\big(\frac{y_1+y_2}{2}, \frac{\varsigma^2}{2}\big)$ realizes (G8.26) at that ς, and the deterministic kernel $y \mapsto (y_1+y_2)/\sqrt{2}$ sends $\mathcal{N}(0,\varsigma_1^2)\otimes\mathcal{N}(0,\varsigma_2^2)$ to $\mathcal{N}\big(0, \frac{\varsigma_1^2+\varsigma_2^2}{2}\big)$ for all $\varsigma_1,\varsigma_2 > 0$. [ESTABLISHED]

Proof. For $\omega \in \mathbb{R}^{n'}$ put $k_\omega(y) = \int e^{\mathrm{i}\omega^\top y'}\kappa(y,dy')$, a measurable function with $|k_\omega| \le 1$, and the bounded function $h_\omega(y) = \exp\big(\mathrm{i}\omega^\top(\mathsf{H}y+\mathsf{b}) - \frac{1}{2}\omega^\top(\Sigma_1 - \mathsf{H}\Sigma_0\mathsf{H}^\top)\omega\big)$. The characteristic function of $\mathcal{N}(\mu,\Sigma_0)$ at $\mathsf{H}^\top\omega$ gives $\int h_\omega d\mathcal{N}(\mu,\Sigma_0) = \exp(\mathrm{i}\omega^\top(\mathsf{H}\mu+\mathsf{b}) - \frac{1}{2}\omega^\top\Sigma_1\omega)$, which by (G8.25) equals $\int k_\omega d\mathcal{N}(\mu,\Sigma_0)$, so the bounded function $k_\omega - h_\omega$ has zero convolution with the $\mathcal{N}(0,\Sigma_0)$ density. In tempered distributions its Fourier transform is annihilated by multiplication with a nowhere-vanishing

Gaussian. The quotient of any compactly supported smooth test function by that Gaussian is again compactly supported and smooth, so the transform vanishes on all such tests. It is therefore zero as a distribution, hence as a tempered distribution, and $k_\omega = h_\omega$ almost everywhere. Then $|h_\omega| \le 1$ on a nonempty set forces $\omega^\top(\Sigma_1 - \mathsf{H}\Sigma_0\mathsf{H}^\top)\omega \ge 0$, and for ω in a countable dense set the characteristic functions of $\kappa(y,\cdot)$ and of $\mathcal{N}(\mathsf{H}y + \mathsf{b}, \Sigma_1 - \mathsf{H}\Sigma_0\mathsf{H}^\top)$ agree for almost every y, hence everywhere by continuity. For (G8.26), $\Sigma_0 = \varsigma^2\mathrm{I}_2$, $\mathsf{H} = (\frac{1}{2}, \frac{1}{2})$, $\mathsf{b} = 0$, and $\Sigma_1 = \varsigma^2$ force $\kappa(y,\cdot) = \mathcal{N}\big(\frac{y_1+y_2}{2}, \frac{\varsigma^2}{2}\big)$ almost everywhere, and two values of ς give incompatible laws on a set of full measure; the joint barycenter restricted to $\varsigma_1 = \varsigma_2$ is (G8.26). For the realizations, the output means are $\frac{\mu_1+\mu_2}{2}$ and 0, and the output variances are $\frac{1}{4}(2\varsigma^2) + \frac{\varsigma^2}{2} = \varsigma^2$ and $\frac{1}{2}(\varsigma_1^2 + \varsigma_2^2)$. □

G8.4 Quotient volume for hard membership

Proposition G8.6 (Translation-quotient metric of hard membership). Let $\widehat{S} \in \{0,1\}^{N\times|\mathcal{P}|}$ be the membership matrix of a partition $\mathcal{P}$ with nonempty cluster sizes n_I, write $\mathbf{n} = (n_I)_I$, so that $N = \sum_I n_I$, and let pdet denote the product of the positive eigenvalues of a positive-semidefinite matrix. On the translation quotient, the metric induced from the fine Euclidean quotient is

$$\mathsf{G}_\mathcal{P} = \mathrm{diag}(n_I) - \frac{\mathbf{n}\mathbf{n}^\top}{N}, \qquad \ker \mathsf{G}_\mathcal{P} = \mathrm{span}(\mathbf{1}), \qquad \mathrm{pdet}\, \mathsf{G}_\mathcal{P} = \frac{|\mathcal{P}| \prod_I n_I}{N}. \tag{G8.27}$$

For K channels, let ν_{std} be Lebesgue measure on the representatives in $\mathbb{R}^{|\mathcal{P}|K}$ orthogonal to $\mathbf{1}\otimes\mathbb{R}^K$, and let ν_{ind} be the Riemannian volume, equivalently the Hausdorff measure, of $\mathsf{G}_\mathcal{P} \otimes I_K$. Then

$$\nu_{\mathrm{ind}} = \mathrm{Jac}_\mathcal{P}\, \nu_{\mathrm{std}}, \qquad \mathrm{Jac}_\mathcal{P} = \Big(\frac{|\mathcal{P}| \prod_I n_I}{N}\Big)^{K/2}. \tag{G8.28}$$

If $\mathrm{range}(S) \cap \ker \Lambda = \mathbf{1}_N \otimes \mathbb{R}^K$, the quotient Gaussian with precision $\Lambda_\mathrm{c} = S^\top \Lambda S$ is proper, and its determinant relative to the quotient metric is

$$\det_{\mathsf{G}_\mathcal{P}\otimes I_K}(\Lambda_\mathrm{c}) := \frac{\mathrm{pdet}\, \Lambda_\mathrm{c}}{(\mathrm{pdet}\, \mathsf{G}_\mathcal{P})^K} = \frac{\mathrm{pdet}\, \Lambda_\mathrm{c}}{\mathrm{Jac}_\mathcal{P}^2}. \tag{G8.29}$$

For the unnormalized factor $\exp(-x^\top \Lambda_\mathrm{c} x/2)$ and $d_q = (|\mathcal{P}| - 1)K$, the normalizers are

$$\begin{aligned} Z_{\mathrm{std}} &= (2\pi)^{d_q/2} (\mathrm{pdet}\, \Lambda_\mathrm{c})^{-1/2}, \\ Z_{\mathrm{ind}} &= (2\pi)^{d_q/2} \det_{\mathsf{G}_\mathcal{P}\otimes I_K}(\Lambda_\mathrm{c})^{-1/2} = \mathrm{Jac}_\mathcal{P}\, Z_{\mathrm{std}}. \end{aligned} \tag{G8.30}$$

The factor cancels from the normalized law, because the measure changes by the same factor, while raw partition functions differ by $\log \mathrm{Jac}_\mathcal{P}$; a partition score therefore needs a fixed volume convention, an exact correction, or a compensating prior. [ESTABLISHED]

Proof. For a coarse class $[x]$, minimizing over a common translation $\tau \in \mathbb{R}$ gives $\min_\tau \|\widehat{S}x + \tau \mathbf{1}_N\|^2 = \sum_I n_I x_I^2 - (\mathbf{n}^\top x)^2/N = x^\top \mathsf{G}_\mathcal{P} x$. Deleting the row and column of one cluster, the matrix determinant lemma gives the principal cofactor $\prod_I n_I/N$. Since $\mathsf{G}_\mathcal{P}\mathbf{1} = 0$ and the kernel is one-dimensional, $\mathrm{adj}(\mathsf{G}_\mathcal{P})\mathsf{G}_\mathcal{P} = 0$ forces $\mathrm{adj}(\mathsf{G}_\mathcal{P}) = \mathrm{c}\mathbf{1}\mathbf{1}^\top$, and its trace, the sum of the principal cofactors, equals

the product of the nonzero eigenvalues; hence $\operatorname{pdet} \mathsf{G}_{\mathcal{P}}$ is $|\mathcal{P}|$ times any principal cofactor, which proves (G8.27). In an orthonormal chart of the representatives the metric $\mathsf{G}_{\mathcal{P}} \otimes I_K$ has determinant $(\operatorname{pdet} \mathsf{G}_{\mathcal{P}})^K$, which gives (G8.28). The kernel hypothesis makes Λ_{c} positive definite on the quotient, and the two normalizers are the Gaussian integrals in the two measures. □

Physicist summary. Hard aggregation of a Gaussian network is exact Galerkin restriction: internal springs disappear, cut springs and self terms add, and the coarse law is the fine law conditioned on zero internal disagreement, equivalently a precision-dependent linear pushforward. A twisted internal link leaves a residual self term that only a singular weight can hide, and hard aggregation of a cluster keeps whatever its connection energy leaves flat. Marginalization keeps the hidden normalization and closes only in specific regimes. Independent tests at one fixed anchor force common congruence axes; on a fixed flat graph, closure for every positive assignment holds exactly when each hidden component touches at most two terminals, ground included, so that with a prior on every agent, marginalizing a relay between two retained agents generically leaves the positive family, while a hidden component hanging on one agent only stiffens that agent's anchor. Nonflat degree-two paths and cycles close by composing links and covariances with their loop holonomy retained, and anchored ones close with new self terms but no canonical link. Branching can produce a proper Gaussian whose marginal has no positive two-node presentation, although its action is still of Möbius order two. Relaxation under a frozen metric is an exact natural-gradient flow on a fixed-covariance mean family, the holonomy-constrained Gaussian parent is the Haar-projected moment match, and quotient volumes change raw normalizers without supplying an intrinsic blocking scale.

Chapter G9

Operator Coarse-Graining

What does a hard aggregation of agents do to the operator parameters of the interaction family, and when is its output again a normalized Gaussian law? The chapter answers both questions for one finite step and its finite compositions. On the parameters (A_i, W_{ij}) of Definition G3.1, aggregation is a positive linear semigroup action that sums self terms and cut weights and annihilates internal edges (Proposition G9.2). On the interior of the interaction cone, with Lebesgue reference measure and a declared information-vector rule, a step rescaled by $\zeta_\ell > 0$ is a normalized Gaussian law step, which one parameter-independent kernel realizes on a fixed-precision mean fiber exactly when $\zeta_\ell \leq 1$ (Proposition G9.3). The only positive rules that depend additively on scalar node masses are the biadditive ones, which are closed under every hard partition (Proposition G9.5). The operator step is the Gaussian instance of the exact coarse channels of Chapters 10 and 12; it is not their definition, because a law-level step also needs a coarse reference measure, an information-vector rule, and a finite normalizer, which the interior supplies and the singular boundary does not. Hard aggregation is one admissible coarse arrow among several: the effective-support, sheaf, and Kron reductions of Chapter G7 give others on narrower domains. The pencil invariants that type the boundary are in Section G5.3.

The chapter treats the belief and recognition operator sector at c_*; a model-channel step is a separate construction with its own dimension, links, and parameters. It does not assert a statement about infinite hierarchies, limiting rays, or the behavior of iterated steps as the depth grows, or that a law-level step exists at the boundary member of the biadditive family without a declared repair. [NOT-CLAIMED] The monotonicity question beyond one fixed precision is Open Problem G9.4, and a gauge-compatible nondegenerate partition selector is Open Problem 10.20. Balanced cut links admit no continuous extension to all singular cuts (Proposition 11.28).

G9.1 The operator arrow

Definition G9.1 (Operator arrow). At depth ℓ, let $\Xi_\ell = ((A_i^{(\ell)}), (W_{ij}^{(\ell)}))$ with $A_i^{(\ell)}, W_{ij}^{(\ell)} \in \mathbb{S}_+^K$, and assemble the depth-ℓ interaction precision of Definition G3.1,

$$(\Lambda_\ell)_{ii} = A_i^{(\ell)} + \sum_{j \neq i} W_{ij}^{(\ell)}, \qquad (\Lambda_\ell)_{ij} = -W_{ij}^{(\ell)}; \tag{G9.1}$$

a precision assembled from the directed model enters through Hypothesis G3.13. For a hard partition of the agents into clusters I, let $\widehat{S}_\ell$ be its $0/1$ membership matrix and $S_\ell = \widehat{S}_\ell \otimes I_K$. The unrescaled aggregation map acts on precisions and on parameters by

$$\mathcal{A}_{S_\ell}(\Lambda_\ell) = S_\ell^\top \Lambda_\ell S_\ell, \qquad \mathcal{A}_{S_\ell}(\boldsymbol{\Xi}_\ell) = \left(\left(\sum_{i \in I} A_i^{(\ell)} \right)_I , \left(\sum_{\substack{i \in I \\ j \in J}} W_{ij}^{(\ell)} \right)_{I \neq J} \right), \tag{G9.2}$$

and a coarse-graining step declares a positive rescaling $\zeta_\ell > 0$ and sets

$$\Lambda_{\ell+1} = \zeta_\ell^{-1} \mathcal{A}_{S_\ell}(\Lambda_\ell). \tag{G9.3}$$

Proposition G9.2 (Aggregation is a positive semigroup action). (i) The two maps of (G9.2) agree: assembling $\mathcal{A}_S(\boldsymbol{\Xi})$ by (G9.1) gives $S^\top \Lambda S$. (ii) If S_1 maps agents to intermediate clusters and S_2 maps those clusters to coarse clusters, then

$$\mathcal{A}_{S_2} \mathcal{A}_{S_1} = \mathcal{A}_{S_1 S_2}. \tag{G9.4}$$

(iii) Each $\mathcal{A}_S$ is linear and positive on the product cone of self terms and edge weights, and it is not injective when some cluster contains an edge, because every edge internal to a cluster is annihilated. (iv) If every cluster has b agents, then $S^\top S = bI$ and, for every symmetric Λ,

$$b\lambda_{\min}(\Lambda) \leq \lambda_{\min}(S^\top \Lambda S), \qquad \lambda_{\max}(S^\top \Lambda S) \leq b\lambda_{\max}(\Lambda). \tag{G9.5}$$

[ESTABLISHED]

Proof. (i) The (I, J) block of $S^\top \Lambda S$ is $\sum_{i \in I, j \in J} \Lambda_{ij}$. For $I \neq J$ this is $-\sum_{i \in I, j \in J} W_{ij}$. For $I = J$ it is $\sum_{i \in I} \big(A_i + \sum_{j \neq i} W_{ij}\big) - \sum_{i \neq i' \in I} W_{ii'} = \sum_{i \in I} A_i + \sum_{i \in I, j \notin I} W_{ij}$, which is the diagonal block that (G9.1) assembles from $\mathcal{A}_S(\boldsymbol{\Xi})$. (ii) Congruence gives $S_2^\top (S_1^\top \Lambda S_1) S_2 = (S_1 S_2)^\top \Lambda (S_1 S_2)$, and the parameter formula follows by associativity of finite sums. (iii) Finite sums preserve positive semidefiniteness, and changing an internal weight while holding every cut weight and self term fixed leaves $\mathcal{A}_S(\boldsymbol{\Xi})$ unchanged. (iv) Congruence by S preserves the Loewner inequalities $\lambda_{\min}(\Lambda) I \preceq \Lambda \preceq \lambda_{\max}(\Lambda) I$, and $S^\top S = bI$. □

Source and target of (G9.3) have different dimensions. A finite sequence of steps is therefore a composition of positive maps between different spaces, one arrow of the scale diagram of Section 10.6 per step, and not an autonomous dynamical system on one parameter space. The bounds (G9.5) are finite-level statements; divergence or convergence along a sequence of steps would require an infinite hierarchy and a declared normalization, which this supplement does not declare. The parameter formula of (G9.2) is written in the chart in which Definition G3.1 is imposed; nonflat data must first satisfy the effective-support conditions of Section G8.1 or be treated by the rectangular endpoint maps of Section 11.6, and the full treatment of hard aggregation is Section G7.2.

Proposition G9.3 (Interior law-level step). Let $\Lambda_\ell \succ 0$, let S_ℓ be injective, let $\zeta_\ell > 0$, and write $\Lambda_c = S_\ell^\top \Lambda_\ell S_\ell$. Declare Lebesgue reference measure on each coarse coordinate space and the information-vector rule $\mathsf{h}_{\ell+1} = S_\ell^\top \mathsf{h}_\ell$. Then $(\mathsf{h}_{\ell+1}, \Lambda_{\ell+1}) = (S_\ell^\top \mathsf{h}_\ell, \zeta_\ell^{-1}\Lambda_c)$ is the natural parameter of a normalized Gaussian with finite log normalizer $\mathsf{A}(\mathsf{h}_{\ell+1}, \Lambda_{\ell+1})$, so every iterate of (G9.3) started in the interior stays in the Gaussian natural domain. At $\zeta_\ell = 1$ the step is the pushforward of the fine law under P_{Λ_ℓ} of Proposition G7.5. For general ζ_ℓ, the step sends $\mathcal{N}(\mu, \Lambda_\ell^{-1})$ to $\mathcal{N}(\zeta_\ell P_{\Lambda_\ell}\mu, \zeta_\ell \Lambda_c^{-1})$, and on the fixed-precision mean fiber of Λ_ℓ it is realized by a parameter-independent Markov kernel if and only if $\zeta_\ell \le 1$, for instance by $y \mapsto \mathcal{N}(\zeta_\ell P_{\Lambda_\ell} y, \zeta_\ell(1-\zeta_\ell)\Lambda_c^{-1})$. The alternative rule $\mathsf{h}_{\ell+1} = \zeta_\ell^{-1/2} S_\ell^\top \mathsf{h}_\ell$, with the same coarse precision, is the pushforward under $y \mapsto \zeta_\ell^{1/2} P_{\Lambda_\ell} y$ for every $\zeta_\ell > 0$. [ESTABLISHED]

Proof. Injectivity of S_ℓ and $\Lambda_\ell \succ 0$ give $\Lambda_c \succ 0$, hence $\zeta_\ell^{-1}\Lambda_c \succ 0$, and Proposition G2.4 gives a finite normalizer; induction gives the statement about iterates. The case $\zeta_\ell = 1$ is Proposition G7.5(i). In general the coarse mean is $\zeta_\ell \Lambda_c^{-1} S_\ell^\top \mathsf{h}_\ell = \zeta_\ell P_{\Lambda_\ell}\mu$ with $\mathsf{h}_\ell = \Lambda_\ell \mu$. For $\zeta_\ell \le 1$ the displayed kernel produces the covariance $\zeta_\ell^2 P_{\Lambda_\ell}\Lambda_\ell^{-1} P_{\Lambda_\ell}^\top + \zeta_\ell(1-\zeta_\ell)\Lambda_c^{-1} = \zeta_\ell \Lambda_c^{-1}$. Conversely, if a parameter-independent kernel realizes the step, the data-processing inequality Lemma 3.12 applied to the fine laws with means μ and $\mu + S_\ell v$ gives $\frac{1}{2} v^\top \Lambda_c v \ge \frac{1}{2}\zeta_\ell v^\top \Lambda_c v$, the right side being the relative entropy of the two coarse laws, so $\zeta_\ell \le 1$. The last rule is the image of Proposition G7.5(i) under multiplication of the coarse coordinates by $\zeta_\ell^{1/2}$. □

Open Problem G9.4 (Monotonicity beyond one fixed precision). On one fixed-precision mean fiber the unrescaled step is the pushforward of one parameter-independent kernel (Proposition G7.5), and after a rescaling with $\zeta_\ell \le 1$ it is still realized by one kernel (Proposition G9.3), so the Fisher metric of the mean family contracts there by Theorem 7.8. Three cases are unsettled. First, along a family of precisions on which $\mathrm{range}(\Lambda S)$ varies, no single kernel exists (Proposition G7.5(v)); a monotone functional would have to be supplied and its decrease proved directly, for instance as a Lyapunov functional of the step. Second, at the boundary member $A = 0$ of the biadditive family (G9.6), Λ is singular and P_Λ need not be defined; a declared quotient, pin, or mass repair with its own reference measure, normalizer, and pushforward identity would settle it. Third, under iteration to unbounded depth, Proposition G7.5(iii) controls every finite composition, but no limit object of the composed kernels has been constructed; a projective-limit theorem for the composed kernels would settle it.

G9.2 The biadditive closed family

Let each agent carry a mass $x_i \ge 0$, and fix $A, M \in \mathbb{S}_+^K$. The biadditive family is

$$A_i = x_i A, \qquad W_{ij} = x_i x_j M, \tag{G9.6}$$

and its image under a hard partition, with cluster masses $x_I = \sum_{i \in I} x_i$, is

$$A_I = x_I A, \qquad W_{IJ} = x_I x_J M. \tag{G9.7}$$

Proposition G9.5 (Positive additive rules are forced to be linear). (i) The family (G9.6) is closed under every hard partition, with image (G9.7). (ii) Suppose $\alpha : \mathbb{R}_+ \to \mathbb{S}_+^K$ is additive and $w : \mathbb{R}_+^2 \to \mathbb{S}_+^K$ is symmetric and separately additive. Then there are $A, M \in \mathbb{S}_+^K$ with

$$\alpha(x) = xA, \qquad w(x, y) = xyM. \tag{G9.8}$$

Hence every positive rule that depends only on additive scalar masses and obeys these additivity hypotheses has the form (G9.6), which is the unique such family. [ESTABLISHED]

Proof. (i) By Proposition G9.2(i), $A_I = \sum_{i\in I} x_i A = x_I A$ and $W_{IJ} = \sum_{i\in I, j\in J} x_i x_j M = x_I x_J M$. (ii) For $v \in \mathbb{R}^K$, the scalar function $f(x) = v^\top \alpha(x) v$ is additive and nonnegative, so $f(0) = 0$, $f(qx) = qf(x)$ for rational $q \geq 0$, and f is monotone, since $f(y) - f(x) = f(y - x) \geq 0$ for $y \geq x$. For $x \geq 0$ and rationals $q \leq x \leq q'$, $qf(1) \leq f(x) \leq q'f(1)$, so $f(x) = xf(1)$. Polarization of the symmetric matrices $\alpha(x)$ gives $\alpha(x) = x\alpha(1)$. For fixed y, the same argument gives $w(x, y) = xw(1, y)$, and applied to the second argument it gives $w(1, y) = yw(1, 1)$. Take $A = \alpha(1)$ and $M = w(1, 1)$. □

Closure is not selection: nothing in the proposition makes the family attract other data. Its boundary member $A = 0$ is a boundary operator of the interaction representation. There $\Lambda = L$, every configuration constant across agents lies in the kernel (Proposition G5.7), the pencil (L, Λ) is irregular (Proposition G5.6), and no full-space normalized Gaussian density exists. Pinning, quotienting the kernel, and retaining an external positive mass are different declared repairs and define different probability problems; a pseudoinverse inserted into the original Lebesgue density is none of them. Under $\Gamma_b \supseteq \mathrm{GL}^+(K)$ the matrix factor M carries no continuous label beyond its rank (Proposition G5.10).

Physicist summary. The interaction precision has a clean positive coarse map: cut weights and self terms add, internal edges drop out, and a declared scalar restores the chosen units. Nested aggregations compose, so the unrescaled map is a semigroup action, but its source and target differ in dimension and no autonomous system is defined without further identifications. On the interior of the interaction cone the unrescaled step is the pushforward of the fine Gaussian law through the precision-dependent Galerkin map, and a rescaled step is still a normalized Gaussian law, realized by one kernel on a fixed-precision fiber exactly when the rescaling does not increase the Fisher information of the mean family. The only positive rules additive in scalar node masses are biadditive, $W_{ij} = x_i x_j M$ and $A_i = x_i A$; their matrix factor carries only its rank as a congruence label under orientation-preserving frames, and their member without self terms is a boundary operator that carries no normalized law until a repair is declared.

Chapter G10

Reciprocal Pairs, Anchored Stars, and Loop Holonomy

When does a reciprocal Gaussian agreement between two agents define a law, and what replaces it when it does not? A tower of scales in which every level's prior is supplied by the level above has a top without a level above it, and it is tempting to close the tower by letting two agents supply each other's priors. The general form of that question is treated in Section 13.7; this chapter answers its Gaussian realization. For two latents joined by two parallel edge copies traversed in opposite directions, the kernel of the assembled precision is, up to a transport, the fixed space of the loop holonomy, and the determinant factors through $\det(I - \mathrm{Hol}(p))$ (Propositions G10.1 and G10.8). The flat unanchored fold is therefore singular; a proper anchor or holonomy without eigenvalue one restores definiteness, and in either case the global normalizer depends on the links and must be retained (Proposition G10.3). The standard conjugate replacement, an apex latent with a proper prior and conditionally independent constituents, has a mean-field schedule that converges geometrically to the exact posterior mean at a rate equal to the largest squared canonical correlation between apex and constituents (Theorem G10.5). Finally, Proposition G10.9 locates the conflict between a nonflat reciprocal cycle and the strict no-new-anchor coarse family of Definition 11.23, a conflict that the default coarse family of this supplement, which admits newly generated positive-semidefinite self terms, does not have.

Every statement of the chapter concerns a Gaussian realization. The singular fold is a statement about one flat unanchored reciprocal potential and not about cyclic models, which are legitimate laws whenever they are globally normalized (Section 4.2). Graph links follow Definition 2.11; the reciprocal pair uses two distinct parallel edge copies, so its two link values need not be inverse to one another, while the reversal of each copy carries the inverse link. The chapter does not assert that the star schedule converges under delayed, noisy, asynchronous, or inexact updates (Open Problem G10.6). [NOT-CLAIMED] Balanced cut links admit no continuous extension to all singular cuts (Proposition 11.28).

G10.1 The flat unanchored reciprocal fold

Let two latents $y_1, y_2 \in \mathbb{R}^K$ be joined by distinct edge copies $e : 2 \to 1$ and $e' : 1 \to 2$ with links $\Theta_e, \Theta_{e'} \in \Gamma_b$ and residual covariances $\Sigma_e, \Sigma_{e'} \succ 0$, so that y_2 supplies a penalty for y_1 and y_1 one

for y_2. The two residuals assemble the precision

$$\Lambda = \delta_e^\top \Sigma_e^{-1} \delta_e + \delta_{e'}^\top \Sigma_{e'}^{-1} \delta_{e'}, \qquad \delta_e = \begin{pmatrix} I & -\Theta_e \end{pmatrix}, \qquad \delta_{e'} = \begin{pmatrix} -\Theta_{e'} & I \end{pmatrix}. \tag{G10.1}$$

Let p be the closed walk that traverses e and then e', based at the second vertex; its holonomy is $\mathrm{Hol}(p) = \Theta_{e'}\Theta_e$. The reversals of e and e' do not occur in p.

Proposition G10.1 (Kernel of a reciprocal pair). For all residual covariances $\Sigma_e, \Sigma_{e'} \succ 0$,

$$\ker \Lambda = \big\{(\Theta_e w, w) : w \in \ker(\mathrm{Hol}(p) - I)\big\}, \qquad \dim \ker \Lambda = \dim \ker(\mathrm{Hol}(p) - I). \tag{G10.2}$$

In particular the kernel does not depend on the residual covariances. [ESTABLISHED]

Proof. Λ is a sum of two positive semidefinite terms, and for $\Sigma \succ 0$ the form $v \mapsto v^\top \delta^\top \Sigma^{-1} \delta v$ vanishes exactly when $\delta v = 0$, so $\ker \Lambda = \ker \delta_e \cap \ker \delta_{e'}$. For $v = (y_1, y_2)$ the two conditions are $y_1 = \Theta_e y_2$ and $y_2 = \Theta_{e'} y_1$, which together say $y_1 = \Theta_e y_2$ and $\mathrm{Hol}(p) y_2 = y_2$. The map $w \mapsto (\Theta_e w, w)$ is injective. □

The precision (G10.1) is the connection Laplacian of the two-vertex multigraph with edge copies e and e' and weights Σ_e^{-1} and $\Sigma_{e'}^{-1}$, and the proposition is its instance of the kernel-holonomy isomorphism of Proposition 11.19.

Corollary G10.2 (A flat parallel-edge fold is singular). (i) If the two copies give the same flat endpoint comparison, $\Theta_{e'} = \Theta_e^{-1}$, then $\mathrm{Hol}(p) = I$ and

$$\ker \Lambda = \big\{(\Theta_e w, w) : w \in \mathbb{R}^K\big\}, \qquad \dim \ker \Lambda = K, \tag{G10.3}$$

so the assembled precision is singular for all residual covariances, and its kernel is the set of transport-consistent configurations. (ii) For anchors $A_1, A_2 \succ 0$,

$$\Lambda_{\mathrm{anch}} = \Lambda + \mathrm{diag}(A_1, A_2) \succ 0, \tag{G10.4}$$

so for every information vector h the anchored reciprocal energy defines the normalized Gaussian law with information parameters $(\mathsf{h}, \Lambda_{\mathrm{anch}})$ and finite log normalizer $\mathsf{A}(\mathsf{h}, \Lambda_{\mathrm{anch}})$. The same holds for Gaussian edge energies on any finite multigraph with a positive-definite anchor at every vertex. [ESTABLISHED]

Proof. (i) is Proposition G10.1 with $\ker(I - I) = \mathbb{R}^K$. (ii) A sum of Gaussian edge energies is a positive semidefinite form, and adding a positive-definite block-diagonal form makes it positive definite; Proposition G2.4 gives the normalizer. □

The flat fold charges nothing on the configurations in which the two latents agree under transport, which is its intended agreement direction, so it supplies no coercivity there; and an energy whose precision has a K-dimensional kernel defines no normalized Lebesgue density, so the evidence identity of Chapter 5 has no joint to apply to. Definiteness returns in two independent ways. A

declared anchor works under flat holonomy. Alternatively, by (G10.2) the unanchored pair is definite exactly when $1 \notin \operatorname{spec}(\mathrm{Hol}(p))$. That requires $\mathrm{Hol}(p) \neq I$, so the two copies cannot have the coboundary form of Proposition 2.13, which pointwise frame comparisons at one context always have; at least one copy must carry a link declared independently of the frames in the sense of Definition 2.11. This route leaves the fixed-K flat interaction family: a nonflat cycle inside a cluster leaves a self residue (Lemma G3.5), and a cut between clusters needs the rectangular endpoint maps of Theorem 11.27 or the anchored presentations of Lemma 11.24.

G10.2 Anchors and the global normalizer

An anchor restores definiteness but not the freedom to treat the normalizer as a constant. At zero information vector the log normalizer (G2.4) of a precision $\Lambda \succ 0$ on $\mathbb{R}^n$ reduces to

$$\mathsf{A}(0, \Lambda) = -\tfrac{1}{2} \log\det \Lambda + \tfrac{n}{2} \log 2\pi, \tag{G10.5}$$

so every dependence of Λ on the links passes into the normalizer. In (G10.1) the links sit inside δ_e and $\delta_{e'}$, and their contribution to the determinant does not cancel.

Proposition G10.3 (The global normalizer depends on the model graph link). Take $K = 1$, unit residual variances, flat link values $\Theta_e = \mathsf{g}$ and $\Theta_{e'} = \mathsf{g}^{-1}$ with $\mathsf{g} > 0$, and the anchor $a_0 I$ with $a_0 > 0$. Then

$$\det\big(\Lambda + a_0 I\big) = a_0^2 + a_0\big(\mathsf{g} + \mathsf{g}^{-1}\big)^2, \tag{G10.6}$$

a nonconstant function of g, uniquely minimized on $\mathrm{GL}^+(1) = (0, \infty)$ at $\mathsf{g} = 1$, with value $a_0^2 + 4a_0$. At zero information vector the log normalizer $\mathsf{A}(\mathsf{g}) = -\frac{1}{2} \log\det(\Lambda + a_0 I) + \log 2\pi$ satisfies

$$\mathsf{A}'(1) = 0, \qquad \mathsf{A}''(1) = -\frac{4}{a_0 + 4} < 0. \tag{G10.7}$$

[ESTABLISHED]

Proof. With $\delta_e = (1, -\mathsf{g})$ and $\delta_{e'} = (-\mathsf{g}^{-1}, 1)$, the two rank-one contributions sum to

$$\Lambda = \begin{pmatrix} 1 + \mathsf{g}^{-2} & -(\mathsf{g} + \mathsf{g}^{-1}) \\ -(\mathsf{g} + \mathsf{g}^{-1}) & 1 + \mathsf{g}^2 \end{pmatrix},$$

whose determinant is $(1 + \mathsf{g}^{-2})(1 + \mathsf{g}^2) - (\mathsf{g} + \mathsf{g}^{-1})^2 = 0$, as Corollary G10.2 requires. Adding $a_0 I$ gives $a_0^2 + a_0(2 + \mathsf{g}^2 + \mathsf{g}^{-2})$, and $2 + \mathsf{g}^2 + \mathsf{g}^{-2} = (\mathsf{g} + \mathsf{g}^{-1})^2$. Since $\mathsf{g} + \mathsf{g}^{-1} \geq 2$ with equality only at $\mathsf{g} = 1$, the minimizer is unique. Writing $D(\mathsf{g})$ for (G10.6), $D'(1) = 0$ and $D''(1) = 8a_0$, so $\mathsf{A}''(1) = -\frac{1}{2} D''(1)/D(1) = -4/(a_0 + 4)$. □

The identity link is thus a strict maximum of the isolated log-normalizer contribution. The quadratic and information-vector terms also vary with g, so no direction follows for the complete objective; what the witness shows is that the global term $-\frac{1}{2} \log\det(\Lambda + a_0 I)$ cannot be absorbed into per-

agent constants or dropped from a normalized joint. The variation is a change of model and not of coordinates: it varies the declared links while holding $a_0 I$, the residual variances, and the Lebesgue coordinates fixed, whereas under a reframing all of these transform together and Proposition 4.9 leaves the law and the evidence invariant.

G10.3 The anchored Gaussian star

The standard conjugate replacement of the fold is an apex latent inside the joint. Let $y_0 \in \mathbb{R}^K$ carry the proper prior $\mathcal{N}(0, A_0^{-1})$, $A_0 \succ 0$, and let N constituents be conditionally independent given it,

$$y_0 \sim \mathcal{N}(0, A_0^{-1}), \qquad y_i \mid y_0 \sim \mathcal{N}(\Theta_i y_0, \Sigma_i), \qquad i = 1, \dots, N, \tag{G10.8}$$

with $\Theta_i \in \Gamma_b$ and $\Sigma_i \succ 0$. The scale graph is a tree, so no cycle forms. Write $\Lambda_\star$ for the joint precision of $(y_0, y_1, \dots, y_N)$. Its constituent blocks are Σ_i^{-1}, its apex-constituent blocks are $-\Theta_i^\top \Sigma_i^{-1}$, and its apex block is

$$\Lambda_{00} = A_0 + \sum_{i=1}^{N} \Theta_i^\top \Sigma_i^{-1} \Theta_i. \tag{G10.9}$$

This is the conditional precision of y_0 given the constituents in the linear-Gaussian form of Bayes' theorem [5, Sec. 2.3.3], and it is the precision of the optimal mean-field factor of the apex, which for a Gaussian joint is the corresponding diagonal block of the joint precision [5, Sec. 10.1.2]. The constituents enter the apex through the single statistic $\sum_i \Theta_i^\top \Sigma_i^{-1} y_i$, whose conditional covariance given y_0 is $\Lambda_{00} - A_0$. The star is one conjugate hierarchical Gaussian model among many; the computations below use only its tree form and its proper root.

Proposition G10.4 (The declared prior supplies exactly the missing coercivity). For all invertible Θ_i and all $\Sigma_i \succ 0$, $\Lambda_\star \succ 0$. On every transport-consistent configuration $w = (v, \Theta_1 v, \dots, \Theta_N v)$,

$$w^\top \Lambda_\star w = v^\top A_0 v \tag{G10.10}$$

for every $v \in \mathbb{R}^K$. [ESTABLISHED]

Proof. The quadratic form is $y_0^\top A_0 y_0 + \sum_i (y_i - \Theta_i y_0)^\top \Sigma_i^{-1} (y_i - \Theta_i y_0)$, a sum of positive semidefinite terms. It vanishes only if $y_0 = 0$, because $A_0 \succ 0$, and then $y_i = \Theta_i y_0 = 0$. On a transport-consistent configuration every residual vanishes and only $v^\top A_0 v$ remains. □

The contrast with the fold is exact: by Corollary G10.2 the flat pair charges nothing on its transport-consistent directions, and by (G10.10) the star charges exactly $v^\top A_0 v$ there.

Theorem G10.5 (Exact Gaussian-star fixed point and contraction). Let $A_0 \succeq 0$, $O_i \succeq 0$, and let $r_0, r_1, \dots, r_N \in \mathbb{R}^K$. Consider the Gaussian target on $(y_0, y_1, \dots, y_N)$ whose log density is, up to

a constant,

$$-\tfrac{1}{2}y_0^\top A_0 y_0 - \tfrac{1}{2}\sum_i (y_i - \Theta_i y_0)^\top \Sigma_i^{-1}(y_i - \Theta_i y_0) - \tfrac{1}{2}\sum_i y_i^\top O_i y_i + r_0^\top y_0 + \sum_i r_i^\top y_i;$$

the terms in O_i and r_i are Gaussian observations of the constituents, and $A_0 \succ 0$, $O_i = 0$, $r = 0$ recover (G10.8). Keep Λ_{00} of (G10.9), write $\Lambda_{ii} = \Sigma_i^{-1} + O_i$, and put

$$B_\star = \sum_{i=1}^N \Theta_i^\top \Sigma_i^{-1}\Lambda_{ii}^{-1}\Sigma_i^{-1}\Theta_i, \qquad S_\star = \Lambda_{00} - B_\star, \qquad M = \Lambda_{00}^{-1}B_\star, \qquad \bar r = r_0 + \sum_{i=1}^N \Theta_i^\top \Sigma_i^{-1}\Lambda_{ii}^{-1} r_i. \tag{G10.11}$$

Assume $S_\star \succ 0$, which holds whenever $A_0 \succ 0$ and is equivalent to positive definiteness of the target precision. Consider the mean-field schedule that updates every constituent factor from the current apex factor, in parallel or in any order, and then updates the apex factor. The factor covariances become Λ_{ii}^{-1} and Λ_{00}^{-1} on their first updates, and the means obey

$$\mu_i^{(t+1)} = \Lambda_{ii}^{-1}\left(\Sigma_i^{-1}\Theta_i \mu_0^{(t)} + r_i\right), \tag{G10.12}$$

$$\mu_0^{(t+1)} = \Lambda_{00}^{-1}\left(r_0 + \sum_i \Theta_i^\top \Sigma_i^{-1}\mu_i^{(t+1)}\right) = \Lambda_{00}^{-1}\bar r + M\mu_0^{(t)}. \tag{G10.13}$$

There is a unique fixed point,

$$\mu_0^\star = S_\star^{-1}\bar r, \qquad \mu_i^\star = \Lambda_{ii}^{-1}\left(\Sigma_i^{-1}\Theta_i\mu_0^\star + r_i\right), \tag{G10.14}$$

which is the exact mean of the target, and in the norm $\|x\|_{\Lambda_{00}} = (x^\top \Lambda_{00} x)^{1/2}$,

$$\|\mu_0^{(t+1)} - \mu_0^\star\|_{\Lambda_{00}} \le \varrho_\star \|\mu_0^{(t)} - \mu_0^\star\|_{\Lambda_{00}}, \qquad \varrho_\star = \lambda_{\max}\left(\Lambda_{00}^{-1/2} B_\star \Lambda_{00}^{-1/2}\right) < 1. \tag{G10.15}$$

The rate $\varrho_\star$ is the largest squared canonical correlation between y_0 and $(y_1, \dots, y_N)$ under the target. The relative entropy of the apex factor to its fixed-point factor contracts by $\varrho_\star^2$ per sweep, and the product factor converges to the product of the fixed-point factors in relative entropy. When every $O_i = 0$, one has $B_\star = \Lambda_{00} - A_0$ and $S_\star = A_0$, so the hypothesis is $A_0 \succ 0$ and the fixed point is $\mu_0^\star = A_0^{-1}(r_0 + \sum_i \Theta_i^\top r_i)$, $\mu_i^\star = \Theta_i\mu_0^\star + \Sigma_i r_i$; if instead $A_0 \succeq 0$ is singular and every $O_i = 0$, then $\varrho_\star = 1$ and neither uniqueness nor contraction holds. [ESTABLISHED]

Proof. The target precision $\Lambda_\star$ has apex block Λ_{00}, constituent blocks Λ_{ii}, apex-constituent blocks $-\Theta_i^\top \Sigma_i^{-1}$, and no block between distinct constituents. Its constituent part $\operatorname{diag}(\Lambda_{ii})$ is positive definite, and the Schur complement of that part is $S_\star$, so $\Lambda_\star \succ 0$ exactly when $S_\star \succ 0$. If $A_0 \succ 0$, then $S_\star - A_0 = \sum_i \Theta_i^\top(\Sigma_i^{-1} - \Sigma_i^{-1}\Lambda_{ii}^{-1}\Sigma_i^{-1})\Theta_i \succeq 0$, because $\Lambda_{ii}^{-1} \preceq \Sigma_i$. The optimal mean-field factor of one block is Gaussian with precision the diagonal block of $\Lambda_\star$ and with the information vector of the target minus the off-diagonal blocks applied to the current means of the other factors [5, Secs. 10.1.1–10.1.2]; this gives the covariances, (G10.12), and the first expression in (G10.13), and substitution gives the affine recursion. Since $I - M = \Lambda_{00}^{-1}S_\star$ is invertible, the recursion has the

unique fixed point (G10.14); the two block equations it satisfies are the two block rows of $\Lambda_\star \mu = r$, so it is the exact mean.

Because $\Lambda_{00} \succeq B_\star + S_\star$ with $S_\star \succ 0$ and $B_\star \succeq 0$, the matrix M is similar to the symmetric matrix $\Lambda_{00}^{-1/2} B_\star \Lambda_{00}^{-1/2} = I - \Lambda_{00}^{-1/2} S_\star \Lambda_{00}^{-1/2}$, whose eigenvalues lie in $[0, 1)$. With $\xi_t = \mu_0^{(t)} - \mu_0^\star$, $\|\xi_{t+1}\|_{\Lambda_{00}} = \|\Lambda_{00}^{-1/2} B_\star \Lambda_{00}^{-1/2} \Lambda_{00}^{1/2} \xi_t\|_2 \le \varrho_\star \|\xi_t\|_{\Lambda_{00}}$, which is (G10.15). Write Λ_{0y}, Λ_{yy}, and $\Sigma_\star = \Lambda_\star^{-1}$ with blocks $\Sigma_{00}, \Sigma_{0y}, \Sigma_{yy}$ for the apex and the stacked constituents. Then $B_\star = \Lambda_{0y} \Lambda_{yy}^{-1} \Lambda_{y0}$, the block identities of $\Lambda_\star \Sigma_\star = I$ give $-\Lambda_{00}^{-1} \Lambda_{0y} = \Sigma_{0y} \Sigma_{yy}^{-1}$ and $-\Lambda_{yy}^{-1} \Lambda_{y0} = \Sigma_{y0} \Sigma_{00}^{-1}$, and so $M = \Sigma_{0y} \Sigma_{yy}^{-1} \Sigma_{y0} \Sigma_{00}^{-1}$, whose eigenvalues are the squared canonical correlations between y_0 and the constituents.

After the first updates every factor has its fixed-point covariance. The apex factor then satisfies $D_{\mathrm{KL}}(q_0^{(t)} \| q_0^\star) = \frac{1}{2} \|\xi_t\|_{\Lambda_{00}}^2$, which contracts by $\varrho_\star^2$ per sweep. The constituent means updated from $\mu_0^{(t)}$ differ from their fixed points by $\Lambda_{ii}^{-1} \Sigma_i^{-1} \Theta_i \xi_t$, so the sum of their divergences is $\frac{1}{2} \xi_t^\top B_\star \xi_t \le \frac{1}{2} \xi_t^\top \Lambda_{00} \xi_t$, and the product factor converges in relative entropy. When every $O_i = 0$, $\Lambda_{ii} = \Sigma_i^{-1}$ gives $B_\star = \sum_i \Theta_i^\top \Sigma_i^{-1} \Theta_i$ and $S_\star = A_0$, and the displayed fixed point follows. If then $A_0 v = 0$ for some $v \neq 0$, $Mv = \Lambda_{00}^{-1}(\Lambda_{00} - A_0)v = v$, so $\varrho_\star = 1$, $I - M$ is singular, and the error along v is never reduced. □

The mean recursion is block Gauss–Seidel on $\Lambda_\star \mu = r$. Convergence of the point Gauss–Seidel iteration for a symmetric positive definite system is classical [14, Thm. 11.2.3], and the same iteration matrix governs the rate of the deterministic-scan Gibbs sampler for a Gaussian target [25]. What the theorem supplies for the star is the closed form of $\varrho_\star$, its identification with a canonical correlation, the extension to observed constituents, and the convergence of the product factor in relative entropy.

Open Problem G10.6 (Robustness of the star schedule). Theorem G10.5 covers the exact block schedule in which every constituent update reads the current apex factor and the apex update reads the updated constituents. Whether the fixed point is reached, and at what rate, when the updates are delayed, noisy, asynchronous, or inexact is open. A convergence theorem for a declared class of perturbed schedules, or a counterexample within such a class, would settle it.

G10.4 Loop holonomy and the reciprocal fold

By Proposition G10.1 the obstruction of Section G10.1 is measured by one number, $\dim \ker(\mathrm{Hol}(p) - I)$, and Corollary G10.2 evaluates it under flat holonomy, where it is largest. The kernel is produced by the flat loop product, not by the cycle.

Corollary G10.7 (Nontrivial holonomy shrinks the reciprocal-pair kernel). Under the hypotheses of Proposition G10.1,

$$\begin{aligned} \dim\ker\Lambda = K &\iff \mathrm{Hol}(p) = I, \\ \mathrm{Hol}(p) \neq I &\implies \dim\ker\Lambda \leq K-1, \\ \ker\Lambda = \{0\} &\iff 1 \notin \mathrm{spec}(\mathrm{Hol}(p)). \end{aligned} \tag{G10.16}$$

Under the specialization $\Gamma_b = \mathrm{GL}^+(K)$, the holonomies at which the last condition fails form the intersection of $\mathrm{GL}^+(K)$ with the zero set of the nonzero polynomial $\det(\mathrm{Hol} - I)$, a relatively closed subset with dense complement and Lebesgue measure zero, so the flat point $\mathrm{Hol}(p) = I$ is the most degenerate member of the family and a generic holonomy gives no kernel. If instead $\Gamma_b = \mathrm{SO}(K)$ with K odd, every holonomy has eigenvalue one and the unanchored pair is singular for every choice of links. [ESTABLISHED]

Proof. By (G10.2), $\dim\ker\Lambda = \dim\ker(\mathrm{Hol}(p) - I)$, which is K exactly when $\mathrm{Hol}(p) - I = 0$, at most $K-1$ otherwise, and zero exactly when one is not an eigenvalue. The polynomial $\det(\mathrm{Hol} - I)$ takes the value 1 at $\mathrm{Hol} = 2I$, and the zero set of a nonzero polynomial is closed, nowhere dense, and Lebesgue null. An element of $\mathrm{SO}(K)$ has determinant one and eigenvalues of modulus one in conjugate pairs; for odd K the number of real eigenvalues is odd and their product is one, so at least one of them equals one. □

Proposition G10.8 (Graph-link dependence of the determinant factors through holonomy). For the reciprocal pair (G10.1),

$$\det\Lambda = \frac{\big(\det(I - \mathrm{Hol}(p))\big)^2}{\det\Sigma_e \det\Sigma_{e'}}. \tag{G10.17}$$

The links enter the determinant only through the loop holonomy, Λ is singular exactly when $\det(I - \mathrm{Hol}(p)) = 0$, and where $\Lambda \succ 0$ the link-dependent part of the zero-information log normalizer (G10.5) is

$$-\tfrac{1}{2}\log\det\Lambda = -\log\big|\det(I - \mathrm{Hol}(p))\big| + \tfrac{1}{2}\log\det\Sigma_e + \tfrac{1}{2}\log\det\Sigma_{e'}. \tag{G10.18}$$

The singular locus $\{\det(I - \mathrm{Hol}(p)) = 0\}$ strictly contains the flat point: for $K = 2$, $\Theta_e = I$, $\Theta_{e'} = \mathrm{diag}(1, 2)$, and $\Sigma_e = \Sigma_{e'} = I$, one has $\mathrm{Hol}(p) \neq I$ and $\det\Lambda = 0$. [ESTABLISHED]

Proof. Stack the residual maps as $\delta = \left(\begin{smallmatrix} I & -\Theta_e \\ -\Theta_{e'} & I \end{smallmatrix}\right)$, so that $\Lambda = \delta^\top \mathrm{diag}(\Sigma_e, \Sigma_{e'})^{-1}\delta$ by (G10.1). Hence $\det\Lambda = (\det\delta)^2/(\det\Sigma_e \det\Sigma_{e'})$, and the Schur formula gives $\det\delta = \det(I - \Theta_{e'}\Theta_e)$. Taking logarithms gives (G10.18). In the witness $\mathrm{Hol}(p) = \mathrm{diag}(1, 2)$ and $\det(I - \mathrm{Hol}(p)) = 0$. □

Holonomy without eigenvalue one removes exactly one defect of this architecture, the kernel of the unanchored pair: the reciprocal energy with its global normalizer then defines a Gaussian law, and the transport-consistent zero modes collapse to the origin. The statement concerns the two-residual precision (G10.1) and no other cyclic potential. Holonomy does not turn the reciprocal energy into

a pair of locally normalized directed kernels; it defines an undirected Gaussian potential whose global partition function is part of the model, and by (G10.18) that term is a nonconstant function of the links that diverges on the singular locus, where the unanchored law ceases to exist. Holonomy also does not remove declarations. A closed walk with nontrivial holonomy requires at least one link declared independently of the frames, because links of the coboundary form of Proposition 2.13 telescope to the identity around every closed walk, and pointwise frame comparisons at one context (Section 2.2.1) have that form; replacing a declared prior by a declared link relocates the declaration. Both Proposition G10.3 and (G10.18) vary declared links while holding the other coordinate parameters fixed, so neither is a statement about motion along a gauge orbit. With latent links in the exponential chart, the zero-information log normalizer of the pair is minus the induced action of Proposition G11.5 at the given links, and an unanchored cycle admits no joint law of generators and matter, because the partition function diverges like $|\det(I - \mathrm{Hol}(p))|^{-1}$ across a hypersurface of generators (Corollary G11.6).

The strict no-new-anchor pair subfamily of Definition 11.23, in which a blocking step creates no self term beyond the summed self terms and the internal residues, conflicts with an unanchored reciprocal pair. Its exact fixed-K aggregation needs clusters whose links can all be removed by a choice of frames, and exact closure of each cut by one bare coarse link, while the unanchored pair needs definiteness:

$$\begin{aligned} \text{strict flat coarse-graining needs} \quad & \text{trivial holonomy on every closed walk in its domain,} \\ \text{the unanchored reciprocal pair needs} \quad & 1 \notin \mathrm{spec}(\mathrm{Hol}(p)). \end{aligned} \tag{G10.19}$$

Proposition G10.9 (A nonflat cycle leaves the strict flat coarse-graining). Let a finite sequence of nested hard partitions of the agents be given in which every cluster at every level is trivializing, that is, the links inside it take the coboundary form of Proposition 2.13. Suppose that the coarsest level is either one cluster or two clusters whose cut, with positive-definite weights, is reproduced exactly by one coarse link without a new self term, as in the subfamily of Definition 11.23. Then every closed walk of the interaction graph has trivial holonomy. In particular, no such sequence exists for a population that contains an unanchored reciprocal pair with $1 \notin \mathrm{spec}(\mathrm{Hol}(p))$. [ESTABLISHED]

Proof. If the coarsest level is one cluster, that cluster is the whole population and is trivializing by hypothesis. If it is two clusters I and J, choose frames in which every link inside I and inside J is the identity. The cut links become twists $\Theta_e \in \Gamma_b$, and exact reproduction by one coarse link without a new self term is $\Delta = 0$ in (11.29), which for positive-definite weights holds exactly when all twists are equal. Re-expressing every frame of J by this common twist makes every link of the population the identity, so the whole population is trivializing and every closed walk has trivial holonomy. For the reciprocal pair this gives $\mathrm{Hol}(p) = I$, which has eigenvalue one. □

Within the strict subfamily, a nonflat reciprocal cycle can therefore stay outside the flat coarse-graining only by stopping the sequence before it reaches the coarsest levels of the proposition. The default coarse family of this supplement admits newly generated positive-semidefinite self terms, and there the conflict dissolves. For a cut between two trivialized clusters with weights $W_e \succ 0$,

twists Θ_e, and Λ_{II}, Λ_{IJ}, $\bar{\Theta} = \Lambda_{II}^{-1}\Lambda_{IJ}$, and Δ as in (11.29), Lemma 11.24 writes the cut energy as

$$\sum_e (z_I - \Theta_e z_J)^\top W_e (z_I - \Theta_e z_J) = (z_I - \bar{\Theta} z_J)^\top \Lambda_{II} (z_I - \bar{\Theta} z_J) + z_J^\top \Delta z_J, \tag{G10.20}$$

one link $\bar{\Theta}$ of weight Λ_{II} together with the self term Δ at J, and an exact anchored one-link presentation exists if and only if $\det \Lambda_{IJ} \neq 0$, with $\det \Lambda_{IJ} > 0$ for links in $\mathrm{GL}^+(K)$. The admitted self term is exactly the anchor that removes the right-hand demand of (G10.19); the balanced presentation of Proposition 11.25 selects one of the presentations, no presentation at the level of laws is balanced once the cut has excess (Proposition 11.26), and no continuous extension of the balanced links covers all singular and rectangular cuts (Proposition 11.28).

Chapter G11

Latent Lie-Algebra Links in the Gaussian Realization

What do the latent exponential-chart links of Chapter 9 become when every belief law is Gaussian, and what does Gaussian matter do to them when it is integrated out? In this realization the law-level transported divergence is an explicit function of the generator whose log-volume part is linear in it (Proposition G11.1). The expected energy of a linear comparison record gives the delta-rule current, and without a link prior its minimizing transport matrix has rank at most one and, for $K \geq 2$, lies outside the exponential chart, while stationary points in the chart can still sit at critical points of the exponential map (Proposition G11.2); when its trace is negative, no chart link with semisimple negative eigenvalues attains the infimum over the chart, so for $K = 2$ and $K = 3$ the infimum is not attained (Proposition G11.3). Under a product Gaussian recognition law of the generators the expected holonomy of a closed walk that traverses each reversal pair at most once is an ordered product of per-copy exponential moments (Proposition G11.4). Jointly normalized undirected Gaussian matter induces on the generators the action $\mathscr{S}^{\mathrm{ind}} = -\mathrm{A}(\mathsf{h}, \Lambda(\mathring{\omega}))$, one half the log-determinant of a connection-Laplacian precision at zero information vector, bounded below by the self terms (Proposition G11.5); its hopping expansion is a convergent sum of weighted Wilson loops whose leading link-dependent term on a triangle of agents, for isotropic data and isometric links, is a multiple of the loop defect of the triangle (Proposition G11.8). The link-dependent normalizers of Propositions G7.11, G10.3 and G10.8 are values of this induced action at fixed links, and an unanchored cycle, whose law at fixed links exists off a null set, admits no joint law with latent links (Corollary G11.6), while for $K = 1$ unanchored matter on a connected multigraph of cycle rank at least two has one (Proposition G11.7). An explicit spectrum of the linearized link operator on a ring closes the chapter (Proposition G11.9). These are the Gaussian values of Propositions 9.23, 9.27 and 9.34 and corollaries 9.22 and 9.24 of the main manuscript, and the induced action is the quantity against which a loop action of the generator prior of Definition 9.11 can be compared.

Every statement concerns the Gaussian realization with the specialization $\Gamma_b = \mathrm{GL}^+(K)$ of Hypothesis G2.1, one channel, and held model latents. The chapter does not claim that inference of the generators reaches the stationary points it computes, does not select anchors, generator covariances, loop coefficients, or increment precisions, does not identify the induced action with a loop action of the generator prior beyond the leading hopping order, gives no expansion of the induced action over cycle-rooted spanning forests in rank three or more, where a combinatorial interpretation is an open question of Kenyon [20, Sec. 14], and evaluates expected loop defects

as ordered products only for walks without a repeated reversal pair; the ring of Section G11.5 illustrates the linearized operator and sets no direction for the theory. [NOT-CLAIMED] Joint normalization of unanchored matter on multigraphs of cycle rank at least two at a stated $K \geq 2$ is Open Problem G11.10, and attainment of the chart infimum of an unregularized record energy for $K \geq 4$ at a chart link with a non-semisimple negative eigenvalue is Open Problem G11.11.

G11.1 Generator coordinates and reference laws

Throughout the chapter $G = \mathrm{GL}^+(K)$ with ρ_b its defining representation, which realizes $\Gamma_b = \mathrm{GL}^+(K)$ (Hypothesis G2.1); the Lie algebra is $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$, $\mathrm{d}\rho_b$ is the identity, and the exponential is the matrix exponential. Only the belief channel is considered, and its index is dropped. Each oriented edge copy $e : j \to i$ carries a generator $\mathring{\omega}_e \in \mathfrak{g}$ with twist $\mathring{\Theta}_e = \exp(\mathring{\omega}_e)$ and $\mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e$ (Definition 9.2); E_+ contains one orientation of each reversal pair and $\mathscr{T} = \mathfrak{g}^{E_+}$. A closed walk $p = (e_0, \dots, e_{r-1})$ with $e_a : i_{a+1} \to i_a$ and $i_r = i_0$ has the reference-coordinate holonomy $\mathring{\Theta}_p = \mathring{\Theta}_{e_0} \cdots \mathring{\Theta}_{e_{r-1}}$, and $\mathrm{Hol}(p) = U_{i_0} \mathring{\Theta}_p U_{i_0}^{-1}$ by (2.33).

Laws, self terms, weights, observations, and information vectors are written in reference coordinates: the belief law of agent i stands for $\rho_b(U_i)^{-1}_{\#} q_i$. In these coordinates the link of $e : j \to i$ acts as $\exp(\mathring{\omega}_e)$, because the passive re-expression $g_i = U_i^{-1}$ makes every frame the reference frame, leaves every twist unchanged (Proposition 2.12), and pushes both arguments of every transported divergence forward by one bijection, which preserves relative entropy. A Gaussian energy in local coordinates becomes one in reference coordinates by the block congruence of Remark G3.6, which changes a log-determinant by the link-independent constant $2 \sum_i \log|\det U_i|$, the Jacobian bookkeeping of Proposition G7.22.

Matrices are paired by the trace pairing $\langle \mathbf{S}, \mathbf{T} \rangle = \mathrm{Tr}(\mathbf{S}^\top \mathbf{T})$, and gradients are taken for this pairing; the gradient for the inner product $\langle \cdot, \cdot \rangle_{\mathsf{G}}$ of Chapter 9 is $\mathsf{G}^{-1} \nabla \mathsf{G}$, with $\mathsf{G} = \mathsf{G}^{\mathrm{fib}}$. Kronecker products have the coordinates $(\mathbf{S} \otimes \mathbf{T})_{(a,b),(c,d)} = \mathbf{S}_{ac} \mathbf{T}_{bd}$. The Fréchet derivative of the exponential is $\mathbf{L}(\mathbf{B}, \mathbf{T}) = \int_0^1 e^{t\mathbf{B}} \mathbf{T} e^{(1-t)\mathbf{B}} dt$ (Lemma 9.19), and cyclicity of the trace under the integral makes $\mathbf{L}(\mathbf{B}^\top, \cdot)$ its adjoint for the trace pairing. The second derivative $\mathbf{L}^{(2)}(\mathbf{B}; \mathbf{T}) = \frac{d^2}{dt^2} e^{\mathbf{B} + t\mathbf{T}}\big|_{t=0}$ is used once. The loop defect of a closed walk is $\mathfrak{Q}_p = \frac{1}{2} \langle \mathring{\Theta}_p - I, \mathring{\Theta}_p - I \rangle_{\mathsf{G}}$ (Definition 9.8). Bold capitals $\mathbf{B}, \mathbf{S}, \mathbf{T}$ denote generic matrices inside one statement or proof, and upright Roman letters denote statement-local real constants.

G11.2 Law-level and record energies of one copy

Proposition G11.1 (Law-level transported divergence in generator coordinates). Let $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and $q_j = \mathcal{N}(\mu_j, \Sigma_j)$ be nondegenerate Gaussian laws in reference coordinates, $e : j \to i$,

and $\mathring{\omega} = \mathring{\omega}_e \in \mathfrak{g}$. (i) The transported divergence is

$$D_{\mathrm{KL}}\big(q_i \| \exp(\mathring{\omega})_{\#} q_j\big) = \tfrac{1}{2}\operatorname{Tr}\big(\Sigma_j^{-1} e^{-\mathring{\omega}} \Sigma_i e^{-\mathring{\omega}^\top}\big) + \tfrac{1}{2}\big(e^{-\mathring{\omega}}\mu_i - \mu_j\big)^\top \Sigma_j^{-1}\big(e^{-\mathring{\omega}}\mu_i - \mu_j\big) - \tfrac{K}{2} + \tfrac{1}{2}\log\frac{\det\Sigma_j}{\det\Sigma_i} + \operatorname{Tr}\mathring{\omega}, \tag{G11.1}$$

and it equals $D_{\mathrm{KL}}(q_i \| (\Theta_e)_{\#} q_j)$ computed in local coordinates for every choice of frames. The log-volume part $\operatorname{Tr}\mathring{\omega}$ is linear in the generator, and every other dependence on $\mathring{\omega}$ passes through $e^{-\mathring{\omega}}$. (ii) The divergence is a real-analytic function of $\mathring{\omega}$ with differential and gradient

$$dD_{\mathrm{KL}}[\mathbf{T}] = \operatorname{Tr}\mathbf{T} - \big\langle \mathbf{A}_e, \mathbf{L}(-\mathring{\omega}, \mathbf{T})\big\rangle, \qquad \nabla_{\mathring{\omega}} D_{\mathrm{KL}} = I - \mathbf{L}(-\mathring{\omega}^\top, \mathbf{A}_e), \qquad \mathbf{A}_e = \Sigma_j^{-1}\big(e^{-\mathring{\omega}}(\Sigma_i + \mu_i\mu_i^\top) - \mu_j\mu_i^\top\big). \tag{G11.2}$$

(iii) At $\mathring{\omega} = 0$ it is the Gaussian relative entropy of q_i to q_j. If the transported source has the receiver's covariance, $\Sigma_i = e^{\mathring{\omega}}\Sigma_j e^{\mathring{\omega}^\top}$, it is the twisted pair energy (G8.1) with $W_e = \Sigma_i^{-1}$, $\frac{1}{2}(\mu_i - e^{\mathring{\omega}}\mu_j)^\top \Sigma_i^{-1}(\mu_i - e^{\mathring{\omega}}\mu_j)$, which at $\mathring{\omega} = 0$ and $\Sigma_i = \Sigma_j$ is the edge relative entropy of Proposition G5.3. If $q_i = q_j = \mathcal{N}(\mu, \Sigma)$ and $\mathring{\omega} = t\mathring{v}$, then as $t \to 0$ the divergence is $\frac{t^2}{2}\mathfrak{k}_q(\mathring{v}, \mathring{v}) + O(t^3)$ with the Gaussian orbit Fisher form

$$\mathfrak{k}_q(\mathring{v}, \mathring{v}) = (\mathring{v}\mu)^\top \Sigma^{-1}\mathring{v}\mu + \operatorname{Tr}(\mathring{v}^2) + \operatorname{Tr}\big(\Sigma^{-1}\mathring{v}\Sigma\mathring{v}^\top\big) \tag{G11.3}$$

of Corollary 7.7. [ESTABLISHED]

Proof. (i) By Proposition G2.11, $\exp(\mathring{\omega})_{\#} q_j = \mathcal{N}(e^{\mathring{\omega}}\mu_j, e^{\mathring{\omega}}\Sigma_j e^{\mathring{\omega}^\top})$. For nondegenerate Gaussian laws on $\mathbb{R}^K$, integrating the difference of the log densities gives $D_{\mathrm{KL}}(\mathcal{N}(\mu_i, \Sigma_i) \| \mathcal{N}(\mu', \Sigma')) = \frac{1}{2}[\operatorname{Tr}(\Sigma'^{-1}\Sigma_i) + (\mu_i - \mu')^\top \Sigma'^{-1}(\mu_i - \mu') - K + \log\det\Sigma' - \log\det\Sigma_i]$. With $\mu' = e^{\mathring{\omega}}\mu_j$ and $\Sigma' = e^{\mathring{\omega}}\Sigma_j e^{\mathring{\omega}^\top}$, the inverse is $e^{-\mathring{\omega}^\top}\Sigma_j^{-1}e^{-\mathring{\omega}}$, cyclicity of the trace moves the outer factors onto Σ_i and onto the mean difference, and $\log\det\Sigma' = \log\det\Sigma_j + 2\log\det e^{\mathring{\omega}}$. Jacobi's formula $\frac{d}{dt}\det e^{t\mathring{\omega}} = \det e^{t\mathring{\omega}}\operatorname{Tr}(e^{-t\mathring{\omega}}\mathring{\omega}e^{t\mathring{\omega}}) = \operatorname{Tr}(\mathring{\omega})\det e^{t\mathring{\omega}}$ integrates to $\det e^{\mathring{\omega}} = e^{\operatorname{Tr}\mathring{\omega}}$, which gives (G11.1). In local coordinates the two laws are $\rho_b(U_i)_{\#} q_i$ and $(U_i e^{\mathring{\omega}} U_j^{-1})_{\#}\rho_b(U_j)_{\#} q_j = \rho_b(U_i)_{\#}\exp(\mathring{\omega})_{\#} q_j$, and relative entropy is invariant under the common bijection $\rho_b(U_i)$. (ii) The right side of (G11.1) is a polynomial in the entries of $\mathring{\Theta}_e^{-1} = e^{-\mathring{\omega}}$ plus $\operatorname{Tr}\mathring{\omega}$, and the exponential is entire. As a function of $\mathring{\Theta}_e^{-1}$, the first two terms have the differential $\langle \mathbf{A}_e, d\mathring{\Theta}_e^{-1}\rangle$: the first contributes $\langle \Sigma_j^{-1}\mathring{\Theta}_e^{-1}\Sigma_i, d\mathring{\Theta}_e^{-1}\rangle$ because Σ_i and Σ_j are symmetric, and the second $\langle \Sigma_j^{-1}(\mathring{\Theta}_e^{-1}\mu_i - \mu_j)\mu_i^\top, d\mathring{\Theta}_e^{-1}\rangle$. By Lemma 9.19, the derivative of $e^{-\mathring{\omega}}$ in the direction $\mathbf{T}$ is $\mathbf{L}(-\mathring{\omega}, -\mathbf{T}) = -\mathbf{L}(-\mathring{\omega}, \mathbf{T})$, and $d\operatorname{Tr}\mathring{\omega}[\mathbf{T}] = \operatorname{Tr}\mathbf{T}$, which gives the differential; the adjoint of $\mathbf{L}(-\mathring{\omega}, \cdot)$ is $\mathbf{L}(-\mathring{\omega}^\top, \cdot)$, which gives the gradient. (iii) The first two cases are substitution. For the third, insert $e^{-t\mathring{v}} = I - t\mathring{v} + \frac{t^2}{2}\mathring{v}^2 + O(t^3)$ into (G11.1) with $\mu_i = \mu_j = \mu$ and $\Sigma_i = \Sigma_j = \Sigma$. The trace term is $\frac{K}{2} - t\operatorname{Tr}\mathring{v} + \frac{t^2}{2}[\operatorname{Tr}(\mathring{v}^2) + \operatorname{Tr}(\Sigma^{-1}\mathring{v}\Sigma\mathring{v}^\top)] + O(t^3)$, because the trace is invariant under cyclic permutation and transposition, so that the first-order terms $\operatorname{Tr}(\Sigma^{-1}\mathring{v}\Sigma)$ and $\operatorname{Tr}\mathring{v}^\top$ both equal $\operatorname{Tr}\mathring{v}$, and the terms $\operatorname{Tr}(\Sigma^{-1}\mathring{v}^2\Sigma)$ and $\operatorname{Tr}(\mathring{v}^\top)^2$ both equal $\operatorname{Tr}(\mathring{v}^2)$; the mean term is $\frac{t^2}{2}(\mathring{v}\mu)^\top\Sigma^{-1}\mathring{v}\mu + O(t^3)$, since $e^{-t\mathring{v}}\mu - \mu = -t\mathring{v}\mu + O(t^2)$; the log-determinant term vanishes, and the linear terms $-t\operatorname{Tr}\mathring{v}$ and $t\operatorname{Tr}\mathring{v}$ cancel. The sum is $\frac{t^2}{2}$ times (G11.3) plus $O(t^3)$. The form is the orbit Fisher form of Corollary 7.7: the moment curve $t \mapsto (e^{t\mathring{v}}\mu, e^{t\mathring{v}}\Sigma e^{t\mathring{v}^\top})$ has velocity $(\mathring{v}\mu, \mathring{v}\Sigma + \Sigma\mathring{v}^\top)$ at $t = 0$, the fundamental vector field of $\mathring{v}$ at q, and the Fisher metric (G5.2)

evaluates it to $(\mathring{v}\mu)^\top\Sigma^{-1}\mathring{v}\mu + \frac{1}{2}\operatorname{Tr}[(\Sigma^{-1}\mathring{v}\Sigma + \mathring{v}^\top)^2]$; expanding the square, the two pure terms each have trace $\operatorname{Tr}(\mathring{v}^2)$ and the two cross terms each $\operatorname{Tr}(\Sigma^{-1}\mathring{v}\Sigma\mathring{v}^\top)$, which is (G11.3). □

The divergence charges the generator in two ways: through $e^{-\mathring{\omega}}$, which pulls the receiver's law back into the source frame, and through the volume term $\operatorname{Tr}\mathring{\omega}$, which penalizes expansion of the source and rewards contraction. For trace-free generators, the isometries of Section G11.4 among them, the volume term vanishes.

The comparison record of Proposition G2.20(iii) with $S_e^Q = I$ and $S_e^K = \mathbf{M}$ is the kernel $o_e \mid z_i, z_j \sim \mathcal{N}(z_i - \mathbf{M}z_j, W_e^{-1})$, $W_e \succ 0$, a Gaussian kernel for every transport matrix $\mathbf{M} \in \mathbb{R}^{K\times K}$, singular ones included; its energy at an observed o_e is

$$\mathsf{E}_e(z_i, z_j; \mathbf{M}) = \tfrac{1}{2}(o_e - z_i + \mathbf{M}z_j)^\top W_e(o_e - z_i + \mathbf{M}z_j) + \tfrac{1}{2}\log\det(2\pi W_e^{-1}), \tag{G11.4}$$

and the latent link sets $\mathbf{M} = e^{\mathring{\omega}_e}$. At $o_e = 0$ its quadratic part is the record energy of Corollary 9.22(a), and $\mathbf{M}$ plays the role of the matrix of Corollary 9.24(a).

Proposition G11.2 (Linear comparison record, delta-rule current, and the unregularized transport). Let $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and $q_j = \mathcal{N}(\mu_j, \Sigma_j)$ be nondegenerate and let matter be recognized by the product $q_i \otimes q_j$. (i) The expected energy (G11.4) of the comparison record with weight $W_e \succ 0$ at an observation $o_e \in \mathbb{R}^K$ is

$$\begin{aligned}\mathbb{E}_{q_i\otimes q_j}\mathsf{E}_e(\cdot;\mathbf{M}) = {}& \tfrac{1}{2}(\mu_i - o_e - \mathbf{M}\mu_j)^\top W_e(\mu_i - o_e - \mathbf{M}\mu_j) + \tfrac{1}{2}\operatorname{Tr}(W_e\Sigma_i)\\ & + \tfrac{1}{2}\operatorname{Tr}\big(W_e\mathbf{M}\Sigma_j\mathbf{M}^\top\big) + \tfrac{1}{2}\log\det(2\pi W_e^{-1}),\end{aligned} \tag{G11.5}$$

and if $\mathbf{M} = e^{\mathring{\omega}_e}$ with $\mathring{\omega}_e$ random and independent of the matter, its expectation over $\mathring{\omega}_e$ is the affine function $\frac{1}{2}(\mu_i - o_e)^\top W_e(\mu_i - o_e) - (\mu_i - o_e)^\top W_e(\mathbb{E}e^{\mathring{\omega}_e})\mu_j + \frac{1}{2}\sum_{a,b,c,d}(W_e)_{ab}(\Sigma_j + \mu_j\mu_j^\top)_{cd}\big(\mathbb{E}[e^{\mathring{\omega}_e}\otimes e^{\mathring{\omega}_e}]\big)_{(b,a),(c,d)} + \frac{1}{2}\operatorname{Tr}(W_e\Sigma_i) + \frac{1}{2}\log\det(2\pi W_e^{-1})$ of the first two exponential moments. (ii) The derivative of (G11.5) in $\mathbf{M}$ is $-\mathbf{J}_e$, with the current

$$\mathbf{J}_e = W_e\big[(\mu_i - o_e - \mathbf{M}\mu_j)\mu_j^\top - \mathbf{M}\Sigma_j\big], \tag{G11.6}$$

a precision-weighted prediction error times the source mean, minus the source covariance transported by $\mathbf{M}$, and its gradient in the generator at $\mathbf{M} = e^{\mathring{\omega}_e}$ is $-\mathbf{L}(\mathring{\omega}_e^\top, \mathbf{J}_e)$; this is the Gaussian form of the current of Corollary 9.22 with an observed offset. (iii) Without a link prior, (G11.5) is a strictly convex quadratic function of $\mathbf{M} \in \mathbb{R}^{K\times K}$ with the unique minimizer

$$\mathbf{M}^\star = (\mu_i - o_e)\mu_j^\top(\Sigma_j + \mu_j\mu_j^\top)^{-1}, \qquad \operatorname{Tr}\mathbf{M}^\star = \frac{\mu_j^\top\Sigma_j^{-1}(\mu_i - o_e)}{1 + \mu_j^\top\Sigma_j^{-1}\mu_j}, \tag{G11.7}$$

of rank at most one. For $K \geq 2$ it is singular and equals no $e^{\mathring{\omega}}$; consequently the gradient in $\mathring{\omega}_e$ vanishes only where $\mathbf{L}(\mathring{\omega}_e, \cdot)$ is singular, that is, where two eigenvalues of $\mathring{\omega}_e$ differ by a nonzero integer multiple of $2\pi\mathrm{i}$. For $K \geq 2$, if $\operatorname{Tr}\mathbf{M}^\star \geq 0$, the infimum over $\mathring{\omega}_e \in \mathfrak{g}$ equals the value at $\mathbf{M}^\star$ and is not attained; if $\operatorname{Tr}\mathbf{M}^\star < 0$, no sequence of exponentials converges to $\mathbf{M}^\star$, and the infimum

over $\mathfrak{g}$ exceeds the value at $\mathbf{M}^\star$. (iv) A stationary point at a critical point of the exponential can be a saddle. For $K=2$, $W_e=I_2$, $\Sigma_j=\varsigma^2 I_2$, $\mu_j=(1,0)^\top$, and $\mu_i-o_e=-(1+2\varsigma^2)e^{\mathrm{a}}(1,0)^\top$ with $\varsigma>0$ and $\mathrm{a}\in\mathbb{R}$, the generator $\mathring{\omega}_0=\mathrm{a}I_2+\pi\mathfrak{J}$, with $\mathfrak{J}=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$, is a stationary point of $\mathring{\omega}\mapsto\mathbb{E}\mathsf{E}_e(\cdot;e^{\mathring{\omega}})$ with $e^{\mathring{\omega}_0}=-e^{\mathrm{a}}I_2$, and it is not a local minimizer. [ESTABLISHED]

Proof. (i) Under $q_i\otimes q_j$ the residual $o_e-z_i+\mathbf{M}z_j$ is Gaussian with mean $-(\mu_i-o_e-\mathbf{M}\mu_j)$ and covariance $\Sigma_i+\mathbf{M}\Sigma_j\mathbf{M}^\top$, and the expectation of a quadratic form is the form at the mean plus the trace against the covariance. Expanding the first and third terms as $\frac{1}{2}(\mu_i-o_e)^\top W_e(\mu_i-o_e)-(\mu_i-o_e)^\top W_e\mathbf{M}\mu_j+\frac{1}{2}\operatorname{Tr}(W_e\mathbf{M}\mathbf{S}\mathbf{M}^\top)$ with $\mathbf{S}=\Sigma_j+\mu_j\mu_j^\top$, and writing $\operatorname{Tr}(W_e\mathbf{M}\mathbf{S}\mathbf{M}^\top)=\sum_{a,b,c,d}(W_e)_{ab}\mathbf{M}_{bc}\mathbf{S}_{cd}\mathbf{M}_{ad}$, shows that the expression is affine in $\mathbf{M}$ and in $\mathbf{M}\otimes\mathbf{M}$; the exponential moments are finite because $\|e^{\mathring{\omega}}\|\le e^{\|\mathring{\omega}\|}$ and Gaussian laws have exponential moments of every order. (ii) The differential of (G11.5) is $-(\mu_i-o_e-\mathbf{M}\mu_j)^\top W_e(d\mathbf{M})\mu_j+\operatorname{Tr}(W_e\mathbf{M}\Sigma_j(d\mathbf{M})^\top)=-\langle\mathbf{J}_e,d\mathbf{M}\rangle$, and the chain rule through Lemma 9.19 with the adjoint $\mathbf{L}(\mathring{\omega}_e^\top,\cdot)$ gives the generator gradient. (iii) The quadratic part $\frac{1}{2}\operatorname{Tr}(W_e\mathbf{M}(\Sigma_j+\mu_j\mu_j^\top)\mathbf{M}^\top)=\frac{1}{2}\|W_e^{1/2}\mathbf{M}(\Sigma_j+\mu_j\mu_j^\top)^{1/2}\|_{\mathrm{F}}^2$ is positive definite in $\mathbf{M}$, so the function is strictly convex and coercive, and $\mathbf{J}_e=0$ is $(\mu_i-o_e)\mu_j^\top=\mathbf{M}(\Sigma_j+\mu_j\mu_j^\top)$, whose unique solution is (G11.7); the trace is $\mu_j^\top(\Sigma_j+\mu_j\mu_j^\top)^{-1}(\mu_i-o_e)$, and the Sherman–Morrison identity gives its second form. Every exponential is invertible, so $e^{\mathring{\omega}}\neq\mathbf{M}^\star$ for $K\ge2$, which the rest of (iii) assumes. With $\operatorname{ad}_{\mathbf{B}}\mathbf{T}=[\mathbf{B},\mathbf{T}]$, the identity $e^{t\mathbf{B}}\mathbf{T}e^{-t\mathbf{B}}=e^{t\operatorname{ad}_{\mathbf{B}}}\mathbf{T}$ writes $\mathbf{L}(\mathbf{B},\mathbf{T})=\big(\int_0^1 e^{t\operatorname{ad}_{\mathbf{B}}}dt\big)(\mathbf{T})e^{\mathbf{B}}$, and right multiplication by $e^{\mathbf{B}}$ is invertible. Simultaneous triangularization of the commuting maps $\mathbf{T}\mapsto\mathbf{B}\mathbf{T}$ and $\mathbf{T}\mapsto\mathbf{T}\mathbf{B}$ shows that the eigenvalues of $\operatorname{ad}_{\mathbf{B}}$ are the differences $\mathring{\lambda}_a-\mathring{\lambda}_b$ of eigenvalues of $\mathbf{B}$, so those of $\int_0^1 e^{t\operatorname{ad}_{\mathbf{B}}}dt$ are $\int_0^1 e^{t(\mathring{\lambda}_a-\mathring{\lambda}_b)}dt$, which vanish exactly when $\mathring{\lambda}_a-\mathring{\lambda}_b\in2\pi\mathrm{i}\mathbb{Z}\setminus\{0\}$; and $\mathbf{B}^\top$ has the eigenvalues of $\mathbf{B}$. Where $\mathbf{L}(\mathring{\omega}_e^\top,\cdot)$ is invertible, a zero gradient forces $\mathbf{J}_e(e^{\mathring{\omega}_e})=0$, hence $e^{\mathring{\omega}_e}=\mathbf{M}^\star$, which is impossible. If $\operatorname{Tr}\mathbf{M}^\star>0$, the rank-one matrix $\mathbf{M}^\star$ has the simple eigenvalue $\operatorname{Tr}\mathbf{M}^\star$ and the eigenvalue 0 with a $(K-1)$-dimensional eigenspace, so $\mathbf{M}^\star=\mathbf{B}\operatorname{diag}(\operatorname{Tr}\mathbf{M}^\star,0,\dots,0)\mathbf{B}^{-1}$ for an invertible $\mathbf{B}$, and $\exp\big(\mathbf{B}\operatorname{diag}(\log\operatorname{Tr}\mathbf{M}^\star,\log\mathrm{c}_0,\dots,\log\mathrm{c}_0)\mathbf{B}^{-1}\big)\to\mathbf{M}^\star$ as $\mathrm{c}_0\to0$. If $\operatorname{Tr}\mathbf{M}^\star=0$ and $\mathbf{M}^\star\neq0$, then $(\mathbf{M}^\star)^2=0$ and $\exp\big((\log\mathrm{c}_0)I+\mathrm{c}_0^{-1}\mathbf{M}^\star\big)=\mathrm{c}_0I+\mathbf{M}^\star\to\mathbf{M}^\star$; if $\mathbf{M}^\star=0$, $\exp((\log\mathrm{c}_0)I)\to0$. In each case continuity carries the values to the infimum, which no exponential attains by strict convexity. If $\operatorname{Tr}\mathbf{M}^\star<0$, it is a simple eigenvalue of $\mathbf{M}^\star$; every real matrix near $\mathbf{M}^\star$ has a simple real negative eigenvalue near it, because a simple real eigenvalue of a real matrix stays real under small real perturbations, and its single Jordan block of size one occurs once, so by Culver [12, Theorem 1] that matrix has no real logarithm. A neighborhood of $\mathbf{M}^\star$ therefore contains no exponential, and outside it the strictly convex coercive function stays above a value larger than its minimum. (iv) At $\mathbf{M}=e^{\mathring{\omega}_0}=-e^{\mathrm{a}}I_2$, $\mu_i-o_e-\mathbf{M}\mu_j=-2\varsigma^2e^{\mathrm{a}}(1,0)^\top$, so $\mathbf{J}_e=-2\varsigma^2e^{\mathrm{a}}\operatorname{diag}(1,0)+\varsigma^2e^{\mathrm{a}}I_2=-\varsigma^2e^{\mathrm{a}}\operatorname{diag}(1,-1)$. The matrix $\operatorname{diag}(1,-1)$ anticommutes with $\mathfrak{J}$, so $e^{\mathrm{b}\mathfrak{J}}\operatorname{diag}(1,-1)=\operatorname{diag}(1,-1)e^{-\mathrm{b}\mathfrak{J}}$ for real b, and with $e^{t\mathring{\omega}_0^\top}=e^{t\mathrm{a}}e^{-t\pi\mathfrak{J}}$ the gradient is $-\mathbf{L}(\mathring{\omega}_0^\top,\mathbf{J}_e)=\varsigma^2e^{2\mathrm{a}}\operatorname{diag}(1,-1)\int_0^1e^{(2t-1)\pi\mathfrak{J}}dt=0$, since $\int_{-1}^1\cos(\pi u)du=\int_{-1}^1\sin(\pi u)du=0$. Along the direction $\operatorname{diag}(1,-1)$, which anticommutes with $\mathfrak{J}$, $(\pi\mathfrak{J}+t\operatorname{diag}(1,-1))^2=-(\pi^2-t^2)I_2$, so with $\mathrm{r}_t=(\pi^2-t^2)^{1/2}$, $e^{\mathring{\omega}_0+t\operatorname{diag}(1,-1)}=e^{\mathrm{a}}\big[\cos\mathrm{r}_tI_2+\mathrm{r}_t^{-1}\sin\mathrm{r}_t(\pi\mathfrak{J}+t\operatorname{diag}(1,-1))\big]=-e^{\mathrm{a}}I_2+\frac{e^{\mathrm{a}}t^2}{2\pi^2}(\pi\mathfrak{J}+t\operatorname{diag}(1,-1))+O(t^4)$. The expected energy is quadratic in $\mathbf{M}$ with derivative $-\mathbf{J}_e$, and $\operatorname{diag}(1,-1)$ is orthogonal to $\mathfrak{J}$ and to I_2 for the trace pairing, so its change is $\varsigma^2e^{2\mathrm{a}}t^3/\pi^2+O(t^4)$, negative for small $t<0$. □

Part (iii) is the Gaussian instance of Corollary 9.24(a) with an observed offset: a single context determines only a rank-one regression, and without the anchor of the generator prior the record drives the link out of the exponential chart. Part (iv) adds to the stationary generator of Corollary 9.24(a) that a stationary point at a critical point of the exponential map can be a saddle of the chart energy. When $\operatorname{Tr}\mathbf{M}^\star < 0$, Proposition G11.3 excludes attainment of the chart infimum at every chart link with semisimple negative eigenvalues, which settles $K = 2$ and $K = 3$; attainment at a chart link with a non-semisimple negative eigenvalue, possible only for $K \geq 4$, is Open Problem G11.11.

Proposition G11.3 (Non-attainment of the chart infimum at semisimple negative eigenvalues). Let $K \geq 2$, let $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and $q_j = \mathcal{N}(\mu_j, \Sigma_j)$ be nondegenerate, let matter be recognized by $q_i \otimes q_j$, and, without a link prior, let $\mathbf{l}(\mathbf{M}) = \mathbb{E}_{q_i \otimes q_j} \mathsf{E}_e(\cdot; \mathbf{M})$ be the expected energy (G11.5) of the comparison record with weight $W_e \succ 0$ at an observation $o_e \in \mathbb{R}^K$, whose minimizer $\mathbf{M}^\star$ of (G11.7) has $\operatorname{Tr}\mathbf{M}^\star < 0$. Call an element of $\exp(\mathfrak{g})$ a chart link, and an eigenvalue semisimple if all its Jordan blocks have size one. Then no chart link $\mathbf{M}$ whose negative eigenvalues are all semisimple satisfies $\mathbf{l}(\mathbf{M}) = \inf_{\exp(\mathfrak{g})} \mathbf{l}$. For $K = 2$ and $K = 3$ every chart link has this property, so $\mathring{\omega} \mapsto \mathbf{l}(e^{\mathring{\omega}})$ does not attain its infimum over $\mathfrak{g}$; for $K = 2$ this is Corollary 9.24(a)(iv) for the matter law of $(z_i - o_e, z_j)$. [ESTABLISHED]

Proof. By Proposition G11.2(iii), $\mathbf{l}$ is a strictly convex quadratic function on $\mathbb{R}^{K\times K}$ whose gradient vanishes only at $\mathbf{M}^\star$, which is singular because its rank is at most one and $K \geq 2$, while every chart link is invertible, since $\det e^{\mathring{\omega}} = e^{\operatorname{Tr}\mathring{\omega}}$. By Culver [12, Theorem 1], a real $K \times K$ matrix is a chart link exactly when it is invertible and every Jordan block belonging to a negative eigenvalue occurs an even number of times. Let a chart link $\mathbf{M}$ with semisimple negative eigenvalues satisfy $\mathbf{l}(\mathbf{M}) = \inf_{\exp(\mathfrak{g})} \mathbf{l}$. It suffices to find a nonempty open set of real matrices $\mathbf{E}$ such that, for each of them, $\mathbf{M} + t\mathbf{E}$ is a chart link for every sufficiently small $|t|$. Indeed, for such $\mathbf{E}$ minimality gives $\mathbf{l}(\mathbf{M} + t\mathbf{E}) \geq \mathbf{l}(\mathbf{M})$ for small t of either sign, so $\langle \nabla\mathbf{l}(\mathbf{M}), \mathbf{E}\rangle = \frac{d}{dt}\mathbf{l}(\mathbf{M} + t\mathbf{E})\big|_{t=0} = 0$; the linear span of a nonempty open set is $\mathbb{R}^{K\times K}$, so $\nabla\mathbf{l}(\mathbf{M}) = 0$ and $\mathbf{M} = \mathbf{M}^\star$, which contradicts the invertibility of $\mathbf{M}$.

If $\mathbf{M}$ has no negative eigenvalue, its spectrum is a compact set at positive distance from $(-\infty, 0]$; by continuity of the roots of the characteristic polynomial every real matrix near $\mathbf{M}$ has no eigenvalue in $(-\infty, 0]$ and is a chart link by the criterion, so every $\mathbf{E}$ qualifies. Otherwise let $\mathrm{c}'_1, \dots, \mathrm{c}'_{\mathrm{r}}$ be the distinct negative eigenvalues of $\mathbf{M}$. The Jordan blocks of c'_l have size one, so their number is its multiplicity n_l, which the criterion makes even. The real eigenspaces of the c'_l and the sum of the real generalized eigenspaces of the other eigenvalues decompose $\mathbb{R}^K$ into invariant subspaces, so that for a real invertible $\mathbf{S}$

$$\mathbf{S}^{-1}\mathbf{M}\mathbf{S} = \operatorname{blockdiag}\big(\mathrm{c}'_1 I_{\mathrm{n}_1}, \dots, \mathrm{c}'_{\mathrm{r}} I_{\mathrm{n}_{\mathrm{r}}}, \mathbf{M}_0\big),$$

with $\mathbf{M}_0$ real and without eigenvalues in $(-\infty, 0]$. For a real $\mathbf{E}$, let $\widetilde{\mathbf{E}}_{ll}$ be the diagonal block of $\mathbf{S}^{-1}\mathbf{E}\mathbf{S}$ at c'_l, the compression of $\mathbf{E}$ to the eigenspace of c'_l along this decomposition, and call $\mathbf{E}$ admissible if every $\widetilde{\mathbf{E}}_{ll}$ has distinct non-real eigenvalues. The admissible matrices form an open set, because each $\mathbf{E} \mapsto \widetilde{\mathbf{E}}_{ll}$ is linear and, by continuity of the roots of the characteristic polynomial, distinct

non-real eigenvalues stay distinct and non-real under small real perturbations; and a nonempty one, because each n_l is even and the block-diagonal matrix with the 2×2 blocks $\left(\begin{smallmatrix} 0 & -\mathrm{k} \\ \mathrm{k} & 0 \end{smallmatrix}\right)$, $\mathrm{k}=1,\dots,\mathrm{n}_l/2$, has the distinct eigenvalues $\pm\mathrm{i}\mathrm{k}$.

Fix an admissible $\mathbf{E}$, and let $\mathrm{r}_0 > 0$ be smaller than the distance between any two distinct eigenvalues of $\mathbf{M}$ and than the distance from the spectrum of $\mathbf{M}_0$ to $(-\infty,0]$. Fix l, order the blocks so that the one at c'_l comes first, let $\mathbf{B}$ be the complementary diagonal block of $\mathbf{S}^{-1}\mathbf{M}\mathbf{S}$, of which c'_l is not an eigenvalue, and split $\mathbf{S}^{-1}\mathbf{E}\mathbf{S}$ accordingly into $\mathbf{E}_{11}=\widetilde{\mathbf{E}}_{ll}$, $\mathbf{E}_{12}$, $\mathbf{E}_{21}$, and $\mathbf{E}_{22}$. For $|\mathring{\lambda}-\mathrm{c}'_l|\le \mathrm{r}_0/2$ and small $|t|$, the matrix $\mathbf{T}(t,\mathring{\lambda})=\mathring{\lambda}I-\mathbf{B}-t\mathbf{E}_{22}$ is invertible, and the Schur complement gives

$$\det(\mathring{\lambda}I-\mathbf{M}-t\mathbf{E})=\det\mathbf{T}(t,\mathring{\lambda})\det\big((\mathring{\lambda}-\mathrm{c}'_l)I-t\mathbf{E}_{11}-t^2\mathbf{E}_{12}\mathbf{T}(t,\mathring{\lambda})^{-1}\mathbf{E}_{21}\big).$$

With $\mathring{\lambda}=\mathrm{c}'_l+t\mathring{\nu}$ the second factor is $t^{\mathrm{n}_l}\mathrm{f}(t,\mathring{\nu})$, where $\mathrm{f}(t,\mathring{\nu})=\det\big(\mathring{\nu}I-\mathbf{E}_{11}-t\mathbf{E}_{12}\mathbf{T}(t,\mathrm{c}'_l+t\mathring{\nu})^{-1}\mathbf{E}_{21}\big)$ is real-analytic in t and holomorphic in $\mathring{\nu}$ near $t=0$, and $\mathrm{f}(0,\cdot)$ is the characteristic polynomial of $\mathbf{E}_{11}$, whose roots $\mathring{\nu}_1,\dots,\mathring{\nu}_{\mathrm{n}_l}$ are simple and non-real. The implicit function theorem at these simple roots gives continuous $\mathring{\nu}_k(t)$ for small $|t|$, with $\mathrm{f}(t,\mathring{\nu}_k(t))=0$ and $\mathring{\nu}_k(0)=\mathring{\nu}_k$, which stay distinct and non-real. For small $t\neq 0$ the numbers $\mathrm{c}'_l+t\mathring{\nu}_k(t)$ are therefore n_l distinct eigenvalues of $\mathbf{M}+t\mathbf{E}$ within $\mathrm{r}_0/2$ of c'_l, and they are non-real, with imaginary parts $t\operatorname{Im}\mathring{\nu}_k(t)\neq 0$. By continuity of the roots of the characteristic polynomial, for small $|t|$ exactly n_l eigenvalues of $\mathbf{M}+t\mathbf{E}$, counted with multiplicity, lie within $\mathrm{r}_0/2$ of c'_l, so these are all of them. The same continuity places every eigenvalue of $\mathbf{M}+t\mathbf{E}$ within $\mathrm{r}_0/2$ of the spectrum of $\mathbf{M}$ for small $|t|$, and those not near any c'_l lie near the spectrum of $\mathbf{M}_0$, off $(-\infty,0]$. Hence, for every small $t\neq 0$ of either sign, $\mathbf{M}+t\mathbf{E}$ has no eigenvalue in $(-\infty,0]$ and is a chart link by the criterion.

For $K\le 3$, a negative eigenvalue of a chart link has even multiplicity by the criterion, hence multiplicity two, and a single Jordan block of size two would occur once, so it has two blocks of size one and is semisimple; for $K=3$ the remaining eigenvalue is real and, as $\det e^{\mathring{\omega}}>0$, positive. The infimum of $\mathring{\omega}\mapsto \mathrm{l}(e^{\mathring{\omega}})$ over $\mathfrak{g}$ is $\inf_{\exp(\mathfrak{g})}\mathrm{l}$, attained at $\mathring{\omega}$ exactly when $e^{\mathring{\omega}}$ attains it. For $K=2$, l is, up to the constant $\frac{1}{2}\log\det(2\pi W_e^{-1})$, the expected record energy of Corollary 9.24(a) for the matter law of (z_i-o_e,z_j), a product law with $\Sigma_j\succ 0$, whose condition $\mu_j^{\top}\Sigma_j^{-1}(\mu_i-o_e)<0$ is $\operatorname{Tr}\mathbf{M}^{\star}<0$ by (G11.7). □

G11.3 Expected loop defects under product recognition

This section computes expected loop defects when the generators are recognized by a product Gaussian law.

Proposition G11.4 (Expected loop defects under product Gaussian recognition). Let the generators be recognized by the product Gaussian law $R=\bigotimes_{e\in E_+}\mathcal{N}(\mathring{\mu}_e,\mathring{\Lambda}_e^{-1})$, with each precision $\mathring{\Lambda}_e$ a positive-definite self-adjoint operator on $(\mathfrak{g},\langle\cdot,\cdot\rangle_{\mathsf{G}})$, as in Proposition 9.18, and for a copy $e\notin E_+$ write $\mathring{\mu}_e=-\mathring{\mu}_{\bar{e}}$ and $\mathring{\Lambda}_e=\mathring{\Lambda}_{\bar{e}}$, so that its generator $\mathring{\omega}_e=-\mathring{\omega}_{\bar{e}}$ has the law $\mathcal{N}(\mathring{\mu}_e,\mathring{\Lambda}_e^{-1})$ under R. Let $p=(e_0,\dots,e_{r-1})$ be a closed walk whose copies belong to pairwise distinct reversal pairs.

(i) The expected reference holonomy and its second moment are the ordered products

$$\mathbb{E}_R\mathring{\Theta}_p=\prod_{a=0}^{r-1}\mathbb{E}_R e^{\mathring{\omega}_{e_a}},\qquad \mathbb{E}_R\big[\mathring{\Theta}_p\otimes\mathring{\Theta}_p\big]=\prod_{a=0}^{r-1}\mathbb{E}_R\big[e^{\mathring{\omega}_{e_a}}\otimes e^{\mathring{\omega}_{e_a}}\big], \tag{G11.8}$$

with factors in the order of the walk. (ii) With $\mathsf{G}=\mathsf{G}^{\mathrm{fib}}$ and the vectors $\mathbf{G}^{\pm}\in\mathbb{R}^K\otimes\mathbb{R}^K$ of coordinates $\mathbf{G}^{+}_{(a,b)}=\mathsf{G}_{ab}$ and $\mathbf{G}^{-}_{(a,b)}=(\mathsf{G}^{-1})_{ab}$, the expected loop defect is the finite expression

$$\mathbb{E}_R\mathfrak{Q}_p=\tfrac{1}{2}(\mathbf{G}^{+})^{\top}\Big(\prod_{a=0}^{r-1}\mathbb{E}_R\big[e^{\mathring{\omega}_{e_a}}\otimes e^{\mathring{\omega}_{e_a}}\big]\Big)\mathbf{G}^{-}-\mathrm{Tr}\Big(\prod_{a=0}^{r-1}\mathbb{E}_R e^{\mathring{\omega}_{e_a}}\Big)+\tfrac{K}{2} \tag{G11.9}$$

in the per-copy first and second moments of exponentials of Gaussian matrices. (iii) Write $\mathring{\omega}_{e_a}=\mathring{\mu}_{e_a}+\mathring{\zeta}_a$, with $\mathring{\mu}_{e_a}$ the mean of the generator of e_a, and let $\mathring{\Theta}_p(\mathring{\mu})$, $\mathring{\Theta}_p^{<a}=\prod_{b<a}e^{\mathring{\mu}_{e_b}}$, and $\mathring{\Theta}_p^{>a}=\prod_{b>a}e^{\mathring{\mu}_{e_b}}$ be evaluated at the means. As the covariances $\mathring{\Lambda}_e^{-1}$ of the copies of p tend to zero with the means held,

$$\begin{aligned}\mathbb{E}_R\mathfrak{Q}_p=\mathfrak{Q}_p(\mathring{\mu})+\tfrac{1}{2}\sum_{a=0}^{r-1}\mathbb{E}\Big[&\big\|\mathring{\Theta}_p^{<a}\mathbf{L}(\mathring{\mu}_{e_a},\mathring{\zeta}_a)\mathring{\Theta}_p^{>a}\big\|_{\mathsf{G}}^2\\&+\big\langle\mathring{\Theta}_p(\mathring{\mu})-I,\mathring{\Theta}_p^{<a}\mathbf{L}^{(2)}(\mathring{\mu}_{e_a};\mathring{\zeta}_a)\mathring{\Theta}_p^{>a}\big\rangle_{\mathsf{G}}\Big]+O\Big(\max_a\|\mathring{\Lambda}_{e_a}^{-1}\|^2\Big),\end{aligned} \tag{G11.10}$$

and at flat means, $\mathring{\mu}_{e_a}=0$ for every a, $\mathbb{E}_R\mathfrak{Q}_p=\frac{1}{2}\sum_a\mathrm{Tr}\,\mathring{\Lambda}_{e_a}^{-1}+O(\max_a\|\mathring{\Lambda}_{e_a}^{-1}\|^2)$, where Tr is the trace of an operator on $(\mathfrak{g},\langle\cdot,\cdot\rangle_{\mathsf{G}})$. (iv) If a walk traverses a reversal pair twice, (i) fails in general: for $K=1$ and the walk $(e,\bar{e})$, $\mathring{\Theta}_p=1$ while $\mathbb{E}_R e^{\mathring{\omega}_e}\mathbb{E}_R e^{-\mathring{\omega}_e}=e^{\mathring{\Lambda}_e^{-1}}\neq 1$. [ESTABLISHED]

Proof. (i) Distinct reversal pairs carry independent generators under R, so the factors $e^{\mathring{\omega}_{e_a}}$ are independent random matrices. They are integrable to every order, because $\|e^{\mathbf{B}}\|\le e^{\|\mathbf{B}\|}$ and Gaussian laws have exponential moments of every order. For independent integrable random matrices $\mathbb{E}[\mathbf{S}\mathbf{T}]=(\mathbb{E}\mathbf{S})(\mathbb{E}\mathbf{T})$ entrywise, which gives the first product, and the mixed-product rule $(\mathbf{B}_0\cdots\mathbf{B}_{r-1})\otimes(\mathbf{B}_0\cdots\mathbf{B}_{r-1})=(\mathbf{B}_0\otimes\mathbf{B}_0)\cdots(\mathbf{B}_{r-1}\otimes\mathbf{B}_{r-1})$ reduces the second to the first. (ii) In the coordinates of Section G11.1, $(\mathbf{G}^{+})^{\top}(\mathbf{B}\otimes\mathbf{B})\mathbf{G}^{-}=\sum_{a,b,c,d}\mathsf{G}_{ab}\mathbf{B}_{ac}\mathbf{B}_{bd}(\mathsf{G}^{-1})_{cd}=\mathrm{Tr}(\mathsf{G}^{-1}\mathbf{B}^{\top}\mathsf{G}\mathbf{B})=\langle\mathbf{B},\mathbf{B}\rangle_{\mathsf{G}}$. Expanding $\frac{1}{2}\langle\mathring{\Theta}_p-I,\mathring{\Theta}_p-I\rangle_{\mathsf{G}}$ with $\langle I,\mathbf{B}\rangle_{\mathsf{G}}=\mathrm{Tr}\,\mathbf{B}$ and $\langle I,I\rangle_{\mathsf{G}}=K$, and taking expectations with (i), gives (G11.9). (iii) At the generators $\mathring{\mu}_{e_a}+t\mathring{\zeta}_a$, the defect $\mathfrak{Q}_p$ is a polynomial in the entries of the factors $e^{\mathring{\mu}_{e_a}+t\mathring{\zeta}_a}$, and the power series of the exponential bounds the k-th derivative of $t\mapsto e^{\mathbf{B}+t\mathbf{T}}$ by $\|\mathbf{T}\|^k e^{\|\mathbf{B}\|+|t|\|\mathbf{T}\|}$; hence, with $\mathring{\zeta}=(\mathring{\zeta}_0,\dots,\mathring{\zeta}_{r-1})$, the fourth t-derivative of the defect is bounded by $\mathrm{c}_1e^{\mathrm{c}_1\|\mathring{\zeta}\|}\|\mathring{\zeta}\|^4$ for $|t|\le 1$, with c_1 depending on the means. Taylor's formula in t to third order with integral remainder at $t=1$, the vanishing of first and third moments of a centered Gaussian, and the bound $\mathbb{E}[e^{\mathrm{c}_1\|\mathring{\zeta}\|}\|\mathring{\zeta}\|^4]=O(\max_a\|\mathring{\Lambda}_{e_a}^{-1}\|^2)$ for bounded covariances give $\mathbb{E}_R\mathfrak{Q}_p=\mathfrak{Q}_p(\mathring{\mu})$ plus one half the expected second t-derivative at $t=0$, up to $O(\max_a\|\mathring{\Lambda}_{e_a}^{-1}\|^2)$. In that second derivative the terms mixing $\mathring{\zeta}_a$ and $\mathring{\zeta}_b$, $a\neq b$, have zero expectation because the two are independent and centered. Along $\mathring{\zeta}_a$ alone, the first and second derivatives of $\mathring{\Theta}_p$ are $\mathring{\Theta}_p^{<a}\mathbf{L}(\mathring{\mu}_{e_a},\mathring{\zeta}_a)\mathring{\Theta}_p^{>a}$ and $\mathring{\Theta}_p^{<a}\mathbf{L}^{(2)}(\mathring{\mu}_{e_a};\mathring{\zeta}_a)\mathring{\Theta}_p^{>a}$, and the second derivative of $\mathfrak{Q}_p$ is the squared G-norm of the first plus the pairing of $\mathring{\Theta}_p-I$ with

the second, which gives (G11.10). At flat means $\mathring{\Theta}_p^{<a} = \mathring{\Theta}_p^{>a} = I$, $\mathbf{L}(0, \mathring{\zeta}_a) = \mathring{\zeta}_a$, and $\mathring{\Theta}_p(0) = I$, so the bracket is $\mathbb{E}\|\mathring{\zeta}_a\|_{\mathsf{G}}^2 = \operatorname{Tr} \mathring{\Lambda}_{e_a}^{-1}$. (iv) For $K = 1$ and $\mathring{\omega}_e \sim \mathcal{N}(\mathring{\mu}_e, \mathring{\Lambda}_e^{-1})$, $\mathbb{E}e^{\pm\mathring{\omega}_e} = e^{\pm\mathring{\mu}_e + \mathring{\Lambda}_e^{-1}/2}$. □

At flat means the recognition spread alone produces curvature energy: each copy of the walk contributes half its generator variance, measured in the fiber metric. For a repeated pair the expectation involves joint moments of one Gaussian generator, such as $\mathbb{E}[e^{\mathring{\omega}_e}\mathbf{T}e^{\mathring{\omega}_e}]$, which the same Taylor argument expands to first order but which do not factor.

G11.4 The action induced by undirected Gaussian matter

On a finite multigraph with agents $V = \{1, \dots, N\}$, let every copy $e : j \to i$ in E_+ carry a weight $W_e \succ 0$, every agent a self term $A_i \succeq 0$, and let $\mathsf{h} \in \mathbb{R}^{NK}$ be an information vector, all in reference coordinates. The matter energy with latent links is

$$\mathsf{E}(z; \mathring{\omega}) = \tfrac{1}{2} \sum_{i \in V} z_i^\top A_i z_i + \tfrac{1}{2} \sum_{e \in E_+} \big(z_i - e^{\mathring{\omega}_e} z_j\big)^\top W_e \big(z_i - e^{\mathring{\omega}_e} z_j\big) - \mathsf{h}^\top z, \tag{G11.11}$$

with precision $\Lambda(\mathring{\omega}) = A_\oplus + L(\mathring{\omega})$, where $A_\oplus = \operatorname{blockdiag}(A_1, \dots, A_N)$ and $L(\mathring{\omega})$ is the connection Laplacian of Definition 11.18 with represented links $e^{\mathring{\omega}_e}$; a loop copy contributes $(I - e^{\mathring{\omega}_e})^\top W_e (I - e^{\mathring{\omega}_e})$ to its agent's block. At $\mathring{\omega} = 0$ the precision is the member of the interaction family (G3.1) with self terms A_i and weights summed over parallel copies, the form adopted in Hypothesis G3.13; for $\mathring{\omega} \neq 0$ it has the nonflat form of Definition G7.18. The generator family carries a base law $e^{-\mathscr{S}}\nu^{\mathscr{T}}$ with $0 < Z_{\mathscr{S}} < \infty$ as in Proposition 5.26, for instance the generator prior of Definition 9.11, and the joint law of generators and matter is proportional to $e^{-\mathscr{S}(\mathring{\omega})}\nu^{\mathscr{T}}(d\mathring{\omega})e^{-\mathsf{E}(z;\mathring{\omega})}dz$, normalized as a whole, the undirected model of Definition 4.5.

Proposition G11.5 (Induced action of undirected Gaussian matter). (i) If every $A_i \succ 0$, then $\Lambda(\mathring{\omega}) \succeq A_\oplus \succ 0$ for every $\mathring{\omega}$, the joint normalizer $\mathfrak{Z}$, as in Proposition 9.27, is finite and positive, and the generator marginal is $\mathfrak{Z}^{-1}e^{-\mathscr{S}(\mathring{\omega}) - \mathscr{S}^{\mathrm{ind}}(\mathring{\omega})}\nu^{\mathscr{T}}(d\mathring{\omega})$ with

$$\mathscr{S}^{\mathrm{ind}}(\mathring{\omega}) = -\mathsf{A}\big(\mathsf{h}, \Lambda(\mathring{\omega})\big) = \tfrac{1}{2}\log\det\Lambda(\mathring{\omega}) - \tfrac{1}{2}\mathsf{h}^\top\Lambda(\mathring{\omega})^{-1}\mathsf{h} - \tfrac{NK}{2}\log 2\pi \geq -\mathsf{A}(\mathsf{h}, A_\oplus). \tag{G11.12}$$

(ii) If every $A_i \succ 0$ and $E_+ \neq \varnothing$, then $e^{-\mathscr{S}^{\mathrm{ind}}}$ is not integrable for Lebesgue measure on $\mathscr{T}$; the induced action does not by itself define a law of the generators. (iii) If the multigraph is a forest (no loop copy, no parallel copies, no cycle), choose a root in every component, let $\mathring{\Theta}_{I \leftarrow i}$ be the product of the twists along the path from i to the root of its component I, the tree transport of Definition 11.4 in reference coordinates, and put $\mathring{\Theta}_\oplus = \operatorname{blockdiag}(\mathring{\Theta}_{I \leftarrow i})$. Then $\Lambda(\mathring{\omega}) = \mathring{\Theta}_\oplus^\top \Lambda^{\mathrm{root}}(\mathring{\omega})\mathring{\Theta}_\oplus$, where $\Lambda^{\mathrm{root}}(\mathring{\omega})$ belongs to the interaction family with self terms $\mathring{\Theta}_{I \leftarrow i}^{-\top} A_i \mathring{\Theta}_{I \leftarrow i}^{-1}$ and weights $\mathring{\Theta}_{I \leftarrow i}^{-\top} W_e \mathring{\Theta}_{I \leftarrow i}^{-1}$ on $e : j \to i$; no holonomy enters $\mathscr{S}^{\mathrm{ind}}$. If moreover $A_i = a_i\mathsf{G}$, $W_e = w_e\mathsf{G}$ with scalars $a_i, w_e > 0$, $\mathsf{h} = 0$, and every link is an isometry, $e^{\mathring{\omega}_e^\top}\mathsf{G}e^{\mathring{\omega}_e} = \mathsf{G}$, then $\mathscr{S}^{\mathrm{ind}}$ takes one value on all such configurations. (iv) For two agents joined by one copy $e : j \to i$ and $\mathsf{h} = 0$, $\det\Lambda(\mathring{\omega}) = \det(A_i + W_e)\det(A_j + e^{\mathring{\omega}_e^\top} W_e^\sharp e^{\mathring{\omega}_e})$ with $W_e^\sharp = W_e(A_i + W_e)^{-1}A_i$. For $\mathsf{G} = I_K$, $A_i = a_i I$ with $a_i > 0$, $A_j = a_j I$, $W_e = w_e I$, $w^\sharp = a_i w_e/(a_i + w_e)$,

and $\mathring{\omega}_e = t\mathring{v}$ with $\mathring{v}$ symmetric,

$$\frac{d}{dt}\mathscr{S}^{\mathrm{ind}}\Big|_{t=0} = \frac{w^\sharp \operatorname{Tr}\mathring{v}}{a_j + w^\sharp}, \qquad \frac{d^2}{dt^2}\mathscr{S}^{\mathrm{ind}}\Big|_{t=0} = \frac{2w^\sharp a_j \|\mathring{v}\|_{\mathrm{F}}^2}{(a_j + w^\sharp)^2}, \tag{G11.13}$$

while $\mathscr{S}^{\mathrm{ind}}$ is constant along skew-symmetric $\mathring{v}$: noncompact generator directions receive edge-local terms without any loop, and along the trace direction the action decreases as the link contracts. (v) At fixed links the induced action takes the values of earlier chapters. With every coordinate hidden it is the scalar $\mathfrak{c}^{\mathrm{eff}} - \mathfrak{c}$ of (G7.32) at the links $e^{\mathring{\omega}}$. For an unanchored cycle of $m \geq 2$ distinct agents, with copies $e_a : i_{a+1} \to i_a$, $i_m = i_0$, and every $A_i = 0$,

$$\det \Lambda(\mathring{\omega}) = \det\big(I - \mathring{\Theta}_p\big)^2 \prod_{a=0}^{m-1} \det W_{e_a}, \qquad p = (e_0, \dots, e_{m-1}), \tag{G11.14}$$

which for $m = 2$ and $W_e = \Sigma_e^{-1}$ is (G10.17). For $K = 1$, the reciprocal pair of Proposition G10.3 with unit weights, anchors $a_0 > 0$, and the flat slice $\mathring{\omega}_{e'} = -\mathring{\omega}_e$ has $\mathscr{S}^{\mathrm{ind}} = \frac{1}{2}\log(a_0^2 + 4a_0\cosh^2\mathring{\omega}_e) - \log 2\pi$ at $\mathsf{h} = 0$, which is (G10.6) with $\mathsf{g} = e^{\mathring{\omega}_e}$. [ESTABLISHED]

Proof. (i) Every edge term of $L(\mathring{\omega})$ is positive semidefinite, so $\Lambda(\mathring{\omega}) \succeq A_\oplus \succ 0$. By Proposition G2.4, $\int e^{-\mathsf{E}(z;\mathring{\omega})}dz = \exp \mathsf{A}(\mathsf{h}, \Lambda(\mathring{\omega}))$. The Loewner inequality gives $\log\det\Lambda(\mathring{\omega}) \geq \log\det A_\oplus$ and $\Lambda(\mathring{\omega})^{-1} \preceq A_\oplus^{-1}$, so $\mathsf{A}(\mathsf{h}, \Lambda(\mathring{\omega})) \leq \mathsf{A}(\mathsf{h}, A_\oplus)$, which is the lower bound. The map $\mathring{\omega} \mapsto \mathsf{A}(\mathsf{h}, \Lambda(\mathring{\omega}))$ is continuous, so by Tonelli's theorem $\mathfrak{Z} = \int e^{-\mathscr{S}} e^{\mathsf{A}(\mathsf{h},\Lambda)} d\nu^{\mathscr{T}} \in (0, Z_{\mathscr{S}} e^{\mathsf{A}(\mathsf{h},A_\oplus)}]$, and integrating out z gives the generator marginal. (ii) Fix $e_\star \in E_+$ and consider the generator families with $\mathring{\omega}_{e_\star} = tI + \mathring{\xi}$, $t \leq 0$, $\|\mathring{\xi}\| \leq 1$, and $\|\mathring{\omega}_e\| \leq 1$ for $e \neq e_\star$, in the operator norm. On this set every link lies in a compact set of matrices: $e^{\mathring{\omega}_e}$ in the image of the unit ball, and $e^{tI+\mathring{\xi}} = e^t e^{\mathring{\xi}}$ among the multiples $\mathrm{a}\mathbf{B}$ with $\mathrm{a} \in [0,1]$ and $\mathbf{B}$ in that image. The precision is a polynomial in the links and is $\succeq A_\oplus$ at every matrix value of the links, singular ones included, so $\frac{1}{2}\log\det\Lambda$ is bounded above on the set, and $\mathscr{S}^{\mathrm{ind}} \leq \frac{1}{2}\log\det\Lambda - \frac{NK}{2}\log 2\pi$. The set contains the disjoint sets on which $\|\mathring{\omega}_{e_\star} + kI\| \leq \frac{1}{3}$ and $\|\mathring{\omega}_e\| \leq 1$ for $e \neq e_\star$, $k = 1, 2, \dots$, of equal positive Lebesgue measure, so it has infinite measure, and $e^{-\mathscr{S}^{\mathrm{ind}}}$ is bounded below by a positive constant on it. (iii) The transports exist and are unique given the roots, because each copy of a forest joins a vertex to its parent and the relation $\mathring{\Theta}_{I\leftarrow i} e^{\mathring{\omega}_e} = \mathring{\Theta}_{I\leftarrow j}$ on $e : j \to i$ determines the child's transport from the parent's in either orientation. With $z = \mathring{\Theta}_\oplus^{-1} z^{\mathrm{root}}$, every residual becomes $z_i - e^{\mathring{\omega}_e} z_j = \mathring{\Theta}_{I\leftarrow i}^{-1} z_i^{\mathrm{root}} - e^{\mathring{\omega}_e}\mathring{\Theta}_{I\leftarrow j}^{-1} z_j^{\mathrm{root}} = \mathring{\Theta}_{I\leftarrow i}^{-1}(z_i^{\mathrm{root}} - z_j^{\mathrm{root}})$, so $\mathring{\Theta}_\oplus^{-\top}\Lambda(\mathring{\omega})\mathring{\Theta}_\oplus^{-1}$ is the interaction-family precision Λ^{root} with the stated self terms and weights. Isometric links make each $\mathring{\Theta}_{I\leftarrow i}$ an isometry of G, so the self terms and weights of Λ^{root} are A_i and W_e, and $(\det\mathring{\Theta}_{I\leftarrow i})^2 = 1$ because $\det(\mathring{\Theta}_{I\leftarrow i}^\top \mathsf{G}\mathring{\Theta}_{I\leftarrow i}) = \det\mathsf{G}$; hence $\det\Lambda(\mathring{\omega}) = \det\Lambda^{\mathrm{root}}(0)$. (iv) With $\mathring{\Theta}_e = e^{\mathring{\omega}_e}$, the blocks are $\Lambda_{ii} = A_i + W_e$, $\Lambda_{ij} = -W_e\mathring{\Theta}_e$, and $\Lambda_{jj} = A_j + \mathring{\Theta}_e^\top W_e \mathring{\Theta}_e$, and the Schur complement of Λ_{ii} is $A_j + \mathring{\Theta}_e^\top(W_e - W_e(A_i+W_e)^{-1}W_e)\mathring{\Theta}_e$, where $W_e - W_e(A_i+W_e)^{-1}W_e = W_e(A_i+W_e)^{-1}(A_i+W_e-W_e) = W_e(A_i+W_e)^{-1}A_i$, with $A_i + W_e \succ 0$ because $W_e \succ 0$; Lemma G2.6(iv) gives the determinant. In the isotropic case $\mathscr{S}^{\mathrm{ind}} = \frac{1}{2}\log\det\mathbf{B}_t$ plus a constant, with $\mathbf{B}_t = a_j I + w^\sharp e^{2t\mathring{v}}$, which commutes with $\mathring{v}$; the derivatives $\frac{d}{dt}\log\det\mathbf{B}_t = \operatorname{Tr}(\mathbf{B}_t^{-1}\mathbf{B}_t')$ and $\frac{d^2}{dt^2}\log\det\mathbf{B}_t = \operatorname{Tr}(\mathbf{B}_t^{-1}\mathbf{B}_t'' - \mathbf{B}_t^{-1}\mathbf{B}_t'\mathbf{B}_t^{-1}\mathbf{B}_t')$ at $t = 0$, where $\mathbf{B}_0 = (a_j + w^\sharp)I$, $\mathbf{B}_0' = 2w^\sharp\mathring{v}$, and $\mathbf{B}_0'' = 4w^\sharp\mathring{v}^2$, give (G11.13). For skew $\mathring{v}$, $e^{t\mathring{v}^\top}e^{t\mathring{v}} = I$. (v) With the pivot Λ, $n_{\mathrm{h}} = NK$, and

no retained coordinate, (G7.32) gives $\mathfrak{c}^{\mathrm{eff}} - \mathfrak{c} = \frac{1}{2}\log\det\Lambda - \frac{NK}{2}\log 2\pi - \frac{1}{2}\mathsf{h}^\top\Lambda^{-1}\mathsf{h}$. For the cycle, $z^\top\Lambda z = (\mathbf{T}z)^\top W_\oplus(\mathbf{T}z)$ with $W_\oplus = \mathrm{blockdiag}(W_{e_a})$ and the square block matrix $\mathbf{T}$ whose block row a has I in column a and $-\mathring{\Theta}_{e_a}$ in column $a+1$ modulo m, so $\det\Lambda = (\det\mathbf{T})^2\det W_\oplus$. Deleting block row and column 0 of $\mathbf{T}$ leaves a block upper bidiagonal matrix with identity diagonal blocks, of determinant one, whose inverse has the block $\mathring{\Theta}_{e_1}\cdots\mathring{\Theta}_{e_{m-2}}$ in its upper right corner (the identity when $m = 2$); the Schur complement of that matrix in $\mathbf{T}$ is $I - \mathring{\Theta}_{e_0}\mathring{\Theta}_{e_1}\cdots\mathring{\Theta}_{e_{m-1}} = I - \mathring{\Theta}_p$, which gives (G11.14). The last statement is substitution into (G10.6), since $(e^{\mathring{\omega}} + e^{-\mathring{\omega}})^2 = 4\cosh^2\mathring{\omega}$. □

Part (ii) is the Gaussian counterpart of the requirement that the generator prior be normalized: the induced action is bounded below but stays bounded on a set of infinite volume toward contracting links, so it confines no generator direction by itself, and the base law $\nu^{\mathscr{T}}$ must. Parts (iii) and (iv) separate the two kinds of generator direction. Compact directions enter $\mathscr{S}^{\mathrm{ind}}$ only through loops, while noncompact directions change the diagonal blocks of Λ on a single copy; along the trace direction the induced action decreases as links contract, toward the singular boundary that Proposition G11.2(iii) meets without a prior.

Corollary G11.6 (An unanchored cycle admits no joint law with latent links). Let $m \geq 2$ distinct agents form a cycle $p = (e_0, \ldots, e_{m-1})$ of copies from distinct reversal pairs, with weights $W_{e_a} \succ 0$, every self term zero, any information vector h, and latent generators of the cycle copies. Let ν be a finite measure on $\mathfrak{g}^m$ whose density is bounded below by a positive constant near some point at which $\det(I - \mathring{\Theta}_p)$ vanishes with nonzero differential; for every $K \geq 1$ the point with $\mathring{\omega}_{e_1} = \mathrm{diag}(0, \log 2, \ldots, \log K)$ and every other generator zero is such a point, and every Gaussian law with positive-definite covariance qualifies. Then $\int \nu(d\mathring{\omega})\int e^{-\mathsf{E}(z;\mathring{\omega})}dz = +\infty$, so no law proportional to $\nu(d\mathring{\omega})e^{-\mathsf{E}(z;\mathring{\omega})}dz$ exists, although for every generator family off the Lebesgue-null set $\{\det(I - \mathring{\Theta}_p) = 0\}$ the matter law at fixed links exists. With positive-definite self terms at every agent the joint law exists for every finite base measure (Proposition G11.5(i)). [ESTABLISHED]

Proof. Write $\mathrm{D}_p(\mathring{\omega}) = \det(I - \mathring{\Theta}_p)$. Off its zero set, (G11.14) gives $\Lambda(\mathring{\omega}) \succ 0$, and since $\mathsf{h}^\top\Lambda^{-1}\mathsf{h} \geq 0$, $\int e^{-\mathsf{E}}dz = e^{\mathsf{A}(\mathsf{h},\Lambda)} \geq (2\pi)^{mK/2}(\det\Lambda)^{-1/2} = (2\pi)^{mK/2}\prod_a(\det W_{e_a})^{-1/2}|\mathrm{D}_p(\mathring{\omega})|^{-1}$; on the zero set the inner integral is $+\infty$. The function D_p is real-analytic and not identically zero, so its zero set is Lebesgue-null. At a point where $\mathrm{D}_p = 0$ and $d\mathrm{D}_p \neq 0$, the submersion theorem gives local coordinates on a neighborhood whose first coordinate is D_p, with Jacobian bounded above and below on a smaller neighborhood, and the density of ν is bounded below there, so the integral of $|\mathrm{D}_p|^{-1}$ is bounded below by a positive multiple of $\int_{-\mathrm{a}}^{\mathrm{a}}|\mathrm{u}|^{-1}d\mathrm{u} = +\infty$ for some $\mathrm{a} > 0$. At the stated point, along $\mathring{\omega}_{e_0} = tI$, $\mathring{\Theta}_p = e^t\,\mathrm{diag}(1, 2, \ldots, K)$ and $\mathrm{D}_p = \prod_{k=1}^K(1 - ke^t)$, which vanishes at $t = 0$ with derivative $-\prod_{k=2}^K(1-k) \neq 0$. A Gaussian law with positive-definite covariance has a continuous positive density. □

At fixed links, Proposition G10.1 and corollary G10.7 show that holonomy without eigenvalue one restores definiteness of the unanchored pair, so its law exists off a null set of links; the corollary shows that this does not survive integration over the links, because the partition function diverges like $|\det(I - \mathring{\Theta}_p)|^{-1}$ across a hypersurface. For $K = 1$ more cycles restore integrability (Proposition G11.7); for $K \geq 2$ whether they do is Open Problem G11.10.

Proposition G11.7 (Joint normalization of unanchored scalar matter). Let $K = 1$, so that generators and links are real numbers, and in (G11.11) let the multigraph be connected, every weight $W_e = w_e$ positive, every self term zero, and the information vector zero. Let the generators carry a Gaussian base law ν on $\mathscr{T} = \mathbb{R}^{E_+}$ with positive-definite covariance and no action, for instance a product of nondegenerate Gaussian laws of the generators. With the value $+\infty$ for $\det \Lambda(\mathring{\omega})^{-1/2}$ where $\Lambda(\mathring{\omega})$ is singular, the joint normalizer of Proposition 9.27 is

$$\mathfrak{Z} = \int \nu(d\mathring{\omega}) \int e^{-\mathsf{E}(z;\mathring{\omega})} dz = (2\pi)^{N/2} \int \nu(d\mathring{\omega}) \det \Lambda(\mathring{\omega})^{-1/2}. \tag{G11.15}$$

(i) If the cycle rank $|E_+| - N + 1$ of the multigraph, the number of its copies in E_+ outside a spanning tree, is at least two, then $\Lambda(\mathring{\omega}) \succ 0$ off a Lebesgue-null set and $0 < \mathfrak{Z} < \infty$, so generators and matter have the joint law $\mathfrak{Z}^{-1} \nu(d\mathring{\omega}) e^{-\mathsf{E}(z;\mathring{\omega})} dz$. (ii) If the cycle rank is one, then $\mathfrak{Z} = +\infty$ and no law proportional to $\nu(d\mathring{\omega}) e^{-\mathsf{E}(z;\mathring{\omega})} dz$ exists; for a cycle of distinct agents this is Corollary G11.6 at $K = 1$. [ESTABLISHED]

Proof. Where $\Lambda(\mathring{\omega}) \succ 0$, Proposition G2.4 gives $\int e^{-\mathsf{E}} dz = e^{\mathsf{A}(0,\Lambda)} = (2\pi)^{N/2} \det \Lambda^{-1/2}$; where Λ is singular, E is constant along a kernel vector and the inner integral is $+\infty$; Tonelli's theorem gives (G11.15). Let $\mathbf{T}(\mathring{\omega})$ be the $|E_+| \times N$ matrix whose row at a copy $e : j \to i$ has the entry 1 in column i, the entry $-e^{\mathring{\omega}_e}$ in column j, and zeros elsewhere, the row of a loop copy having the single entry $1 - e^{\mathring{\omega}_e}$. Then $2\mathsf{E}(z;\mathring{\omega}) = \sum_{e \in E_+} w_e ((\mathbf{T} z)_e)^2$, so $\Lambda = \mathbf{T}^\top \operatorname{diag}(w_e) \mathbf{T}$, and the Cauchy–Binet formula gives

$$\det \Lambda(\mathring{\omega}) = \sum_{E'} \Big(\prod_{e \in E'} w_e \Big) \big(\det \mathbf{T}_{E'}(\mathring{\omega}) \big)^2,$$

the sum over the sets $E' \subseteq E_+$ of N copies, with $\mathbf{T}_{E'}$ the square submatrix of the rows in E'; every term is nonnegative.

Fix a spanning tree, with copy set $E_{\mathrm{tr}} \subseteq E_+$ of $N - 1$ copies, let $e'_1, \dots, e'_{\mathrm{r}}$ be the other copies of E_+, with $\mathrm{r} = |E_+| - N + 1$, and for each k let p_k be the closed walk that traverses e'_k in its orientation in E_+ and returns along the path of the tree between its endpoints ($p_k = (e'_k)$ for a loop copy), with the value $\mathring{s}_k = (\mathbf{C}\mathring{\omega})_{p_k}$ of the cycle map (9.8), the sum of the generators along p_k under the reversal convention. The copies $E'_k = E_{\mathrm{tr}} \cup \{e'_k\}$ form a connected subgraph with N vertices and N copies, the cycle of p_k with trees attached. While an attached tree remains, it has a vertex off the cycle that lies on exactly one copy e of the subgraph; the column of that vertex has the single nonzero entry 1 or $-e^{\mathring{\omega}_e}$, in the row of e, and expanding the determinant along it leaves, up to that entry and a sign, the determinant of the matrix of the subgraph without that vertex and copy. When only the cycle remains, the row of a copy $e : j \to i$ that p_k traverses against its orientation in E_+ is $-e^{\mathring{\omega}_e}$ times the row of the reversed copy $\bar{e}$, which has the entry 1 at j and $-e^{\mathring{\omega}_{\bar{e}}} = -e^{-\mathring{\omega}_e}$ at i. After these replacements, and with the m vertices and the m copies e_a of the cycle ordered along p_k, the matrix has the diagonal entries 1 and the entries $-e^{\mathring{\omega}_{e_a}}$ at the positions $(a, a+1)$ modulo m, or the single entry $1 - e^{\mathring{\omega}_{e'_k}}$ when $m = 1$. For $m \geq 2$ only the identity and the cyclic shift, of sign $(-1)^{m-1}$, contribute to its determinant, which is therefore $1 - e^{\mathring{s}_k}$, as it is for $m = 1$; for $m \geq 2$ its square is (G11.14) at $K = 1$ with unit weights. So $|\det \mathbf{T}_{E'_k}|$ is $|1 - e^{\mathring{s}_k}|$ times a product of factors $e^{\mathring{\omega}_e}$ over

distinct copies e, and

$$e^{-\sum_{e\in E_+}|\mathring{\omega}_e|}|1-e^{\mathring{s}_k}| \le |\det \mathbf{T}_{E'_k}(\mathring{\omega})| \le e^{\sum_{e\in E_+}|\mathring{\omega}_e|}|1-e^{\mathring{s}_k}|.$$

The linear map $\mathring{\omega} \mapsto (\mathring{s}, \mathring{\omega}_{\mathrm{tr}})$, with $\mathring{s} = (\mathring{s}_k)_k \in \mathbb{R}^{\mathrm{r}}$ and $\mathring{\omega}_{\mathrm{tr}} = (\mathring{\omega}_e)_{e\in E_{\mathrm{tr}}}$, is invertible, because p_k traverses e'_k once and no other copy outside the tree, so that $\mathring{s}_k$ is $\mathring{\omega}_{e'_k}$ plus a signed sum of tree generators.

(i) Keeping the terms $E' = E'_k$ of the Cauchy–Binet sum, with $w_{\min} = \min_e w_e$,

$$\det \Lambda(\mathring{\omega}) \ge w_{\min}^N e^{-2\sum_e|\mathring{\omega}_e|} \sum_{k=1}^{\mathrm{r}} \big(1-e^{\mathring{s}_k}\big)^2 \ge w_{\min}^N e^{-2-2\sum_e|\mathring{\omega}_e|} \min\big(|\mathring{s}|^2, 1\big),$$

because $|1-e^{\mathrm{u}}| \ge e^{-1}\min(|\mathrm{u}|,1)$ for real u, by the mean value theorem for $|\mathrm{u}| \le 1$ and by $|1-e^{\mathrm{u}}| \ge 1-e^{-1} > e^{-1}$ otherwise, and because $\sum_k \min(\mathring{s}_k^2, 1) \ge \min(|\mathring{s}|^2, 1)$, the left side being at least one if some $|\mathring{s}_k| \ge 1$ and equal to $|\mathring{s}|^2$ otherwise. So $\Lambda(\mathring{\omega}) \succ 0$ off the subspace $\{\mathring{s} = 0\}$, which has codimension $\mathrm{r} \ge 1$ and is Lebesgue-null, and there $\det \Lambda^{-1/2} \le w_{\min}^{-N/2} e^{1+\sum_e|\mathring{\omega}_e|} \max(|\mathring{s}|^{-1}, 1)$. Since the map above is a linear isomorphism and ν is Gaussian with positive-definite covariance, there are constants $\mathrm{c}_1, \mathrm{c}_2, \mathrm{c}_3 > 0$ such that the density of ν is at most $\mathrm{c}_1 e^{-\mathrm{c}_2(|\mathring{s}|^2+|\mathring{\omega}_{\mathrm{tr}}|^2)}$ and $\sum_e|\mathring{\omega}_e| \le \mathrm{c}_3(|\mathring{s}| + |\mathring{\omega}_{\mathrm{tr}}|)$. In the coordinates $(\mathring{s}, \mathring{\omega}_{\mathrm{tr}})$, whose Jacobian is constant, $\int \nu(d\mathring{\omega}) \det \Lambda^{-1/2}$ is therefore at most a constant times

$$\int_{\mathbb{R}^{N-1}} e^{-\mathrm{c}_2|\mathring{\omega}_{\mathrm{tr}}|^2 + \mathrm{c}_3|\mathring{\omega}_{\mathrm{tr}}|} d\mathring{\omega}_{\mathrm{tr}} \int_{\mathbb{R}^{\mathrm{r}}} e^{-\mathrm{c}_2|\mathring{s}|^2 + \mathrm{c}_3|\mathring{s}|} \max\big(|\mathring{s}|^{-1}, 1\big) d\mathring{s},$$

the first factor being absent when $N = 1$. The first integral is finite, and so is the part of the second over $|\mathring{s}| > 1$; on $|\mathring{s}| \le 1$ the second integrand is at most $e^{\mathrm{c}_3}|\mathring{s}|^{-1}$, whose integral there is the area of the unit sphere of $\mathbb{R}^{\mathrm{r}}$ times $\int_0^1 \mathrm{u}^{\mathrm{r}-2} d\mathrm{u}$, finite because $\mathrm{r} \ge 2$. So $\mathfrak{Z} < \infty$, and $\mathfrak{Z} > 0$ because its integrand is positive. (ii) If $\mathrm{r} = 1$, then $|E_+| = N$, the Cauchy–Binet sum has the single term $E' = E_+ = E'_1$, and $\det \Lambda \le \big(\prod_e w_e\big) e^{2\sum_e|\mathring{\omega}_e|}(1-e^{\mathring{s}_1})^2$. On the bounded set where $|\mathring{s}_1| \le 1$ and $|\mathring{\omega}_{\mathrm{tr}}| \le 1$, the mean value theorem gives $|1-e^{\mathring{s}_1}| \le e|\mathring{s}_1|$, the sum $\sum_e|\mathring{\omega}_e|$ is bounded, and the density of ν is bounded below by a positive constant, so $\mathfrak{Z}$ is at least a positive multiple of $\int_{-1}^1 |\mathrm{u}|^{-1} d\mathrm{u} = +\infty$. □

Proposition G11.8 (Hopping expansion of the induced action). In the setting of Proposition G11.5 with every $A_i \succ 0$ and $\mathsf{h} = 0$, let $\Lambda^{\mathrm{diag}}(\mathring{\omega}) = \mathrm{blockdiag}(\Lambda_i^{\mathrm{diag}})$, where $\Lambda_i^{\mathrm{diag}}$ is A_i plus W_e for every $e \in E_+$ ending at i and $e^{\mathring{\omega}_e^\top} W_e e^{\mathring{\omega}_e}$ for every $e \in E_+$ starting at i, a loop counting in both sums, and let $\Lambda^{\mathrm{hop}}(\mathring{\omega}) = \Lambda^{\mathrm{diag}}(\mathring{\omega}) - \Lambda(\mathring{\omega})$. For a copy $e \notin E_+$ put $W_e = e^{\mathring{\omega}_{\bar{e}}^\top} W_{\bar{e}} e^{\mathring{\omega}_{\bar{e}}}$, the reversal law of Proposition 11.19(i). (i) For every $\mathring{\omega}$, $\mathrm{spr}((\Lambda^{\mathrm{diag}})^{-1}\Lambda^{\mathrm{hop}}) \le 1 - \min_i \min_{z\neq 0} z^\top A_i z / z^\top \Lambda_i^{\mathrm{diag}} z < 1$, with spr the spectral radius. (ii) The induced action has the absolutely convergent expansion

$$\mathscr{S}^{\mathrm{ind}}(\mathring{\omega}) = \tfrac{1}{2}\log\det\Lambda^{\mathrm{diag}}(\mathring{\omega}) - \tfrac{1}{2}\sum_{r\ge 1}\frac{1}{r}\mathrm{Tr}\big[\big((\Lambda^{\mathrm{diag}})^{-1}\Lambda^{\mathrm{hop}}\big)^r\big] - \tfrac{NK}{2}\log 2\pi. \qquad \text{(G11.16)}$$

(iii) The r-th trace is a sum over the closed walks $(e_1,\dots,e_r)$ of length r, with $e_b : i_b \to i_{b-1}$, $i_r = i_0$, and copies of either orientation, loops included,

$$\operatorname{Tr}\big[\big((\Lambda^{\mathrm{diag}})^{-1}\Lambda^{\mathrm{hop}}\big)^r\big] = \sum_{(e_1,\dots,e_r)} \operatorname{Tr}\Big((\Lambda^{\mathrm{diag}}_{i_0})^{-1} W_{e_1} e^{\mathring{\omega}_{e_1}} (\Lambda^{\mathrm{diag}}_{i_1})^{-1} W_{e_2} e^{\mathring{\omega}_{e_2}} \cdots (\Lambda^{\mathrm{diag}}_{i_{r-1}})^{-1} W_{e_r} e^{\mathring{\omega}_{e_r}}\Big), \tag{G11.17}$$

a weighted Wilson loop: the trace of the reference holonomy $e^{\mathring{\omega}_{e_1}}\cdots e^{\mathring{\omega}_{e_r}}$ of the walk with a weight inserted at every vertex. (iv) If $A_i = a_i \mathsf{G}$ and $W_e = w_e \mathsf{G}$ with scalars $a_i, w_e > 0$, and every link is an isometry of G, then $\Lambda^{\mathrm{diag}}_i = a^{\mathrm{diag}}_i \mathsf{G}$, with a^{diag}_i equal to a_i plus the sum of w_e over the copies $e \in E_+$ incident to i, a loop counted twice; Λ^{diag} does not depend on the links, and the walk $(e_1,\dots,e_r)$ contributes $\big(\prod_b w_{e_b}/a^{\mathrm{diag}}_{i_{b-1}}\big)\operatorname{Tr}(e^{\mathring{\omega}_{e_1}}\cdots e^{\mathring{\omega}_{e_r}})$. A walk that reduces to the trivial walk by cancelling backtracks $(e,\bar e)$ contributes a link-independent constant. (v) For the triangle, three agents i_0, i_1, i_2 joined by three copies that form one cycle p and by no other copy, in the isotropic isometric setting of (iv) and with $w^{\mathrm{hop}} = \max_e w_e / \min_i a^{\mathrm{diag}}_i < \frac{1}{2}$,

$$\mathscr{S}^{\mathrm{ind}}(\mathring{\omega}) - \mathscr{S}^{\mathrm{ind}}(0) = \frac{w_{e_0} w_{e_1} w_{e_2}}{a^{\mathrm{diag}}_{i_0} a^{\mathrm{diag}}_{i_1} a^{\mathrm{diag}}_{i_2}} \mathfrak{Q}_p(\mathring{\omega}) + \mathfrak{o}(\mathring{\omega}), \qquad |\mathfrak{o}(\mathring{\omega})| \le \frac{K}{5}\Big(\frac{(2w^{\mathrm{hop}})^5}{1-2w^{\mathrm{hop}}} + \frac{2(w^{\mathrm{hop}})^5}{1-w^{\mathrm{hop}}}\Big), \tag{G11.18}$$

where $\mathfrak{Q}_p = K - \operatorname{Tr}\mathring{\Theta}_p$ for isometric holonomy. [ESTABLISHED]

Proof. (i) Each energy term $(z_i - e^{\mathring{\omega}_e} z_j)^\top W_e (z_i - e^{\mathring{\omega}_e} z_j)$ contributes W_e and $e^{\mathring{\omega}_e^\top} W_e e^{\mathring{\omega}_e}$ to the diagonal blocks, collected in Λ^{diag}, and $-W_e e^{\mathring{\omega}_e}$ and $-e^{\mathring{\omega}_e^\top} W_e$ to the blocks (i,j) and (j,i), so Λ^{hop} has blocks $W_e e^{\mathring{\omega}_e}$ and $e^{\mathring{\omega}_e^\top} W_e$. The same terms with the sign of every link reversed, $(z_i + e^{\mathring{\omega}_e} z_j)^\top W_e (z_i + e^{\mathring{\omega}_e} z_j)$, assemble $\Lambda^{\mathrm{diag}} + \Lambda^{\mathrm{hop}}$. Both forms are sums of positive semidefinite terms plus $z^\top A_\oplus z$, so $\Lambda^{\mathrm{diag}} \pm \Lambda^{\mathrm{hop}} \succeq A_\oplus$. With the symmetric matrix $\breve{\Lambda}^{\mathrm{hop}} = (\Lambda^{\mathrm{diag}})^{-1/2}\Lambda^{\mathrm{hop}}(\Lambda^{\mathrm{diag}})^{-1/2}$ and $\breve{A}_\oplus = (\Lambda^{\mathrm{diag}})^{-1/2} A_\oplus (\Lambda^{\mathrm{diag}})^{-1/2}$, this reads $-(I - \breve{A}_\oplus) \preceq \breve{\Lambda}^{\mathrm{hop}} \preceq I - \breve{A}_\oplus$, and $(\Lambda^{\mathrm{diag}})^{-1}\Lambda^{\mathrm{hop}} = (\Lambda^{\mathrm{diag}})^{-1/2}\breve{\Lambda}^{\mathrm{hop}}(\Lambda^{\mathrm{diag}})^{1/2}$ has the spectrum of $\breve{\Lambda}^{\mathrm{hop}}$. The block-diagonal matrix $\breve{A}_\oplus$ has smallest eigenvalue $\min_i \min_{z\neq 0} z^\top A_i z / z^\top \Lambda^{\mathrm{diag}}_i z$, by the Rayleigh quotient in the variable $(\Lambda^{\mathrm{diag}}_i)^{1/2} z$, and it lies in $(0,1]$ because $\Lambda^{\mathrm{diag}}_i \succeq A_i \succ 0$. (ii) $\Lambda = (\Lambda^{\mathrm{diag}})^{1/2}(I - \breve{\Lambda}^{\mathrm{hop}})(\Lambda^{\mathrm{diag}})^{1/2}$, and for the symmetric matrix $\breve{\Lambda}^{\mathrm{hop}}$ with eigenvalues $\mathring{\lambda}_k \in (-1,1)$, $\log\det(I - \breve{\Lambda}^{\mathrm{hop}}) = \sum_k \log(1-\mathring{\lambda}_k) = -\sum_{r\ge 1}\frac{1}{r}\operatorname{Tr}(\breve{\Lambda}^{\mathrm{hop}})^r$, absolutely convergent because $\sum_r \sum_k |\mathring{\lambda}_k|^r / r < \infty$; similarity gives $\operatorname{Tr}(\breve{\Lambda}^{\mathrm{hop}})^r = \operatorname{Tr}((\Lambda^{\mathrm{diag}})^{-1}\Lambda^{\mathrm{hop}})^r$, and (G11.12) at $\mathsf{h} = 0$ gives (G11.16). (iii) The block (i,j) of $(\Lambda^{\mathrm{diag}})^{-1}\Lambda^{\mathrm{hop}}$ is $(\Lambda^{\mathrm{diag}}_i)^{-1}$ times the sum of $W_e e^{\mathring{\omega}_e}$ over the copies $e : j \to i$ of either orientation: for $e \in E_+$ it was computed in (i), and for the reversal of $e \in E_+$ the block $e^{\mathring{\omega}_e^\top} W_e$ equals $W_{\bar e} e^{\mathring{\omega}_{\bar e}}$ with $W_{\bar e} = e^{\mathring{\omega}_e^\top} W_e e^{\mathring{\omega}_e}$ and $\mathring{\omega}_{\bar e} = -\mathring{\omega}_e$. The diagonal blocks of the r-th power are sums over vertex sequences of products of these blocks, which is (G11.17). (iv) An isometric link satisfies $e^{\mathring{\omega}_e^\top} W_e e^{\mathring{\omega}_e} = w_e \mathsf{G}$, and so does every reversed weight, which gives $\Lambda^{\mathrm{diag}}_i$ and the factors $(\Lambda^{\mathrm{diag}}_{i_{b-1}})^{-1} W_{e_b} e^{\mathring{\omega}_{e_b}} = (w_{e_b}/a^{\mathrm{diag}}_{i_{b-1}}) e^{\mathring{\omega}_{e_b}}$. A backtrack contributes $e^{\mathring{\omega}_e} e^{-\mathring{\omega}_e} = I$ to the product for every link value. (v) On the triangle every closed walk winds $\mathbf{k} \in \mathbb{Z}$ times around p, and after cancelling backtracks its holonomy is conjugate to $\mathring{\Theta}_p^{\mathbf{k}}$, so it contributes its weight times $\operatorname{Tr}\mathring{\Theta}_p^{\mathbf{k}}$. A walk of length r has $r \equiv \mathbf{k} \pmod 2$ and $r \ge 3|\mathbf{k}|$, so in $\mathscr{S}^{\mathrm{ind}}(\mathring{\omega}) - \mathscr{S}^{\mathrm{ind}}(0)$, where Λ^{diag} cancels, the walks with $\mathbf{k} = 0$ cancel, no walk of length four contributes, and the walks of length three are the six traversals of p, three starting points in two orientations, all of weight $w_{e_0} w_{e_1} w_{e_2} / (a^{\mathrm{diag}}_{i_0} a^{\mathrm{diag}}_{i_1} a^{\mathrm{diag}}_{i_2})$. Their term in (G11.16) is $-\frac{1}{2}\cdot\frac{1}{3}\cdot 6$ times

that weight times $\operatorname{Tr}\mathring{\Theta}_p - K$, using $\operatorname{Tr}\mathring{\Theta}_p^{-1} = \operatorname{Tr}(\mathsf{G}^{-1}\mathring{\Theta}_p^{\top}\mathsf{G}) = \operatorname{Tr}\mathring{\Theta}_p$ for an isometry, and for an isometry $\frac{1}{2}\langle\mathring{\Theta}_p - I, \mathring{\Theta}_p - I\rangle_{\mathsf{G}} = K - \operatorname{Tr}\mathring{\Theta}_p$. The remaining walks have length $r \geq 5$; the triangle has $2^r + 2(-1)^r \leq 2^r + 2$ closed walks of length r, the eigenvalues of its adjacency matrix being $2, -1, -1$, each of weight at most $(w^{\mathrm{hop}})^r$, and $|\operatorname{Tr}\mathring{\Theta}_p^{\mathbf{k}} - K| \leq 2K$ because an isometry has eigenvalues of modulus one. Summing $\frac{1}{2r}(2^r + 2)(w^{\mathrm{hop}})^r \cdot 2K \leq \frac{K}{5}((2w^{\mathrm{hop}})^r + 2(w^{\mathrm{hop}})^r)$ over $r \geq 5$ gives the bound. □

The expansion is the Gaussian-matter form of the induced gauge action of Kazakov and Migdal [19, Sec. 2, Eqs. (2)–(4)], in which integrating out a field coupled to the links gives one half the logarithm of the determinant of its quadratic operator and an expansion over closed lattice loops; here the field is the belief fiber, and the leading curvature term is the triangle's loop defect with the product of the hopping ratios $w_{e_a}/a_{i_a}^{\mathrm{diag}}$ along the cycle as coefficient. For the connection Laplacian of a line bundle the determinant, and for a rank-two bundle with links in SL_2 its quaternion determinant, has a finite expansion over cycle-rooted spanning forests [20, Theorem 1 and Theorem 9]; Kenyon lists a combinatorial interpretation in higher rank as an open question [20, Sec. 14], and no finite expansion of $\det\Lambda(\mathring{\omega})$ for $K \geq 3$ is stated here.

G11.5 An explicit spectrum of the linearized link operator

On a ring of agents with isotropic data, the linearized operator of Proposition 9.34 has an explicit spectrum.

Proposition G11.9 (The linearized operator on a ring of agents and episodes). Take a ring of $N \geq 2$ agents $i \in \mathbb{Z}_N$, with copies $e_i : i+1 \to i$, $E_+ = \{e_0, \dots, e_{N-1}\}$, and the closed walk $p = (e_0, \dots, e_{N-1})$, which spans the cycle space of the ring, of dimension one; the learning episodes of Definition 13.26, indexed here by $\mathring{n} = 0, \dots, \mathring{n}_1$ with $\mathring{n}_1 \geq 0$; and generator histories $(\mathring{\omega}_{e_i}^{(\mathring{n})})$ in the history space $\mathbb{R}^N \otimes \mathbb{R}^{\mathring{n}_1+1} \otimes \mathfrak{g}$ of edge index, episode index, and generator, with the inner product $\sum_{\mathring{n},i}\langle\cdot,\cdot\rangle_{\mathsf{G}}$. Let $\mathrm{a} > 0$ and $\mathrm{b}, \mathrm{c}, \mathbf{m} \geq 0$, and take the isotropic data of the anchor $\mathring{\alpha} = 0$ with prior covariance $\mathring{\Sigma}_{e_i} = \mathrm{a}^{-1}\mathrm{Id}$, the loop coefficient c of p, and a matter mass b times the identity at every episode, and the increment precision $\mathbf{m}$ between successive episodes, with the history action

$$\mathscr{S}^{\mathrm{hist}}(\mathring{\omega}) = \sum_{\mathring{n}=0}^{\mathring{n}_1}\Big[\tfrac{\mathrm{a}+\mathrm{b}}{2}\sum_i \big\|\mathring{\omega}_{e_i}^{(\mathring{n})}\big\|_{\mathsf{G}}^2 + \mathrm{c}\mathfrak{Q}_p\big(\mathring{\omega}^{(\mathring{n})}\big)\Big] + \tfrac{\mathbf{m}}{2}\sum_{\mathring{n}=1}^{\mathring{n}_1}\sum_i \big\|\mathring{\omega}_{e_i}^{(\mathring{n})} - \mathring{\omega}_{e_i}^{(\mathring{n}-1)}\big\|_{\mathsf{G}}^2, \tag{G11.19}$$

which for $\mathbf{m} > 0$ is, up to a constant, the negative logarithm of the density of the history prior of Definition 9.31 with the Gaussian increment as kinetic term, plus the isotropic second-order matter term. Let $\mathbf{K}$ be its Hessian at $\mathring{\omega} = 0$, the operator of Proposition 9.34 for these data with the matter operator bId. (i) As $\mathring{\omega} \to 0$, $\mathfrak{Q}_p(\mathring{\omega}) = \frac{1}{2}\|\mathbf{C}\mathring{\omega}\|_{\mathsf{G}}^2 + O(\|\mathring{\omega}\|^3)$ with the cycle map $\mathbf{C}\mathring{\omega} = \sum_i \mathring{\omega}_{e_i}$, so the loop part of $\mathbf{K}$ is $\mathrm{c}\mathbf{C}^*\mathbf{C}$, which acts on the edge index as the rank-one matrix $\mathrm{c}\mathbf{1}\mathbf{1}^{\top}$ and not as a Laplacian of the ring. (ii) The operator is

$$\mathbf{K} = \Big[(\mathrm{a}+\mathrm{b})I_N \otimes I_{\mathring{n}_1+1} + \mathrm{c}\mathbf{1}\mathbf{1}^{\top} \otimes I_{\mathring{n}_1+1} + \mathbf{m}I_N \otimes L_{\mathring{n}_1}\Big] \otimes \mathrm{Id}_{\mathfrak{g}}, \tag{G11.20}$$

with $L_{\mathring{n}_1}$ the Laplacian of the path on $\{0,\dots,\mathring{n}_1\}$, equal to $2\mathrm{v}_{\mathring{n}} - \mathrm{v}_{\mathring{n}-1} - \mathrm{v}_{\mathring{n}+1}$ in the interior and to $\mathrm{v}_0 - \mathrm{v}_1$ and $\mathrm{v}_{\mathring{n}_1} - \mathrm{v}_{\mathring{n}_1-1}$ at the ends. Its eigenvalues are

$$\mathrm{a}+\mathrm{b}+\mathrm{c}N+4\mathbf{m}\sin^2\frac{\pi l}{2(\mathring{n}_1+1)} \quad \text{and} \quad \mathrm{a}+\mathrm{b}+4\mathbf{m}\sin^2\frac{\pi l}{2(\mathring{n}_1+1)}, \qquad l=0,\dots,\mathring{n}_1, \tag{G11.21}$$

and for each l, with the episode profile $\mathrm{v}^{(l)} \in \mathbb{R}^{\mathring{n}_1+1}$ of entries $\mathrm{v}^{(l)}_{\mathring{n}} = \cos(\pi l(\mathring{n}+\frac{1}{2})/(\mathring{n}_1+1))$, $\mathbf{K}$ acts by the first value on the edge-uniform subspace $\mathbf{1}\otimes\mathrm{v}^{(l)}\otimes\mathfrak{g}$, of dimension K^2, and by the second on the zero-edge-sum subspace $\mathbf{1}^\perp\otimes\mathrm{v}^{(l)}\otimes\mathfrak{g}$, of dimension $(N-1)K^2$. These $2(\mathring{n}_1+1)$ subspaces are orthogonal and span the history space, so the multiplicity of an eigenvalue of $\mathbf{K}$ is the sum of the dimensions of the subspaces on which $\mathbf{K}$ acts by it, larger than K^2 or $(N-1)K^2$ when values coincide, as when $\mathrm{c}=0$ or $\mathbf{m}=0$. The discrete Fourier basis of $\mathbb{Z}_N$ is one orthonormal eigenbasis in the edge index. (iii) Every eigenvalue is at least $\mathrm{a}+\mathrm{b}\geq\mathrm{a}>0$, attained by profiles constant in the episode index whose edge sum vanishes; there is no zero or negative mode. (iv) For $\mathbf{m}>0$ and either spatial sector, with q its constant part $\mathrm{a}+\mathrm{b}+\mathrm{c}N$ or $\mathrm{a}+\mathrm{b}$, the homogeneous episode recursion $\mathbf{m}(2\mathrm{v}_{\mathring{n}}-\mathrm{v}_{\mathring{n}-1}-\mathrm{v}_{\mathring{n}+1})+\mathrm{q}\mathrm{v}_{\mathring{n}}=0$ has the characteristic roots $1+\frac{\mathrm{q}}{2\mathbf{m}}\pm\big((1+\frac{\mathrm{q}}{2\mathbf{m}})^2-1\big)^{1/2}$, real, positive, and reciprocal, so its solutions are combinations of one increasing and one decreasing geometric sequence and none oscillates. [ESTABLISHED]

Proof. In (9.51) with these data, the base law $\mathcal{N}(0,\mathrm{a}^{-1}\mathrm{Id})$ of each copy contributes $\frac{\mathrm{a}}{2}\|\mathring{\omega}^{(\mathring{n})}_{e_i}\|^2_{\mathsf{G}}$ to the negative logarithm of the density up to a constant, the action contributes $\mathrm{c}\mathfrak{Q}_p$ at every episode, and the Gaussian increment with mass $\mathbf{m}$ contributes the last sum of (G11.19), which gives the identification for $\mathbf{m}>0$. (i) $e^{\mathbf{B}}=I+\mathbf{B}+O(\|\mathbf{B}\|^2)$ gives $\mathring{\Theta}_p - I=\sum_i\mathring{\omega}_{e_i}+O(\|\mathring{\omega}\|^2)$, and $\mathfrak{Q}_p$ is half its squared G-norm; the Hessian of $\frac{\mathrm{c}}{2}\|\sum_i\mathring{\omega}_{e_i}\|^2_{\mathsf{G}}$ is $\mathrm{c}\mathbf{C}^*\mathbf{C}$, which sends $(\mathring{\omega}_{e_i})_i$ to the constant profile $\sum_i\mathring{\omega}_{e_i}$. (ii) The anchor and matter terms contribute $(\mathrm{a}+\mathrm{b})\mathrm{Id}$, the loop term $\mathrm{c}\mathbf{1}\mathbf{1}^\top$ in the edge index at every episode, and the increment term $\mathbf{m}L_{\mathring{n}_1}$ in the episode index for every edge, which is (G11.20); these are the summands $\mathring{\Sigma}^{-1}$, $\mathrm{c}\mathbf{C}_p^\top\mathbf{C}_p$, and $\mathbf{m}(-\Delta_n)$ of the operator of Proposition 9.34, with bId in place of its matter operator. The three Kronecker summands commute. The matrix $\mathbf{1}\mathbf{1}^\top$ has eigenvalue N on $\mathbf{1}$ and 0 on $\mathbf{1}^\perp$, and it is circulant, so the Fourier basis diagonalizes it. For $\phi=\pi l/(\mathring{n}_1+1)$ and $\mathrm{v}_{\mathring{n}}=\mathrm{v}^{(l)}_{\mathring{n}}=\cos(\phi(\mathring{n}+\frac{1}{2}))$, the identity $2\cos(\phi(\mathring{n}+\frac{1}{2}))-\cos(\phi(\mathring{n}-\frac{1}{2}))-\cos(\phi(\mathring{n}+\frac{3}{2}))=4\sin^2(\phi/2)\cos(\phi(\mathring{n}+\frac{1}{2}))$ gives the interior rows; at $\mathring{n}=0$, $\cos\frac{\phi}{2}-\cos\frac{3\phi}{2}=4\sin^2\frac{\phi}{2}\cos\frac{\phi}{2}$, and at $\mathring{n}=\mathring{n}_1$ the same identity holds after the substitution $\mathring{n}\mapsto\mathring{n}_1-\mathring{n}$, because $\phi(\mathring{n}_1+1)=\pi l$ multiplies both terms by $(-1)^l$. These $\mathring{n}_1+1$ nonzero vectors, with $\mathrm{v}^{(l)}_0>0$, are eigenvectors for distinct eigenvalues of the symmetric $L_{\mathring{n}_1}$, hence orthogonal, so they exhaust its spectrum, and eigenvalues of commuting summands add on the subspaces $\mathbf{1}\otimes\mathrm{v}^{(l)}\otimes\mathfrak{g}$ and $\mathbf{1}^\perp\otimes\mathrm{v}^{(l)}\otimes\mathfrak{g}$, which are orthogonal because $\mathbf{1}\perp\mathbf{1}^\perp$ and the $\mathrm{v}^{(l)}$ are orthogonal, and whose dimensions add up to $N(\mathring{n}_1+1)K^2$. (iii) Every term of (G11.21) beyond $\mathrm{a}+\mathrm{b}$ is nonnegative and vanishes for $l=0$ on $\mathbf{1}^\perp$, which is nonzero for $N\geq2$. (iv) The roots solve $\mathrm{u}^2-(2+\mathrm{q}/\mathbf{m})\mathrm{u}+1=0$, whose discriminant $(2+\mathrm{q}/\mathbf{m})^2-4$ is positive for $\mathrm{q}>0$; the product of the roots is 1 and their sum is positive. □

The ring is a validation instance, not a model: it shows that the operator of Proposition 9.34 is positive with an explicit gap, that the loop term of a ring is a single collective mode, and that the

episode index enters with the same sign as the anchor, so the most probable history relaxes monotonically between its ends.

G11.6 Open problems of the Gaussian link sector

Open Problem G11.10 (Joint normalization of unanchored matter with latent links). With positive-definite self terms the joint law of generators and undirected Gaussian matter exists for every finite base measure (Proposition G11.5), and without self terms an unanchored cycle admits no joint law (Corollary G11.6), because the partition function diverges like $|\det(I - \mathring{\Theta}_p)|^{-1}$. For $K = 1$, with positive weights, zero self terms, and zero information vector, the partition function of a connected multigraph of cycle rank at least two is finite under every Gaussian base law with positive-definite covariance (Proposition G11.7). On a connected multigraph of cycle rank at least two, the matter precision without self terms is singular exactly when the holonomy group has a nonzero common fixed vector (Proposition 11.19), and for $K \geq 2$ whether its partition function is integrable is not settled. It is settled, at a stated $K \geq 2$, for connected multigraphs of cycle rank at least two with positive-definite weights, zero self terms, zero information vector, and a product Gaussian base law of the generators with positive-definite covariances, by a proof that $\int \nu(d\mathring{\omega}) \det \Lambda(\mathring{\omega})^{-1/2} < \infty$ for every such multigraph and law, or by a multigraph and law in this class for which the integral diverges.

Open Problem G11.11 (Attainment of the chart infimum without a link prior). Without a link prior and for $K \geq 2$, the expected record energy l of Proposition G11.3 has its infimum over the exponential chart at the value of the rank-one minimizer $\mathbf{M}^\star$, unattained, when $\operatorname{Tr} \mathbf{M}^\star \geq 0$, and strictly above it when $\operatorname{Tr} \mathbf{M}^\star < 0$ (Proposition G11.2(iii)), where stationary points in the chart can sit at critical points of the exponential map without being minimizers. In the second case Proposition G11.3 excludes attainment at every chart link whose negative eigenvalues are semisimple, which covers every chart link for $K = 2$ and $K = 3$ and, for $K = 2$, agrees with Corollary 9.24(a)(iv). A chart link outside its reach has a negative eigenvalue with a Jordan block of size at least two, which occurs an even number of times by Culver [12, Theorem 1], so that $K \geq 4$; its proof perturbs semisimple negative eigenvalues to first order and does not reach such a link. It is settled, for $K \geq 4$, $W_e, \Sigma_j \succ 0$, and data with $\operatorname{Tr} \mathbf{M}^\star < 0$, by a proof that no chart link having a negative eigenvalue with a Jordan block of size at least two minimizes l over the chart links, which with Proposition G11.3 gives non-attainment for every K, or by data in this class and such a chart link that minimizes l over the chart links.

Appendix GA

Decision Metatheory

When is the question whether every admitted fine Gaussian has an admitted coarse presentation of its exact marginal decidable, and what does a decision procedure deliver? For families given by finite semialgebraic descriptions, real quantifier elimination decides it and constructs the exact good and failure loci (Theorem GA.4); the same holds for singular Gaussian laws on a declared quotient reference with variable rank (Theorem GA.12). The decision does not extend to families given by programs that enumerate them (Proposition GA.7); selectors are measurable but in general neither continuous nor equivariant (Proposition GA.6); and a change of rank destroys statistical regularity (Proposition GA.13). The appendix supplies the last two rows of Table G7.1 and the only treatment of singular Gaussian laws in this supplement; no result of the main manuscript depends on it. Notation follows Chapter G7; in addition x and y denote tuples of real variables of first-order formulas, x for the fine family and y for a retained presentation.

This appendix claims no complexity bound useful at any practical scale, no numerical sign tolerance, no decision procedure for parameters known only through approximations, no closed-form catalog of closed families, and no limit or fixed-point theorem for iterated compositions. The classification of families outside finite encodings is open (Open Problem GA.14). [NOT-CLAIMED]

GA.1 Finite semialgebraic classification

Definition GA.1 (Finite semialgebraic description). A finite semialgebraic description of a Kron problem fixes finitely many labeled nodes with finite coordinate spaces and coordinate Lebesgue references, a finite list of oriented edge copies and loops, and a finite list of retained coordinate sets. The fine parameters $x \in \mathbb{R}^p$ range over the set defined by a finite formula $\mathrm{Adm}(x)$, a Boolean combination of polynomial equalities and strict or weak inequalities, possibly with existentially quantified auxiliary variables, whose coefficients are real algebraic numbers, each given by a defining integer polynomial and a rational isolating interval. The fine precision $\Lambda(x)$ and source $\mathsf{h}(x)$ have entries that are rational functions of x on an explicitly specified nonzero-denominator domain, and every admitted $\Lambda(x)$ is SPD. A retained presentation is a tuple $y \in \mathbb{R}^q$ with a semialgebraic admissibility formula $\mathrm{Adm}_{\mathrm{c}}(y)$, rational coarse precision and source $\Lambda_{\mathrm{c}}(y)$ and $\mathsf{h}_{\mathrm{c}}(y)$, and a finite semialgebraic compatibility formula $\mathrm{Comp}(x, y)$. Correlated weights, nonconvex domains, variable anchors, nonflat links, polynomial path identities, and finite unions of graph inventories are allowed; singular or rank-changing supports are treated in Appendix GA.2.

Proposition GA.2 (Real quantifier elimination). There is an algorithm that transforms every first-order formula in the language of ordered fields with integer coefficients into a quantifier-free formula in the same free variables that is equivalent to it in every real closed field. In particular the truth of every such closed sentence over $\mathbb{R}$, and over the field of real algebraic numbers, is decidable, and the two truth values agree. [STANDARD]

This is Tarski's decision method for elementary algebra [29]; Basu et al. [2, pp. 533–562] treat quantifier elimination algorithmically, and Cohen and Mahboubi [11, Sec. 6] give a formally verified proof over real closed fields with decidable comparison. The decision theorem below uses only termination.

Lemma GA.3 (Polynomial encoding of the exact Gaussian marginal). Under Definition GA.1, with $M = \Lambda_{\mathrm{hh}}(x)$ for a nonempty hidden coordinate set, put

$$\begin{aligned} \widetilde{\Lambda}_{\mathrm{r}}(x) &= (\det M)\Lambda_{\mathrm{rr}} - \Lambda_{\mathrm{rh}}(\operatorname{adj} M)\Lambda_{\mathrm{hr}}, \\ \widetilde{\mathsf{h}}_{\mathrm{r}}(x) &= (\det M)\mathsf{h}_{\mathrm{r}} - \Lambda_{\mathrm{rh}}(\operatorname{adj} M)\mathsf{h}_{\mathrm{h}}, \\ (\det M)\Lambda_{\mathrm{c}}(y) = \widetilde{\Lambda}_{\mathrm{r}}(x), &\qquad (\det M)\mathsf{h}_{\mathrm{c}}(y) = \widetilde{\mathsf{h}}_{\mathrm{r}}(x), \end{aligned} \tag{GA.1}$$

and for an empty hidden set use $\det M = 1$, $\widetilde{\Lambda}_{\mathrm{r}} = \Lambda$, and $\widetilde{\mathsf{h}}_{\mathrm{r}} = \mathsf{h}$. Let $\mathrm{Rep}(x, y)$ be the conjunction of $\mathrm{Adm}_{\mathrm{c}}(y)$, $\mathrm{Comp}(x, y)$, the SPD condition on $\Lambda_{\mathrm{c}}(y)$, all denominator admissions, and the last line of (GA.1). After finitely many auxiliary variables are introduced, Rep and Adm are finite first-order formulas with integer coefficients, equivalent to the originals at every real parameter. For admitted x, $\mathrm{Rep}(x, y)$ holds exactly when $(\Lambda_{\mathrm{c}}(y), \mathsf{h}_{\mathrm{c}}(y))$ are the natural parameters of the exact normalized marginal of $\mathcal{N}(\Lambda(x)^{-1}\mathsf{h}(x), \Lambda(x)^{-1})$ on the retained coordinates and the presentation is admitted; that marginal precision is SPD,

$$z^{\top}\Lambda^{\mathrm{eff}}z = \min_{u}(z, u)^{\top}\Lambda(z, u) > 0, \qquad u_{\min} = -M^{-1}\Lambda_{\mathrm{hr}}z, \qquad z \neq 0. \tag{GA.2}$$

[ESTABLISHED]

Proof. The adjugate identity $M^{-1} = \operatorname{adj} M / \det M$ with Proposition G7.11 shows that the last line of (GA.1) is equality of marginal precision and source, and for nonsingular Gaussian laws this is equality of laws, since covariance determines precision and $\mathsf{h} = \Lambda\mu$ then determines the source, while the explicit densities give the converse. The admissions have polynomial encodings. A symmetric matrix is SPD exactly when its leading principal minors are positive: completing the square in the first coordinate gives a positive first pivot and a Schur block whose leading minors are the subsequent leading minors divided by that pivot, and induction proves the claim. A symmetric matrix is PSD exactly when it equals $TT^{\top}$ for some T, by orthogonal diagonalization and $v^{\top}TT^{\top}v = \|T^{\top}v\|^2$. Invertibility of a link is $(\det \Theta)^2 > 0$. A denominator q is replaced by a variable u with $uq = 1$, which keeps its nonzero domain; multiplying an equation by a nonzero denominator is valid, whereas an inequality needs this variable or a split into the two sign cases. An algebraic coefficient with defining polynomial f and isolating interval (a_0, b_0) is replaced by an existentially quantified variable v with $f(v) = 0$ and $a_0 < v < b_0$, and clearing the rational denominators leaves

integer coefficients; the equivalence holds at every real parameter because the constant is the unique root in its interval. Finally, the minimum in (GA.2) is attained at $u_{\min}$, and its minimizing vector has nonzero retained part. □

Theorem GA.4 (Exact finite semialgebraic Kron classification). Under Definition GA.1, there is an algorithm producing a quantifier-free formula $\mathrm{Good}(x)$ for exactly the admissible fine parameters whose exact marginal has an admitted retained presentation, and the failure locus, and deciding universal closure and nonemptiness:

$$\begin{aligned} \mathrm{Good}(x) &\iff \mathrm{Adm}(x) \wedge \exists y\big[\mathrm{Rep}(x,y)\big], \\ \mathrm{Fail}(x) &= \mathrm{Adm}(x) \wedge \neg\mathrm{Good}(x), \\ \forall x\big(\mathrm{Adm}(x) &\Rightarrow \exists y\big[\mathrm{Rep}(x,y)\big]\big), \qquad \exists x\big[\mathrm{Adm}(x)\big]. \end{aligned} \tag{GA.3}$$

The witnesses include the auxiliary inverse and Gram variables, and nonemptiness is decided separately, since vacuous closure is not an inhabited construction. A nonempty failure locus contains a point with real algebraic coordinates. The raw scalar is carried outside the formula by

$$\mathfrak{c}_{\mathrm{c}} = \mathfrak{c} + \frac{1}{2}\log\det M - \frac{\dim z_{\mathrm{h}}}{2}\log(2\pi) - \frac{1}{2}\mathsf{h}_{\mathrm{h}}^{\top} M^{-1}\mathsf{h}_{\mathrm{h}}, \tag{GA.4}$$

and the observed experiment of a linear Gaussian sensor with algebraic matrix data and SPD noise is included by keeping the observation coordinates among the retained ones. For a fixed finite acyclic diagram of retained coordinate sets, with one tuple y_v at each object and all arrow relations, path identities, and structural conditions collected in one formula $\mathrm{Rep}_{\mathrm{diag}}$, the same algorithm decides

$$\forall x\Big(\mathrm{Adm}(x) \Rightarrow \exists (y_v)_v \mathrm{Rep}_{\mathrm{diag}}\big(x,(y_v)_v\big)\Big). \tag{GA.5}$$

For a fixed finite list of rational operations on one declared domain, totality and universal one-step domain invariance imply admission of every finite composition. [ESTABLISHED]

Proof. By Lemma GA.3, Adm and Rep are finite integer-coefficient formulas, and $\mathrm{Rep}(x,y)$ holds exactly when y is an admitted presentation of the exact marginal, so Proposition GA.2 produces Good and Fail and decides the two closed sentences. The existential sentence $\exists x\big[\mathrm{Fail}(x)\big]$ has the same truth value over $\mathbb{R}$ and over the real algebraic numbers by the same proposition, so a nonempty failure locus has an algebraic point, which enumeration of finite tuples of algebraic numbers with exact sign evaluation of Fail eventually finds; the point need not be rational or small. The raw scalar is the third line of (G7.32); it is a transcendental identity rather than an ordered-field constraint, and conditions on logarithms, exponentials, or normalizers enter the decision only after a valid polynomial reduction has been supplied. For a linear sensor with SPD noise covariance Σ_o, the joint of latent and observation coordinates is a proper Gaussian with rational natural parameters, its quadratic being the prior quadratic plus $(o - \mathsf{C}z)^{\top}\Sigma_o^{-1}(o - \mathsf{C}z)$, so the classification applies to it; pushing the whole observed joint preserves its evidence and the conditional-KL identity of Section 10.2. For a diagram, one formula with shared variables is decided, whereas separate witnesses for each arrow would not give a compatible value at a shared

object; coordinate marginalization composes by the equality of iterated Gaussian integrals, sources and raw scalars included. The composition statement is induction on the number of operations, undefined denominators counting as failures of totality. □

Bounding an unmarked positive presentation. An unrestricted number of independently scalable unmarked summands can also be reduced to finite input. Let $\mathbb{R}^n = \bigoplus_{i\in V} \mathbb{R}^{d_i}$ with coordinate projections pr_i. An anchor contributes $\mathrm{pr}_i^\top A_i \,\mathrm{pr}_i$ with $A_i \succeq 0$, and for nodes of equal dimension a link contributes

$$\mathsf{Q}_e = (\mathrm{pr}_i - \Theta_e \,\mathrm{pr}_j)^\top W_e (\mathrm{pr}_i - \Theta_e \,\mathrm{pr}_j), \qquad W_e \succ 0, \quad \det \Theta_e \neq 0. \tag{GA.6}$$

Allowed endpoint types form a fixed finite list, every generator may be independently rescaled by a positive number, and unmarked copies may be deleted.

Proposition GA.5 (Finite positive pair-presentation bound). If $Q \in \mathbb{S}^n$ is a finite sum of the anchors and links just declared, it is such a sum with at most $n(n+1)/2$ nonzero summands, and with unrestricted PSD anchors it has at most one anchor at each node and no self-loops; every weight stays SPD and every link invertible. Membership in this unmarked presentation class is equivalent to one finite disjunction of existential polynomial systems, whose projection onto the entries of Q is semialgebraic and decidable at exact algebraic inputs. The scaling and deletion hypotheses are material: in one scalar dimension, generators restricted to the value 1 represent 2 with two copies but not within the bound 1. The class need not be closed, and replacing SPD weights by PSD ones changes membership: for two two-dimensional nodes and $u = (1,0)^\top$, the matrix

$$Q_\partial = \begin{pmatrix} u \\ -u \end{pmatrix} \begin{pmatrix} u \\ -u \end{pmatrix}^\top \tag{GA.7}$$

is a limit of single links with $\Theta = I$ and $W = \mathrm{diag}(1, \delta)$, $\delta \downarrow 0$, but is not a sum of admitted generators. [ESTABLISHED]

Proof. A loop is the PSD anchor $\mathrm{pr}_i^\top (I - \Theta_e)^\top W_e (I - \Theta_e)\,\mathrm{pr}_i$. Absorb unmarked loops into anchors, sum the anchors at each node, and delete zero terms. Reversing a non-loop orientation replaces (Θ, W) by $(\Theta^{-1}, \Theta^\top W \Theta)$ and preserves the residual quadratic, so one orientation suffices when the link domain is the whole general linear group; a restricted domain keeps its finite orientation types. Write the remaining sum as $Q = \sum_{r=1}^m t_r \mathsf{Q}_r$, initially with $t_r = 1$. If $m > n(n+1)/2 = \dim \mathbb{S}^n$, choose a nonzero dependence $\sum_r \xi_r \mathsf{Q}_r = 0$; every nonzero PSD matrix has positive trace, so the ξ_r have both signs, and with

$$\bar{t} = \min_{\xi_r > 0} \frac{t_r}{\xi_r}, \qquad t_r' = t_r - \bar{t}\xi_r, \tag{GA.8}$$

all $t_r' \geq 0$, one vanishes, and $\sum_r t_r' \mathsf{Q}_r = Q$. Repetition terminates, and each surviving coefficient is absorbed into its A_i or W_e without changing the admissions. This is Carathéodory's theorem [26, Thm. 17.1] applied to $Q/\operatorname{Tr} Q$ in the convex hull of the normalized generators $\mathsf{Q}_r/\operatorname{Tr} \mathsf{Q}_r$, which lie in an affine hyperplane of dimension $n(n+1)/2 - 1$. Only finitely many endpoint and anchor

patterns have at most $n(n+1)/2$ terms; for each, with symmetric anchor and weight variables and unrestricted link variables, impose

$$\begin{aligned} Q &= \sum_i \mathrm{pr}_i^\top A_i \,\mathrm{pr}_i + \sum_e (\mathrm{pr}_i - \Theta_e \,\mathrm{pr}_j)^\top W_e (\mathrm{pr}_i - \Theta_e \,\mathrm{pr}_j), \\ A_i &= F_i F_i^\top, \qquad \mathrm{Tr}(F_i F_i^\top) > 0, \\ &\text{all leading principal minors of } W_e \text{ positive}, \qquad (\det \Theta_e)^2 > 0, \end{aligned} \tag{GA.9}$$

with the factors F_i existential. By the encodings of Lemma GA.3 every solution is a permitted presentation, and the reduction puts every finite presentation into one listed pattern, so the disjunction is necessary and sufficient; quantifier elimination makes its projection semialgebraic. The scalar example has $\dim \mathbb{S}^1 = 1$. The matrix Q_∂ has rank one and a nonzero cross block, while every non-loop generator on those nodes has rank two, because $(I, -\Theta)$ has full row rank and $W \succ 0$; PSD summands of Q_∂ vanish on its kernel, so their ranges lie in its one-dimensional range, which excludes every non-loop term, and anchors and loops have zero cross blocks. The links with $W = \mathrm{diag}(1, \delta)$ converge to Q_∂. □

If $Q \succ 0$, repacking Q while carrying $\mathfrak{h}$, $\mathfrak{c}$, and the reference unchanged keeps the complete Gaussian action, its mass, and its normalized law, whereas independently normalized local factors introduce determinant scalars that must be compensated when raw equality is required. Individually marked edges, their inventories, and prescribed loop products cannot be discarded; such problems keep their finite inventory in Theorem GA.4, and the proper example of Proposition G7.17 excludes even the larger class of PSD two-node forms.

Proposition GA.6 (Measurable selection and structural obstructions). For the good locus of Theorem GA.4 there is a semialgebraic, hence Borel, selector s with $\mathrm{Rep}(x, s(x))$ for every x in the locus. Neither a continuous selector nor a gauge-equivariant structural selector follows from these hypotheses, even when every accompanying Gaussian law is proper and identical. For the parameter $(\cos t, \sin t) \in S^1$ and the polynomial condition $O \in \mathrm{SO}(2)$, $O^2 = R_t$, each fiber has two points and no continuous selection exists, while the state law may stay $\mathcal{N}(0, I_2)$. For the input covariance I_2 and a factor O with $OO^\top = I_2$, acted on by $O \mapsto gO$, $g \in \mathrm{SO}(2)$, no equivariant selector exists. [ESTABLISHED]

Proof. The semialgebraic sets are the definable sets of the real field, an o-minimal expansion of an ordered group, which has definable choice [32, Ch. 6, pp. 93–94]: the semialgebraic family of fibers $\{y : \mathrm{Rep}(x, y)\}$, x in the good locus, has a semialgebraic selector. A semialgebraic map is Borel, since preimages of rational open boxes are semialgebraic and such boxes form a countable base. The solutions of $O^2 = R_t$ in $\mathrm{SO}(2)$ are $\pm R_{t/2}$, so along $t \in [0, 2\pi]$ a continuous selection is $\sigma R_{t/2}$ with $\sigma \in \{I, -I\}$ continuous, hence constant, and then its value at 2π is minus its value at 0 although the parameter has returned. The covariance I_2 is fixed by every $g \in \mathrm{SO}(2)$, so an equivariant selector would satisfy $O = gO$ for every g, and $g = -I_2$ forces $O = 0$, contradicting $OO^\top = I_2$. □

The selected tuple is a measurable presentation of the exact marginal; it is not a smooth frame, an equivariant section, or a construction compatible with an unstated global composition law. Finite diagram compatibility is handled by (GA.5) with its shared variables.

Proposition GA.7 (Obstruction for computably enumerated domains). There is no terminating algorithm which, from arbitrary code for a total computable sequence of rational parameters, decides whether every member of the sequence has a positive pairwise coarse presentation. This remains true when the topology and node dimensions are fixed, every fine law is proper, and the range is promised to lie in one fixed two-point set; the obstruction concerns the input description, although every such range is finite and semialgebraic as a set. [ESTABLISHED]

Proof. Use the four-agent, two-dimensional star of Proposition G7.17. Let x_{bad} be its rational edge data with retained anchors δI, $\delta = 1/20000$, whose proper marginal has no positive one-node or two-node presentation. Let x_{good} have the same topology and anchors and all three star weights equal to I; the hidden pivot is $3I$, the retained blocks are $2I/3$ on the diagonal and $-I/3$ off it, and the complete retained graph with weights $I/3$ together with the anchors is an exact positive presentation. For a program $\mathtt{p}$ define

$$x_k(\mathtt{p}) = \begin{cases} x_{\mathrm{bad}}, & \mathtt{p} \text{ halts within } k \text{ steps}, \\ x_{\mathrm{good}}, & \text{otherwise}. \end{cases} \tag{GA.10}$$

Computing a term takes k simulation steps, so this is a total computable rational sequence with range in $\{x_{\mathrm{good}}, x_{\mathrm{bad}}\}$, and all its members admit positive pairwise presentations exactly when $\mathtt{p}$ never halts. An algorithm deciding the property would therefore decide halting. No machine decides whether an arbitrary machine ever prints a given symbol [31, Sec. 8], and a program that simulates a machine and halts when the machine first prints that symbol reduces this problem to halting, so no such algorithm exists. □

There is no conflict with Theorem GA.4: an explicit semialgebraic formula for the range is different input from code that enumerates it. The proposition does not exclude a theorem about a particular infinite family, a nonalgorithmic classification, or exact full-Gaussian marginalization.

GA.2 Finite Gaussian quotients with variable support

The full-SPD assumption of Theorem GA.4 can be replaced by an explicit quotient reference, which gives a finite classification that includes variable ranks.

Definition GA.8 (Quotient Gaussian data). Quotient Gaussian data on $\mathbb{R}^n$ consist of a symmetric PSD form Q, an SPD reference metric $\mathsf{G}_{\mathrm{ref}}$, a covector $\mathsf{h} \in \operatorname{range} Q$, an affine origin $z_0 \in \mathbb{R}^n$, and a real scalar $\mathfrak{c}$. The representative slice and the action on it are

$$\mathcal{S} = (\ker Q)^{\perp_{\mathsf{G}_{\mathrm{ref}}}}, \qquad \mathsf{E}(z_0 + s) = \mathfrak{c} + \frac{1}{2} s^{\top} Q s - \mathsf{h}^{\top} s, \quad s \in \mathcal{S}, \tag{GA.11}$$

with the volume induced by $\mathsf{G}_{\mathrm{ref}}$ on $\mathcal{S}$, translated to $z_0 + \mathcal{S}$, and unit mass on the point in dimension zero. Since Q is symmetric, $\operatorname{range} Q = (\ker Q)^{\perp}$, so both energies descend to $\mathbb{R}^n / \ker Q$. The metric, origin, and volume specify a law on a representative slice, not a normalization of a singular ambient Lebesgue density; a law on the discarded null coordinates would define a different completion.

Proposition GA.9 (Polynomial quotient covariance). For every such Q and $\mathsf{G}_{\mathrm{ref}}$ there is exactly one symmetric Σ_Q with

$$Q\Sigma_Q Q = Q, \qquad \Sigma_Q Q \Sigma_Q = \Sigma_Q, \qquad \mathsf{G}_{\mathrm{ref}} \Sigma_Q Q = Q \Sigma_Q \mathsf{G}_{\mathrm{ref}}. \tag{GA.12}$$

With the whitened forms

$$\widehat{Q} = \mathsf{G}_{\mathrm{ref}}^{-1/2} Q \mathsf{G}_{\mathrm{ref}}^{-1/2}, \qquad \widehat{\Sigma} = \mathsf{G}_{\mathrm{ref}}^{1/2} \Sigma_Q \mathsf{G}_{\mathrm{ref}}^{1/2}, \tag{GA.13}$$

it is

$$\Sigma_Q = \mathsf{G}_{\mathrm{ref}}^{-1/2} \widehat{Q}^{+} \mathsf{G}_{\mathrm{ref}}^{-1/2}, \tag{GA.14}$$

where $\widehat{Q}^{+}$ inverts the positive eigenvalues and vanishes on the kernel. Σ_Q is PSD with range $\mathcal{S}$, and $\Pi_Q = \Sigma_Q Q$ is the $\mathsf{G}_{\mathrm{ref}}$-orthogonal projection onto $\mathcal{S}$ with kernel $\ker Q$. The equations (GA.12) are the defining equations of the weighted Moore–Penrose inverse of Q with weights $\mathsf{G}_{\mathrm{ref}}^{-1}$ and $\mathsf{G}_{\mathrm{ref}}$, because for symmetric Σ_Q and Q both weighted symmetry conditions reduce to the third equation, and they specify the covariance at every rank. [ESTABLISHED]

Proof. The SPD square root of $\mathsf{G}_{\mathrm{ref}}$ exists by orthogonal diagonalization, and in the variables (GA.13) the equations become $\widehat{Q}\widehat{\Sigma}\widehat{Q} = \widehat{Q}$, $\widehat{\Sigma}\widehat{Q}\widehat{\Sigma} = \widehat{\Sigma}$, and $\widehat{\Sigma}\widehat{Q} = \widehat{Q}\widehat{\Sigma}$, with $\widehat{\Sigma}$ symmetric. Decompose into the positive eigenspace of $\widehat{Q}$ and its kernel; commutation makes the off-diagonal blocks vanish, the first equation gives $\widehat{\Sigma} = \widehat{Q}^{-1}$ on the positive block, and the second gives $\widehat{\Sigma} = 0$ on the kernel. This proves uniqueness, (GA.14), and positivity. In whitened coordinates $\widehat{\Sigma}\widehat{Q}$ is the Euclidean orthogonal projection onto $\operatorname{range} \widehat{Q}$, and since $\ker \widehat{Q} = \mathsf{G}_{\mathrm{ref}}^{1/2} \ker Q$, undoing the isometry $s \mapsto \mathsf{G}_{\mathrm{ref}}^{1/2} s$ gives the stated projection and range. For the weighted conditions, multiplying $(\mathsf{G}_{\mathrm{ref}}^{-1} Q \Sigma_Q)^{\top} = \mathsf{G}_{\mathrm{ref}}^{-1} Q \Sigma_Q$ on both sides by $\mathsf{G}_{\mathrm{ref}}$ gives the third equation, and $(\mathsf{G}_{\mathrm{ref}} \Sigma_Q Q)^{\top} = \mathsf{G}_{\mathrm{ref}} \Sigma_Q Q$ is the third equation itself. □

The PSD and SPD admissions have the finite encodings of Lemma GA.3, and $\mathsf{h} \in \operatorname{range} Q$ is the existential equation $Qv = \mathsf{h}$; the square root and the spectral inverse fix the meaning of the equations and are not operations of the polynomial input.

Theorem GA.10 (Normalized quotient Gaussian law). Let $r = \operatorname{rank} Q$ and let $\det_{+}(\widehat{Q})$ be the product of the positive eigenvalues of $\widehat{Q}$, with empty product one. Relative to the volume on $z_0 + \mathcal{S}$, the action (GA.11) has positive finite mass and defines the Gaussian probability measure with

ambient mean μ, covariance Σ_Q, and characteristic function

$$\begin{aligned} Z_Q &= e^{-\mathfrak{c}}(2\pi)^{r/2}\det{}_+(\widehat{Q})^{-1/2}\exp\Big(\frac{1}{2}\mathsf{h}^\top\Sigma_Q\mathsf{h}\Big),\\ \mu &= z_0+\Sigma_Q\mathsf{h},\\ \widehat{\mathbb{P}}_Q(\xi) &= \exp\Big(i\xi^\top\mu-\frac{1}{2}\xi^\top\Sigma_Q\xi\Big). \end{aligned} \tag{GA.15}$$

For $Q=0$ these give $\mathsf{h}=\Sigma_Q=0$, $Z_Q=e^{-\mathfrak{c}}$, and the point mass at z_0. Under an invertible passive change $z'=gz$ with all data transformed,

$$Q'=g^{-\top}Qg^{-1},\quad \mathsf{G}'_{\mathrm{ref}}=g^{-\top}\mathsf{G}_{\mathrm{ref}}g^{-1},\quad \mathsf{h}'=g^{-\top}\mathsf{h},\quad z'_0=gz_0, \tag{GA.16}$$

one has $\Sigma'_Q=g\Sigma_Qg^\top$, $\mu'=g\mu$, and $\mathbb{P}'=g_\#\mathbb{P}$, with Z_Q and $\mathfrak{c}$ unchanged. The reference metric must transform with the data: for $Q=\mathrm{diag}(1,0)$, $\mathsf{G}_{\mathrm{ref}}=I_2$, and $g=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, the pushed covariance is $\mathrm{diag}(1,0)$, while resetting the transformed metric to I_2 gives

$$Q'=\begin{pmatrix}1&-1\\-1&1\end{pmatrix},\qquad (Q')^+_{\mathrm{Euclidean}}=\frac{1}{4}\begin{pmatrix}1&-1\\-1&1\end{pmatrix}, \tag{GA.17}$$

a different slice and law. [ESTABLISHED]

Proof. In the coordinates $y=\mathsf{G}_{\mathrm{ref}}^{1/2}s$ the slice becomes $\operatorname{range}\widehat{Q}$ and its volume becomes Euclidean volume; the restriction of $\widehat{Q}$ there is positive definite, and the source $\mathsf{G}_{\mathrm{ref}}^{-1/2}\mathsf{h}$ lies in that range because it annihilates $\ker\widehat{Q}$. Diagonalizing and completing r scalar squares gives the first line of (GA.15); transforming the mean and covariance back gives $\Sigma_Q\mathsf{h}$ and Σ_Q, and translation adds z_0. The same one-dimensional integrations give the characteristic function, and in dimension zero the integral is over one point, so $0<Z_Q<\infty$ in all cases. Substituting (GA.16) into (GA.12) and using uniqueness gives $\Sigma'_Q=g\Sigma_Qg^\top$; g is an isometry between the two representative metrics, so it pushes their volume measures and leaves Z_Q unchanged. The counterexample is direct computation. □

Proposition GA.11 (Pushforwards of quotient Gaussian laws). (i) Every affine map $z\mapsto \mathsf{F}z+\mathfrak{b}$, including rectangular or rank-deficient F, pushes the law of Theorem GA.10 to the Gaussian with

$$\mu_{\mathrm{push}}=\mathsf{F}(z_0+\Sigma_Q\mathsf{h})+\mathfrak{b},\qquad \Sigma_{\mathrm{push}}=\mathsf{F}\Sigma_Q\mathsf{F}^\top. \tag{GA.18}$$

Two Gaussian measures on one finite ambient space, singular or not, agree exactly when their means and covariances agree, so target data $(Q_{\mathrm{c}},\mathsf{G}_{\mathrm{ref,c}},\mathsf{h}_{\mathrm{c}},z_{0,\mathrm{c}})$ with covariance Σ_{c} represent this push exactly when

$$\Sigma_{\mathrm{c}}=\mathsf{F}\Sigma_Q\mathsf{F}^\top,\qquad z_{0,\mathrm{c}}+\Sigma_{\mathrm{c}}\mathsf{h}_{\mathrm{c}}=\mathsf{F}(z_0+\Sigma_Q\mathsf{h})+\mathfrak{b}, \tag{GA.19}$$

as equalities of probability measures, supports included. (ii) A map $[z] \mapsto [\mathsf{F}z]$ between the energy quotients is well defined exactly when $\mathsf{F} \ker Q \subseteq \ker Q_{\mathrm{c}}$, that is,

$$Q_{\mathrm{c}}\mathsf{F}(I - \Sigma_Q Q) = 0. \tag{GA.20}$$

With $\Pi_{\mathrm{c}} = \Sigma_{\mathrm{c}} Q_{\mathrm{c}}$ and a declared quotient offset $\mathfrak{b}_0$, the target representative map $z_0 + s \mapsto z_{0,\mathrm{c}} + \Pi_{\mathrm{c}}(\mathsf{F}s + \mathfrak{b}_0)$ pushes the law to the target law exactly when

$$\Sigma_{\mathrm{c}} = \Pi_{\mathrm{c}}\mathsf{F}\Sigma_Q\mathsf{F}^{\top}\Pi_{\mathrm{c}}^{\top}, \qquad \Sigma_{\mathrm{c}}\mathsf{h}_{\mathrm{c}} = \Pi_{\mathrm{c}}(\mathsf{F}\Sigma_Q\mathsf{h} + \mathfrak{b}_0); \tag{GA.21}$$

choosing a map on one slice alone does not prove descent. (iii) For three scalar nodes with two unit edges, no anchors, zero source, and $\mathsf{G}_{\mathrm{ref}} = I_3$,

$$Q_{\mathrm{f}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad \Sigma_{\mathrm{f}} = \frac{1}{9}\begin{pmatrix} 5 & -1 & -4 \\ -1 & 2 & -1 \\ -4 & -1 & 5 \end{pmatrix}, \tag{GA.22}$$

and F selecting the first and third coordinates, with $\mathsf{G}_{\mathrm{ref,c}} = I_2$,

$$Q_{\mathrm{c}} = \Sigma_{\mathrm{c}} = \Pi_{\mathrm{c}} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \mathsf{F}\Sigma_{\mathrm{f}}\mathsf{F}^{\top} = \frac{1}{9}\begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}; \tag{GA.23}$$

the map descends and $\Pi_{\mathrm{c}}\mathsf{F}\Sigma_{\mathrm{f}}\mathsf{F}^{\top}\Pi_{\mathrm{c}}^{\top} = \Sigma_{\mathrm{c}}$, so the coarse edge of weight $1/2$ is an exact quotient push with zero sources and offsets, whereas the unprojected endpoint covariance differs from Σ_{c}. [ESTABLISHED]

Proof. (i) Substitute $\mathsf{F}^{\top}\xi$ in the characteristic function and multiply by $e^{i\xi^{\top}\mathfrak{b}}$. Equal means and covariances give equal characteristic functions, and the converse follows from finite first and second moments. (ii) Replacing z by $z + k$ with $k \in \ker Q$ shows necessity of $\mathsf{F} \ker Q \subseteq \ker Q_{\mathrm{c}}$, and the same calculation shows sufficiency; $I - \Sigma_Q Q$ projects onto $\ker Q$ by Proposition GA.9, which gives (GA.20). The representative map is affine, so (i) applied to it gives (GA.21), the target mean being $z_{0,\mathrm{c}} + \Sigma_{\mathrm{c}}\mathsf{h}_{\mathrm{c}}$. (iii) Multiplication gives $Q_{\mathrm{f}}\Sigma_{\mathrm{f}} = I_3 - \mathbf{1}\mathbf{1}^{\top}/3$ and the covariance equations; the fine common-translation kernel maps into the coarse one, and direct multiplication gives the projected covariance. Equivalently, the fine residuals $z_1 - z_2$ and $z_2 - z_3$ are independent standard normals, and their sum has variance two. □

A target metric or projection that varies with a model parameter does not by itself define one parameter-independent statistical channel, so Fisher contraction for a family requires that admission separately.

Theorem GA.12 (Finite variable-support Gaussian classification). Fix finite ambient dimensions, finitely many declared affine or intrinsic quotient maps, and finite semialgebraic input and target predicates over effectively represented real algebraic coefficients, with rational matrix expressions on explicit nonzero-denominator domains. At each object require $Q \succeq 0$, $\mathsf{G}_{\mathrm{ref}} \succ 0$, and

$\mathsf{h} \in \operatorname{range} Q$, with reference and origin as in Definition GA.8; ranks and supports may vary with the parameters. Then an algorithm constructs the exact good and failure loci for quotient-Gaussian representability,

$$\begin{aligned} \mathrm{Good}_{\mathrm{quot}}(x) &\Longleftrightarrow \mathrm{Adm}(x) \wedge \exists y \big[\mathrm{Rep}_{\mathrm{quot}}(x, y)\big], \\ \mathrm{Fail}_{\mathrm{quot}}(x) &= \mathrm{Adm}(x) \wedge \neg \mathrm{Good}_{\mathrm{quot}}(x), \end{aligned} \tag{GA.24}$$

and decides universal closure, separate nonemptiness, and any finite simultaneous diagram with its supplied polynomial structural compatibilities; for affine maps it uses (GA.19), and for intrinsic maps both (GA.20) and (GA.21). The asserted equivalence is equality of the normalized measures on their representative spaces. The raw finite measures are carried by the analytic normalizer at each object: with

$$\begin{aligned} \mathsf{A}_i^{\mathrm{mass}} &= \frac{r_i}{2} \log(2\pi) - \frac{1}{2} \log \det{}_+(\widehat{Q}_i) + \frac{1}{2} \mathsf{h}_i^\top \Sigma_{Q_i} \mathsf{h}_i, \\ \mathfrak{c}_{\mathrm{c}} &= \mathfrak{c}_{\mathrm{f}} + \mathsf{A}_{\mathrm{c}}^{\mathrm{mass}} - \mathsf{A}_{\mathrm{f}}^{\mathrm{mass}}, \end{aligned} \tag{GA.25}$$

a free target scalar gives $Z_{\mathrm{c}} = Z_{\mathrm{f}}$, and equality of the normalized laws then gives equality of the pushed raw finite measures. An observed experiment is included by keeping its observation coordinates in the finite joint and requiring the representative map to leave them unchanged, which preserves its observation marginal, singular or not. [ESTABLISHED]

Proof. Introduce a symmetric covariance variable at each object and impose (GA.12), $Qv = \mathsf{h}$, and the PSD and SPD admissions; uniqueness in Proposition GA.9 ensures that an existential covariance cannot select a different law. Denominators are handled as in Lemma GA.3, so the map equations are finite polynomial relations, and by Proposition GA.11 they are necessary and sufficient for the normalized pushforward, descent included when an intrinsic map is requested. A prescribed rank $r > 0$ is the vanishing of all $(r+1)$-minors and positivity of the sum of squares of the r-minors, rank zero is the vanishing of all entries, and the finitely many ranks form a finite disjunction; $\operatorname{range} U \subseteq \operatorname{range} U'$ is the existential equation $U = U'C$, and affine support relations add membership of the displacement. Proposition GA.2 then produces (GA.24) and decides the closed sentences. For a diagram, one target tuple and its unique covariance at each object carry all arrow equations simultaneously, with descent and the declared identities of the induced quotient maps for quotient arrows; equality modulo a target null direction is encoded as such, and composable representative maps compose their moment transformations by affine substitution. No nonsingular inverse or common support is assumed. The raw-scalar choice gives $Z_{\mathrm{c}} = Z_{\mathrm{f}}$ by (GA.15), and a common positive mass factor cancels from a normalized law, so raw equality is a separate requirement that this choice meets. □

A preferred positive pair factorization is an extra target condition, expressible by (GA.9) when its hypotheses hold, or through a supplied marked inventory; variable support does not relax positive definiteness of edge weights, erase holonomy marks, or enlarge a prescribed target domain. Logarithms, positive pseudodeterminants, and rank-dependent normalizers enter the polynomial decision only through such a reduction.

Proposition GA.13 (Rank change obstructs statistical regularity). Finite semialgebraic feasibility across ranks does not imply total-variation continuity, differentiability in quadratic mean, or a smooth positive Fisher geometry, even for a smooth PSD coefficient path whose quotient law is normalized at every parameter. Even the smooth covariance path $\mathrm{diag}(1, t^2)$ has total-variation distance one from its line-supported law at $t = 0$, although these laws converge weakly. [ESTABLISHED]

Proof. Take

$$Q(t) = \mathrm{diag}(1, t^2), \qquad \mathsf{G}_{\mathrm{ref}} = I_2, \qquad \mathsf{h} = z_0 = 0. \tag{GA.26}$$

For $t \neq 0$ the covariance is $\mathrm{diag}(1, t^{-2})$ and the support is $\mathbb{R}^2$; at zero the covariance is $\mathrm{diag}(1, 0)$ and the support is the first coordinate axis, which has probability zero for every $t \neq 0$ and one at zero, so with $\|\mathbb{P} - \mathbb{P}'\|_{\mathrm{TV}} = \sup_B |\mathbb{P}(B) - \mathbb{P}'(B)|$ the distance is exactly one. For every finite R_0,

$$\mathbb{P}_t(|X_2| \leq R_0) = \mathbb{P}(|\varsigma| \leq R_0|t|) \longrightarrow 0, \qquad \varsigma \sim \mathcal{N}(0, 1), \tag{GA.27}$$

so there is no weak probability limit as $t \to 0$: the bounded continuous positive function $e^{-X_2^2}$ has expectation $|t|/\sqrt{t^2 + 2} \to 0$, whereas its integral against any probability measure on $\mathbb{R}^2$ is positive. On $t > 0$ the second-coordinate log density has derivative $1/t - tX_2^2$, with mean zero and variance $2/t^2$, the Fisher information. Two-dimensional Lebesgue measure plus the one-dimensional measure on the axis dominates all these laws. Differentiability in quadratic mean at zero would require a square-integrable score s_0 with $\|\sqrt{p_t} - \sqrt{p_0} - \frac{t}{2}\mathrm{s}_0\sqrt{p_0}\|_2 = o(|t|)$; the triangle inequality would give $\|\sqrt{p_t} - \sqrt{p_0}\|_2 \to 0$, and Cauchy–Schwarz applied to $(\sqrt{p_t} - \sqrt{p_0})(\sqrt{p_t} + \sqrt{p_0})$ would give $\|\mathbb{P}_t - \mathbb{P}_0\|_{\mathrm{TV}} \leq \|\sqrt{p_t} - \sqrt{p_0}\|_2 \to 0$, contradicting the distance-one calculation. For the covariance path $\mathrm{diag}(1, t^2)$ the same support argument gives distance one, and the coupling $(\varsigma_1, t\varsigma_2)$ with dominated convergence gives weak convergence; its quotient precisions satisfy (GA.12), so weak convergence alone does not supply a regular statistical geometry. □

Open Problem GA.14 (Kron classification beyond finite encodings). Finite algebraically encoded full-SPD problems are decided by Theorem GA.4, and declared metric quotient laws with variable rank and support by Theorem GA.12; correlated, nonconvex, variable-anchor, and branching nonflat domains are included when their constraints fit the finite input, and the scalar formulas are exact analytic outputs outside the polynomial formulas. Universal continuity and equivariance of selectors (Proposition GA.6), smooth passage between support strata (Proposition GA.13), and a universal decision procedure for enumerated inputs (Proposition GA.7) are refuted. Unproved is any classification, closed form or algorithmic, of families outside finite encodings, such as nonalgebraic domains, changing-dimension limits, and alternative completions of the null coordinates, and any construction with a global regularity target, such as a smooth, equivariant, or functorial structural selection. For a stated family, a declaration of its domain and compatibility conditions together with a proof of the requested invariance, limit, or global construction would settle it, as would a reduction of the family to a finite encoding covered by the two theorems.

Appendix GB

Notation, Open Problems, and Numerical Index

GB.1 Notation

This supplement uses the symbols of the main manuscript (Table D.1) with their meanings there. Table GB.1 lists the symbols that recur across the chapters of this supplement and the local symbols of Chapter G11; any other symbol used inside one chapter is declared at its opening. The word pullback means restriction of a tensor along an injective map of parameters, as in Corollary G2.9.

Table GB.1: Notation of Supplement G.

Symbol	Meaning	Introduced
K, d_m; $n = NK$	dimensions of the belief and model sample fibers, not assumed equal; dimension of the stacked belief latent	chapter G0 and section G3.1
$\Gamma_b = \rho_b(G)$, $\Gamma_m = \rho_m(G)$	represented frame groups, with the specialization $\Gamma_b = \mathrm{GL}^+(K)$ where it is named	hypothesis G2.1
ν, T, λ, A(λ), $\boldsymbol{\mu}_\lambda$	reference measure, statistic, natural parameter, log partition, and mean parameter of a canonical family	definition G1.1
$\mathcal{W}$	closed convex operator cone preserved by the algebraic maps under study	definition G1.6
(h, Λ); $\mu = \Lambda^{-1}\mathsf{h}$, $\Sigma = \Lambda^{-1}$; A(h, Λ)	information vector and precision of a Gaussian law; its mean and covariance; its log normalizer	section G2.2
F_i^b, F_i^m, B$_i$, C$_i^b$, C$_i^m$	gains, model-to-belief bridge, and observation matrices of the directed linear-Gaussian forest	definition G2.13
ξ_i^b, ξ_i^m, ξ_i^o; Σ_i^b, Σ_i^m, Σ_i^o; $\mathrm{par}(i)$	offsets and receiving covariances of the belief, model, and observation factors; the parent of a nonroot agent	definition G2.13
Ω_i	transition $F_i\Theta_{i\,\mathrm{par}(i)}$ of a nonroot agent	proposition G2.16
$\rho_i^b = \rho_b(g_i)$, $\rho_i^m = \rho_m(g_i)$	represented passive re-expressions of agent i, the maps R_i^b and R_i^m of the main manuscript	proposition G2.17
$\mathcal{I}(V)$; A_i, $W_{ij} = W_{ji}$, L	interaction family on the agents; self term, edge weight, and matrix-weighted Laplacian part, with $\Lambda_{ii} = A_i + \sum_{j\neq i} W_{ij}$ and $\Lambda_{ij} = -W_{ij}$	definition G3.1
O_i	observation precision added to the self term of agent i	proposition G3.8
$z_i = \rho_b(U_i)^{-1}\mu_i$	mean of agent i in link-trivializing coordinates	proposition G2.12

continued on the next page

Symbol	Meaning	Introduced
$\mathbf{P} = \mathrm{diag}(\mathbf{P}_1, \dots, \mathbf{P}_N)$, $\mathbf{P}_i \in \Gamma_b$	block reframing from new to old coordinates, under which a precision becomes $\mathbf{P}^\top \Lambda \mathbf{P}$	definition G5.4
S, n_c; n_I	aggregation matrix of a hard partition into n_c clusters; size of the cluster I	proposition G7.3
$P_\Lambda = (S^\top \Lambda S)^{-1} S^\top \Lambda$	coarse coordinate map carrying the Galerkin law; SP_Λ is the Λ-orthogonal projector onto cluster-constant configurations	proposition G7.5
$\widehat{\Lambda}$, $\widehat{\mathsf{h}}$	precision and information vector of the joint law of (o, Y) given the structural data	proposition G4.6
E	energy of a law proportional to $e^{-\mathsf{E}}$	definition G7.1
$\mathcal{W}_H$, M_0	congruence-diagonal cone with frame H, $HH^\top = M_0$; an SPD order unit of a cone	definition G7.9
$\mathbf{\Xi}_\ell$, $\mathcal{A}_S$, ζ_ℓ	operator parameters $((A_i^{(\ell)}), (W_{ij}^{(\ell)}))$ at depth ℓ, the unrescaled aggregation map, and the block rescaling	definition G9.1
L_0^2, b; $\mathscr{I}_b$, $\mathscr{L}_b$	centered square-integrable scores of a law, block size; score replication lift and score operator of a block of b copies	definition G6.1 and theorem G6.4
He_k, $\widehat{\mathrm{He}}_k = \mathrm{He}_k/\sqrt{k!}$	probabilists' Hermite polynomial and the normalized Hermite basis	theorem G6.8
$\mathbf{L}^{(2)}(\mathbf{B}; \mathbf{T})$	second derivative at $t = 0$ of $t \mapsto e^{\mathbf{B}+t\mathbf{T}}$	section G11.1
$\mathbf{A}_e$	matrix in the gradient of the law-level divergence in generator coordinates	proposition G11.1
E_e; $\mathsf{E}(z; \mathring{\omega})$	energy of the comparison record of the copy e at its observation o_e; matter energy with latent links	eqs. (G11.4) and (G11.11)
$\mathbf{l}$	expected energy of the comparison record as a function of the transport matrix	proposition G11.3
$\mathbf{G}^{\pm}$	coordinate vectors of $\mathsf{G}^{\mathrm{fib}}$ and of its inverse in $\mathbb{R}^K \otimes \mathbb{R}^K$	proposition G11.4
$\mathring{\zeta}_a$; $\mathring{\Theta}_p^{<a}$, $\mathring{\Theta}_p^{>a}$	centered part of the generator of the a-th copy of a walk; partial reference holonomies at the generator means	proposition G11.4
$\Lambda(\mathring{\omega}) = A_\oplus + L(\mathring{\omega})$, $A_\oplus$	precision of undirected Gaussian matter with latent links; block-diagonal self terms	section G11.4
$\mathring{\Theta}_{I \leftarrow i}$, $\mathring{\Theta}_\oplus$; Λ^{root}, z^{root}	tree transport in reference coordinates and its block-diagonal assembly; precision and coordinates transported to the roots of a forest	proposition G11.5
$W_e^\sharp$, $w^\sharp$	series combination $W_e(A_i + W_e)^{-1}A_i$ of an edge weight with a self term, and its isotropic scalar	proposition G11.5
Λ^{diag}, Λ^{hop}; a_i^{diag}, w^{hop}; $\mathfrak{o}$	diagonal and hopping parts of the matter precision; isotropic diagonal coefficient and bound on the hopping ratios; remainder of the triangle expansion	proposition G11.8
$\mathring{n}$, $\mathring{n}_1$	learning-episode index and last episode, decorated because $n = NK$ in this supplement	section G11.5

GB.2 Open problems

Table GB.2 lists the open problems of this supplement, each with the statement at which it arises and a condition that would settle it. The first three arise across chapters and are stated here. The Gaussian chapters also bear on open problems of the main manuscript, among them the choice of a nondegenerate recognition structure (Open Problem 5.14) and of a natural partition (Open Problem 10.20). The Gaussian coarse families also meet the established obstruction to a continuous extension of balanced cut links (Proposition 11.28).

Open Problem GB.1 (Bayesian renormalization and Gaussian aggregation). Aggregation of interaction parameters is a positive semigroup that sums cut weights and self terms (Proposition G9.2), and a rescaled step on the interior maps normalized Gaussian laws to normalized Gaussian laws (Proposition G9.3). In its late-data regime Bayesian renormalization works with a Gaussian posterior whose covariance is the scale parameter times the inverse Fisher information (Section 16.5). Neither this supplement nor the main manuscript relates the two constructions, and in particular neither shows that they run in opposite directions; similar Gaussian formulas do not supply such a relation. The problem is settled by a common state space carrying both constructions, an explicit map between the Bayesian scale parameter and a declared block ratio with its rescaling, the Fisher tensors transported along that map, and a proof that a transported functional is monotone along both constructions, or by a Gaussian hierarchy in a stated class for which no such functional exists.

Open Problem GB.2 (Transfer of a model-space Fisher scale). At a fixed precision Λ the Fisher metric of the Gaussian mean family is Λ (Proposition G5.1), and restriction and marginalization give the two metrics compared in Corollary G2.9. The information scale of Bayesian renormalization is built from a Fisher metric on the parameters of a generative model and is written with its inverse (Section 16.5). Whether that construction applies to Λ read as the Fisher metric of a recognition mean is not settled by the common name: a coarse recognition mean is not a parameter of the generative law, and the construction does not treat a singular metric. The problem is settled by a declaration that makes the coarse recognition mean a parameter of the generative law fitted against the evidence bound, a proof that the Fisher metric in that parameter is nondegenerate on the declared family, and then the transferred construction with its proof, or a declared family in which that metric is singular.

Open Problem GB.3 (Stochastic inversion of aggregation). Hard aggregation annihilates every edge internal to a cluster (Proposition G7.3), so the aggregation map $\mathcal{A}_S$ on operator parameters is not injective (Proposition G9.2). A refinement kernel that assigns to each coarse parameter tuple a law on its preimage under $\mathcal{A}_S$ assigns the same law to every fine tuple with that image, so it redraws fine parameters compatible with the coarse ones, and on a preimage with more than one point it returns the realized fine tuple almost surely for at most one of them. What is open is a stochastic inverse of a hierarchy of aggregations. It is settled by priors for the internal edge weights discarded at every step, refinement kernels that are right inverses of the aggregation maps and

compose across depths, the chart data that identify the levels, and a projective-limit theorem for the refined laws, or by a proof that no compatible family exists for a stated class of hierarchies.

Table GB.2: Open problems of Supplement G.

Item	Locator	Settling condition
Bayesian renormalization and Gaussian aggregation	open problem GB.1	A common state space, a map between the Bayesian scale parameter and a declared block ratio with its rescaling, transported Fisher tensors, and a transported functional proved monotone along both constructions; or a Gaussian hierarchy in a stated class with no such functional.
Transfer of a model-space Fisher scale	open problem GB.2	A declaration making the coarse recognition mean a parameter of the generative law fitted against the evidence bound, nondegeneracy of the Fisher metric in that parameter, and the transferred construction with proof; or a declared family in which that metric is singular.
Stochastic inversion of aggregation	open problem GB.3	Priors for the discarded internal edge weights, refinement kernels that are right inverses of the aggregation maps and compose across depths, chart data, and a projective-limit theorem; or a proof that no compatible family exists for a stated class of hierarchies.
Scaling of a retained mass sector	open problem G1.14	For a given scale map, an explicit induced law on the mass coordinate together with a proof, under stated hypotheses on the map, that its orbit stays in the natural domain or reaches its boundary.
Commutation of the information projection	open problem G1.18	An intertwining theorem with hypotheses on a gauge action or a coarse parameter map under which the affine information projection commutes with it, or a counterexample within a regular minimal family.
Monotonicity along aggregation with varying precision	open problem G9.4	A Lyapunov functional with a proof of its decrease for families on which $\mathrm{range}(\Lambda S)$ varies; a declared quotient, pin, or mass repair of the boundary member of the bi-additive family, with its own reference measure, normalizer, and pushforward identity; and a projective-limit theorem for composed kernels at unbounded depth.
Kron classification beyond finite encodings	open problem GA.14	For a stated family outside finite semialgebraic or quotient encodings, a declaration of its domain and compatibility conditions with a proof of the requested invariance, limit, or global construction; or a reduction of the family to a finite encoding covered by Theorems GA.4 and GA.12.
Robustness of the star schedule	open problem G10.6	A convergence theorem, with a rate, for a declared class of delayed, noisy, asynchronous, or inexact updates of the anchored star, or a counterexample within such a class.

continued on the next page

Item	Locator	Settling condition
Joint normalization of unanchored matter with latent links	open problem G11.10	At a stated $K \geq 2$, for connected multigraphs of cycle rank at least two with positive-definite weights, zero self terms, zero information vector, and product Gaussian base laws of the generators with positive-definite covariances, a proof that $\int \nu(d\mathring{\omega}) \det \Lambda(\mathring{\omega})^{-1/2} < \infty$ for every such multigraph and law; or a multigraph and law in the class for which the integral diverges.
Attainment of the chart infimum without a link prior	open problem G11.11	For $K \geq 4$, $W_e, \Sigma_j \succ 0$, and data with $\operatorname{Tr} \mathbf{M}^\star < 0$, a proof that no chart link having a negative eigenvalue with a Jordan block of size at least two minimizes the expected record energy over the chart links; or data in the class and such a chart link that minimizes it.

GB.3 Numerical index

Table GB.3 indexes the computations recorded for this supplement, in the columns fixed for every numerical index. No statement of this supplement has the status NUMERICAL. Each row is a regression check of established statements, which rest on their proofs; the checks corroborate the declared finite instances and prove nothing beyond them.

Table GB.3: Numerical index of Supplement G.

Claim	Check	What is tested	Control	Seed
corollary G4.11, (G4.24)	`CHK-CG-FACTOR-GAP-STRESS-3138`	The factorization gap computed in binary64 through canonical correlations, against an evaluation of the same stored inputs at 100 decimal digits, on 3,138 deterministic cases of dimensions 2 to 16 with nominal condition numbers from 1 to 10^{14}.	18 designated inputs evaluated exactly at 100 decimal digits.	20260803
proposition G10.4 and corollary G10.2, (G10.9)	`CHK-OBS-STAR-FOLD-NEW-PROTOCOL`	Anchored-star and flat reciprocal-pair instances: the apex precision-addition identity and the contrast of the two forms on transport-consistent configurations.	Flat unanchored pairs as comparison instances.	20260728
propositions G10.3 and G10.8 and corollary G10.7	`CHK-OBS-HOLONOMY-DET-KERNEL`, `CHK-OBS-NORMALIZER-WITNESS`	Reciprocal pairs with explicit transports and covariances: the determinant identity, the prescribed dimensions of the kernel, and the normalizer witness.	A normalized anchored instance.	20260728

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